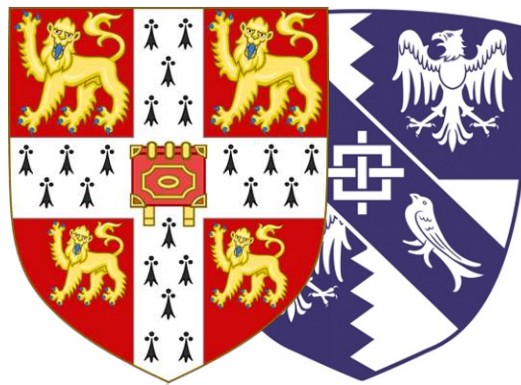

EVALUATION AND ADAPTATION TO FREQUENCY-TO-PLACE FACTORS IN COCHLEAR IMPLANT USERS



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Doctor of Philosophy

Declaration

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text.

Louis Villejoubert
Cambridge, April 2026

Abstract

Frequency-to-place factors - frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) - are recognised as limiting factors in cochlear implant (CI) outcomes, yet their perceptual consequences at individual electrodes remain poorly characterised across perceptual domains and stages of adaptation. This thesis presents four studies investigating how these factors shape perception in vocoder simulation and in three CI populations: experienced users implanted after language acquisition (post-lingually implanted late cochlear implant user (Post-LCI)), experienced users implanted before language acquisition (pre-lingually implanted early cochlear implant user (Pre-ECI)), and a newly implanted user tested longitudinally during adaptation.

In a noise-vocoder study with 23 normal-hearing listeners, greater FTPM degraded perceived sound quality, particularly at low-frequency electrodes, and inter-electrode variability independently reduced quality, establishing a causal baseline for electrode-level investigation.

An original vowel identification and confidence task administered to 10 Pre-ECI and 14 Post-LCI users alongside 10 normal-hearing listeners showed that Post-LCI users preferred lower F2 frequencies whilst Pre-ECI users weighted F1 more heavily. None of the three factors predicted residual deviations from normal hearing at the electrode level, suggesting compensation through central adaptation.

An original chord paired-comparison paradigm administered to the same 24 CI users showed that FTPM-induced interval distortion affected chord preference in Post-LCI users, and that narrower CS and reduced FTPM predicted highest electrode-level preference, providing the first evidence that these factors shape music quality perception at the electrode level dependent on implantation history.

Longitudinal application of both tasks across four sessions spanning four months post-activation revealed a two-phase adaptation: an early phase in which NR facilitates preference formation, giving way to a late phase dominated by FTPM, with a transition around three to four weeks post-activation.

Taken together, these findings characterise FTPM as real but centrally modulated, NR as an early scaffold rapidly accommodated, and CS as the most persistent predictor. The paradigms provide a foundation for individually targeted frequency remapping.

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Abbreviations

AIC	Akaike information criterion
CF	Characteristic frequency
CI	Cochlear Implant
CS	Current Spread
CT	Computed Tomography
ECAP	Electrically Evoked Compound Action Potential
EEG	Electroencephalography
ENI	Electrode-Neuron Interface
F1	Formant 1
F2	Formant 2
F3	Formant 3
F4	Formant 4
F5	Formant 5
FIF	Frequency importance function
FTPM	Frequency-to-Place Mismatch
GLLM	Generalized linear mixed model
HA	Hearing Aid
LMM	Linear Mixed Model
MCL	Most Comfortable Level
MMN	Mismatch negativity
MUSHRA	MUltiple Stimuli with Hidden Reference and Anchor
NH	Normal Hearing
NR	Neural Responsiveness
OLS	Ordinary least squares

PCA Principal Component Analysis

PECAP Panoramic Electrically Evoked Compound Action Potential

Post-LCI post-lingually implanted late cochlear implant user

Pre-ECI pre-lingually implanted early cochlear implant user

Preprint – not peer-reviewed

Chapter 1

General introduction

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1.1 Cochlear Implant Fundamentals

1.1.1 History

The first reports of external electrical stimulation of the auditory system date to the late 18th century (Figure 1.1 provides a timeline of major milestones). Alessandro Volta's self-experiments in 1800, applying electrical current to his own ears, produced auditory sensations that he described as a "kind of crackling" and the sound of "boiling thick soup" - but the experiments were followed by headaches and convulsions that discouraged him from further investigation (Volta, 1800).

The modern era of cochlear implant (CI) development really began with Djourno and Eyriès in 1957. During a cholesteatoma surgery, they placed an electrode on the auditory nerve of a deaf patient; when electrical stimulation was applied, the patient reported hearing sounds (Djourno and Eyriès, 1957). In the 1960s, William House built upon this discovery to develop the first CI prototype. Using a single-channel device placed inside the cochlea, current was modulated according to the temporal envelope of speech, with the goal of supplementing lipreading (House and Urban, 1973). In the following decade, Graeme Clark in Melbourne pursued multi-channel stimulation (*i.e.*, with more than one electrode), each channel conveying temporal information derived from different frequency bands (Clark, 2006), while the San Francisco group led by Michael Merzenich and Robin Michelson independently developed multi-electrode approaches that would later form the basis for Advanced Bionics devices (Merzenich et al., 1973; Loeb et al., 1983).

This "single-channel versus multi-channel" debate dominated the 1970s and was ultimately resolved in favour of multi-channel devices. Indeed, they exploited the cochlea's tonotopic organisation and proved to be capable of restoring speech understanding without visual cues, whereas single-channel devices could not (Bilger et al., 1977; Tong et al., 1982).

In the following decades, numerous innovations were introduced: electrode number was increased, wearable external processors were developed, new coding strategies emerged (developed in Section 1.2.1), signal processing improved, surgical techniques were refined, electric-acoustic stimulation (EAS) was introduced, bilateral implantation became possible and features such as bidirectional telemetry enabled objective measurements of neural responses.

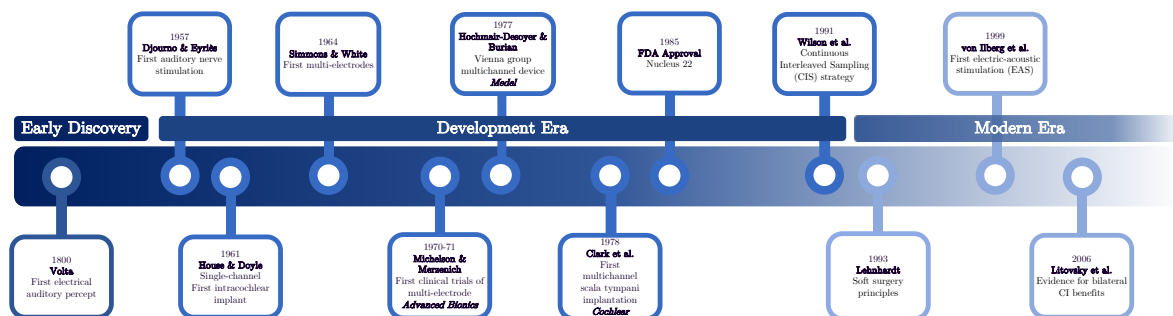


Figure 1.1: Historical milestones and major advances in the cochlear implant development.

These advances led to improvements in CI outcomes, making them one of the most successful medical devices to date. CIs address a historical unmet need for individuals with severe-to-profound sensorineural hearing loss. These individuals cannot benefit adequately from conventional hearing aids, which rely on residual hair cell function to transduce amplified sound into neural signals. The CI bypasses this limitation entirely by delivering electrical pulses directly to the auditory nerve, restoring hearing even in the complete absence of functional hair cells. This approach has proven transformative: over one million devices have been implanted worldwide as of 2024 (National Institute on Deafness and Other Communication Disorders, 2024; Zeng et al., 2008), and the CI is widely regarded as the most successful neural prosthesis in clinical use (Wilson and Dorman, 2008).

1.1.2 Basic Principles

To achieve the complete translation from acoustic signal to auditory nerve stimulation, the CI architecture is nowadays divided into two main parts: the processor (*i.e.*, the external part) and the implant (*i.e.*, the internal part).

The external part includes the following components:

- **Microphone(s):** One or more microphones (typically two) capture the acoustic signal from the environment (shown as part of the processor in **1** in Figure 1.2). Multiple directionality modes are typically available, and the active strategy can vary depending on the CI manufacturer, clinician settings, or patient preference.
- **Sound processor:** The processor serves as the “brain” of the system, converting incoming sound into electrical stimulation patterns through a series of signal processing steps. The acoustic signal is first split into a number of frequency bands corresponding to the number of active electrodes. The envelope of each band is then extracted via rectification and low-pass filtering, and used to modulate a series of charge-balanced biphasic current pulses delivered to the corresponding electrode. Additional functionalities may include volume controls, program selection, and accessory connectivity. The processor is powered by rechargeable batteries and can be worn behind the ear (shown in **1** in Figure 1.2) - similar in appearance to hearing aids - or, in some systems, attached directly to the head as an “off-the-ear” unit.
- **Transmitter coil (antenna):** The transmitter coil is held against the skin by magnetic attraction to the internal receiver (indicated as **2** in Figure 1.2). It communicates with the internal device transcutaneously via inductive coupling - a technique that allows power and data to be transferred across intact skin through a shared magnetic field, avoiding any physical penetration of the skin. Both power (to run the internal components) and data (encoding the stimulation parameters) are transmitted via a radio-frequency carrier signal. The coil also supports bidirectional telemetry, enabling the internal device to send information back to the processor, such as electrode impedances or neural response measurements.

The internal part includes the following components:

- **Receiver-stimulator:** Placed beneath the skin and secured to the skull, the receiver-stimulator demodulates the transmitted signal to recover power and data.

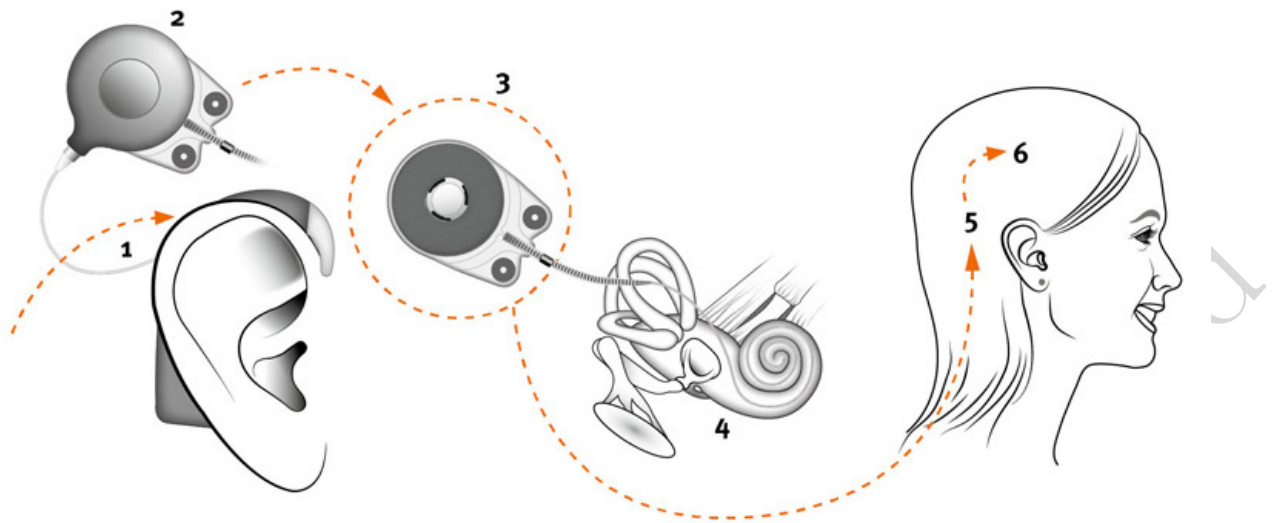


Figure 1.2: **Schematic overview of a cochlear implant system.** The numbered labels indicate the signal pathway from sound to neural stimulation: (1) **microphone(s)** and **sound processor** capture and encode the acoustic signal; (2) **transmitter coil (antenna)** sends power and data transcutaneously; (3) internal **receiver-stimulator** decodes the signal and generates stimulation pulses; (4) **electrode array** delivers electrical stimulation within the cochlea; (5) auditory nerve fibres are activated; (6) neural signal propagates to the auditory cortex. Reproduced from *Oticon Medical*.

It contains electronic circuitry to generate stimulation pulses according to the incoming signal, backup capacitors to maintain operation during brief signal interruptions, and telemetry circuitry for bidirectional communication with the external processor (indicated as **3** in Figure 1.2).

- **Electrode array:** The electrode array is surgically inserted into the scala tympani of the cochlea. It contains between 12 and 22 electrode contacts distributed along its length. The design exploits the cochlea's tonotopic organisation: electrodes positioned near the base stimulate neural populations tuned to high frequencies, while those closer to the apex target low-frequency regions. Each contact can be independently activated, allowing the processor to deliver frequency-specific information across the array (indicated as **4** in Figure 1.2). The resulting electrical stimulation activates auditory nerve fibres (**5** in Figure 1.2), and the neural signal propagates to the auditory cortex (**6**).
- **Ground/reference electrode:** Extracochlear electrode, often located in the temporalis muscle or on the receiver-stimulator case, serves as the return path for electrical current.

1.2 Clinical Outcomes and Limitations

1.2.1 Technological Progress and Clinical Gains

Stimulation strategy. Several key technological changes have shaped the landscape of CI performance since the device’s introduction in the late 1970s. Among the earliest was the development of multichannel electrode arrays, with the first multichannel implantation performed by Clark and colleagues in 1978 (Tong et al., 1979) and subsequent refinements by the Vienna group in the late 1970s (Hochmair-Desoyer and Hochmair, 1980). However, simultaneous stimulation of adjacent electrodes introduced channel interactions that degraded spectral resolution, a limitation that constrained the practical benefit of multichannel stimulation. The pivotal breakthrough came with the development of Continuous Interleaved Sampling (CIS) in 1991 by Wilson and colleagues, which addressed this by delivering non-simultaneous pulses across electrodes - substantially reducing channel interaction while preserving temporal envelope cues (Wilson et al., 1991). CIS became the foundation upon which all modern processing strategies were built, and its core principle of sequential, non-overlapping stimulation remains the basis of current coding approaches.

Surgical technics. Another major breakthrough was the refinement of surgical techniques and electrode array design toward minimising intracochlear trauma. The introduction of “soft surgery” principles in 1993 (Lehnhardt, 1993), combined with the development of thinner, more flexible electrode arrays (Dhanasingh and Jolly, 2017), substantially improved the preservation of residual acoustic hearing following implantation. This development was consequential far beyond surgical safety: by demonstrating that cochlear structures could be preserved during electrode insertion, it directly enabled the relaxation of candidacy criteria to include individuals with substantial residual hearing who would previously have been excluded. As a result, the audiological profile of CI candidates has broadened considerably over decades, encompassing individuals with far greater residual hearing than early eligibility criteria would have permitted. Critically, hearing preservation also opened the way to EAS, in which low-frequency acoustic hearing is retained alongside high-frequency electrical stimulation in the same ear (von Ilberg et al., 1999), and to bimodal stimulation (CI in one ear, hearing aid in the other) - both providing access to pitch and fine structure cues that purely electric stimulation cannot reliably deliver.

Candidacy criteria. These technological advances have been accompanied by a gradual evolution in clinical guidelines. Candidacy criteria, initially restricted to profound bilateral hearing loss, have progressively expanded to encompass single-sided deafness and, importantly, bilateral implantation. The latter has been shown to confer significant benefits in sound localisation and speech perception in noise through binaural redundancy (Litovsky et al., 2006).

CI benefits. The CI is nowadays considered as the standard of care for rehabilitation of severe-to-profound sensorineural hearing loss. On average, word perception improves from approximately 8% preoperatively to 54% postoperatively, and 82% of postlingually

deafened adults achieve speech perception gains of 15 percentage points or more (Boisvert et al., 2020b). Beyond speech perception, meta-analyses have demonstrated positive effects of implantation on quality of life (McRackan et al., 2018), communication abilities (Hirschfelder et al., 2008), social participation (Mosnier et al., 2015), and emotional well-being (Olze et al., 2011).

1.2.2 Current Limitations

The Outcome Plateau Despite these benefits, a performance plateau has been observed over the last decades. A systematic review examining outcomes from 1983 to 2019 found no significant association between year of implantation and 12-month postoperative sentence recognition scores (Brown et al., 2025). This plateau has been attributed to the absence of fundamental changes in coding strategies: despite numerous refinements, all current approaches remain based on the temporal envelope extraction principles established with CIS, and no modification to signal processing has produced a step change in outcomes (Carlyon and Goehring, 2021).

Moreover, this plateau appears paradoxical when considered alongside the expansion of candidacy criteria. Over the past decades, indications have broadened to include patients with greater residual hearing, shorter durations of deafness, and higher preoperative speech recognition (Gifford et al., 2010; Carlson et al., 2018). Crucially, the shift toward hearing preservation surgery and less traumatic electrode insertion was itself premised on the expectation that maintaining intracochlear structures would yield a healthier electrode-neuron interface (ENI) and thereby improve stimulation efficacy. Yet despite both the inclusion of better-performing candidates and advances aimed at preserving the neural substrate, mean outcomes have not improved correspondingly. This suggests that the quality of the ENI - shaped not only by surgical trauma but by pre-existing neural degeneration and individual variability in spiral ganglion survival - may represent a more fundamental bottleneck than device technology or patient selection alone.

Interindividual Variability The most striking feature of CI outcomes is their enormous variability. Studies report that 10-50% of adult users are still classified as “poor performers” (Moberly et al., 2016), and this variability persists across device generations, manufacturers, and coding strategies. Beyond audiological variables, socio-economic status, home language environment, educational level, and access to rehabilitation services have all been shown to contribute to outcome variability, particularly in children (Niparko et al., 2010; Geers et al., 2017). Understanding which factors drive this variability - and whether any of them are clinically modifiable - is a central challenge in the field.

1.2.3 Individual Factors

Substantial research effort has been directed at identifying the sources of this variability, yet even the most comprehensive predictive models leave the majority of outcome variance unaccounted for.

The largest demographic models, including a nine-factor general linear model fitted in 2,251 recipients across 15 international centres (Lazard et al., 2012) and a 21-variable

predictive model in 2,735 recipients (Goudey et al., 2021), consistently explain no more than 22% of speech recognition variance, regardless of sample size.

Preoperative factors - age at onset of hearing loss, aetiology, duration of moderate and severe-to-profound hearing loss, and bilateral hearing aid use - account for the largest share of this explained variance, while peri-operative factors such as surgical approach and insertion angle fail to reach significance in large multicentre datasets, likely because crude imaging metrics underestimate their true contribution (Lazard et al., 2012).

When electrode position is measured precisely from postoperative computed tomography imaging, the peri-operative contribution rises substantially: in a single-centre simultaneous model ($n = 118$), insertion depth and scalar translocation explained 20% of sentence recognition variance independently of patient demographics, raising total explained variance to 60% (James et al., 2019).

A further step change is achieved by replacing indirect demographic proxies with direct electrophysiological measures of cochlear neural substrate: preoperative electrocochleography total response (ECochG-TR) combined with cognitive status explained 65% of speech-in-noise variance at three months post-activation (Harrison et al., 2022), outperforming all demographic models despite a substantially smaller sample ($n = 35$).

One factor systematically absent from all these models is the degree of frequency-to-place mismatch (FTPM) between the CI frequency allocation and the patient individual cochlear tonotopy, which can exceed one octave at apical electrodes (Landsberger et al., 2015) and may impose a perceptual distortion that limits adaptation even after extended device experience (Canfarotta et al., 2020b; Mertens et al., 2022).

Incorporating ECochG-derived FTPM alongside cochlear health in a simultaneous model explained 62% of word recognition variance (Walia et al., 2025), with FTPM contributing uniquely beyond the neural health predictor alone, suggesting it is an independent determinant of speech perception outcomes whose influence on auditory adaptation and neural representation is the central concern of the present thesis.

Demographic and clinical predictors consistently explain no more than 22% of outcome variance regardless of sample size, yet precision measurement of electrode position and cochlear neural substrate raises this figure to 60–65%. This reveals that the apparent unpredictability of CI outcomes reflects measurement limitations rather than fundamental biological noise. The residual unexplained variance, and the systematic absence of FTPM from existing models, represent both a clinical challenge and a scientific opportunity. The following section examines the structure that mediates between electrode stimulation and auditory perception - the electrode-neuron interface - and the capacity of the auditory system to adapt to the distortions it imposes.

1.3 The Electrode-Neuron Interface

The preceding section established that demographic and audiological predictors leave the majority of outcome variance unexplained, and that precision measurement of electrode position and cochlear neural substrate can raise explained variance to 60–65%. These observations point to a common structure: the electrode-neuron interface (ENI) - the functional connectivity between each electrode and the associated neural interface. The delivery of electrical stimulation across the ENI is affected by characteristics of the electrode array itself, its positioning within the cochlea, any insertion trauma, and patient-specific pathologies that influence neural firing (Bierer, 2010).

Although CIs are capable of providing important spectral and temporal cues for speech perception, performance on speech tests is highly variable across listeners, and psychophysical measures obtained from individual participants can be equally variable across implant channels. Only a small portion of this variability can be accounted for by aetiology, duration of deafness, and age at onset of hearing loss (Gantz et al., 1993; Gfeller et al., 2008), and it is likely that much of the remaining variability reflects deviations in ENI quality (Bierer, 2010). One direct consequence of poor ENI is channel interaction: if threshold and comfortable levels are measured in a cochlear region where the ENI is degraded, acoustic information delivered to an electrode in that region will be detected by adjacent neural populations (off-frequency listening), increasing channel interaction - an effect known to be associated with poorer speech recognition (Boëx et al., 2003; Throckmorton and Collins, 1999).

The most common ENI pathology is that of cochlear dead regions, referring to sections of the cochlea where there is atrophy of either inner hair cells or auditory neurones (Makary et al., 2011; Moore, 2004). Assessing ENI quality in routine clinical practice is complicated, primarily because behavioural methods - such as channel-to-channel variability in threshold, psychophysical tuning curves, or loudness growth functions - are not sufficiently sensitive when using monopolar coupling. It is equally difficult to predict patterns of neural survival from demographic factors: post-mortem analysis of temporal bones reveals neurone counts ranging from less than one third to almost 80% of those in age-matched healthy ears (Hinojosa and Marion, 1983; Nadol, 1997), and patterns of neural loss vary from segment to segment within the cochlea (Fayad and Linthicum, 2006; Khan et al., 2005).

Kalkman et al. (2014) developed a computational model to investigate pitch percepts in the implanted human cochlea and found that pitch can be described using the Greenwood function (Greenwood, 1990) in the first cochlear turn. Beyond the first turn, however, pitch becomes increasingly unpredictable, with apical contacts exciting spectrally broad and overlapping neural regions. This finding is supported by the observation that, with long electrode arrays, some users obtain good spectral information from all electrodes whilst others have difficulty discriminating the most apical contacts (Landsberger et al., 2015). Optimising the mapping of low-frequency information - for example by accounting for the impact of neural dead regions on electrode-level perception - may therefore represent a viable route to improving outcomes.

The ENI refers to the functional connectivity between each implant electrode and the local auditory neurone population it stimulates. ENI quality varies substantially

across electrodes and across individuals, driven by differences in neural survival, electrode positioning, and insertion trauma. Poor ENI increases channel interaction and degrades spectral selectivity, yet current clinical methods cannot reliably assess ENI at the individual electrode level. Characterising the ENI is therefore a prerequisite for understanding why outcomes vary so widely across CI users, and for determining whether and how frequency allocation should be adjusted. The following section examines the specific frequency-to-place factors that shape the ENI in detail.

1.4 Frequency-to-Place Factors

In a CI, the incoming acoustic signal is decomposed into frequency bands and each band is assigned to a specific electrode. The perceptual outcome of this process depends on how faithfully the frequency content delivered to each electrode matches the characteristic frequency of the neurons it stimulates. Three factors govern this relationship, and we refer to them collectively as **frequency-to-place factors** because each modifies, at the level of individual electrodes, the mapping between the allocation frequency of the incoming signal and the characteristic frequency of the neuronal population being stimulated.

The first and most widely studied is the mismatch imposed by electrode array insertion depth, usually referred to as **frequency-to-place mismatch (FTPM)** or tonotopic mismatch. Insertion depth determines where the electrodes physically sit along the cochlea, and any discrepancy between the allocation frequency assigned to an electrode and the characteristic frequency at that cochlear location constitutes a baseline spectral shift.

The second factor is **current spread (CS)**, corresponding to the spatial extent of the electrical field generated by each electrode. When current spreads broadly, neurons tuned to frequencies different from those assigned to the electrode are recruited, effectively blurring the spectral information delivered to that stimulation site and adding a layer of complexity to the mismatch imposed by electrode position alone.

The third factor is **neural responsiveness (NR)**, encompassing spiral ganglion survival, peripheral process integrity, and local neural responsiveness. NR determines which neurons within the electrical field actually respond: if neurons near the electrode are degenerated or absent (*e.g.*, as in cochlear dead regions), the signal may be transduced by surviving neurons at more distant locations, further shifting the effective site of stimulation.

These three factors were selected as the focus of this thesis because they share a critical property: each operates at the level of individual electrodes and each, in principle, presents an opportunity for clinical re-mapping. Re-mapping refers to the manipulation of the frequency allocation across electrodes - that is, how the incoming frequency information is distributed to different stimulation sites. In current clinical practice, re-mapping most commonly involves deactivating individual electrodes due to technical failure (*e.g.*, abnormal impedance indicating short circuit) or extracochlear positioning; clinicians may also reduce stimulation levels on poorly performing channels, or the clinical software may redistribute frequency information across the remaining active electrodes to maintain coherent coverage of the speech spectrum. However, deliberate re-allocation of frequency boundaries to compensate for individual tonotopic mismatch remains rare in routine practice.

The primary reason is not technical limitation but rather a lack of knowledge: we do not know exactly under which conditions re-mapping would be beneficial, nor how to determine the optimal frequency allocation for a given patient. One goal of this thesis is to understand these individual frequency-to-place factors and their perceptual consequences at the electrode level, so that re-mapping decisions can ultimately be informed by individual electrode-specific data rather than population-level assumptions.

1.4.1 Frequency-to-Place Mismatch

General Concept

FTPM refers to the discrepancy between the frequency content delivered to a CI electrode, or allocation frequency, and the characteristic frequency of the neural population it stimulates. In normal hearing (NH), the cochlea performs a precise tonotopic decomposition: each location along the basilar membrane responds maximally to a specific frequency, with high frequencies represented at the base and low frequencies at the apex. CIs attempt to replicate this organisation by assigning frequency bands to electrodes according to their position on the electrode array. However, the correspondence between incoming frequency and the actual cochlear place stimulated is systematically inaccurate in current clinical devices. The primary cause is the incomplete coverage of the cochlea by the electrode array.

The quantification of FTPM induced by the incomplete insertion of the electrode

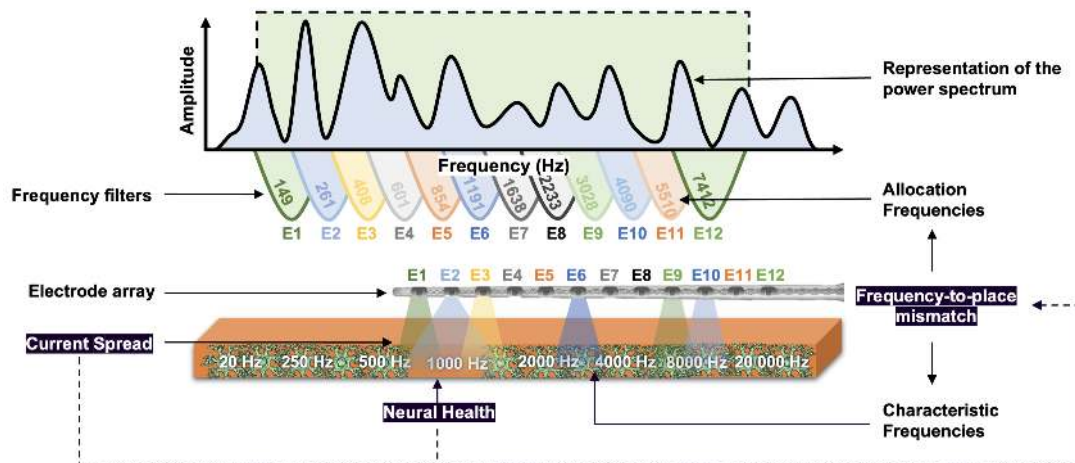


Figure 1.3: Schematic illustration of frequency-to-place factors in cochlear implants. The incoming acoustic signal (top) is decomposed into frequency bands and assigned to electrodes E1-E12 according to the processor allocation frequencies. Within the schematic representation of the cochlea (bottom), three factors determine the relationship between allocated and perceived frequency. First, electrode array insertion depth sets the baseline FTPM by imposing a shift between characteristic frequencies and allocation frequencies. Second, current spread (coloured triangles) extends stimulation beyond the target site, recruiting neurons tuned to adjacent frequencies; its extent is influenced by local neural health. Third, neural health (illustrated by the absence of neurons) varies along the cochlea; where spiral ganglion neurons are absent, the effective site of stimulation may shift to more distant surviving populations.

array is achieved by comparing the allocation frequency of each electrode with the actual characteristic frequency associated with the cochlear place stimulated by that electrode (see Equation 1.1).

$$\text{FTPM}(i) = \frac{\log_{10} \left(\frac{AF(i)}{CF(i)} \right)}{\log_{10}(2)} \quad (1.1)$$

where i denotes the electrode number, $CF(i)$ is the characteristic frequency of the cochlear location stimulated by electrode i (in Hz), and $AF(i)$ is the centre frequency allocated to electrode i by the speech processor (in Hz). A positive FTPM value indicates a basal shift (stimulated place has a higher characteristic frequency than allocated), while a negative value indicates an apical shift.

Whereas the determination of allocation frequency is directly available from the clinical software, determining the characteristic frequency at each electrode location presents a greater challenge. This requires knowledge of both the electrode's physical position within the cochlea and the use of the appropriate frequency-place function to convert that position into frequency.

Electrode Array Insertion Estimation

Before estimating the characteristic frequency, the estimation of angular insertion depth (AID) associated with each electrode is necessary. For this to be possible, estimation of cochlear anatomical landmarks is necessary.

Such determinations are possible using one of two imaging modalities: computed tomography (CT) or plain radiography (X-rays). It is worth noting that postoperative imaging is not universally performed as part of routine clinical care; practice varies considerably across countries and centres, and in many settings electrode position is not routinely assessed after implantation. Both modalities have advantages and limitations. CT offers three-dimensional visualisation and superior soft-tissue contrast, enabling identification of scalar position (scala tympani versus scala vestibuli) and proximity to the modiolus; however, it involves higher radiation exposure and cost. Plain radiographs provide sufficient resolution for angular insertion depth measurements at a lower cost and radiation dose, but typically require a specific angle of view, and may lack the contrast needed to determine electrode array position and anatomical landmarks clearly - particularly when electrodes from different turns overlap.

Regardless of imaging modality, the standard approach involves identifying anatomical landmarks - typically the round window and the modiolar axis - to establish a coordinate system. The AID of each electrode is then measured as the cumulative angle from the round window around the modiolar axis. This angular position is converted to cochlear duct length using spiral equations such as the Escudé formula, which relates AID to distance along the cochlea based on individual cochlear dimensions. Finally, the appropriate frequency-place function is applied to derive the characteristic frequency at each electrode location - the choice of which function (Greenwood versus Stakhovskaya) has important

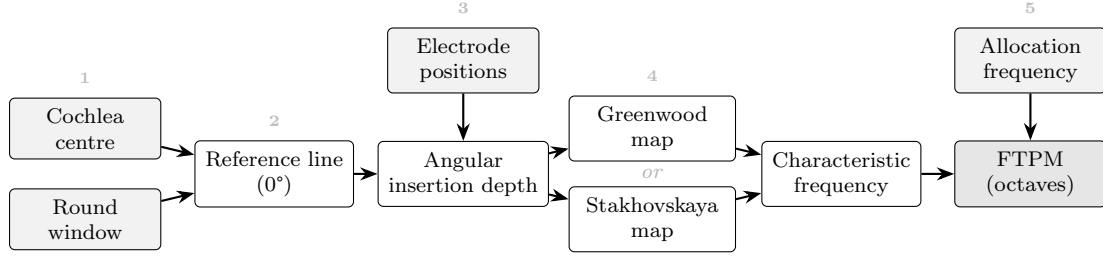


Figure 1.4: **Overview of the FTPM estimation pipeline.** The main stages to estimate FTPM are described in the present pipeline. (1) Anatomical landmarks - the cochlea centre and round window - are identified from imaging. (2) A reference axis at 0° is established between these landmarks. (3) Using the identified electrode positions, the angular insertion depth (AID) is calculated for each electrode relative to this axis. (4) AID values are converted to characteristic frequencies using a frequency-place map (Greenwood or Stakhovskaya). (5) FTPM is computed by comparing each electrode characteristic frequency with its processor allocation frequency.

implications for the resulting FTPM estimate, as discussed in the following subsection. The full estimation pipeline is illustrated in Figure 1.4.

Even though they are not used in the present thesis, it is worth noting that emerging approaches seek to complement or refine imaging-based FTPM assessment. Electrophysiological measures - including electrically evoked compound action potential (ECAP) amplitude growth functions, which reflect the number and synchrony of responding neurons at each electrode (Dillier et al., 2002), and cortical auditory evoked potentials, which index central auditory processing of electrode-specific stimulation - may provide functional estimates of the neural populations activated by each electrode. Pitch-matching experiments in recipients with residual acoustic hearing offer behavioural validation of frequency-place estimates. These physiological and behavioural approaches remain largely investigational but hold promise for individualising FTPM assessment beyond what anatomical imaging alone can provide.

Frequency-Place Models and Characteristic Frequency Estimation

Two anatomical structures are relevant when considering the target of electrical stimulation: the organ of Corti, which usually contains the hair cells, and the spiral ganglion, where the cell bodies of the auditory nerve fibres are located. Although in theory CIs bypass the damaged hair cells to stimulate the auditory nerve directly, the question of which structure is actually stimulated by the current remains debated (Rattay et al., 2001; Briaire and Frijns, 2005), and has important implications for FTPM calculation.

The classical Greenwood function, derived from psychoacoustic measurements in NH listeners, describes the frequency-place relationship along the organ of Corti. This function relates characteristic frequency to position as $f = 165.4 \times (10^{0.06x} - 1)$, where x represents the distance from the apex in millimetres.

However, Stakhovskaya et al. (2007a) demonstrated that the spiral ganglion follows a slightly different spatial organisation, particularly in the apical region (see Figure 1.5 for a histological illustration of how spiral ganglion neurons are distributed along the cochlea and Figure 1.6 for the comparison between the Greenwood and Stakhovskaya frequency-

place models). By tracing the radial nerve fibres connecting the organ of Corti to the spiral ganglion in human temporal bones, they derived a frequency-place function specific to the ganglion. As shown in Figure 1.6, the two functions start to diverge beyond 360° insertion depth, with the spiral ganglion map predicting lower characteristic frequencies at equivalent angular positions. At the apex (720°), the organ of Corti function predicts approximately 150 Hz, while the spiral ganglion function yields only 58 Hz (a difference of around 1.5 octaves).

The choice between these models has direct clinical relevance. Modern perimodiolar electrode arrays are designed to sit close to the modiolus, placing them in closer proximity to spiral ganglion neurons than to the basilar membrane. For such electrodes, the Stakhovskaya function may provide more accurate FTPM estimates. Conversely, lateral-wall electrodes, which maintain greater distance from the modiolus, may stimulate neurons whose peripheral processes extend toward the organ of Corti, potentially making the Greenwood function more applicable. Current consensus favours the spiral ganglion map for perimodiolar arrays, though the optimal model likely depends on electrode design, insertion depth, and the extent of peripheral hair-cell survival¹.

Typical FTPM degree

With default frequency allocations, a mismatch between allocated and characteristic frequencies is present across virtually all CI users, as multiple studies have documented (Landsberger et al., 2015; Canfarotta et al., 2020a; Venail et al., 2015b). Landsberger et al. (2015) compared four electrode arrays from three manufacturers and found that, for insertion angles up to approximately 270°, allocated frequencies were typically one octave below the characteristic frequency predicted by the spiral ganglion map (Landsberger et al., 2015). Beyond 270°, deviations increased further, with longer arrays (MED-EL Standard and Flex28) demonstrating smaller mismatches than shorter electrode array designs (Advanced Bionics HiFocus 1J and Nucleus Contour Advance).

The degree of FTPM varies substantially between individuals due to three interacting factors: cochlear anatomy, electrode array length, and insertion depth. Cochlear duct length varies considerably across the population, ranging from approximately 30 to 42 mm (Alexiades et al., 2015; Meng et al., 2016). Combined with array lengths spanning between 16 and 31.5 mm across manufacturers, this anatomical variability produces wide-ranging insertion depths even with identical surgical technique (Venail et al., 2015b), as illustrated in Figure 1.7.

Critically, substantial variability persists even for the same electrode array. Canfarotta et al. (2020) documented significant differences in FTPM among recipients implanted with identical arrays, attributing this to individual differences in cochlear size and basal insertion depth (Canfarotta et al., 2020a), as shown in Figure 1.8.

¹A detailed comparison of frequency-place maps falls outside the scope of this thesis. The Stakhovskaya map was adopted throughout on the basis of its greater anatomical relevance to the electrode arrays studied here. Moreover, the insertion depths evaluated in the present work did not exceed approximately 440°, a range over which the Greenwood and Stakhovskaya functions yield negligible differences in characteristic frequency estimation.

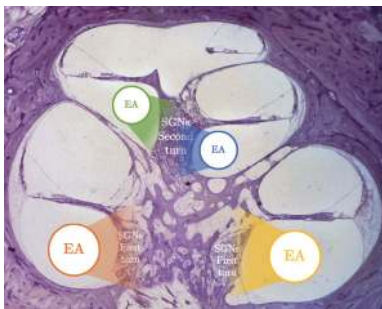


Figure 1.5: **Histological view of the cochlea with annotation of how the electrode array stimulates different pools of spiral ganglion neurons.** This figure represents a histological cross-section of the cochlea. For each turn, the spiral ganglion neurons (SGNs) have been annotated. The electrode array (EA) is represented in two turns, with theoretical current stimulation oriented towards the SGNs. While the SGNs are clearly grouped into distinct clusters in the basal turn, they become increasingly pooled together in the second turn. Such convergence is likely to distort spectral selectivity when these apical regions are electrically stimulated. Adapted from Nadol (1997).

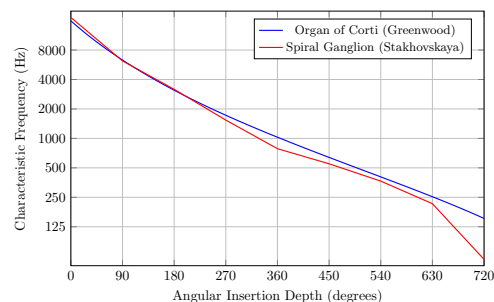


Figure 1.6: **Frequency-position functions for the organ of Corti (Greenwood function) and the spiral ganglion (Stakhovskaya function).** While the two functions follow nearly the same trend at the base of the cochlea (between 0° and 360°), they start to diverge beyond this point. At approximately 630° , the divergence becomes clearly distinguishable, explained by the convergence of SGN dendrites projecting from a broad region of the organ of Corti onto a compressed portion of the spiral ganglion, effectively mapping a wider frequency range onto fewer spatially distinct neural populations.

Effect of Frequency-to-Place Mismatch on Speech perception

Early investigations were conducted using CI simulation (*i.e.*, vocoder), in which different FTPM degrees were presented to NH listeners to evaluate the effect on speech perception in acute conditions (*i.e.*, without acclimation). In such investigations, Baskent and Shannon (2004) observed magnitude-dependent performance decrements: consonant recognition declined from approximately 70% at baseline to 55% with 3 mm shifts, 45% with 4 mm shifts, and 35% with 6 mm shifts (Baskent and Shannon, 2004). The direction of mismatch also influenced outcomes, with compression (mapping lower frequencies to more basal locations) producing greater impairment than expansion.

Experimental manipulation of frequency maps in real CI users confirms these effects: Fu and Galvin (2003) reported that introducing a 3 mm basalward shift reduced consonant recognition from 77% to 45%, with partial but incomplete recovery after several weeks of adaptation (Fu and Galvin, 2003).

The standardisation of postoperative imaging in clinics - with routine CT or X-ray assessment of electrode position - has enabled systematic investigation of FTPM effects in large clinical cohorts. These studies consistently demonstrate negative correlations between mismatch magnitude and speech perception during the early post-activation period. Canfarotta et al. (2020) analysed 101 recipients implanted with lateral wall arrays and

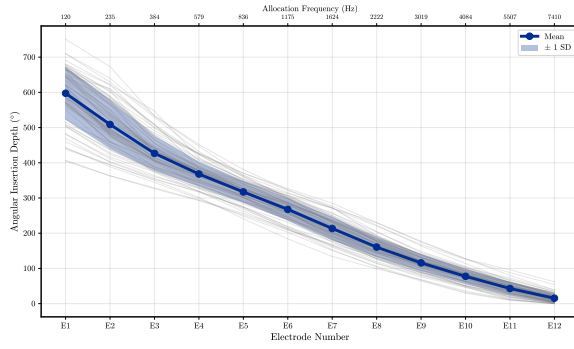


Figure 1.7: **Relationship between angular insertion depth (AID) and allocation frequencies associated with electrode number.** Data from 80 CI recipients collected by Canfarotta et al. (2020a). Each grey line represents one individual, the blue curve indicates the mean, and the shaded area the standard deviation. The variability in AID increases towards the apical electrodes, reflecting differences in insertion depth across individuals.

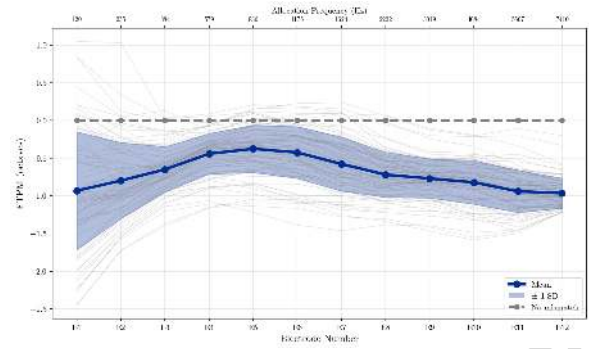


Figure 1.8: **FTPM profiles (in octaves) as a function of allocation frequency and electrode number.** Same individuals as in Figure 1.7, from Canfarotta et al. (2020a). Each grey line represents one individual, the blue curve the mean, and the shaded area the standard deviation. The horizontal grey line indicates zero mismatch. The largest FTPM values are observed at basal and apical electrodes, with a reduction between E4 and E7. The variability in insertion depth seen in Figure 1.7 translates into substantial FTPM variability at the most apical electrodes.

found that mismatch at 1500 Hz significantly predicted word recognition at six months ($r = -0.32$, $p = 0.006$), accounting for approximately 10% of outcome variance (Canfarotta et al., 2020a). Recipients with mismatch below 4 semitones achieved mean CNC scores of 62%, compared to 46% for those exceeding 7 semitones - a relationship that persisted when controlling for age, duration of deafness, and preoperative hearing. Similarly, Venail et al. (2015) reported that each 100 Hz increase in average mismatch was associated with a 20 percentage point decrease in maximum word recognition ($\beta = -19.59$, $p = 0.009$), and that mismatch prolonged rehabilitation: recipients with mismatch below 200 Hz reached plateau performance in 3.2 months compared to 10.3 months for those exceeding 400 Hz (Venail et al., 2015b). These findings indicate that FTPM not only limits ultimate performance but also delays the trajectory toward stable outcomes.

The relationship between FTPM and speech perception appears to weaken beyond twelve months post-activation. Mertens et al. (2022) found a significant correlation at six months ($r = -0.34$, $p = 0.035$), with recipients showing less than 300 Hz mismatch achieving 72% sentence recognition compared to 58% for those exceeding 500 Hz (Mertens et al., 2022). However, this correlation was no longer significant at twelve months ($r = -0.19$, $p = 0.24$), with the high-mismatch group demonstrating greater gains during this interval. DeFreese et al. (2025) reported similar patterns in the largest study to date (498 implants): for straight lateral wall arrays, FTPM did not significantly predict word recognition across the first year ($\beta = -0.23$, $p = 0.12$), with test interval and daily device use emerging as the primary predictors instead (DeFreese et al., 2025).

FTPM correlates with speech perception during the first six to twelve months of CI use, but this relationship weakens over time as central adaptation compensates for the spectral shift. Adaptation is aided by the fact that speech understanding tolerates substantial spectral distortion, relying heavily on temporal envelope cues, linguistic context, and phonetic redundancy. However, perceptual demands that depend on precise spectral representation - such as music perception, sound quality, and integration of electric and acoustic hearing - may reveal persistent deficits that speech tests do not capture.

Effect of Frequency-to-Place Mismatch on Other Perceptual Domains

Music perception provides the clearest evidence that FTPM consequences persist beyond what speech tests capture. Unlike speech, music relies critically on precise pitch relationships, accurate spectral envelopes for timbre, and preserved harmonic structure - all of which FTPM distorts. Reiss et al. (2007) found that CI users with deeper electrode insertion demonstrated more ordered pitch ranking and broader pitch ranges, whereas those with shallower insertion showed compressed and irregular pitch percepts (Reiss et al., 2007). Schatzer et al. (2014) confirmed that deeper insertion predicted better interval discrimination, with effects strongest in the low-frequency range where apical mismatch is typically greatest (Schatzer et al., 2014). Direct evidence linking FTPM magnitude to music outcomes comes from Goupell et al. (2019), who reported pitch shifts of one to three octaves in bimodal users and found that melody recognition and instrument identification correlated inversely with mismatch magnitude (Goupell et al., 2019). Training studies further suggest that FTPM limits rehabilitation potential: Galvin et al. (2007) demonstrated that CI listening participants whose baseline pitch-ranking functions deviated substantially from the expected monotonic pattern - with rank-order correlations below 0.5, indicating pitch reversals or compressions consistent with substantial mismatch - showed minimal improvement despite extensive training, whereas those with ordered pitch percepts (rank-order correlations above 0.8) achieved meaningful gains (Galvin et al., 2007).

For bilateral CI users, interaural asymmetry in FTPM creates additional challenges for spatial hearing: when electrodes at corresponding positions in each ear represent different frequencies due to differences in insertion depth or cochlear anatomy, binaural processing is likely disrupted. Kan et al. (2015) demonstrated that interaural place mismatch exceeding 3–4 mm of cochlear distance substantially degraded sensitivity to interaural time differences, a critical cue for sound localisation (Kan and Litovsky, 2015). Goupell et al. (2013) found that electrode pairs with matched pitches between ears produced strong binaural fusion, whereas mismatched pairs often produced double images or reduced fusion (Goupell et al., 2013). At the functional level, Francart et al. (2011) reported that bilateral CI participants with more symmetric insertion depths showed better localisation accuracy and greater spatial release from masking compared to those with asymmetric insertions (Francart et al., 2011). These findings indicate that minimising interaural FTPM asymmetry is particularly important for optimising binaural benefits.

Subjective sound quality and listening effort represent domains where FTPM effects are theoretically compelling but empirical evidence remains indirect. Many CI users describe their auditory experience as “robotic”, “tinny”, or “unnatural” even when speech

recognition is adequate (Spahr et al., 2012; Croghan et al., 2017). FTPM-induced spectral envelope distortion provides a plausible mechanism: familiar sounds may not match stored representations, producing a persistent sense of artificiality. Supporting this, some CI participants in frequency reallocation studies prefer shifted maps over their defaults, suggesting their original programming contained uncorrected mismatch (Goupell et al., 2016). Listening effort similarly appears elevated in CI users beyond what speech accuracy would predict. Hughes et al. (2018) documented substantial individual variability in self-reported listening effort that did not correlate with speech scores, and participants describing high effort showed decrements on concurrent cognitive tasks (Hughes et al., 2018). Critically, adaptation appears incomplete: Reiss et al. (2018) found that sound quality ratings improved during the first year of CI use but then plateaued at levels substantially below normal-hearing benchmarks, with participants continuing to report unnatural sound quality for complex signals (Reiss et al., 2018).

The most direct evidence that FTPM impacts patient-reported outcomes beyond speech recognition comes from Sturm et al. (2024), who examined quality of life as a function of estimated mismatch (Sturm et al., 2024). Using the Nijmegen Cochlear Implant Questionnaire, they found that participants with larger FTPM reported significantly lower scores in domains of sound quality, listening effort, self-esteem, and social interaction compared to those with smaller mismatch - despite achieving similar speech recognition scores in quiet.

FTPM affects perceptual domains well beyond speech intelligibility: music perception, spatial hearing, subjective sound quality, and quality of life are all negatively associated with mismatch magnitude, even in CI users who achieve adequate speech recognition. Standard audiological assessment fails to capture these broader consequences, indicating that FTPM creates perceptual and cognitive burdens that persist despite apparent adaptation on speech tasks.

Alleviating Frequency-to-Place Mismatch with Anatomy-Based Fitting Strategies

The rationale for anatomy-based fitting (ABF) follows directly from the preceding evidence: if FTPM contributes to perceptual limitations, then correcting it should improve outcomes. ABF strategies aim to realign the frequency allocation of each electrode with the characteristic frequency of the neural population it stimulates, thereby compensating for individual differences in cochlear size, insertion depth, and electrode position (Yang et al., 2022; Noble et al., 2013).

Determination of anatomy-based frequency allocation maps builds directly on the FTPM estimation pipeline described above (see Figure 1.4). Once the characteristic frequency of each electrode contact has been established, the frequency allocation table is adjusted to align filter bands with these calculated frequencies (Canfarotta et al., 2020a; Jiam et al., 2019; Creff et al., 2024; Mertens et al., 2022). Although ABF is technically feasible with any manufacturer’s software that permits manual adjustment of frequency boundaries, the degree of flexibility varies: MED-EL clinical software provides direct access to individual electrode frequency boundaries, whereas Cochlear and Advanced Bionics systems impose greater constraints on how allocation tables can be modified. In strictly place-based mappings, low-frequency information may be lost if the electrode array does

not reach the corresponding apical region; hybrid approaches address this by aligning low-to-mid frequencies tonotopically while compressing higher frequencies toward the basal contacts (Dillon et al., 2021).

ABF consistently achieves its technical objective of reducing calculated mismatch. Across multiple studies, the absolute difference between allocated and tonotopic frequencies is substantially smaller with ABF than with default fittings - typically less than half the mismatch observed with conventional maps (Lassaletta et al., 2023). For MED-EL arrays, median mismatch values below 10% have been reported for mid-array electrodes, although residual deviations persist at the apical and basal extremes due to hardware bandwidth limitations (Lassaletta et al., 2023; Wangchuk et al., 2024). However, the appropriate target for correction remains uncertain. The Greenwood and Stakhovskaya functions were derived from threshold-level measurements and histological data, yet recent work suggests that cochlear tonotopy shifts basally at suprathreshold levels typical of conversational speech (Sagi and Svirsky, 2025). Moreover, electrocochleography-derived best-frequency maps diverge from both anatomical models, raising questions about which reference represents the functionally relevant target (Walia et al., 2025).

Despite consistent reduction in calculated mismatch, clinical outcomes with ABF are inconsistent. Several studies report positive results. Creff et al. (2024) found that tonotopic fitting produced better speech recognition in noise and superior music perception compared to default maps in a randomised crossover trial, with 92% of participants electing to retain the tonotopic map (Creff et al., 2024). Jiam et al. (2019) demonstrated improved pitch-ranking accuracy with image-guided allocations (Jiam et al., 2019), and Kurz et al. (2023) reported better speech perception in quiet and noise for experienced bilateral participants switched to ABF (Kurz et al., 2023). A systematic review found that 79% of participants across studies preferred ABF maps and elected to retain them (Yang et al., 2022). However, other rigorous trials have failed to confirm these benefits. Lambriks et al. (2023), in a daily-randomised controlled trial with new implant users, found that standard fitting achieved superior speech outcomes and was preferred by 12 of 14 participants (Lambriks et al., 2023a). Most strikingly, Walia et al. (2024) reported that electrocochleography-derived frequency maps predicted better speech outcomes when electrodes were shifted *basally* from the calculated best-frequency locations - the opposite direction from what anatomical correction would prescribe (Walia et al., 2025). Paediatric data similarly show no significant correlation between mismatch magnitude and outcomes after several years of device use, consistent with robust developmental adaptation (Alahmadi et al., 2025).

Several factors may explain these inconsistencies. The frequency-place functions used to calculate mismatch may be systematically miscalibrated for suprathreshold stimulation, leading ABF to correct toward an inappropriate target (Sagi and Svirsky, 2025; Walia et al., 2025). Insertion depth also appears to moderate ABF benefit: participants with deeper insertions, whose apical electrodes actually reach low-frequency regions, show more consistent improvement than those with shallower insertions, for whom ABF primarily redistributes mismatch rather than eliminating it (Kurz et al., 2023). The capacity for central adaptation may also play a greater role than peripheral correction for some individuals, particularly those with longer CI experience whose auditory cortex has already reorganised around the existing frequency-place mapping. The observation that ABF benefits appear more consistent in experienced participants switched from default maps than in new participants fitted with ABF from activation supports this interpretation

(Kurz et al., 2022, 2023).

Anatomy-based fitting reliably reduces calculated FTPM but does not consistently improve perceptual outcomes. This dissociation suggests that correcting insertion-depth mismatch alone is insufficient: the perceptual consequence of a given electrode depends not only on where it sits relative to the tonotopic map, but also on how broadly its current spreads to neighbouring neural populations and on the health of the neurons it stimulates. These two additional frequency-to-place factors - NR and CS - are presented in the following sections.

1.4.2 Modulating Factors

Neural Responsiveness

Direct assessment of neural health in living humans remains infeasible, necessitating indirect electrophysiological measures. The electrically evoked compound action potential (ECAP) - a synchronous neural response recorded directly from the electrode array following electrical stimulation (Dillier et al., 2002) - provides the foundation for most objective neural health estimates. Several ECAP-derived methods exist to estimate neural status (*e.g.*, amplitude growth functions, the effect of interphase gap on ECAP offset, and spread-of-excitation measures), alongside behavioural approaches such as focused thresholds and polarity sensitivity, though these show variable convergence and may reflect different aspects of neural function (Brochier et al., 2021; Pfingst et al., 2011b; Schwartz-Leyzac and Pfingst, 2016). Critically, most of these approaches require specialised equipment, lengthy recording sessions, or subjective responses from the participant, limiting their applicability in routine clinical practice.

Among automated approaches, Frijns and colleagues in Leiden developed spread-of-excitation measurements using forward-masking paradigms to characterise channel interaction along the electrode array (Frijns et al., 2002), and Cooper and colleagues at Cochlear explored clinically integrated spread-of-excitation protocols (Cooper et al., 2008).

The Panoramic ECAP (PECAP) method, adopted in the present thesis, overcomes many of the practical limitations described above by providing electrode-specific estimates that are thought to reflect neural responsiveness, derived from a single, fully automated recording session that requires no active participation from the user (Garcia et al., 2021). PECAP records forward-masked electrically evoked compound action potentials for all combinations of masker and probe electrodes, then applies an optimisation algorithm that separates what are interpreted as the contributions of NR and CS, as illustrated in Figure 1.9. The method was selected for the present thesis on the basis of its clinical feasibility (fully automated, no subjective responses required), its provision of electrode-specific rather than global estimates, and its simultaneous characterisation of both NR and CS within a single measurement session.

Several observations in the literature support the relevance of NR as a moderator of FTPM effects. The variable benefit of anatomy-based fitting may partly reflect differences in underlying neural substrate (Lambriks et al., 2023a; Long et al., 2014). Walia et al.'s (2025) finding that basally-shifted maps predicted better outcomes than anatomically-aligned maps has been interpreted as possibly reflecting superior neural health in basal

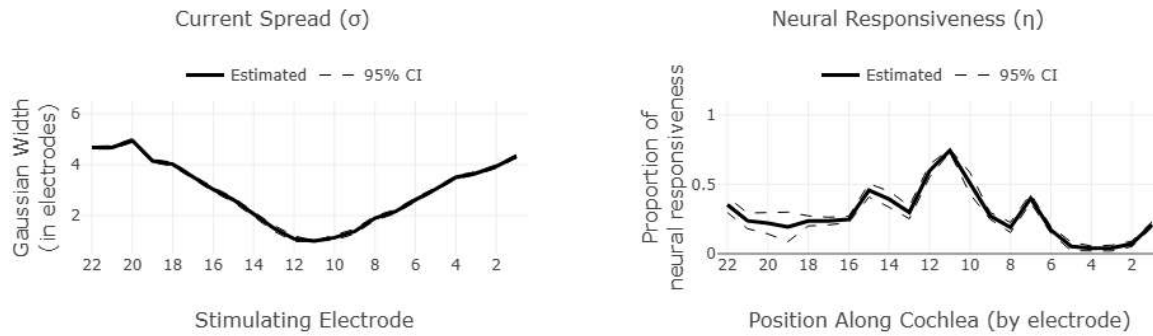


Figure 1.9: **Example of current spread and neural responsiveness estimation using PECAP.** The left panel shows the estimated current spread and the right panel the estimated neural responsiveness for an individual CI user, both derived from the PECAP method. Electrode numbering runs from the most apical (E22) to the most basal (E1).

cochlear regions (Walia et al., 2025), although this interpretation warrants caution given that histological evidence suggests dead regions are at least as prevalent basally as apically (Nadol, 1997); alternative explanations include suprathreshold tonotopic shifts or interactions between electrode position and current spread patterns. The differential effect of array type on FTPM-outcome correlations reported by DeFreese et al. (2025) may likewise reflect array-specific patterns of insertion trauma affecting neural survival (DeFreese et al., 2025).

NR varies substantially along the cochlea due to patterns of spiral ganglion degeneration that differ across individuals and aetiologies. Where NR is degraded, the effective site of stimulation may shift away from the target location, compounding the mismatch imposed by electrode positioning. The PECAP method provides electrode-specific NR estimates from a single automated recording session, enabling joint assessment of NR and FTPM. FTPM and neural health should therefore be considered together rather than in isolation when predicting outcomes or planning re-mapping interventions.

Current Spread

Even under hypothetically ideal conditions - frequency allocation perfectly aligned with cochlear tonotopy and neural survival uniform across the array - the spectral resolution achievable with CIs would still be limited by CS. CS refers to the spatial dispersion of electrical charge through the conductive perilymph and surrounding tissues when an electrode is activated. Stimulation from a single electrode can activate neurons spanning several millimetres of cochlear length, creating substantial overlap between the excitation patterns of adjacent electrodes (Cohen et al., 2003; Jolly et al., 1996). This overlap degrades channel independence: information nominally assigned to distinct spectral bands is delivered to overlapping neural populations, reducing effective spectral resolution below the number of physical electrodes.

The magnitude of CS varies across individuals and across electrodes within individuals, influenced by electrode-modiolus distance, local tissue impedance, and the distri-

bution of surviving neurons. Perimodiolar arrays, positioned closer to SGNs, typically achieve narrower excitation patterns than lateral-wall arrays (Cohen et al., 2006; Srinivasan et al., 2013). Regions of poor neural survival can paradoxically increase functional CS: when local neurons are absent, current must travel further to reach surviving populations, broadening the excitation pattern and increasing overlap with adjacent channels (Bierer and Nye, 2014; DeVries and Arenberg, 2018).

CS also complicates the interpretation of FTPM effects. Correcting the frequency allocation at a given electrode may yield limited perceptual benefit if CS causes that electrode’s signal to excite neurons across a wide range of characteristic frequencies (Srinivasan et al., 2013). Conversely, broad CS might partially compensate for FTPM by activating neurons at the correct tonotopic location even when the electrode is physically mispositioned. Such compensation, however, comes at the cost of reduced selectivity, effectively trading one form of spectral degradation for another.

Critically, the PECAP method estimates CS and NR simultaneously from a single measurement session (Figure 1.9), providing electrode-specific characterisation of both factors (Garcia et al., 2021). This makes it possible to assess the interplay between physical mismatch, neural substrate, and stimulation selectivity within a unified framework.

CS determines the spatial extent of neural excitation from each electrode. Broad CS degrades spectral resolution by activating neurons across a wide range of characteristic frequencies, reducing channel independence below the number of physical electrodes. CS varies across electrodes within individuals, is influenced by electrode-modiolus distance and local neural survival, and can either amplify or partially compensate for FTPM depending on the geometry of the electrode-neuron interface. Without knowledge of how broadly each electrode stimulates, the true spectral mismatch experienced by CI users cannot be determined from electrode position alone.

1.5 Rationale for The thesis

CIs are the most successful neural prosthesis to date, having restored hearing to over one million individuals worldwide. However, despite major technological breakthroughs during the first decade of CI development - and notwithstanding the progressive relaxation of candidacy criteria and improvements in hearing preservation that might have been expected to raise mean performance - average speech perception outcomes have largely plateaued over the past three decades. At the same time, striking variability in performance persists between CI users, even among those with similar audiological profiles and years of device experience, and the sources of this variability remain poorly understood. There is therefore a pressing need to move beyond “one-size-fits-all” approaches and to identify the individual factors that drive these differences and that could be targeted through individualised mapping and rehabilitation strategies.

Three frequency-to-place factors have been identified as candidates for clinical intervention through re-mapping. The first is **frequency-to-place mismatch (FTPM)** - the discrepancy between the frequency allocation of each electrode and the characteristic frequency of the cochlear place it stimulates. The remaining two factors are directly related to the quality of the ENI: **current spread (CS)**, which determines the spatial extent of neural excitation from each electrode, and **neural responsiveness (NR)**, which reflects the health and density of the local neural population.

However, before any re-mapping strategy can be planned, it is first necessary to establish whether these factors are genuinely detrimental to perception, and if so, which electrodes are most affected. The main goal of this thesis is therefore to develop testing procedures capable of tracking the perceptual effects of these frequency-to-place factors at the electrode level.

To this end, we address three interrelated questions:

1. How is sound quality affected for different speech and music-based stimulus types affected by FTPM induced by electrode array insertion? (Chapter 2)
2. What are the long-term perceptual consequences of frequency-to-place factors on speech and music perception at the electrode level? (Chapters 3 and 4)
3. How does a newly implanted listener adapt to frequency-to-place factors in the early months of CI use for speech and music perception at the electrode level? (Chapter 5)

Chapter 2 - Sound Quality and FTPM

Research Question: How is sound quality for different speech and music-based natural stimulus types affected by FTPM profile shapes?

Context: Many studies have examined FTPM effects using speech recognition tasks. However, these tasks constitute a poor proxy for evaluating potential effects of FTPM beyond the first year after CI activation. Sound quality is presumed to play a larger role on a long-term basis. However, very little is known concerning the effect of FTPM profiles on sound quality and how it varies according to the signal used. Different types of FTPM shapes have been vocoded, for 12 different realistic music and speech signals, and presented for sound quality assessment to NH listeners using a MUSHRA interface.

Results: We found that FTPM had a detrimental impact on sound quality for all signals used. That degradation was more pronounced when FTPM was present in low frequencies. Moreover, the variability of FTPM within each profile also had a negative impact on sound quality. No clear link was established between acoustic features and the magnitude of FTPM-induced quality degradation.

Presented in Chapter 2.

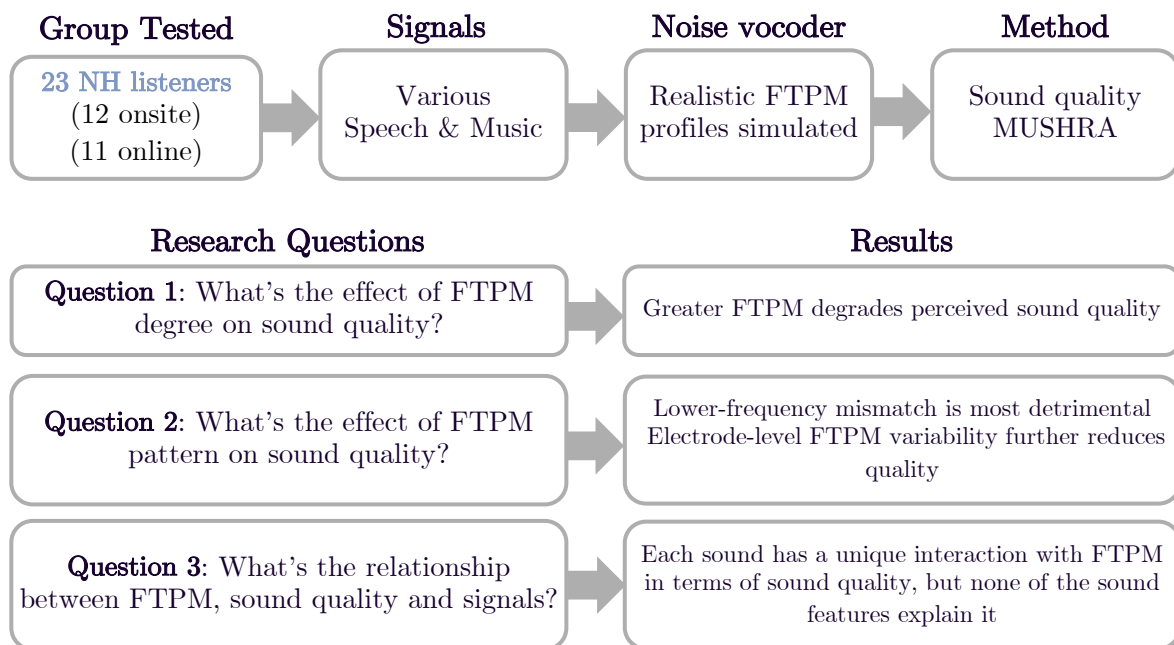


Figure 1.10: Summary of the Chapter 2 study.

Chapter 3 & Chapter 4 - Long-term Effects of Frequency-to-Place Factors on Speech and Music Perception

Research Question: What are the long-term perceptual consequences of frequency-to-place factors on speech and music perception in CI users with different auditory histories?

Context: These chapters constitute the core of the project, deploying two complementary perceptual tasks to assess the effects of frequency-to-place factors on speech and music perception.

In this part, we focus on long-term effects. Two groups of CI users were tested: Post-LCI and Pre-ECI. Because Pre-ECI users have no prior acoustically-derived tonotopic representation, they are not expected to require the same adaptation to frequency-to-place factors unlike Post-LCI, for whom frequency-to-place factors conflict with previously established internal representations.

Comparing these two populations therefore allows us to disentangle how frequency-to-place factors interact differently with perception depending on prior auditory experience.

Vowel Task

The task was a vowel identification and confidence experiment in which two-formant stimuli were presented to CI users to specifically stimulate a pair of electrodes. Participants were asked to select the perceived vowel and rate their confidence in their choice. From these responses, we created vowel perceptual maps, which were compared to equivalent perceptual maps from NH listeners and to frequency-to-place factors (*i.e.*, FTPM, NR, and CS) associated with each electrode.

Chord Task

In this task, pairs of inverted chords were presented, designed to differentially stimulate two electrodes. Participants were asked to select the most pleasant chord in each pair. Responses were then compared to frequency-to-place factors (*i.e.*, FTPM, NR, and CS) associated with each electrode.

Results:**Vowel Task Results**

Vowel representations of CI users were modestly displaced from those of NH listeners overall, with Post-LCI showing a preference for lower F2 frequencies and Pre-ECI weighting F1 cues more heavily in their confidence judgements. None of the three frequency-to-place factors accounted for these small deviations at the electrode level, suggesting either that experienced users had fully adapted to these factors, or that the task lacked sufficient sensitivity to detect their residual influence.

Chord Task Results

Root-position chords were strongly preferred across all participants irrespective of implantation history, most plausibly reflecting the acoustic structure of the stimuli rather than individual FTPM profiles. In Post-LCI, FTPM-induced interval distortion predicted chord preference, but only when root-position and distortion cues were aligned, consistent with a cue integration account. CS was the most consistent electrode-level predictor of perceptual preference weight. A significant $\text{FTPM} \times \text{Group}$ interaction confirmed that Post-LCI and Pre-ECI relate to their mismatch profiles in opposing directions.

Vowel task presented in Chapter 3. Chord task presented in Chapter 4.

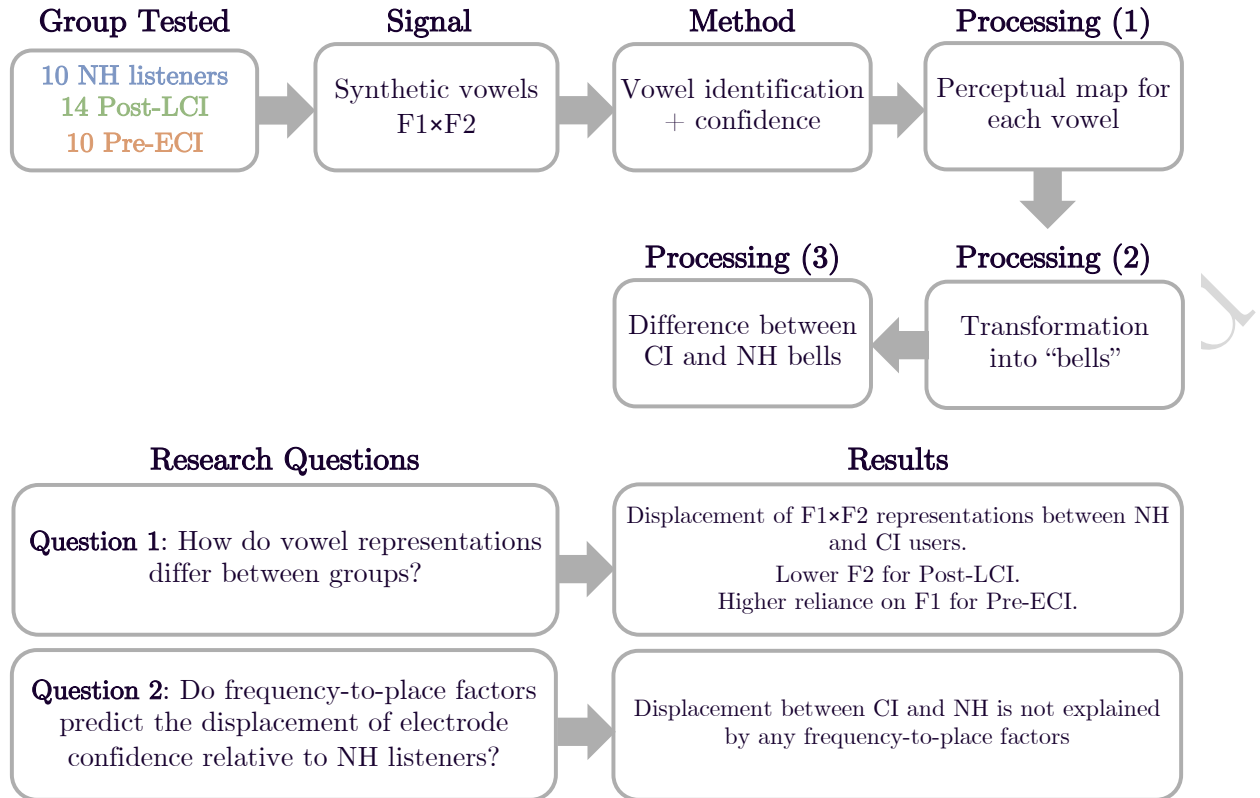


Figure 1.11: Summary of the Chapter 3 - Vowel study.

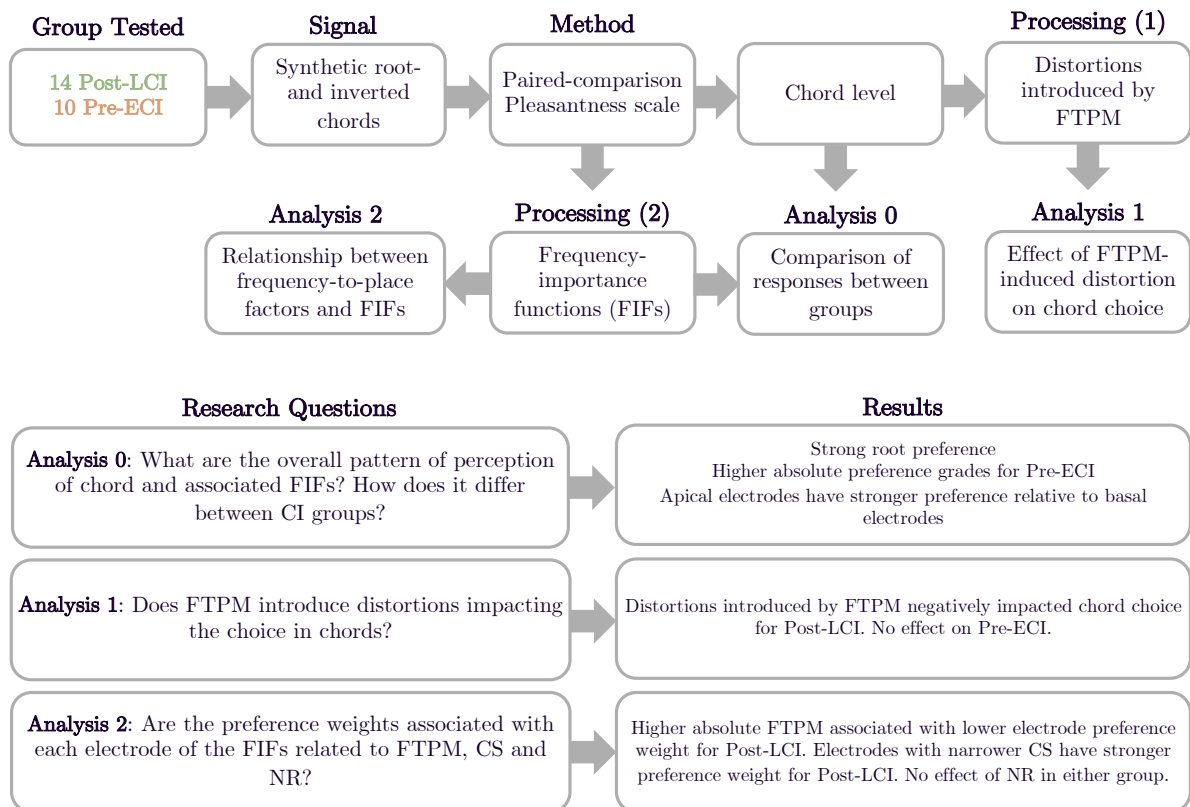


Figure 1.12: Summary of the Chapter 4 - Chord study.

Chapter 5 - Adaptation to Frequency-to-Place Factors

Research Question: How do newly implanted listeners adapt to frequency-to-place factors in the early months of CI use?

Context: The period following CI activation is critical, with most performance improvement occurring during the first six months. In this chapter, we present a longitudinal case study of a newly implanted Post-LCI user, tested four times using the Vowel and Chord tasks (described in Chapters 3 and 4) across Days 6 to 134 post-activation. Adaptation strategies were analysed in relation to FTPM, NR, and CS at the electrode level.

Results:

Vowel Task Results

F2 confidence drifted monotonically away from the NH reference towards more basal cochlear positions across sessions - opposite to the convergence expected from adaptation. One vowel ([3:]) showed independent F1 movement consistent with genuine electrode-to-category learning. The F1/F2 reliance ratio showed no temporal trend, with the Day 6 spectral weighting profile already characteristic of Post-LCI users. At the electrode level, F2 electrodes with greater absolute FTPM, narrower CS, and lower NR attracted more confidence early on, with this pattern reversing gradually over time.

Chord Task Results

FTPM-induced distortion sensitivity was present at Days 6 and 20 but had disappeared by Day 41, giving way to electrode-level FTPM-driven preference by Day 134. NR facilitated preference formation in the early sessions, whilst FTPM emerged as the dominant organisational factor in the extended phase. Across both tasks, the period between Days 20 and 41 marked a qualitative transition from acoustic template matching to electrode-level organisation, characterising a two-phase model of early CI adaptation.

Presented in Chapter 5.

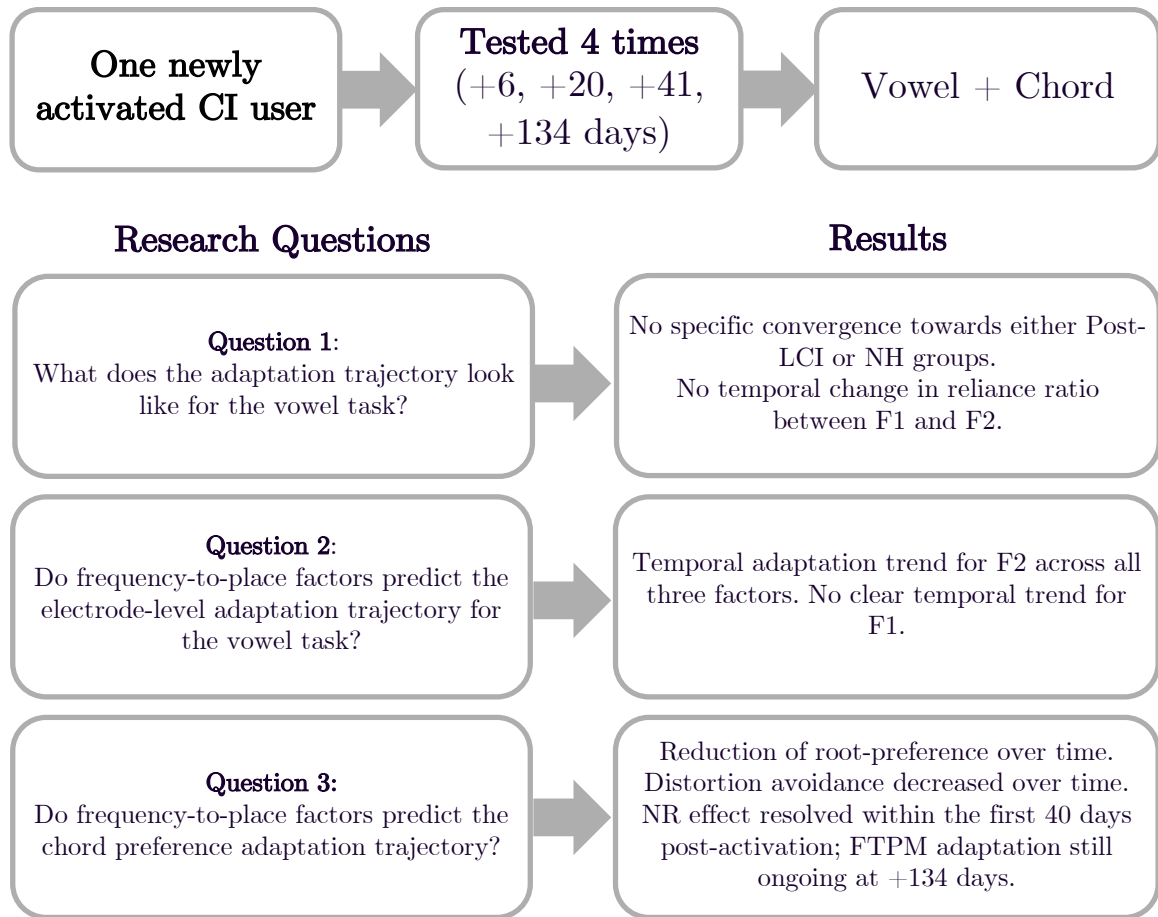


Figure 1.13: Summary of the Chapter 5 study.

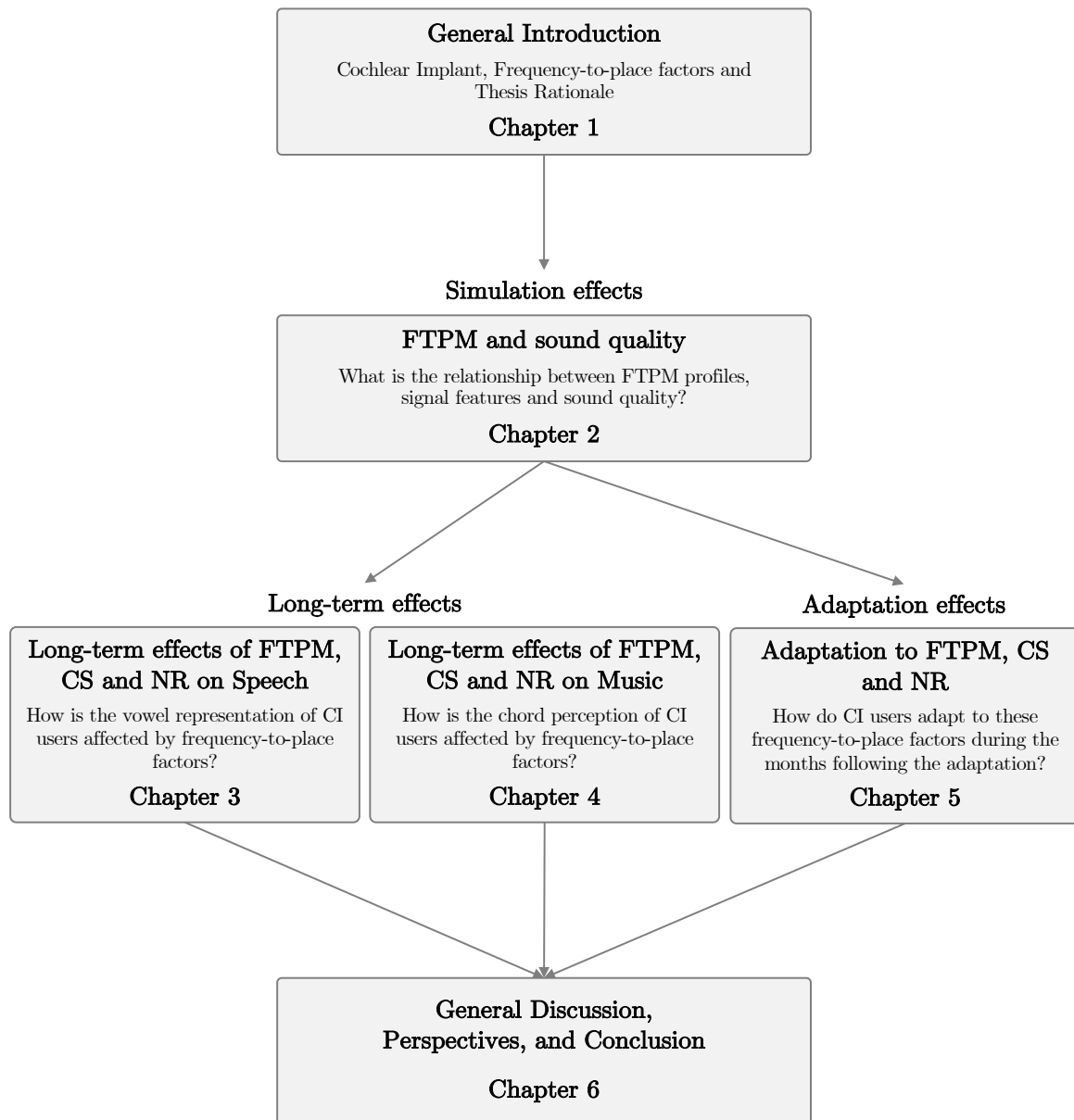


Figure 1.14: Flow diagram of thesis organisation.

Chapter 2

FTPM and Sound Quality

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Preface¹

This chapter is an adaptation of the paper published in the *Journal of the Acoustical Society of America*, under the title “**Effect of frequency-to-place mismatch on speech and music sound quality in acoustic cochlear implant simulation**”.

The original ambition of this work was to design an experiment capable of probing the degree of FTPM - or, more precisely, the degree of adaptation to FTPM - in individual CI users. The idea was to present a frequency-shifted version of a signal in order to compensate acoustically for the mismatch, and to establish a relationship between the optimal compensatory shift and the individual’s residual FTPM. We aimed to use ecologically relevant signals such as continuous speech and music to make the task as meaningful as possible. Three main challenges were identified during piloting, each of which shaped the design of the final study.

Simulating the compensatory shift. Two options were considered for the frequency shift: linear and non-linear. A linear shift is straightforward to implement and easy to optimise, as it introduces only a single free parameter. However, it inevitably reduces gain at the spectral extremes, which could bias sound quality judgements independently of FTPM. A non-linear shift is more principled - FTPM profiles are rarely linear, with greater mismatch typically present at the apex and base - but it introduces a large combinatorial search space and, even with sophisticated algorithms, risks adding undesired distortions.

Tracking the best sound quality. Several methods exist for evaluating sound quality. A Likert scale is simple to administer but produces highly variable ratings and complicates inter-stimulus comparisons. A paired-comparison approach is intuitive, though the number of required comparisons grows rapidly with the number of stimuli; this can be partly mitigated through Bayesian post-hoc analysis. The MUSHRA paradigm, which presents a reference alongside multiple test stimuli on a continuous scale, is widely used in hearing science and CI research and offers the most robust framework for detecting intermediate levels of quality degradation.

Selecting the ideal signal. Both speech and music are spectrally variable over time, making it difficult to identify which part of the signal drives any given quality judgement. Speech is somewhat more tractable, as its voice-frequency content is constrained within a relatively narrow range; music, by contrast, carries information across the full spectrum, may include lyrics, and varies considerably from moment to moment.

Piloting across these dimensions revealed large within- and between-subject variability, and the relationship between FTPM, sound quality, and signal type could not be reliably characterised. Rather than pursuing the original compensatory-shift approach, we therefore focused the study on a more controlled question: **how does the degree and distribution of FTPM affect perceived sound quality across different signal types?** This became the central aim of the work presented in this chapter.

¹Each empirical chapter of this thesis opens with a Preface contextualising the study within the thesis, and closes with a Postface section reflecting on the findings and their implications for subsequent chapters.

Effect of frequency-to-place mismatch on speech and music sound quality in acoustic cochlear implant simulation

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Abstract

Sound quality perception for cochlear implant (CI) users has become increasingly important. While many CI users achieve near-normal speech recognition in quiet, they often report poor sound quality, particularly for music. One factor contributing to this degradation is frequency-to-place mismatch (FTPM), which occurs when electrode positions do not align with the cochlea's characteristic frequency map. This study aimed to better understand the impact of FTPM on sound quality in CI simulations across different signal types and configurations. Twenty-three normal-hearing participants were tested (online or onsite) using an adapted MUSHRA paradigm. Different FTPM configurations were simulated with a noise vocoder to assess their influence on speech and music sound quality. Results showed that greater FTPM caused noticeable degradation, especially when lower frequencies were affected. Variability in FTPM across electrodes also significantly reduced perceived quality. Furthermore, the impact of FTPM depended on the type of stimulus, with speech and music showing distinct sensitivity patterns. Online assessments closely matched onsite results, confirming the reliability of remote testing. Together, these findings clarify why sound quality perception differs between CI users and contexts, and highlight new opportunities to develop strategies for alleviating mismatch effects in CIs.

2.1 Introduction

Cochlear implants (CIs) are auditory prostheses that allow individuals with profound deafness to regain a level of speech understanding close to that of normal-hearing (NH) listeners in quiet environments (Firszt et al., 2004; Wilson and Dorman, 2007). However, CI users often perform worse than their NH peers in complex tasks such as speech understanding in noise (Fu and Galvin, 2003), sound quality perception (Lassaletta et al., 2008; Roy et al., 2012), sound localisation (Litovsky, 2011; Litovsky et al., 2009), or music perception (McDermott, 2004). While some of these performance deficits can be attributed to the limitations of CI technology, which degrades and restricts the proper transmission of both spectral and temporal cues, substantial variability exists among CI users in these tasks (Blamey et al., 2013a; Holden et al., 2013a; Lazard et al., 2012), suggesting that individual factors additionally contribute to these degradations.

FTPM is one factor suspected to contribute to this performance variability (Lands-

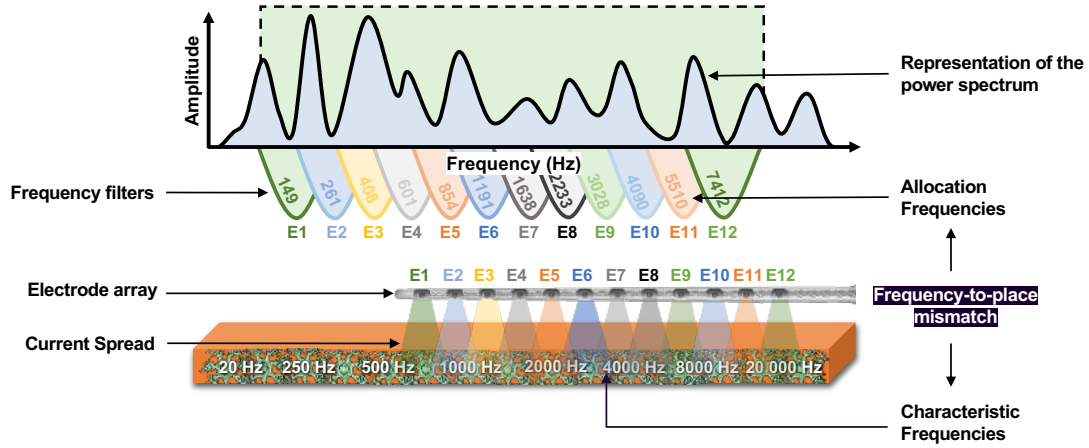


Figure 2.1: **Representation of the FTPM in CI processing.** In this simplified illustration, sound is segmented using frequency filters and the temporal envelope cues in each region extracted. Twelve frequency filters are shown here, each defined by its centre frequency and associated with one electrode. The extracted amplitude is used to modulate the stimulation current of the corresponding electrode, with the goal of stimulating a specific location in the cochlea and the auditory neurons. This figure shows how the incomplete insertion of the electrode array results in a mismatch between the frequency information (allocation frequencies) extracted from the sound and the frequency of the cochlear place stimulated (characteristic frequency).

berger et al., 2015; Yang et al., 2022). FTPM refers to the difference between the frequency information delivered by a given electrode and the frequency typically processed at the cochlear location stimulated by that electrode - the characteristic frequency. This mismatch arises from the interaction between the electrode array’s insertion depth and standard CI mapping strategies: most electrode arrays are not fully inserted into the cochlea, leading to incomplete coverage of the apical region, where low-frequency information is normally processed. To address this, conventional mapping strategies distribute frequency information along the electrode array in a logarithmic fashion rather than aligning it precisely with the cochlear frequency-place map (see Figure 2.1). This approach preserves transmission of a broad frequency range, including low frequencies, but imposes a new pattern of spectral stimulation to which the auditory pathways must adapt.

Currently, there is no clear consensus on the effect of FTPM on CI performance. Some studies report that a greater degree of FTPM increases the time required from device activation to achieve acceptable speech understanding in noise, particularly in the early months following implantation, but that this effect disappears beyond that period (Mertens et al., 2022; Venail et al., 2015b). For example, Mertens et al. (2022) found a significant negative correlation between FTPM and speech perception in noise at 6 months post-activation in 39 adult CI recipients, though this effect was no longer significant at 12 months. The potential benefits of alleviating FTPM through anatomy-based fitting remain unclear for speech recognition, with studies reporting contradictory outcomes. On one hand, one study in 14 adult participants found that image-guided fitting did not improve speech intelligibility on average and was even associated with poorer outcomes in most participants (Lambriks et al., 2023a). Similarly, a longitudinal study of 8 experienced adult CI users found that anatomy-based fitting led to degraded vowel and consonant perception over a 6-month period (Wangchuk et al., 2024). On the other hand, some

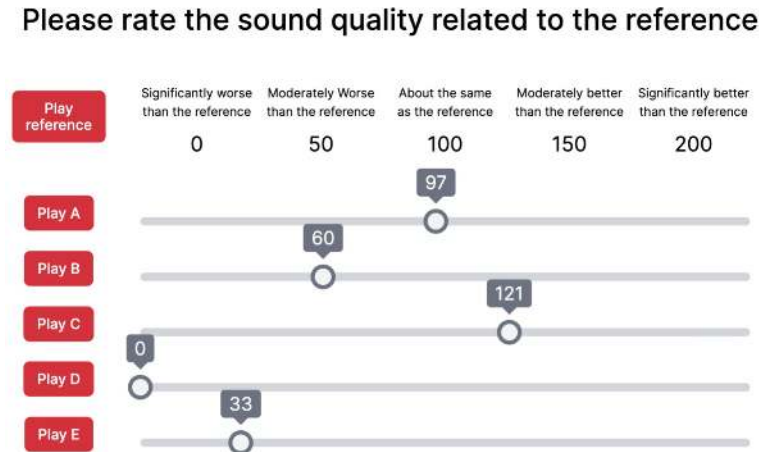


Figure 2.2: **The adapted MUSHRA interface used during the experiments.** Participants were asked to listen to the reference (‘Play Reference’) and then to the different sounds (here labelled from A to E). The goal was to quantify the degree of difference of each sound relative to the reference. This difference could be rated as either a degradation (between 0 and 100) or an improvement (between 100 and 200). If participants considered that no difference was noticeable, they could rate the sound quality as similar (100).

studies report detrimental effects of FTPM on speech perception (DeFreese et al., 2024), implying that alleviating the mismatch should be beneficial. Most studies, however, have not found a significant effect of FTPM on speech recognition (Sturm et al., 2024; Zanetti et al., 2021).

For other aspects of perception, such as music-related tasks, the detrimental effect of FTPM appears more consistent (Creff et al., 2024; Lassaletta et al., 2023; Sturm et al., 2024), and anatomy-based fittings are generally preferred by CI users in terms of sound quality (Canfarotta et al., 2020a; Creff et al., 2024; Lassaletta et al., 2023). For example, Creff et al. (2024) found that 92% of participants chose to retain the tonotopic map after the study period, and Sturm et al. (2024) reported that greater FTPM was associated with lower CI-specific quality of life scores.

These different findings suggest that speech recognition tests may be limited in their ability to capture the effect of FTPM on perception, or that auditory adaptation processes are sufficient to overcome the spectral distortion after a sufficient adaptation period. The impact of FTPM on sound quality and music enjoyment may, however, be greater. Sound quality refers here to the overall subjective impression of a sound, encompassing attributes such as timbral naturalness and pitch accuracy. FTPM is expected to affect sound quality by shifting spectral information to incorrect cochlear places, thereby altering the perceived frequency organisation of sounds. This degradation may occur even when speech intelligibility remains adequate, since speech perception can rely on temporal envelope cues that are relatively unaffected by place mismatch.

The main goal of the present study was therefore to clarify the relationship between FTPM and sound quality. To this end, different FTPM profiles were simulated using a noise vocoder (Williges and Jürgens, 2019) and presented to 23 NH listeners in an adapted version of the MUSHRA paradigm (ITU-R – Radiocommunication Sector of ITU, 2014). The MUSHRA paradigm is well suited to this purpose, as it allows multiple stimuli

to be compared directly and is sensitive to intermediate levels of quality degradation. Participants rated the sound quality of different FTPM profiles relative to a vocoded reference without FTPM, and were free to rate a stimulus as either worse or better than that reference.

Several subsidiary goals were also pursued. The first was to quantify how the effect of FTPM on sound quality depended on which electrodes were affected, by testing three configurations: realistic full-array FTPM profiles, FTPM restricted to apical electrodes (120-579 Hz), and FTPM restricted to basal electrodes (1624-7410 Hz). The second was to determine whether the relationship between FTPM and sound quality varied by signal type, using speech and music stimuli. The third was to identify FTPM profile characteristics and acoustic features that predicted sound quality ratings. The fourth was to evaluate the adapted MUSHRA interface for online use by comparing response quality between online and onsite testing conditions.

2.2 Methods

2.2.1 Participants

The study received ethical approval from the Cambridge Psychology Research Ethics Committee (application number PRE.2019.093). Twenty-three students (12 females, 11 males) aged between 19 and 30 years (mean 26.9, SD = 2.8, median 27) were recruited via online advertisement at the University of Cambridge. Eleven participants were tested online and 12 were tested onsite.

All participants had normal hearing. For online participants, normal hearing was self-reported using the Hearing Handicap Inventory for the Elderly Screening (HHIES) (Lichtenstein et al., 1988). This questionnaire comprises 10 questions, each describing a listening situation to which participants respond as difficult (4 points), sometimes difficult (2 points), or not difficult (0 points). Points are summed to yield a score ranging from 0 (no handicap) to 40 (maximum handicap). Only participants with scores between 0 and 8 were included; scores for the online participants ranged from 0 to 4 (mean = 0.50, SD = 1.41). For onsite participants, normal hearing was confirmed by audiometric screening at ≤ 20 dB HL from 0.25 to 8 kHz (mean = 5.73 dB HL, SD = 5.16 dB HL); all thresholds were required to be within range for inclusion.

2.2.2 MUSHRA Paradigm

An adapted MUSHRA paradigm was used to assess the effect of FTPM on sound quality. In its original version, the MUSHRA paradigm limits judgements to sound quality degradation, presenting a scale from 0 (significant degradation) to 100 (no degradation). In the present study, we aimed to avoid guiding ratings towards a systematic equivalence or degradation, and therefore extended the scale beyond 100 up to 200.

Specifically, a score of 100 indicated that the participant considered the sound quality equivalent to the reference. Scores between 0 and 100 reflected a degradation relative to the reference, and scores between 100 and 200 indicated that the participant judged the sound quality as better than the reference. Category labels were added along the

scale to assist ratings: 0 corresponded to “significantly worse than the reference”, 50 to “moderately worse”, 100 to “similar to the reference”, 150 to “moderately better”, and 200 to “significantly better” (see Figure 2.2).

During the MUSHRA test, participants evaluated the sound quality of different FTPM profiles relative to a vocoded but non-shifted reference. In addition to the FTPM-simulated sounds, the block included a hidden reference (a sound identical to the reference) and an anchor (a sound with a large supposed degradation, corresponding here to a linear FTPM of -1.5 octaves; see Section 2.2.5 for further details).

2.2.3 Sound Selection

Two speech materials were selected from the Quality Assessment Questionnaire (QAQ) database of Advanced Bionics, consisting of stories read aloud by a female and a male speaker respectively. Five music excerpts were selected on the basis of signal complexity (e.g., number of instruments), genre, vocalist gender, and release period, with familiarity ensured by restricting the selection to well-known songs from the Billboard 100 charts spanning 1960 to 2020.

Paired original and instrumental versions of each song were obtained to allow direct quality comparisons. Music quality was estimated using OpenL3 (Cramer et al., 2019), an open-source deep audio embedding model based on self-supervised L3-Net, from which a proxy Mean Opinion Score (MOS) was computed as the arithmetic mean of human quality ratings on a 1-to-5 scale. Comparing MOS between original and instrumental versions yielded an average difference of 0.045 (SD = 0.049), indicating only a minor perceived degradation in instrumental tracks. Steady excerpts were also verified in terms of amplitude and complexity prior to inclusion.

In summary, two speech materials, five songs, and their instrumental equivalents were presented to participants, for a total of twelve sounds (see Table 2.3 for a description of the material).

2.2.4 Sound Processing

Prior to vocoding, all sounds were processed in Audacity. The left and right channels were averaged to mono, a 500 ms fade-in and fade-out were applied at the beginning and end of each sound, and each sound was normalised to the range $[-1, 1]$ to prevent clipping.

2.2.5 FTPM Simulations

Realistic FTPM configurations were simulated using profiles drawn from the database described in Canfarotta et al. (2020a). Specifically, profiles were collected with the FlexSOFT standard electrode array (MED-EL), and only fully inserted arrays were included, yielding 44 FTPM profiles in total.

ID	Song	Artist	Year	Tempo	Gender	Type	Instruments
A	Speech Female		–	–	F	Speech	–
B	Speech Male		–	–	M	Speech	–
C D	Back to Black	Amy Winehouse	2006	123	F	Instrumental Original	D, B, P, V, H
E F	Dancing Queen	ABBA	1976	100	F	Instrumental Original	D, B, P/K, S, G
G H	Hung Up	Madonna	2005	125	F	Instrumental Original	Sy, D, B, G, S
I J	Let It Be	The Beatles	1970	71	M	Instrumental Original	P, D
K L	Think	Aretha Franklin	1968	125	F	Instrumental Original	P, D, B

Figure 2.3: **Overview of selected songs, speech samples, and instrumental tracks used in the analysis.** Gender refers to the gender of the main vocalist (F = Female, M = Male). Instrument codes: D = Drums, B = Bass, P = Piano, K = Keyboard, V = Violins, H = Horns, S = Strings, G = Guitar, Sy = Synthesisers.

Condition 1

The goal of Condition 1 was to investigate the effect of realistic FTPM profiles on sound quality. Five representative profiles were selected from the 44 available; the selection process is summarised in Figure 2.4. The selection was guided by MUSHRA recommendations (ITU-R – Radiocommunication Sector of ITU, 2014), which suggest that test stimuli should span quality degradation judgements between 20 and 80, with the reference scoring 100 and the anchor scoring 0. To operationalise this, the 44 profiles were summarised by taking the mean FTPM across the 12 electrodes and normalising these means to a 0–100 scale. Two assumptions were made: that quality judgements would vary linearly with FTPM level, and that FTPM would induce quality degradation. Based on these assumptions, the profiles with mean values closest to 20%, 35%, 50%, 65%, and 80% were selected.

Each MUSHRA block in Condition 1 therefore presented seven sounds: the reference (no FTPM), five FTPM profiles (-0.26 , -0.43 , -0.63 , -0.81 , and -0.98 octave shift), and an anchor. The anchor was initially set at 100% but was adjusted to 120% (linear FTPM of -1.5 octave shift) following pilot testing, in which it was frequently confused with the 80% profile (see Figure 2.4A).

Condition 2

The goal of Condition 2 was to assess the effect of FTPM restricted to the most apical electrodes, which typically exhibit greater mismatch than middle electrodes, with a knee point at approximately 650 Hz (Landsberger et al., 2015). FTPM variations were therefore applied only to the four electrodes with allocation frequencies below this threshold. The reference was set to the mean of the 44 FTPM profiles. Four test profiles were constructed by subtracting 0.5, 1, 1.5, and 2 standard deviations from the mean

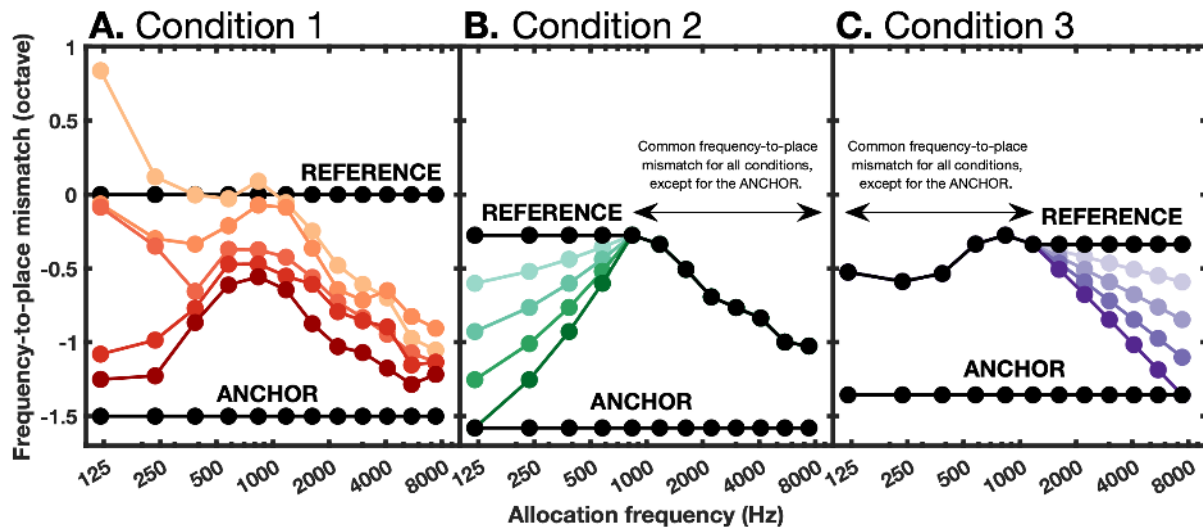


Figure 2.4: **FTPM profiles simulated in the three conditions.** Each panel shows the different levels of mismatch presented in each condition. Each point represents one electrode; the x -axis shows its allocation frequency (centre frequency of the filter) and the y -axis shows the degree of mismatch applied in octaves. A negative value indicates a shift to higher frequencies (output frequency is higher than input frequency); a positive value indicates a shift to lower frequencies. Panel A: realistic FTPM profiles (Condition 1). Panel B: apical-restricted profiles (Condition 2). Panel C: basal-restricted profiles (Condition 3).

FTPM of the most apical electrode, with intermediate values interpolated to the fifth most apical electrode. The anchor was created by applying the greatest FTPM linearly across the apical electrodes.

Each MUSHRA block in Condition 2 presented six sounds: the reference, four FTPM profiles (-0.48 , -0.68 , -0.89 , and -1.09 octave shift), and an anchor (-1.58 octave shift) (see Figure 2.4B).

Condition 3

The goal of Condition 3 was to assess the effect of FTPM restricted to the most basal electrodes, which also exhibit greater mismatch than middle electrodes. The same approach as Condition 2 was applied, with variations from the sixth most basal electrode down to the most basal electrode. The reference was based on the mean FTPM of the seventh electrode, and standard deviation fractions were subtracted progressively from this value. Each MUSHRA block in Condition 3 presented six sounds: the reference, four FTPM profiles, and an anchor (see Figure 2.4C).

2.2.6 Vocoder

A noise vocoder (Bräcker et al., 2009) was employed to simulate electric hearing. This vocoder replicates the signal processing of CI devices using a continuous interleaved sampling (CIS)-like strategy (Wilson et al., 1991), transforming an audio signal into 12 frequency channels corresponding to the electrodes of MED-EL devices. A third-order

gammatone filterbank decomposes the audio signal into these channels, each with a bandwidth of one equivalent rectangular bandwidth (ERB) and centre frequencies ranging from 120 to 7410 Hz.

The output of each channel is sampled at 4583 Hz to produce an envelope-weighted, pulse train-like signal. To simulate the spatial spread of the electric field in the perilymph, each pulse is multiplied by a two-sided exponentially decaying function and transferred to adjacent channels. An auralization step follows, in which the signal from each channel is filtered using two third-order gammatone filterbanks with slightly different centre frequencies to simulate its transfer to the corresponding position along the cochlear partition. The second gammatone filterbank reduces sideband modulations, and the frequency channels are summed to produce a mono audio signal. Finally, RMS scaling is applied to compensate for level changes due to pulsatile sampling or bandpass filtering. The resulting output signal comprises narrow-bandpass filtered carriers with stochastic, noise-like fine structure, spectrally smeared envelopes, and carriers slightly shifted in frequency according to the simulated FTPM profile.

2.2.7 Calibration and Presentation Level

For online participants, calibration was performed by each participant independently by adjusting their computer volume to a comfortable level before starting each condition. They were permitted to adjust the volume if sounds became uncomfortable during the experiment. None of the eleven online participants reported adjusting the volume during testing.

For onsite testing, a calibration was performed to achieve an output level of 65 dB SPL, using a Tektronix MDO3024 Mixed Domain Oscilloscope with headphone sensitivity data to calculate the required voltage. The volume was kept constant throughout the entire onsite testing session.

2.2.8 Procedure

Online testing used the Gorilla Condition Builder platform (www.gorilla.sc). Sounds were converted to MP3 format to meet platform requirements. Sound quality equivalence between the original WAV and MP3 formats was verified using the PEAQ algorithm (Thiede et al., 2000); no differences in quality were found.

For onsite testing, the experiment was conducted in a quiet room. Sounds were presented monaurally to the left ear via a MacBook Pro, an RME Fireface UCX II sound card, and a Sennheiser HD 600 headset, controlled by MATLAB R2023a (The MathWorks Inc., 2023).

Each participant completed 36 MUSHRA blocks, one per condition per sound (e.g., “Condition 1 – Speech Male”). Each condition was evaluated once per participant. The presentation order of conditions was randomised for each participant, as was the presentation order of sounds within each MUSHRA block. Online participants were permitted to pause after each block and resume later, with no restriction on the delay between sessions. Onsite participants completed the experiment in a single session, with breaks permitted between blocks.

2.2.9 Data Post-screening

Post-screening of MUSHRA responses was based on the scoring of the hidden reference and the anchor. A block was rejected if the hidden reference was not scored between 90 and 110, or if the anchor was not assigned the lowest score in the block. The standard MUSHRA post-screening criterion involves rejecting all data from participants who rate the hidden reference below 90 for more than 15% of test items (ITU-R – Radiocommunication Sector of ITU, 2014); however, we adopted a block-level rejection strategy instead. This decision was motivated by the limited sample size ($N = 23$), for which complete participant rejection would have caused substantial data loss and reduced statistical power. Block-level rejection also accommodates temporary lapses in attention or fatigue without penalising the participant’s entire dataset, thereby preserving the maximum amount of reliable data.

After post-screening, scores were re-scaled to the range $[-100, 100]$ for visualisation. Within each MUSHRA block, sounds were then converted to rank positions, with rank 1 assigned to the sound with the best score and higher ranks assigned to progressively worse scores (up to rank 7 depending on the condition).

2.2.10 Statistical Analysis

MUSHRA responses produce ordinal data that typically violate normality assumptions, as the rating process more closely resembles a ranking procedure than an absolute scale measure (Mendonça and Delikaris-Manias, 2018). Our dataset likewise violated the normality assumption.

Raw MUSHRA scores were therefore converted to within-block ranks (see Section 2.2.9). Rank transformation is particularly appropriate for MUSHRA data as it preserves the relative ordering of stimuli within each trial - the core information provided by listeners - whilst mitigating the influence of individual differences in scale usage and avoiding the assumption of equal scale intervals. This approach aligns with recent recommendations that MUSHRA data, being fundamentally ordinal rather than interval-scaled, should be analysed with methods that do not assume equal distances between scale points (Mendonça and Delikaris-Manias, 2018). Transforming to within-block ranks also reduces the influence of between-subject variability in scale interpretation, focusing instead on the consistent within-trial comparisons that form the basis of the MUSHRA methodology.

These ranks were treated as the response variable in generalised linear mixed-effects models (GLMMs), with participant as a random effect. The distribution of the response variable was evaluated for each model and found to follow a Gamma distribution, which was retained for all statistical analyses.

In the first analysis, two GLMMs were fitted to the full dataset to examine how ‘FTPM Mean’ (the FTPM averaged across the 12 electrodes, used as the primary predictor) interacted with condition (Conditions 1, 2, or 3) and with content type (speech, original music, or instrumental music) respectively. The dataset was then split by condition, and additional GLMMs were fitted to assess how content type interacted with FTPM Mean within each condition. Finally, a separate set of GLMMs was fitted to examine how each of the 12 individual sounds interacted with FTPM Mean. In all models, post-hoc comparisons assessed whether intercepts (the baseline rank at FTPM Mean = 0)

and slopes (the rate of change in rank with FTPM Mean) differed among the relevant levels. p -values from these comparisons were adjusted using the Benjamini–Hochberg false discovery rate (BH–FDR) procedure to control Type I error whilst maintaining statistical power.

In the second analysis, we investigated the contributions of FTPM profile characteristics, original sound features (i.e., extracted before vocoding), and vocoded sound features to the rank outcome. This analysis was restricted to Condition 1. Specific features were extracted for each category (for instance, channel-by-channel FTPM variability or the power spectrum of each channel). Principal component analyses (PCAs) were performed to reduce these features to a subset of principal components (PCs) capturing at least 95% of the variance, yielding two PCs for FTPM profile characteristics, three for original sound features, and five for vocoded sound features. Each PC was tested in a GLMM; only those significant at $p < .05$ were retained. The retained PCs were then combined into a single model, from which non-significant effects were iteratively removed until all fixed effects were statistically significant. This final model identified which aspects of FTPM profile characteristics, original sound features, and vocoded sound features most strongly influenced rank-based quality judgements in Condition 1.

2.3 Results

2.3.1 FTPM and Sound Quality

Figure 2.5 shows the relationship between FTPM Mean and sound quality rankings across the three conditions. Figures 2.6–2.8 present the same relationship for Conditions 1, 2, and 3 respectively, with each sound depicted individually. Each value on the x -axis corresponds to a specific simulated FTPM profile; values are negative by convention to indicate that the FTPM Mean was shifted towards higher frequencies (i.e., the output frequency was higher than the input frequency).

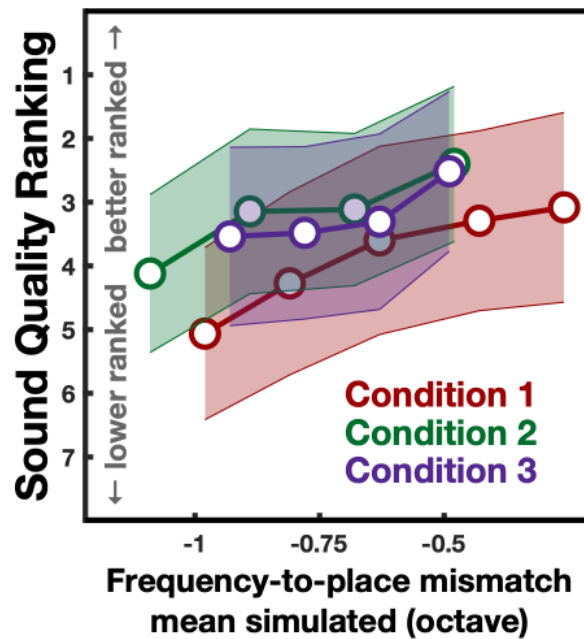


Figure 2.5: **Sound quality rankings as a function of FTPM Mean across conditions.** Sound quality ranking is plotted as a function of the mean electrode FTPM across all sounds. Shaded areas represent standard deviations. Red: Condition 1 (realistic FTPM profiles); green: Condition 2 (FTPM restricted to low-frequency electrodes); purple: Condition 3 (FTPM restricted to high-frequency electrodes).

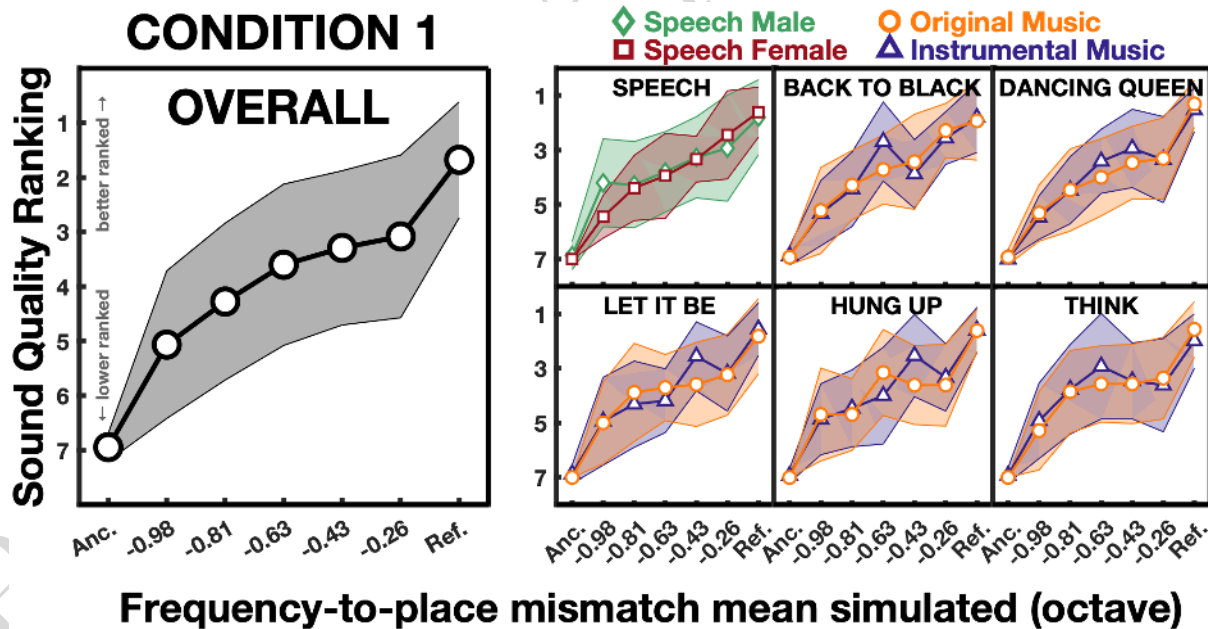


Figure 2.6: **Condition 1: FTPM Mean against sound quality rankings for realistic FTPM profiles.** Shaded areas represent standard deviations. The leftmost panel shows ranking as a function of mean electrode FTPM across all sounds. Right panels show results for each individual stimulus: the top-left panel reports the two speech signals; remaining panels show the music samples.

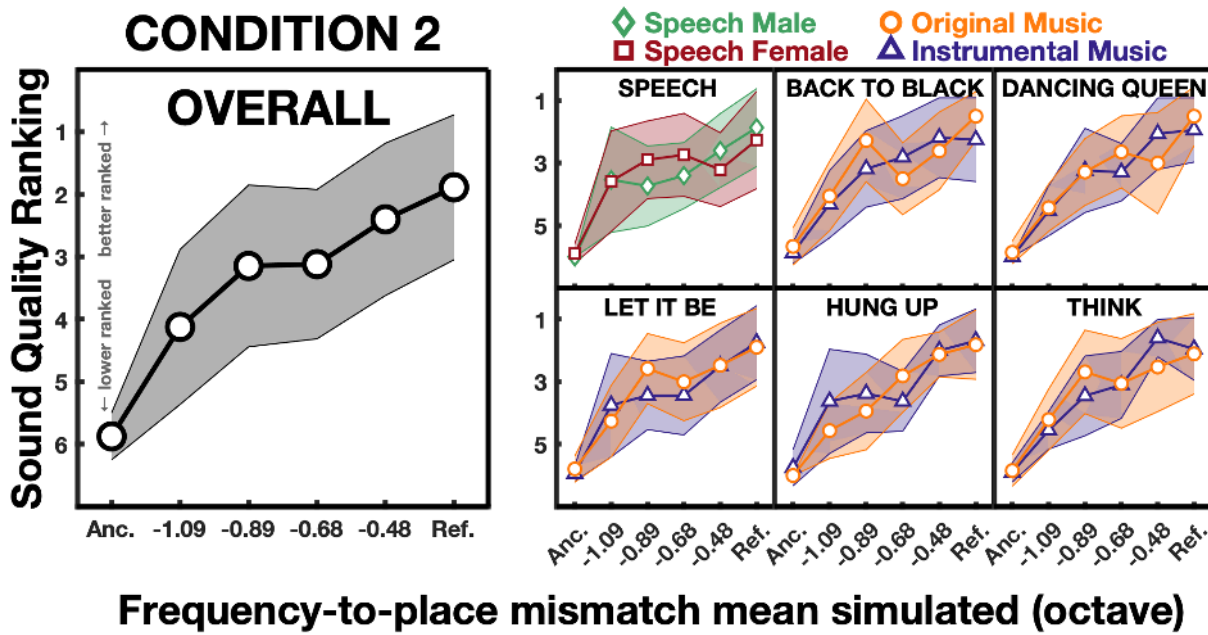


Figure 2.7: Condition 2: FTPM Mean against sound quality rankings for low-frequency FTPM profiles. Same layout as Figure 2.6. The FTPM Mean here corresponds to the mean mismatch across the low-frequency electrodes (120–579 Hz).

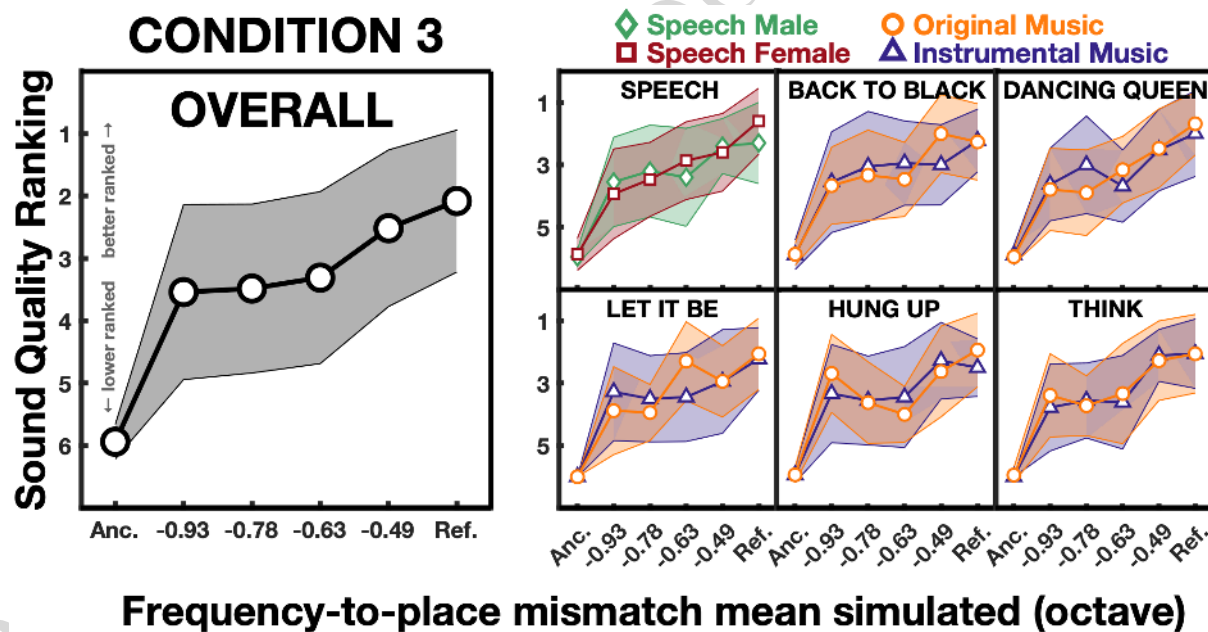


Figure 2.8: Condition 3: FTPM Mean against sound quality rankings for high-frequency FTPM profiles. Same layout as Figure 2.6. The FTPM Mean here corresponds to the mean mismatch across the high-frequency electrodes (1624–7410 Hz).

Relationship Between FTPM and Sound Quality Ranking

Across the eight GLMMs computed, the main effect of FTPM Mean on rank was consistently positive, indicating that higher FTPM Mean values tended to be associated

with higher ranks (i.e., poorer sound quality). However, this effect did not reach statistical significance after correction for multiple comparisons in any model.

Comparison Between Conditions

Differences between the three conditions were examined in Model 1 (Supplementary Table S1); a visual representation is provided in Figure 2.5. None of the interaction terms reached significance after correction, indicating that the influence of FTPM Mean on ranking was similar across all three conditions. There were, however, notable differences in intercept relative to the Condition 1 baseline.

Specifically, both Condition 2 ($p_{\text{adj}} < .001$, estimate = -0.975) and Condition 3 ($p_{\text{adj}} = .046$, estimate = -0.515) had significantly lower intercepts than Condition 1, indicating that at FTPM Mean = 0, participants assigned worse quality rankings in those two conditions than in Condition 1. No significant intercept difference was found between Condition 2 and Condition 3 ($p_{\text{adj}} = .085$).

Differences Between Speech, Original Music, and Instrumental Music

A first model was computed using data from all three conditions together (Model 2) to investigate the effect of stimulus type. No significant differences in intercept or slope were found between stimulus types, indicating that rank did not vary by content at FTPM Mean = 0 and that the rate of change in rank with FTPM Mean was similar across content types.

Three further models were then fitted, each restricted to a single condition (Models 3, 5, and 7). No significant differences in intercepts or slopes were found between stimulus types in Condition 1 (Model 3) or Condition 3 (Model 7). In Condition 2 (Model 5), however, the speech condition significantly differed from the instrumental condition in both intercept ($p_{\text{adj}} = .0005$) and slope ($p_{\text{adj}} = .0003$). Pairwise comparisons confirmed that sound quality degradation was significantly lower for speech than for instrumental music (estimate = 1.67, $p_{\text{adj}} = .0005$) and original music (estimate = 1.19, $p_{\text{adj}} = .0115$). With respect to slopes, the speech condition showed a steeper relationship between FTPM Mean and rank than both instrumental (estimate = 2.14, $p_{\text{adj}} = .0003$) and original music (estimate = 1.49, $p_{\text{adj}} = .0094$), indicating that perceived quality degradation was more strongly driven by apical FTPM for speech than for music.

Differences Between Sounds

One model per condition was fitted with individual sounds as fixed parameters to examine their effect on rank and their interaction with FTPM Mean.

For Condition 1, individual sound coefficients were not significant and no pairwise comparisons of intercepts or slopes survived BH-FDR correction.

For Condition 2, Speech Female showed a significantly elevated intercept relative to baseline (estimate ≈ 2.10 , $p_{\text{adj}} = .020$), indicating lower sound quality degradation at FTPM Mean = 0. Speech Male showed a trend-level elevation (estimate ≈ 1.58 , $p_{\text{adj}} = .084$) that did not reach significance at $\alpha = .05$. After BH-FDR correction, 7 of the

66 intercept pairwise comparisons remained significant ($\sim 10.6\%$) and 9 of the 66 slope comparisons remained significant ($\sim 13.6\%$), with effects dominated by contrasts involving Speech Female, which showed consistently more favourable intercepts and steeper slopes relative to several music conditions.

For Condition 3, no individual sound coefficients were significant and no pairwise comparisons of intercepts or slopes survived correction.

2.3.2 Analysis of FTPM Profiles and Sound Features in Condition 1

A PCA-based analysis was performed to investigate which components of the FTPM profiles, original sound features, and vocoded sound features best predicted sound quality rankings in Condition 1. At the end of the predictor selection process, only the first two PCs derived from FTPM profile characteristics ('PC1-FTPM' and 'PC2-FTPM') were statistically significant. With these two components as explanatory variables, the model yielded an adjusted $R^2 = 0.19$.

'PC1-FTPM' was the primary predictor ($t = -14.35$, $df = 902$, $p < .001$; estimate = -0.46 , $SE = 0.03$), accounting for 85% of the variance in the initial PCA. No single feature dominated its loadings (mean loading = 0.19 , $SD = 0.11$), supporting its interpretation as a comprehensive representation of overall FTPM degree.

'PC2-FTPM' contributed more marginally to the model ($t = 2.85$, $df = 902$, $p = .004$; estimate = 0.27 , $SE = 0.09$), accounting for 10% of the PCA variance with greater variability in its feature loadings (mean = -0.01 , $SD = 0.23$). Its most prominent loadings were associated with inter-electrode FTPM variability (loading = 0.38 for the standard deviation across electrode FTPM values; loading = 0.58 for the standard deviation of adjacent electrode FTPM differences).

2.3.3 Testing Procedure

Results concerning the effect of the testing procedure (online versus onsite) and the extension of the rating scale are shown in Figure 2.9. Scores here refer to the raw ratings given by participants before transformation into ranks.

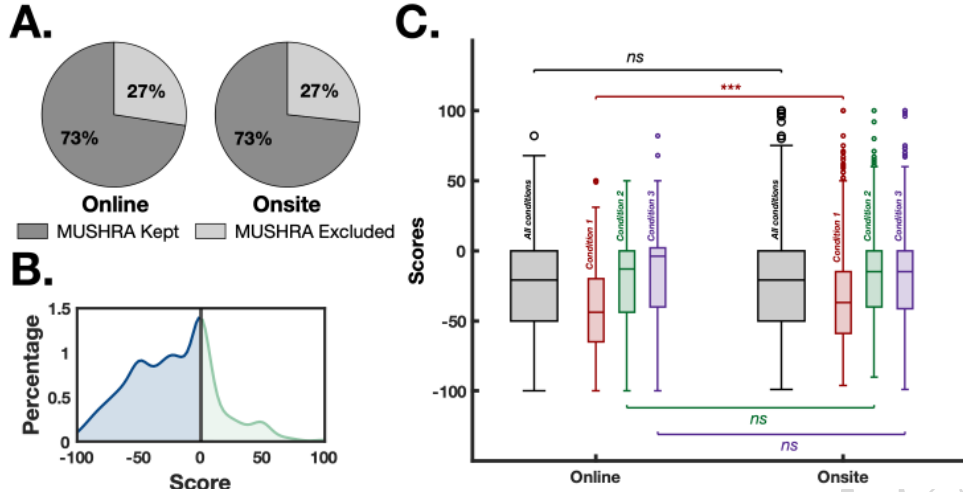


Figure 2.9: **Results for the different test procedures and score distribution over the rating scale.** Panel A: pie charts showing the percentage of MUSHRA responses removed or retained. Panel B: score distribution illustrating the proportion of responses rated as worse than the reference (blue) or better (green). Panel C: boxplots per condition; grey denotes scores pooled across testing modalities, red Condition 1, green Condition 2, purple Condition 3. Outliers are shown as individual dots. Statistical differences between online and onsite conditions are annotated; ‘ns’ = non-significant, ‘***’ = $p < .001$.

Percentage of MUSHRA Blocks Retained

Post-screening excluded blocks in which the hidden reference and/or anchor were incorrectly scored. The percentage of blocks retained was calculated per participant and averaged across the two testing modalities. For online participants, approximately 73% of blocks were retained ($SD = 19.7\%$); for onsite participants, 73% were retained ($SD = 24.1\%$). The difference between modalities was not significant ($t(21) = -0.062$, $p = .952$, 95% CI $[-20.10, 18.95]$).

Scale Adaptation

Extending the rating scale above 100 to allow participants to judge a sound as better than the reference influenced the distribution of responses: approximately 19% of scores fell above 100 (Figure 2.9B).

Online Versus Onsite Testing

Scores averaged across the three conditions were normally distributed for both modalities; individual conditions were not normally distributed. The only significant difference between modalities was found for Condition 1 ($U = 253,700$, $z = 2.94$, $p = .003$, $r = 1.00$); no significant differences were observed for Conditions 2 or 3. Scores close to 200 (indicating a perceived improvement relative to the reference) were more frequent under onsite conditions (0.70% of scores) than online (0.08% of scores).

2.4 Discussion

This study aimed to gain a better understanding of the effect of FTPM on sound quality by examining (i) the effect of FTPM when restricted to specific electrode regions, (ii) the influence of signal type, (iii) the contribution of sound features and FTPM profile characteristics to sound quality rankings, and (iv) the adapted MUSHRA interface and its suitability for online testing.

Condition 1 used realistic full-array FTPM profiles, whilst Conditions 2 and 3 isolated FTPM to apical and basal electrodes respectively. All simulated FTPM configurations were detrimental to sound quality, with greater mismatch leading to greater degradation. This degradation was particularly pronounced in Condition 2, where FTPM was applied to electrodes conveying low frequencies. Although differences were observed between the 12 sounds and their interactions with FTPM Mean, no sound feature adequately explained these differences. Beyond the degree of FTPM, inter-electrode variability in FTPM was additionally associated with degraded sound quality. Participants successfully used the extended rating scale, and online data collection presented both benefits and limitations relative to onsite testing.

2.4.1 Effect of FTPM Degree and Electrode Region on Sound Quality

Higher degrees of simulated FTPM were associated with greater sound quality degradation. This trend has been reported previously: Sturm et al. (2024) used a self-reported questionnaire to evaluate sound quality three months post-implantation and found that CI users with a higher degree of FTPM reported lower perceived sound quality, despite no concurrent effect of FTPM on speech recognition. Roy et al. (2016) indirectly examined the influence of FTPM on sound quality by studying the relationship between electrode array insertion depth and quality ratings, finding that shallower insertions - which are associated with greater FTPM (Landsberger et al., 2015; Venail et al., 2015b) - were associated with lower sound quality.

The three conditions were analysed separately to disentangle the contributions of realistic, apical, and basal FTPM to sound quality. The only statistically significant difference between conditions was in intercept: both Condition 2 and Condition 3 had lower intercepts than Condition 1, indicating that at FTPM Mean = 0, participants assigned poorer quality rankings in those two conditions. One likely explanation is methodological: whilst the FTPM profiles in Condition 1 varied non-linearly across electrodes, those in Conditions 2 and 3 followed a strictly linear degradation. The resulting ranking task was therefore more straightforward in Conditions 2 and 3, which may account for the lower intercepts observed.

The absence of a significant difference between Conditions 2 and 3 may reflect the fact that quality rankings depended not only on the FTPM applied but also on the specific sound signal used. The diversity of sounds may have prevented a homogeneous interaction with the FTPM profiles, a conjecture supported by models comparing individual sounds (Models 4, 6, and 8), in which each sound showed distinct intercept and slope patterns.

This sound-specific effect was particularly pronounced in Condition 2 compared to

Condition 3. The speech condition differed significantly from the other content types in Condition 2 (e.g., speech vs instrumental slope $p_{\text{adj}} = .0003$; original vs speech slope $p_{\text{adj}} = .0094$), and a higher proportion of significant pairwise comparisons was observed in Model 6, suggesting that the impact of FTPM on sound quality was more content- and signal-dependent when mismatch was applied to low-frequency electrodes. Although not fully confirmed by the present results, the critical role of low frequencies in sound quality judgements has been highlighted in earlier work (Arslan et al., 2022; Braida et al., 1979).

2.4.2 Contribution of FTPM Profile Characteristics and Sound Features to Quality

This analysis examined whether a more nuanced characterisation of FTPM profiles, combined with sound features extracted from both original and vocoded signals, could better explain participant rankings. Only two PCs derived from FTPM profile characteristics - conveying information about overall FTPM degree and inter-electrode variability - were statistically significant predictors.

Sound features did not significantly explain participant responses. For features from the original sounds, this is likely attributable to the small number of stimuli tested ($n = 12$), which is insufficient to identify consistent relationships with ranking strategies. For features derived from vocoded sounds, the MUSHRA paradigm itself may have limited explanatory power: within each block, participants rated multiple sounds simultaneously, making it difficult to attribute rankings to individual acoustic features considered in isolation.

‘PC1-FTPM’, which accounted for 85% of the PCA variance and captured a small contribution from all initial features, was the dominant predictor (effect size = -0.46). This supports the use of FTPM Mean as the primary summary predictor in the other analyses. ‘PC2-FTPM’ (10% of PCA variance; effect size = 0.27) was associated with a more limited subset of features, primarily the variability between electrode FTPM values. Greater inter-electrode FTPM variability was associated with worse quality rankings. One possible explanation is that such variability disrupts the harmonic organisation of sounds, degrading the perceived pitch and timbral coherence of both speech and music. This interpretation is consistent with CI studies showing that FTPM has a specific detrimental effect on music tasks, which rely on harmonicity, but not on speech recognition (Creff et al., 2024).

2.4.3 MUSHRA Scale Adaptation and Testing Modality

Scale Expansion

In the present study, the rating scale was extended beyond 100 to allow participants to rate a sound as better than the reference. In its original version, the MUSHRA paradigm is designed to evaluate intermediate quality levels between a reference (the best quality signal) and a highly degraded anchor. This assumption was not applicable here: it would have been scientifically inappropriate to presuppose that FTPM could only degrade and never improve perceived quality relative to the vocoded reference.

As in a previous study (Mo et al., 2022), the scale was therefore extended to 200. Approximately 19% of judgements were rated above 100, confirming that some participants perceived certain FTPM profiles as improving sound quality relative to the reference. Such scale extension is appropriate when the direction of the effect of a sound modification is uncertain, as it avoids biasing participants towards degradation ratings.

Online and Onsite Data Collection

The percentage of valid MUSHRA blocks was comparable across the two modalities (73% retained in both cases). Scores in Condition 1 were significantly higher for onsite participants than online participants ($p < .001$), with a trend towards significance in Condition 3 ($p < .05$); no significant difference was found for Condition 2 ($p = .749$).

The percentage of extreme outliers close to 200 (indicating substantially better sound quality than the reference) was higher for onsite participants (0.70%) than online (0.08%). This difference is most plausibly explained by the listening devices used. Onsite participants used Sennheiser HD 650 headphones (frequency response: 10–39,500 Hz), whilst most online participants used consumer devices such as AirPods (frequency response: 20–20,000 Hz). Onsite participants may have perceived the additional high-frequency information conveyed even by distorted FTPM profiles as improving sound quality compared to the reference.

The reason some online participants also rated distorted sounds above the reference is less clear. One possible explanation is that FTPM may have reduced spectral overlap between adjacent channels by broadening channel bandwidths, thereby inadvertently improving access to certain spectral features or mitigating some perceptual effects of the vocoder.

2.4.4 Clinical Implications

Better Characterising the Perceptual Effect of FTPM

The majority of studies investigating the effect of FTPM on CI performance have used speech-in-noise tasks as their primary outcome measure, and most report no significant effect of FTPM after an adaptation period. This suggests that, once adaptation is established, speech recognition tests may be insensitive to the residual perceptual consequences of FTPM. When studies include additional outcome measures - such as sound quality or music-related tasks - a detrimental effect of FTPM is more consistently found. The higher sensitivity of sound quality measures to FTPM-related degradation was one of the primary motivations for the present study, and the results support the use of sound quality assessment as a more informative tool for characterising the perceptual effects of FTPM.

One important limitation highlighted here is that the impact of FTPM on sound quality depends substantially on the type of signal used, and the present results did not allow this dependency to be fully characterised. This may partly reflect the constraints of the MUSHRA paradigm, which is cognitively demanding and requires training to produce consistent results (Zieliński et al., 2007). The effort required may sometimes reduce the task to a simple ranking exercise, resulting in a loss of information relative to the full continuous scale. Alternative response formats - such as Likert scales or paired-

comparisons - may be easier and more intuitive for participants, particularly for CI users, and could yield more reliable individual estimates.

New Directions for FTPM Alleviation Strategies

The finding that inter-electrode FTPM variability - not only the overall degree of mismatch - contributes to sound quality degradation has direct implications for CI fitting strategies. Current anatomy-based fitting approaches aim to align electrode allocation frequencies with individual cochlear geometry, reducing the overall degree of FTPM and limiting the adaptation gap between previous stimulation and the new tonotopic map. However, this strategy does not explicitly constrain the harmonic organisation of the allocation frequencies across the array, and may limit the transmitted frequency range for users with shallow electrode insertions.

The present results suggest an alternative strategy: reducing inter-electrode FTPM variability whilst preserving the harmonic relationships between adjacent allocation frequencies. Such an approach would have a smaller impact on the transmitted frequency range and would maintain access to low-frequency information - particularly beneficial for CI users with shallow electrode insertions.

Assessment of Sound Quality in CI users

The MUSHRA paradigm, whilst effective in controlled laboratory settings with trained normal-hearing participants, may not be the most appropriate method for assessing sound quality preferences in CI users. The procedure is cognitively demanding and requires a degree of training to yield consistent results (Zieliński et al., 2007). Furthermore, the concept of a reference signal is inherently problematic when evaluating FTPM: the perceptual distortion introduced by mismatch arises relative to each user's internal auditory representation of a sound, not relative to any externally provided reference signal. The use of MUSHRA - at least in its standard reference-anchored form - for this type of assessment warrants critical consideration. Alternative methods such as paired comparisons or rating scales, which do not require an external reference, may be better suited to capturing the perceptual consequences of FTPM in clinical populations.

2.5 Conclusion

The current study provides evidence through CI simulations with NH listeners that FTPM manipulations degrade perceived sound quality relative to a reference without FTPM. Such degradation was particularly marked when FTPM was simulated in electrodes conveying low-frequency information, and its extent depended on the type of signal used. Inter-electrode variability in FTPM was additionally associated with degraded sound quality, independently of overall mismatch degree. This latter finding offers a new perspective on the perceptual consequences of FTPM in CI users, and may motivate fitting strategies that reduce inter-electrode variability - thereby restoring harmonic organisation across the array - alongside strategies that reduce the overall degree of mismatch.

Postface

The study presented in this chapter provided direct evidence that FTPM affects both speech and music quality. That effect varied according to the degree of FTPM and the sound. Moreover, we found that that inter-electrode variability in FTPM has a detrimental effect on sound quality. This latter finding directly motivated the Analysis 1 performed in Chapter 5 of this thesis.

However, the original aim of this study was to identify which sound types show the strongest interaction with FTPM - that is, to determine in which acoustic contexts FTPM is most perceptually disruptive. The absence of a clear relationship between sound features, sound quality, and FTPM can be attributed to several methodological factors.

First, the temporal variability within each sound was a confound: because quality judgements were made across an entire stimulus rather than at a specific temporal moment, it is unclear which part of the sound drove the response, making it difficult to link judgements to specific acoustic features.

Second, the analysis was based on only 12 sounds and restricted to the Condition 1 (*i.e.*, realistic FTPM profiles), which was too limited to draw robust conclusions about the interaction between sound features and FTPM.

Third, the MUSHRA paradigm introduced important constraints: each MUSHRA block is largely independent of the others, making cross-MUSHRA comparisons difficult, and the length of the task limited the number of stimuli and the number of participants tested.

Several alternatives can be proposed to address these limitations.

A 2AFC design would be significantly faster, more reliable, easier to analyse, and less cognitively demanding. As demonstrated in Chapter 5, it also allows the application of Bayesian posterior analysis, which can refine the output and handle missing comparisons gracefully. Extending the number of sounds tested and the number of FTPM profiles is another priority: an online experiment in which FTPM profiles are randomly assigned to many listeners would considerably increase the number of comparisons and the statistical power available. Finally, using sounds with a repetitive or stationary temporal structure - rather than familiar songs, whose selection was somewhat arbitrary - would ensure that quality judgements are made on a consistent acoustic pattern, reducing within-stimulus variability.

One analysis is not reported as part of this study because it was not conclusive: to analyse the relationship between acoustic features, FTPM profiles and sound quality, we perform a machine learning analysis with the aim to predict the sound having the acoustical features interacting the best with FTPM.

With the improvements outlined above - more sounds, more FTPM profiles, a faster paradigm - such an analysis could be considerably more powerful. It could lead to a model capable of identifying the acoustic environments in which FTPM has the greatest impact on sound quality, providing a precise characterisation of the listening difficulties that CI users face in everyday life and potentially informing condition-specific mapping interventions.

Chapter 3

Long-term effects of FTPM, CS and NR on speech

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Preface

This chapter presents a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **“Estimation of internal representation of vowels in experienced cochlear implant users and its relationship with frequency-to-place factors”**.

In this preface, we provide an overview of the motivations that led to the test presented here and the pilots undergone to explore the options related to the test.

The primary motivation for this work was to develop a clinical speech-based task capable of capturing the long-term perceptual consequences of frequency-to-place factors in CI users. Such a task could ultimately be used to identify electrodes for which re-mapping might be beneficial. As discussed in the introduction, persistent effects of these factors (*e.g.*, in experienced CI users) are more likely to manifest as degraded sound quality than as reduced speech recognition. We therefore conducted several studies to evaluate sound quality. Three major concerns guided the design of each pilot:

First, we wanted to use **realistic signals** that could adequately represent potential sound quality difficulties faced by CI users in their daily life.

Second, we wanted to develop a task which would be **suitable for clinical use**. In other words, it needed to be time efficient and easily understood by participants.

Finally, we wanted to be able to relate outcomes to frequency-to-place factors so that the perceptual consequences would ultimately be alleviated or managed through re-mapping. In other words, we wanted to **precisely control which electrode was processing the information**.

These three points are reported in the pilots presented below. They offer an overview of how we developed the final experiment presented in this chapter.

Pilot 1 - Non-linear FTPM compensation

Approach: To present non-linear frequency shifted versions of realistic speech signals with a non-linear shift designed to exactly compensate each participant's theoretical FTPM profile. Paired comparisons were used with an evolutionary algorithm to converge on the optimal non-linear frequency compensation pattern.

Criteria that pilot was assessed against:

- **Realistic signals:** *Satisfied* - realistic speech signals were used throughout, providing ecological validity for the task.
- **Clinical suitability:** *Not satisfied* - the complex pre-processing pipeline required external software (*i.e.*, MaxMSP), with all stimulus configurations computed upstream before testing. This precluded any real-time adaptation and was impractical in a clinical context.
- **Electrode-level control:** *Not satisfied* - non-linear frequency shifts introduced additional spectral distortions beyond the intended manipulation, and the use of multiple realistic signals made it impossible to determine which part of each signal drove participant decisions. Importantly, the non-linear shifts added or removed spectral information unevenly across conditions, making comparisons unfair at the edges of the shift.

Decision: The non-linear frequency processing concept was promising but overly ambitious for a first pilot, as was the use of an evolutionary algorithm. We therefore decided to reduce the complexity of both the signal processing and the data collection method, and to explore simpler experimental designs.

Pilot 2 - Linear shift with MUSHRA

Approach: To present linearly frequency-shifted versions of realistic speech signals within a MUSHRA (MULTiple Stimuli with Hidden Reference and Anchor) interface, allowing participants to rate and compare multiple conditions simultaneously.

Criteria that pilot was assessed against:

- **Realistic signals:** *Satisfied* - realistic speech signals were retained, preserving ecological relevance.
- **Clinical suitability:** *Not satisfied* - the MUSHRA paradigm proved too lengthy and cognitively demanding. Furthermore, the notion of a reference signal was poorly suited to our experiment: most participants treated the reference as an implicit upper bound on quality, preventing them from rating any signal above it, even when such a rating might have been appropriate.
- **Electrode-level control:** *Not satisfied* - linear frequency shifting is inherently asymmetric. Depending on the direction of the shift, spectral information at the edges was either lost or added, confounding the intended manipulation with changes in effective bandwidth, and preventing precise targeting of individual electrodes.

Decision: Abandon the MUSHRA paradigm, which was not adapted for clinical use. Moreover, we identified that the notion of sound quality was not always interpreted equally between participants, so we decided to explore the different components underlying sound quality judgements.

Pilot 3 - Linear shift with paired comparison and different scales

Approach: We presented different linear shifts in paired comparisons and used a Bayesian approach (specifically the Bradley-Terry-Luce method) to classify and extract the shifted configuration with the best sound quality. In the same interface, we included three different rating scales - naturalness, clarity, and overall sound quality - to try to disentangle the different underlying notions of sound quality.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - realistic signals were still used, but the multi-signal approach revealed that results differed across stimuli, raising the question of which temporal or spectral portion of each signal drove participant decisions.
- **Clinical suitability:** *Partially satisfied* - the paired comparison paradigm was faster and simpler than MUSHRA. However, presenting three rating scales simultaneously - though intended to save time - turned out to be redundant, as participants rated naturalness, clarity, and overall quality consistently, suggesting these scales did not capture independent perceptual dimensions (at least when evaluated simultaneously).
- **Electrode-level control:** *Not satisfied* - as in Pilot 2, linear frequency shifts remained asymmetric and did not permit precise control over which electrode received a given spectral component.

Decision: At this stage, we clearly needed a better understanding of which part of the signal was used by the participant. This was therefore the main motivation for the experiment presented in Chapter 2 (*i.e.*, the relationship between FTPM, sound quality, and sound features). We needed to find a signal whose spectral structure would interact maximally with the different types of FTPM profiles.

Pilot 4 - Band importance frequency

Approach: We needed to understand which part of the signal was important in the sound quality judgement. Different methods exist for this; one of them involves removing one or multiple bands from the signal and observing the effect on sound quality. We decided to do this at the electrode level. However, evaluating all possible combinations can be very long and not appropriate for a clinical assessment. We decided to apply such assessment live using an adaptive Bayesian approach: the idea was to keep the paired comparison approach, present two versions of the signal with different parts of the spectrum present or absent, run a Bradley-Terry-Luce algorithm between each iteration, and target the bands for which no differences were observed in order to compare them in the next trial.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - realistic signals were still used, and the band-removal approach was a meaningful step towards understanding their spectral structure. However, it became clear that realistic signals were too complex to isolate the contribution of individual electrodes reliably.
- **Clinical suitability:** *Not satisfied* - processing time increased substantially across trials as the number of paired comparisons grew, with considerable delays between trials towards the end. The adaptive targeting strategy used here also proved inefficient: when participants consistently reported no difference between bands, those same bands were repeatedly selected for comparison.
- **Electrode-level control:** *Partially satisfied* - band removal was applied at the electrode level (using the participant mapping), which represented meaningful progress. However, the adaptive algorithm did not converge reliably, limiting our ability to draw electrode-specific conclusions.

Decision: We decided to abandon the use of fully realistic signals at this stage and focused our efforts on less complex signals, for which we could directly study the effect of individual factors at the electrode and frequency level. However, we kept the use of paired comparisons and the Bayesian post-processing due to their efficiency.

Pilot 5a - First vowel experiment

Approach: A task based on vowel perception was developed using natural speech recordings. The second formant frequency (F2) was manipulated continuously between the first and third formant range. This decision was made because the frequency range between F1 and F3 was large enough to apply modifications. For a single vowel, paired comparisons of different F2 positions were presented, and participants selected the version of better perceptual quality using a rating scale. Responses were expected to correlate with individual frequency-to-place factors.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - moving to isolated vowels represented a meaningful simplification, but the use of real speech recordings still severely constrained the range of manipulations that could be applied without introducing artefacts.
- **Clinical suitability:** *Partially satisfied* - the paired comparison paradigm is fast, simple, and well-suited to Bayesian modelling, which was a clear improvement over earlier approaches. However, the use of natural recordings limited flexibility and reproducibility across participants and sessions.
- **Electrode-level control:** *Not satisfied* - manipulating F2 in a natural vowel inevitably altered vowel identity alongside sound quality, confounding the two perceptual dimensions. The paired comparison paradigm could not disentangle categorical perception effects from genuine quality judgements, and it remained impossible to attribute responses to specific electrodes. Moreover, the range of electrodes studied was limited to the interval between the first and third formant (*i.e.*, where F2 was manipulated).

Decision: The paired comparison paradigm has clear advantages - it is fast, simple, reliable, and well-suited to Bayesian modelling. However, it was not sufficiently adapted to this task. Additionally, the use of recorded speech, even restricted to isolated vowels, severely limited the range of possible signal manipulations.

These iterative refinements motivated the final experimental design adopted in this chapter. We moved from continuous realistic speech to fully synthetic two-formant vowels, which allowed precise control over the spectral content delivered to individual electrodes while remaining perceptually grounded in speech.

To complement this new stimulus approach, we also replaced the MUSHRA and paired comparison paradigms with a two-stage procedure: participants first identify the vowel, then provide a confidence rating for their identification. This confidence rating was thought to serve as our index of sound quality (or sound clarity). The underlying assumption is that, for a given stimulus, reduced confidence reflects a degraded or unnatural percept rather than genuine between-category confusion - that is, the participant can identify the vowel but does not perceive it as sounding “right”.

We acknowledge that vowel identification in F1/F2 space is inherently difficult, with overlap between vowels even in NH. We therefore do not assume that confidence ratings are a pure measure of sound quality, but rather that they capture a combination of perceptual clarity and categorical uncertainty (even if that latter aspect was partly alleviated through the rejection of responses associated with very low confidence on the rating scale). In the following Chapter, we then preferred to refer as “internal representation” instead of “sound quality” to avoid any confusion.

Estimation of internal representation of vowels in experienced cochlear implant users and its relationship with frequency-to-place factors

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Abstract

CI users can achieve near-normal speech understanding in quiet, yet many continue to report limitations in sound quality and spectral clarity even after years of device use. The fidelity with which spectral information is conveyed along the electrode array - determined by factors such as frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) - is suspected to contribute to these residual limitations. We developed an original vowel identification and confidence task in which participants selected the most likely vowel from a grid of $F1 \times F2$ electrode combinations, providing a continuous confidence rating alongside each response. This allowed us to characterise the optimal spectral representation of each participant and examine whether electrode-level frequency-to-place factors predicted deviations from the NH reference. Because implantation history - whether deafness and CI implantation occurred before or after language acquisition - is known to influence adaptation to the CI signal, two groups of experienced adult users were tested: 10 CI users with congenital onset of deafness, implanted early before language acquisition (Pre-ECI), and 14 CI users with post-lingual onset of deafness, implanted late after language acquisition (Post-LCI), alongside 10 NH listeners as a reference. Results indicated that the vowel representations of CI users were modestly displaced from those of NH listeners overall, with Post-LCI users showing a preference for lower $F2$ frequencies and Pre-ECI users weighting $F1$ cues more heavily in their confidence judgements. None of the three frequency-to-place factors accounted for these small deviations at the electrode level, suggesting either that experienced users had fully adapted to these factors at the time of testing, or that the task lacked sufficient sensitivity to detect their residual influence.

3.1 Introduction

Cochlear implantation is the standard of care for restoring auditory function in individuals with severe-to-profound sensorineural hearing loss. Over recent decades, technological improvements, refined surgical techniques, and evolving candidacy criteria have

led to outcomes that were once considered unattainable: the majority of CI users now achieve high levels of speech recognition in quiet after one year of device use (Blamey et al., 2013b; Holden et al., 2013b; Boisvert et al., 2020a).

Despite these encouraging outcomes, perceptual limitations persist beyond the initial adaptation period. One of the most consistently reported residual difficulties concerns sound quality, which many CI users describe as degraded relative to NH even after years of device experience (Caldwell et al., 2017b; Berg et al., 2025a; Dorman et al., 2025). Critically, this degradation is not uniform across recipients: while some users report satisfactory listening experiences across a wide range of everyday situations, others continue to experience substantial perceptual difficulty despite extended implant use (Holden et al., 2013b; Boisvert et al., 2020a). Such inter-individual variability suggests that individual-level factors continue to modulate auditory perception well beyond the initial adaptation period.

One possible source of this variability is the degree of spectral distortion imposed at the level of the ENI, which differs substantially across recipients. CI signal processing maps acoustic frequencies onto a fixed set of electrodes according to a predetermined frequency-to-electrode allocation table. Because the physical position of the electrode array within the cochlea varies across individuals following implantation - determined by insertion depth, array design, surgical approach, and related trauma (Skinner et al., 2002; Noble et al., 2020; Finley et al., 2008) - a systematic mismatch can arise between the frequency a given electrode is intended to represent and the cochlear region it stimulates. The quality of the ENI introduces additional variability: NR and CS differ across electrode sites and across individuals, further shaping the spectral information available to the auditory system and potentially degrading the perceptual quality of individual channels - particularly in cochlear regions affected by neural dead regions (Pfungst et al., 2011a; Bierer, 2007).

The present study focuses on three factors associated with these spectral distortions, collectively referred to here as frequency-to-place factors: FTPM, CS, and NR (for a detailed review of each factor and its known effects on CI perception, see Chapter 1.4). The central question is whether these factors continue to modulate the internal representation of speech in experienced CI users - that is, in individuals who have had sufficient opportunity, due to extensive listening experience, to adapt to the spectral characteristics of their device. As an exploratory extension, we also examine the absolute cochlear position of each electrode - expressed as the characteristic frequency of the cochlear place stimulated (CF_{place}) - to determine whether the physical location of an electrode along the cochlea explains variance in perceptual confidence beyond what is accounted for by its relative FTPM alone. While FTPM captures the deviation between assigned and stimulated frequency, CF_{place} encodes where along the cochlea the electrode resides irrespective of its intended mapping, making the two metrics conceptually distinct despite their mathematical relationship. For example, two electrodes with identical FTPM values (say, -0.5 octaves each) could sit at very different cochlear locations (one apical, one basal), and CF_{place} captures this distinction.

A critical consideration when studying the effect of frequency-to-place factors on speech representation is the listener's auditory history - specifically, listening experience with the CI, duration of deafness prior to implantation, and onset of deafness. We therefore tested two groups. The first group comprised CI users with congenital onset of severe-to-profound deafness, implanted early in childhood before the acquisition of spo-

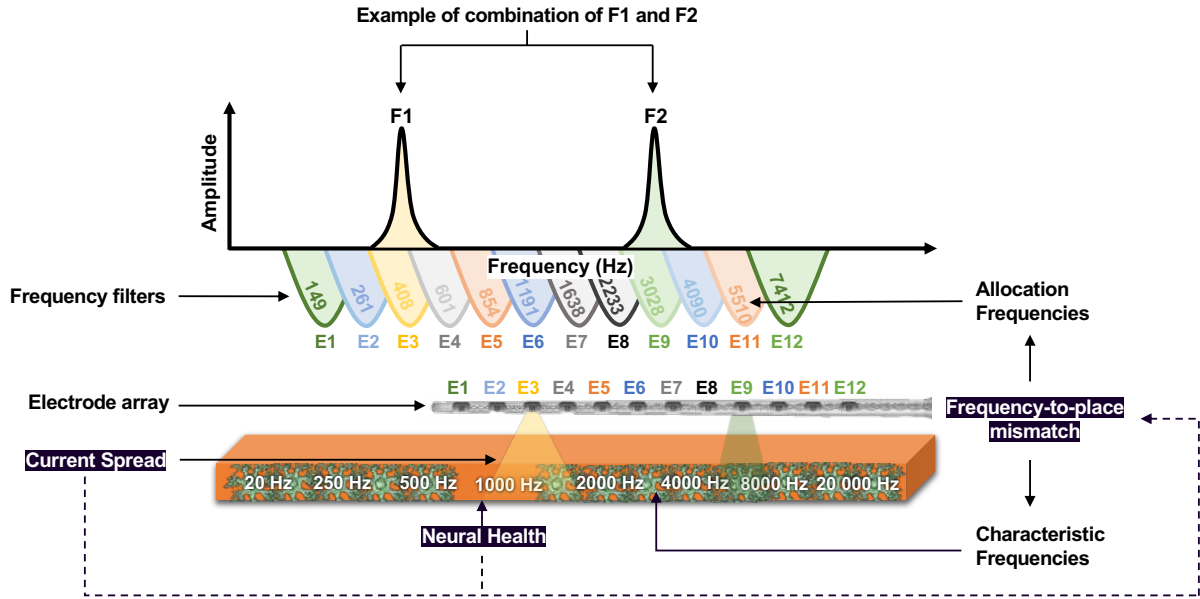


Figure 3.1: **Schematic illustration of the experimental paradigm including the frequency-to-place factors investigated.** The incoming acoustic signal, composed of an F1 and F2 combination, is decomposed into frequency bands and assigned to electrodes E1–E12 as pre-defined by the frequency-to-electrode mapping function, which determines which frequencies are allocated to each electrode. Within the cochlea (bottom), three factors determine the relationship between allocated and perceived frequency. First, electrode array insertion depth sets the baseline FTPM by imposing a shift between characteristic frequencies and allocated frequencies. Second, current spread (coloured triangles) spreads stimulation beyond the target site, recruiting neurons tuned to frequencies outside the frequency band allocated to the electrode being considered; its extent is influenced by local neural health. Third, neural responsiveness varies along the cochlea; where spiral ganglion neurons are absent, the effective site of stimulation may shift to more distant surviving populations. The dotted lines between current spread and neural responsiveness illustrate that these factors are suspected to modulate the primary effect of FTPM.

ken language (Pre-ECI). These individuals developed language and auditory representations exclusively through the CI modality, without ever establishing a natural tonotopic baseline from which to adapt; their internal representation of speech was built entirely around the spectral input provided by their implant, however distorted. The second group comprised CI users with post-lingual onset of deafness, implanted as adults (Post-LCI). These individuals acquired language with natural hearing and subsequently remapped their existing auditory representations onto a spectrally shifted and degraded input once implanted (Peterson et al., 2010; Fu and Galvin, 2007c). These fundamentally different auditory histories may result in different degrees of perceptual adaptation to spectral distortions, and therefore in different relationships between frequency-to-place factors and speech perception representation.

It is worth noting that the speech perception system is inherently flexible: listeners routinely adapt to different speakers’ F1 and F2 realisations of the same vowel category, normalising across substantial acoustic variability (Hillenbrand et al., 1995). This flexibility may facilitate adaptation to the shifted formant representations imposed by the CI, though whether it fully compensates for the distortions introduced by frequency-to-place

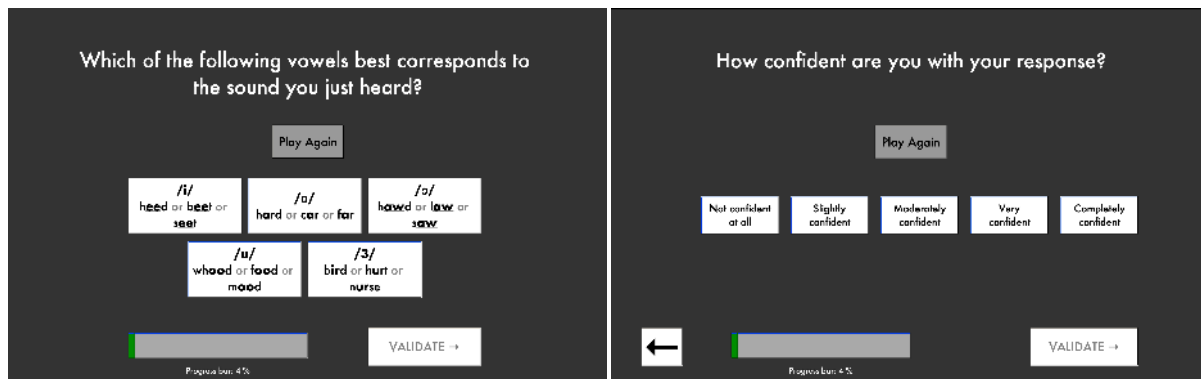


Figure 3.2: **Interface used for the vowel experiment, showing the two response stages.** The left panel shows the first stage, in which the participant selects the perceived vowel from a closed set of five options. The “Play again” button allowed the listener to replay the stimulus. The progress bar indicates the remaining time in the test. Once a vowel is selected, the participant clicks “Validate” to proceed to the second stage. The second stage, shown in the right panel, presents the confidence rating. On a five-point Likert scale ranging from *Not confident at all* to *Completely confident*, participants rated their confidence in their previous identification. They could return to the first stage using the back arrow. Once a confidence rating is selected, they click “Validate” to begin the next trial.

factors in experienced users remains an open question.

The present study examined how the internal representations of vowels differ between NH listeners and experienced CI users - both Pre-ECI and Post-LCI - and how these representations might be shaped by frequency-to-place factors at the individual electrode level.

To probe the internal spectral representation of vowels in experienced CI users, we developed a two-stage vowel identification and confidence rating task. The task exploits the concept of formants - the natural resonant frequencies of the vocal tract, determined by the shape and movement of the vocal organs (lips, teeth, tongue), which produce characteristic spectral peaks in the speech signal. Vowels are defined primarily by the frequencies of their first two formants (F1 and F2) (Peterson and Barney, 1952), and a vowel’s position in the F1-F2 space can be summarised by the spectral centroid of each formant region. Because formant frequencies map directly onto locations along the cochlea via the tonotopic organisation of the auditory system, vowel perception provides a natural and ecologically valid probe for place-dependent spectral distortions: any persistent shift, broadening, or degradation of cochlear place coding should be reflected in the responses of CI users relative to their NH peers.

For CI users, stimuli were constructed according to each participant’s individual frequency-allocation map, using acoustic stimulation with formants intended to activate two independent channels - one for the F1 frequency region and one for F2.

The key idea of the design is that stimuli are defined in terms of electrode positions rather than fixed acoustic frequencies. Each response can therefore be associated with two specific electrodes, and related to the frequency-to-place factors at those electrodes. For example, if a participant identifies a stimulus activating electrodes E3 and E9 as the vowel [ɑ:] with high confidence, this tells us that the spectral information delivered by

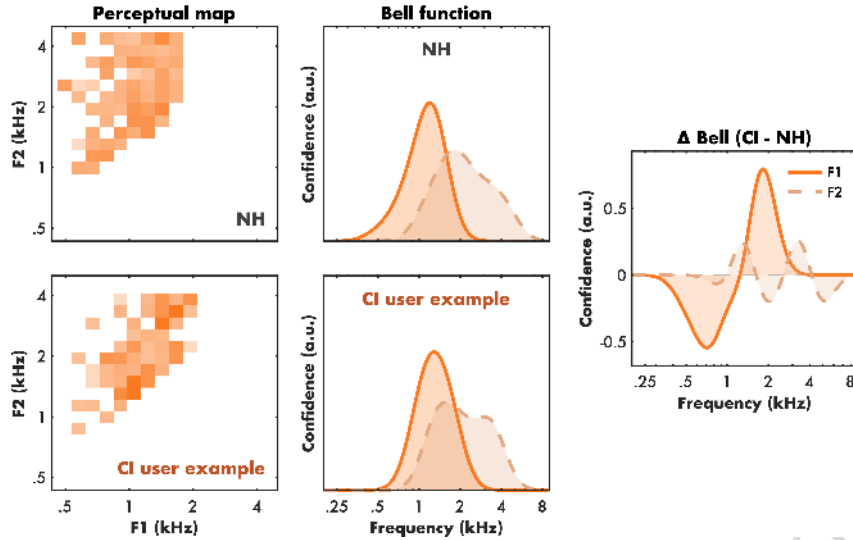


Figure 3.3: **Perceptual map, bell function, and delta bell representation.** The left column shows perceptual maps for vowel [ɑ:] from the NH group (top) and an example CI user (bottom), where colour intensity encodes response confidence across the F1-F2 space. The middle column shows the corresponding bell functions, computed as confidence-weighted kernel density estimates in log-frequency space, with F1 (solid) and F2 (dashed) plotted separately. The right column shows the delta bell (CI – NH), highlighting frequency regions where the CI user perceptual representation deviates from the NH reference.

those two electrode channels is consistent with the internal representation of [ɑ:] for that listener. The design is illustrated in Figure 3.1. For each F1×F2 electrode combination, participants first identified the vowel they perceived from a closed set of five long English vowels (Figure 3.2, left panel), and then rated their confidence of judgement on a five-point scale (Figure 3.2, right panel).

For each participant and each vowel, a Bayesian model was computed to refine the expected confidence response at each F1×F2 electrode combination. The outputs of these models were used to generate perceptual maps across the electrode grid (see left panel of Figure 3.3 for some examples). However, because responses were distributed across multiple competing vowel categories - owing to the perceptual overlap between vowels - some electrode combinations yielded sparse or absent data for a given vowel, even though they would plausibly attract responses under more extensive testing. That is, electrode combinations whose formant frequencies fall within the expected perceptual boundary for a given vowel category may nonetheless receive few responses because participants identified the stimulus as a different, competing vowel on most trials. To address this limitation, perceptual maps were transformed into full distributions of confidence-weighted responses using kernel density estimation, yielding a continuous confidence profile across the electrode array for each participant and each vowel (see middle panel of Figure 3.3). This approach enables a complete comparison along the full acoustical axis rather than at the sparse set of tested electrode combinations.

This primary analytical representation is referred to as the *bell* - which characterises how confidence-weighted perceptual responses are distributed across the acoustical axis for each vowel and participant. The same procedure was applied to NH listeners to construct

a group-level NH reference bell, against which individual CI distributions are compared. Three complementary analyses were then applied to this representation.

First, the overall displacement of each CI user bell from the NH reference was quantified using the Wasserstein distance - a measure of distributional dissimilarity that expresses, in octaves, how far a CI participant's confidence distribution is shifted along the frequency axis relative to the NH reference. This provides a single scalar summary of representational displacement per participant, vowel, and formant region.

Second, the direction of that displacement was characterised through a formant shift analysis: the frequency centroid of each participant's bell was compared to the NH reference frequency centroid, yielding a signed shift in octaves where positive values indicate that the CI user's perceptual representation is shifted towards low frequencies relative to the NH reference, and negative values towards high frequencies.

Third, a reliance ratio was computed to quantify the balance of perceptual confidence between the F1 and F2 electrode regions: $R = \text{peak}(\text{bell}_{F2}) / (\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$, where values above 0.5 indicate that a participant places greater confidence in their F2 electrode responses than in their F1 electrode responses.

At the individual electrode level, the difference between each CI participant's confidence bell and the corresponding NH reference bell was computed to yield a per-electrode delta (Δ). An example of such difference is provided in Figure 3.3. This delta was then entered as the dependent variable in a model to assess whether the three electrode-neuron interface factors - FTPM, CS, and NR - systematically predicted the deviation of CI confidence profiles from the NH reference. As an exploratory extension, absolute cochlear position (CF_{place}) was additionally included as a predictor.

The task was performed by 14 Post-LCI, 10 Pre-ECI, and 10 NH listeners. NH responses served as reference distributions against which CI users' electrode confidence profiles were compared. Wasserstein distances, frequency shifts, and reliance ratios were computed using the responses of all participants. Frequency-to-place factor analyses were performed on a subset of participants for whom these values were available (see Table 3.1).

The study addressed two main questions:

1. **How do vowel representations differ between groups?** Do experienced CI users - both Pre-ECI and Post-LCI - show systematic differences in their internal vowel representations compared to NH listeners, and do Pre-ECI and Post-LCI differ from each other? This question was addressed through three complementary analyses applied to the electrode confidence bell: the Wasserstein distance from the NH reference (overall displacement), the formant centroid shift (direction of displacement), and the F1/F2 reliance ratio (asymmetry of confidence across the cochlear axis).
2. **Do frequency-to-place factors predict the displacement of electrode confidence relative to NH listeners?** Do FTPM, CS, and NR predict the magnitude and direction of the shift between each CI user confidence profile and the NH reference, and do these relationships differ between Pre-ECI and Post-LCI groups? As an exploratory analysis, does absolute cochlear position (CF_{place}) explain additional variance in this displacement beyond FTPM alone?

3.2 Methods

3.2.1 Participants

14 Post-LCI participants (7 female; mean age = 53.7 years, SD = 17.9; mean duration of CI use = 5.9 years, SD = 3.0), 10 Pre-ECI participants (4 female; mean age = 25.8 years, SD = 3.9; mean duration of CI use = 23.4 years, SD = 3.7), and 10 NH listeners (6 female; mean age = 27 years, SD = 4) participated in the study. NH listeners had audiometric thresholds ≤ 20 dB HL at octave frequencies from 250 to 8000 Hz bilaterally. Details about CI users are presented in Table 3.1.

CI participants were recruited from the Emmeline Centre for Hearing Implants at Addenbrooke’s Hospital (Cambridge, UK). All participants were native speakers of British English. The study was approved by the East of England - Cambridge Central Research Ethics Committee (IRAS number: 335150). All participants gave informed consent before the experiment.

Table 3.1: **Participant demographics and data availability.** Act. el. = number of active electrodes in the clinical MAP. Contra. = contralateral ear condition (HA = hearing aid; CI = cochlear implant; None = unaided). Age at testing is derived from age at implantation plus CI experience. FTPM and PECAP tick marks indicate whether electrode-level frequency-to-place factors were available.

Participant	Sex	Device	Act. el.	Side	Contra.	Age impl.	Age test.	CI exp.	FTPM	PECAP
Pre-ECI 1	M	Cochlear	22	L	HA	2	21	19	✓	✓
Pre-ECI 2	F	Cochlear	20	L	CI	3	19	16	✓	
Pre-ECI 3	F	Cochlear	22	R	HA	2	24	22		
Pre-ECI 4	F	AB	16	R	None	4	29	25		
Pre-ECI 5	M	AB	16	R	CI	1	27	26		
Pre-ECI 6	M	Cochlear	22	L	HA	2	26	24		
Pre-ECI 7	F	Med-El	12	R	HA	2	24	22		
Pre-ECI 8	M	Cochlear	22	L	CI	3	31	28		
Pre-ECI 9	M	Cochlear	22	R	CI	3	30	27		
Pre-ECI 10	M	Cochlear	22	L	HA	2	27	25		
Post-LCI 1	F	Cochlear	22	L	HA	67	72	5	✓	
Post-LCI 2	F	Med-El	12	L	None	45	55	10	✓	
Post-LCI 3	F	Cochlear	22	R	HA	20	24	4	✓	✓
Post-LCI 4	M	Cochlear	22	L	HA	19	22	3	✓	
Post-LCI 5	F	AB	16	R	CI	45	50	5	✓	✓
Post-LCI 6	M	Cochlear	22	L	HA	69	73	4		
Post-LCI 7	M	AB	16	L	None	52	55	3		
Post-LCI 8	F	Cochlear	22	R	None	65	72	7		
Post-LCI 9	M	Med-El	12	R	HA	39	43	4		
Post-LCI 10	F	Cochlear	22	L	HA	21	31	10		✓
Post-LCI 11	F	Cochlear	22	L	CI	67	71	4		✓
Post-LCI 12	M	AB	16	R	None	45	58	13		
Post-LCI 13	M	AB	16	L	HA	65	70	5		
Post-LCI 14	M	Cochlear	22	L	HA	50	56	6		

3.2.2 Stimuli

Synthetic vowel stimuli were created using the Parselmouth library (Jadoul et al., 2018), which provides access to Praat functionality within a Python environment. Each stimulus consisted of a harmonic complex with a fundamental frequency fixed at 80 Hz, associated with a first formant (F1) and a second formant (F2). No higher formants

were included. This choice was made to isolate the contribution of F1 and F2 to vowel identification, ensuring that any observed perceptual differences could be attributed to the spectral representation of these two formants rather than to higher-order spectral cues. Each stimulus was 6 s long, with a 500 ms fade-in and fade-out applied at onset and offset. The sampling rate was set at 48 kHz.

NH Stimulus Set

For NH listeners, formant centre frequencies were selected using a Gammatone filter bank. F1 was sampled at 15 frequency values between 234 Hz and 1720 Hz, and F2 at 21 frequency values between 415 Hz and 4093 Hz, with the constraint that F2 was always greater than F1. This yielded 218 unique F1/F2 combinations, each presented four times per participant, for a total of 872 trials.

CI Stimulus Set

For CI users, F1 and F2 values were set to the centre frequencies of the participant's allocated electrodes, as defined by their clinical frequency-allocation map. This ensured that the formants of each stimulus aligned with known electrode stimulation sites, allowing direct comparison between the perceived formant location and the electrode responsible for delivering it. The same constraint as for NH listeners ($F2 > F1$) was applied. Because the number of active electrodes varied across participants and devices, the number of F1/F2 combinations and therefore the number of trials differed between CI users. Each F1/F2 combination was repeated four times.

3.2.3 Experimental Procedure

The experiment employed a two-stage identification and confidence paradigm. On each trial, a single synthetic vowel stimulus was presented monaurally at a comfortable listening level adjusted individually by each participant.

Stage 1: Vowel identification. Following stimulus presentation, participants identified the vowel they perceived by selecting one of five response options displayed on screen (Figure 3.2, left panel): [ɑ:] (as in *start*), [ɜ:] (as in *nurse*), [i:] (as in *seat*), [ɔ:] (as in *thought*), and [u:] (as in *goose*). These representative words were displayed alongside each vowel symbol to aid identification.

Stage 2: Confidence rating. After selecting a vowel, participants rated their confidence in their choice on a five-point Likert scale ranging from “Not confident at all” to “Very confident” (Figure 3.2, right panel).

Before each testing session, a short training block of 30 trials was conducted to ensure task comprehension. Trials were presented in randomised order. Stimuli were presented using a MacBook Pro, an RME Fireface UCX II sound card, and a Sennheiser HD 600 open-back headset. For CI users, the headset was positioned over their clinical processor.

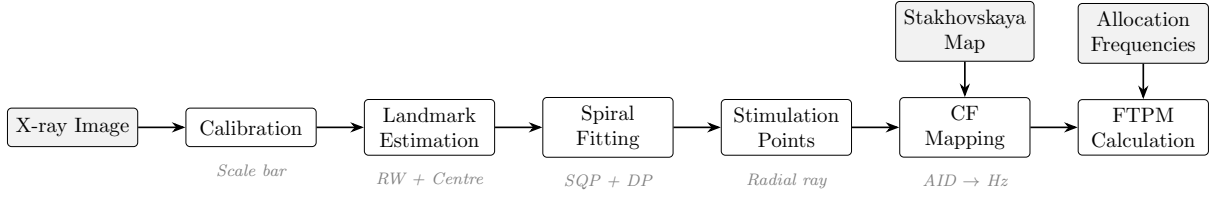


Figure 3.4: **FTPM estimation workflow.** Algorithm for computing electrode-specific frequency-to-place mismatch from post-operative cochlear implant X-rays, combining spiral template fitting with the Stakhovskaya frequency-position function. RW = Round Window, SQP = Sequential Quadratic Programming, DP = Dynamic Programming, AID = Angular Insertion Depth, CF = Characteristic Frequency.

3.2.4 Estimation of Frequency-to-Place Factors

Three electrode-level factors were estimated for a subset of CI participants: FTPM, CS, and NR. Data availability varied across participants, as detailed in Table 3.1. FTPM estimation requires post-operative X-ray imaging; for some individuals, images were either unavailable or of insufficient quality to permit reliable electrode localisation. CS and NR were derived from PECAP recordings, which could not be obtained for all participants due to the absence of a measurable response or device incompatibility. The electrode-level analyses reported in Section 3.2.6 should therefore be interpreted with caution given the reduced and unbalanced availability of these measures.

FTPM Estimation

FTPM was estimated using a semi-automatic algorithm for computing the angular insertion depth of cochlear implant electrodes from post-operative X-ray images (Roberts et al., personal communication). The simplified workflow is shown in Figure 3.4 and representative outputs in Figure 3.5.

After image calibration using the scale bar, electrode contacts were identified manually and key anatomical landmarks - the round window and cochlear centre - were estimated automatically via least-squares circle fitting. A mean cochlear spiral template derived from micro-CT data was then fitted to the electrode positions through 3D rotation optimisation, minimising the sum of squared distances between electrodes and assigned spiral points. For each electrode, the stimulation site was estimated by projecting a line from the cochlear centre through the electrode position to the spiral, and the angular insertion depth relative to the round window (0°) was converted to a CF using the Stakhovskaya spiral ganglion frequency-position function (Stakhovskaya et al., 2007b). The CF at each stimulation site was then compared to the allocated centre frequency of the corresponding electrode to yield the FTPM in octaves for each electrode. The absolute cochlear position of each electrode - expressed as the CF of the place of stimulation (CF_{place}) - was retained from the same estimation pipeline as an exploratory predictor (see Section 3.2.6).

CS Estimation

CS was estimated using the Panoramic ECAP (PECAP) method (Garcia et al., 2021). ECAPs are recorded by delivering a current pulse through a probe electrode and measuring

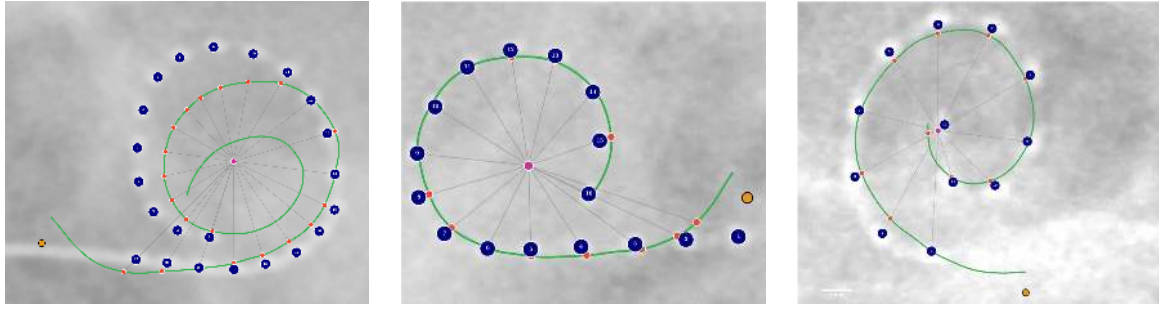


Figure 3.5: **Estimation of FTPM using post-operative X-rays.** Electrodes (blue) are matched to stimulation sites (red) on the fitted cochlear spiral (green) via radial projection from the modiolar axis (magenta). The round window (orange) defines the 0° reference.

the synchronised auditory nerve response at a nearby recording electrode. The PECAP method records ECAPs for all combinations of probe and masker electrodes, yielding an $n \times n$ matrix $\mathbf{M}$. A nonlinear optimisation algorithm then decomposes $\mathbf{M}$ into two vectors: σ , encoding CS, and η , encoding NR. Larger σ values indicate broader current spread and poorer electrode selectivity. All measurements were recorded at maximum comfortable level (MCL) for each active electrode.

NR Estimation

NR was estimated simultaneously with CS from the same PECAP decomposition (Garcia et al., 2021). The η vector encodes the relative responsiveness of auditory nerve fibres along the electrode array, with lower values indicating poorer neural health. Per-electrode η values are used as the NR index throughout.

3.2.5 Data Analysis

Exclusion of [i:] Responses

Although five vowel options were presented during the task, responses for the vowel [i:] were excluded from all subsequent analyses (see Section 3.3.1 for justification). All remaining analyses therefore concern four vowel categories: [ɑ:], [ɜ:], [ɔ:], and [u:].

Electrode Confidence Distribution

The main unit computed here is the *electrode confidence distribution* - hereafter referred to as the *bell* - which characterises how confidence-weighted perceptual responses are distributed across the electrode array for each vowel and participant. Rather than projecting responses onto a formant-frequency grid and computing summary statistics in that space, the bell maps the full distribution of responses directly in the acoustical axis, making it directly comparable with the electrode-level factors FTPM, CS, and NR.

Electrode Allocation Mapping. For each participant, the CI frequency-allocation map was used to associate each F1/F2 stimulus combination with a pair of electrode positions: the electrode whose allocated centre frequency matched F1, and the electrode whose allocated centre frequency matched F2.

Bayesian Confidence Estimation and Perceptual Maps. For each participant, vowel, and F1/F2 combination, a Bayesian model was fitted to estimate and refine the confidence response. The outputs of these models were used to create perceptual maps in the $F1 \times F2$ electrode space.

Confidence-Weighted Kernel Density Estimation. However, responses were sparse in some regions of the perceptual maps: owing to the perceptual overlap between vowels, some combinations yielded little or no data for a given vowel even though they would plausibly attract responses under more extensive testing. To overcome this limitation, perceptual maps were transformed into an aggregated confidence-weighted distribution - the bell - using kernel density estimation (KDE).

For each participant, vowel v , and formant dimension (F1 or F2), responses were aggregated as confidence-weighted counts at each electrode position e :

$$R(e, v) = \sum_{t \in \mathcal{T}(e, v)} c_t \quad (3.1)$$

where $\mathcal{T}(e, v)$ is the set of trials on which vowel v was selected and the corresponding formant stimulus was allocated to electrode e , and $c_t \in \{1, 2, 3, 4, 5\}$ is the confidence rating on trial t . A Gaussian KDE was then applied in log-frequency space over the electrode array to produce a smooth, continuous profile:

$$\text{bell}(f, v) = \frac{1}{Z} \sum_e R(e, v) \cdot \exp \left(-\frac{(\log_{10} f - \log_{10} f_e)^2}{2h^2} \right) \quad (3.2)$$

where f is a frequency grid point, f_e is the centre frequency of electrode e , h is the kernel bandwidth (set to 0.10 in $\log_{10}$ -Hz units, approximately one-third of an octave), and Z is a normalisation constant ensuring that the bell integrates to unity over the frequency grid. Working in log-frequency space ensures that the kernel bandwidth is perceptually uniform across the frequency range. After smoothing, every point on the frequency grid receives a value, providing a complete and continuous profile over the full extent of the cochlear array.

NH Reference Bell. The same KDE procedure was applied to NH listeners, with their Gammatone-spaced stimulus values remapped to the nearest frequency grid points. A group-level NH reference bell $\text{bell}_{\text{NH}}(f, v)$ was computed by averaging individual NH bells across all NH participants, yielding the expected distribution of confident responses for a listener with normal cochlear place coding.

Delta Bell: Displacement Relative to the NH Reference. To quantify the shift between each CI user's confidence profile and the NH reference, a delta bell was computed

for each participant, vowel, and formant dimension as the signed difference between the participant’s normalised bell and the group-level NH reference bell:

$$\Delta\text{bell}(f, v) = \text{bell}_{\text{CI}}(f, v) - \text{bell}_{\text{NH}}(f, v) \quad (3.3)$$

where both bells have been normalised to unit area prior to subtraction, ensuring that differences reflect shape and location rather than overall response count. Positive values at a given frequency f indicate that the CI user assigns greater relative confidence at that cochlear location than the NH reference, and negative values indicate reduced relative confidence. The delta bell therefore captures the direction and magnitude of the perceptual displacement induced by the CI signal relative to normal cochlear place coding, providing the dependent variable for Study Question 2.

3.2.6 Data Extraction and Statistical Analysis

All statistical analyses were conducted in MATLAB R2025a at a significance threshold of $\alpha = .05$. Two families of analyses were conducted, corresponding to the two study questions introduced in Section 3.1.

Study Question 1 - How do Vowel Representations Differ between Groups?

Three complementary analyses were applied to the electrode confidence bell to characterise group differences in vowel representation, each addressing a distinct aspect of the distributional difference between CI users and NH listeners.

Wasserstein Distance. To assess how far CI users’ electrode confidence distributions deviate from those of NH listeners, the Wasserstein distance was computed for each CI participant, vowel, and formant dimension. The Wasserstein distance quantifies the minimum work required to transform one probability distribution into another, and provides a single interpretable scalar - here expressed in octaves - representing how far a CI user bell distribution is displaced from the NH reference along the frequency axis, irrespective of direction. For the NH group, a leave-one-out variant was used: each NH participant’s bell was compared to the mean bell of the remaining NH participants, providing a within-group baseline. This procedure ensures that the NH distances reflect genuine inter-individual variability rather than comparison against a distribution that includes the participant’s own data. A linear mixed model (LMM) was then fitted to the resulting distances:

$$W_{ivj} = \beta_0 + \beta_{\text{group}} \cdot \text{Group}_j + \beta_{\text{vowel}} \cdot \text{Vowel}_i + \beta_{\text{formant}} \cdot \text{Formant}_v + u_j + \varepsilon_{ivj} \quad (3.4)$$

where W_{ivj} is the Wasserstein distance for participant j , vowel i , and formant v ; $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is the participant-level random intercept; and ε_{ivj} is the residual. Group effects were evaluated via Type III ANOVA on the fitted model, and pairwise contrasts between NH, Post-LCI, and Pre-ECI were conducted via model-based contrasts (`coefTest`).

Formant Shift. Whereas the Wasserstein distance captures the overall magnitude of distributional displacement, it does not reveal its direction. To characterise where along the frequency axis CI users' perceptual representations are shifted relative to NH, the weighted centroid of each participant bell was computed as:

$$\hat{f}(v) = 10 \frac{\sum_f \log_{10}(f) \cdot \text{bell}(f, v) \cdot \Delta(\log f)}{\sum_f \text{bell}(f, v) \cdot \Delta(\log f)} \quad (3.5)$$

where the summation is over all frequency grid points f and $\Delta(\log f)$ is the log-spaced integration weight. The formant shift for each participant, vowel, and formant dimension was then defined as:

$$\Delta_{\text{shift}} = \log_2 \left(\frac{\hat{f}_{\text{NH}}(v)}{\hat{f}_{\text{CI}}(v)} \right) \quad (3.6)$$

where $\hat{f}_{\text{NH}}(v)$ is the NH reference centroid for vowel v (computed as the mean centroid across NH participants using the same leave-one-out procedure described above) and $\hat{f}_{\text{CI}}(v)$ is the CI participant's centroid. Positive values indicate that the CI participant's perceptual representation is centred at a more apical electrode location than NH (*i.e.*, the CI user perceives the formant at a lower frequency than expected); negative values indicate a more basal representation. Group differences in formant shift were assessed via Wilcoxon rank-sum tests with Holm-Bonferroni correction for multiple comparisons, applied to per-participant mean shifts (one value per participant, averaged across vowels and formants). Pairwise comparisons were conducted between all three groups (NH, Post-LCI, and Pre-ECI).

F1/F2 Reliance Ratio. To quantify the asymmetry of perceptual confidence between the F1 and F2 electrode regions, a reliance ratio R was computed for each participant and each vowel. For a given vowel, the reliance ratio is defined as:

$$R(v) = \frac{\text{peak}(\text{bell}_{F2}(f, v))}{\text{peak}(\text{bell}_{F1}(f, v)) + \text{peak}(\text{bell}_{F2}(f, v))} \quad (3.7)$$

where $\text{bell}_{F1}(f, v)$ and $\text{bell}_{F2}(f, v)$ are the confidence bells built from trials on which the F1 and F2 electrodes, respectively, were stimulated for vowel v , and $\text{peak}(\cdot)$ denotes the maximum value of the bell over the frequency grid. This formulation uses the peak height of each bell rather than its integral, because both bells are normalised to unit integral by construction; the peak height therefore captures the concentration - and hence the reliability - of each formant's representation. Critically, this measure requires no frequency boundary to separate the F1 and F2 regions: the F1/F2 assignment follows directly from which electrode was stimulated in each trial, making the ratio valid across all vowels regardless of their acoustic formant frequencies. A value of $R > 0.5$ indicates that a participant places greater confidence in their F2 electrode responses than in their F1 electrode responses (F2 dominance); $R < 0.5$ indicates F1 dominance. The overall reliance ratio per participant was obtained by averaging $R(v)$ across vowels. Group differences in R were assessed via Wilcoxon rank-sum tests with Holm-Bonferroni correction.

Study Question 2 - Do Frequency-to-Place Factors Predict the Displacement of Electrode Confidence Relative to Normal-Hearing Listeners?

To test whether FTPM, CS, and NR predict the direction and magnitude of the shift between each CI user's confidence profile and the NH reference, electrode-level LMMs were fitted with the delta bell height as the outcome:

$$\Delta\text{bell}_{iejv} = \beta_0 + \beta_{\text{factor}} \cdot \text{Factor}_e + \beta_{\text{vowel}} \cdot \text{Vowel}_i + u_j + u_e + \varepsilon_{iejv} \quad (3.8)$$

where Δbell_{iejv} is the signed difference between participant j 's normalised bell and the NH reference bell at electrode e , for vowel v and formant dimension i (Equation 3.3); Factor_e denotes one of FTPM_e , CS_e , or NR_e at electrode e ; u_j is the participant random intercept; and u_e is the electrode random intercept, which accounts for the spatial autocorrelation introduced by KDE smoothing (Equation 3.2). A positive coefficient $\hat{\beta}_{\text{factor}}$ indicates that electrodes with higher factor values are associated with a positive displacement relative to the NH reference (greater relative confidence than NH listeners at those cochlear locations); a negative coefficient indicates displacement in the opposite direction. Because the three factors are available for different subsets of participants (see Section 3.2.4), separate univariate models were fitted for each factor to avoid listwise deletion. F1 and F2 were modelled separately. To test whether these relationships differed between Pre-ECI and Post-LCI groups, a Group $\times$ Factor interaction term was added as an extension of each univariate model, evaluated via likelihood ratio test. It should be noted that the small and unequal subsamples available for each factor (see Table 3.1) severely limit the power of this interaction test; the results should therefore be interpreted as exploratory. As an exploratory complement, the absolute cochlear position of each electrode - expressed as $\log_2(\text{CF}_{\text{place}})$, where CF_{place} is the tonotopic characteristic frequency of the stimulated cochlear region - was substituted for FTPM in an otherwise identical LMM. To determine whether absolute cochlear location or relative mismatch is the stronger predictor of confidence displacement, the two models were compared using the Akaike information criterion (AIC): a difference of more than two AIC units in favour of the CF_{place} model was taken as evidence that absolute cochlear location explains variance in the displacement beyond what is captured by relative mismatch alone. This comparison was performed separately for the F1 and F2 frequency regions.

3.3 Results

3.3.1 Overview

Identification and Confidence Distributions

The distribution of vowel responses and the distribution of confidence ratings across the three listener groups are displayed in Figures 3.6 and 3.7, respectively.

For NH listeners, responses were distributed relatively evenly across four of the five vowel categories ([a]: 20.7%, [ɜ]: 23.1%, [u]: 19.8%, [ɔ]: 21.0%), with the exception of [i], which was selected on only 9.0% of trials. A broadly similar pattern was observed for Pre-ECI ([a]: 28.8%, [ɜ]: 22.7%, [u]: 20.9%, [ɔ]: 16.2%, [i]: 11.5%) and Post-LCI ([a]: 31.5%, [ɜ]: 17.9%, [u]: 17.2%, [ɔ]: 23.8%, [i]: 9.3%). No pairwise group differences reached significance for any vowel category (Kruskal-Wallis tests, all $p > .10$).

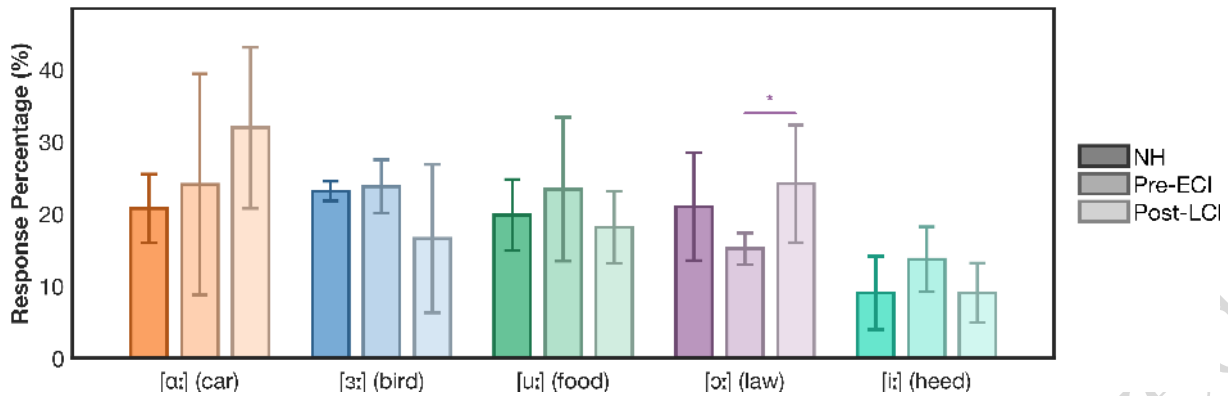


Figure 3.6: **Response distribution across the five vowel categories.** Percentage of trials on which each vowel was selected, shown separately for NH listeners ($n = 10$), Post-LCI ($n = 14$), and Pre-ECI ($n = 10$). Error bars indicate ± 1 SD across participants.

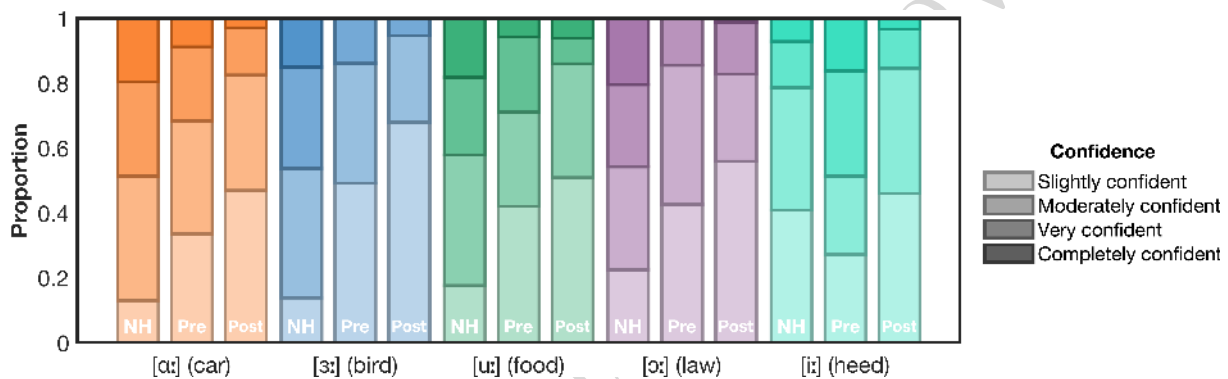


Figure 3.7: **Confidence rating distributions across the five vowel categories.** Proportion of responses in each confidence category, shown separately for the three groups. Both CI groups showed overall lower confidence than NH listeners, with a larger proportion of responses in the lowest confidence category across most vowels.

Visual inspection of confidence ratings confirmed that the distribution was broadly consistent across vowels within each group, with the exception of [i:]. For NH listeners, [i:] responses were characterised by a marked drop in confidence: 40.8% of [i:] responses fell in the lowest confidence category, compared with 12.8–22.3% for the other vowels. Both CI groups showed overall lower confidence than NH listeners across most vowels (Pre-ECI: 25–53%; Post-LCI: 45–65% in the lowest category).

Perceptual Maps and Exclusion of [i:]

Figure 3.8 presents an example of the Bayesian confidence maps for a representative CI participant alongside the mean NH reference, for all five vowel categories. For most vowels, differences between the CI user and the NH groups were subtle, with comparable F1 and F2 ranges and broadly overlapping confidence regions.

The exception was [i:]. Three converging observations motivated its exclusion from subsequent analyses. First, [i:] was selected on fewer than 10% of trials across all three groups, substantially less than the other four vowels (Section 3.3.1). Second, when [i:]

was selected, confidence ratings were markedly lower than for the other vowels in the NH group, with over 40% of responses falling in the lowest confidence category. Third, the perceptual maps revealed a systematic displacement of the F1 representation in CI users: for the NH group, the [i:] confidence map showed a peak at high F2 values (approximately 2000–4000 Hz) and low F1 values (below approximately 500 Hz), consistent with its expected formant structure; for the CI user, the F1 of [i:] responses was displaced upward to approximately 2500 Hz - close to the F2 range - while F2 remained comparable to the NH group. This pattern was observed consistently across CI participants, suggesting that CI users may have perceived the two formants as a single auditory object, or interpreted the stimulus as containing F2 and F3 rather than F1 and F2. The heatmap figure below includes [i:] for completeness, but all quantitative analyses reported in the remainder of this section concern the four retained vowel categories: [ɑ:], [ɜ:], [ɔ:], and [u:].

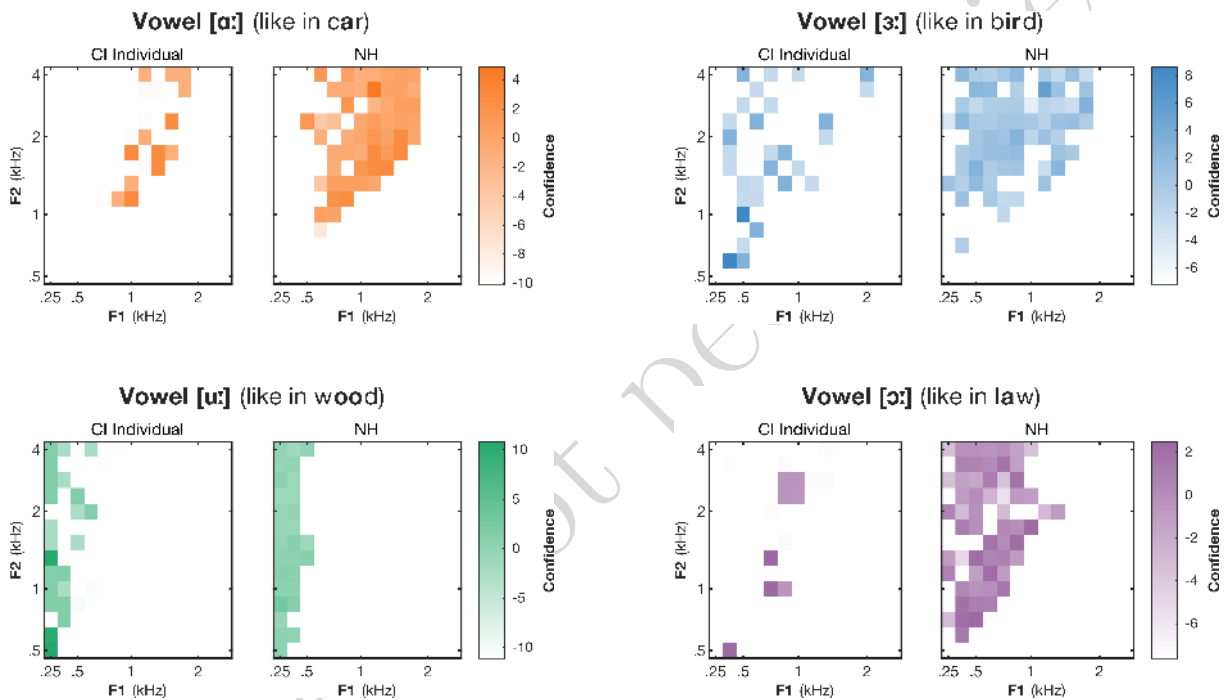


Figure 3.8: **Perceptual maps for a CI participant.** Each panel displays the median confidence surface across the $F1 \times F2$ formant space for one vowel category.

3.3.2 Study Question 1 - How do Vowel Representations Differ between Groups?

The electrode confidence distributions (bells) for each group and vowel are displayed in Figure 3.9. Three complementary analyses were applied to characterise group differences: the Wasserstein distance (overall magnitude of displacement from NH), the formant centroid shift (direction of displacement), and the F1/F2 reliance ratio (asymmetry of confidence across the frequency axis).

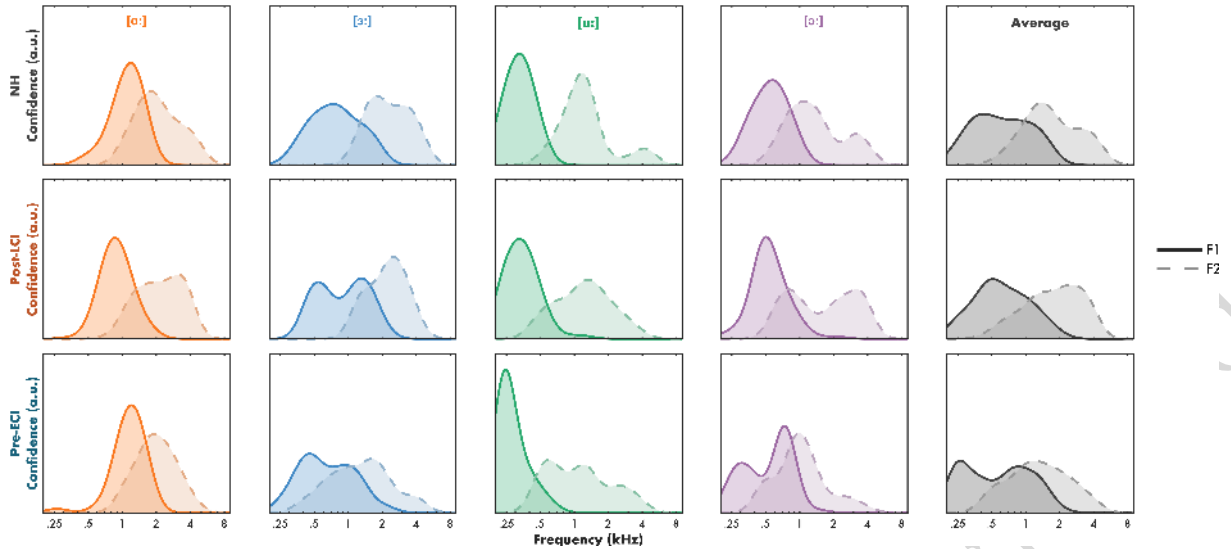


Figure 3.9: **Electrode confidence distributions (bells) for NH, Post-LCI, and Pre-ECI listeners.** Each panel shows the confidence-weighted kernel density estimate over the frequency axis for one group (rows) and one vowel (columns), with F1 (solid) and F2 (dashed) plotted separately. The rightmost column shows the average bell across vowels. The x-axis is in log-frequency (kHz).

Wasserstein distance from the NH reference

Figure 3.10 shows the Wasserstein distances and Table 3.2 summarises the LMM fixed effects. Note that the NH values shown in Figure 3.10 are not zero because they reflect leave-one-out distances: each NH participant’s bell is compared to the mean of the remaining NH participants, providing a within-group variability baseline rather than a fixed reference point.

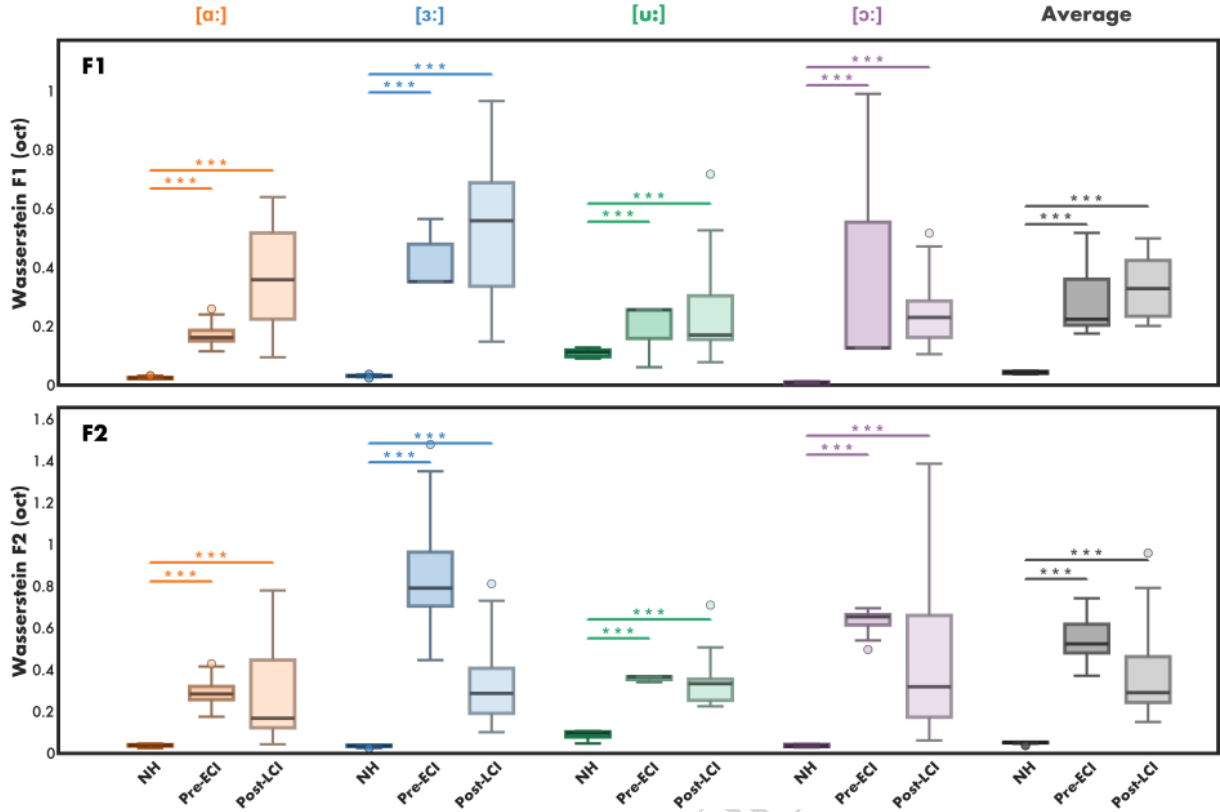


Figure 3.10: **Wasserstein distances from the NH reference bell, per vowel and formant.** Each box shows the distribution across participants for NH (leave-one-out), Post-LCI, and Pre-ECI. Significance bars indicate pairwise contrasts from the LMM ($*p < .05$, $**p < .01$, $***p < .001$, ns: not significant).

Table 3.2: **LMM fixed effects: Wasserstein distance from the NH reference.** Model: $\text{Wasserstein} \sim \text{Group} + \text{Vowel} + \text{Formant} + (1|\text{Participant})$. Reference levels: Group = NH, Vowel = [a:], Formant = F1. $n = 209$ observations from 27 participants. The leave-one-out procedure for NH participants is described in Section 3.2.6.

Term	$\hat{\beta}$	SE	t	p	
Intercept	-0.036	0.033	-1.09	.276	
Group: Post-LCI	+0.306	0.029	+10.46	< .001	***
Group: Pre-ECI	+0.371	0.037	+10.02	< .001	***
Vowel: [ɜ:]	+0.120	0.038	+3.16	.002	**
Vowel: [ɔ:]	+0.059	0.037	+1.62	.108	
Vowel: [u:]	+0.030	0.037	+0.81	.418	
Formant: F2	+0.061	0.026	+2.31	.022	*
<i>ANOVA (Type III):</i>					
Group	$F(2, 201) = 76.00$		< .001		***
Vowel	$F(3, 201) = 3.59$		.015		*
Formant	$F(1, 201) = 5.32$		.022		*

Group was a highly significant predictor of Wasserstein distance ($F(2, 201) = 76.00$, $p < .001$). Both CI groups were substantially displaced from the NH leave-one-out base-

line: Post-LCI ($\hat{\beta} = +0.306$ oct, $p < .001$) and Pre-ECI ($\hat{\beta} = +0.371$ oct, $p < .001$). Crucially, Post-LCI and Pre-ECI did not differ from each other ($F(1, 201) = 2.95$, $p = .088$), indicating that the two groups are displaced from NH to a statistically equivalent degree regardless of their implantation history. Vowel was also significant ($F(3, 201) = 3.59$, $p = .015$), with [ɜ:] showing the greatest displacement ($\hat{\beta} = +0.120$ oct, $p = .002$). Formant was significant ($F(1, 201) = 5.32$, $p = .022$), with F2 displaced further from NH than F1 ($\hat{\beta} = +0.061$ oct).

Formant Centroid Shift

Figure 3.11 displays the frequency centroid shift in octaves for each group, vowel, and formant. Positive values indicate that a participant's confidence distribution is centred at a more apical cochlear location than the NH reference (*i.e.*, the CI user perceives the formant at a lower frequency than expected).

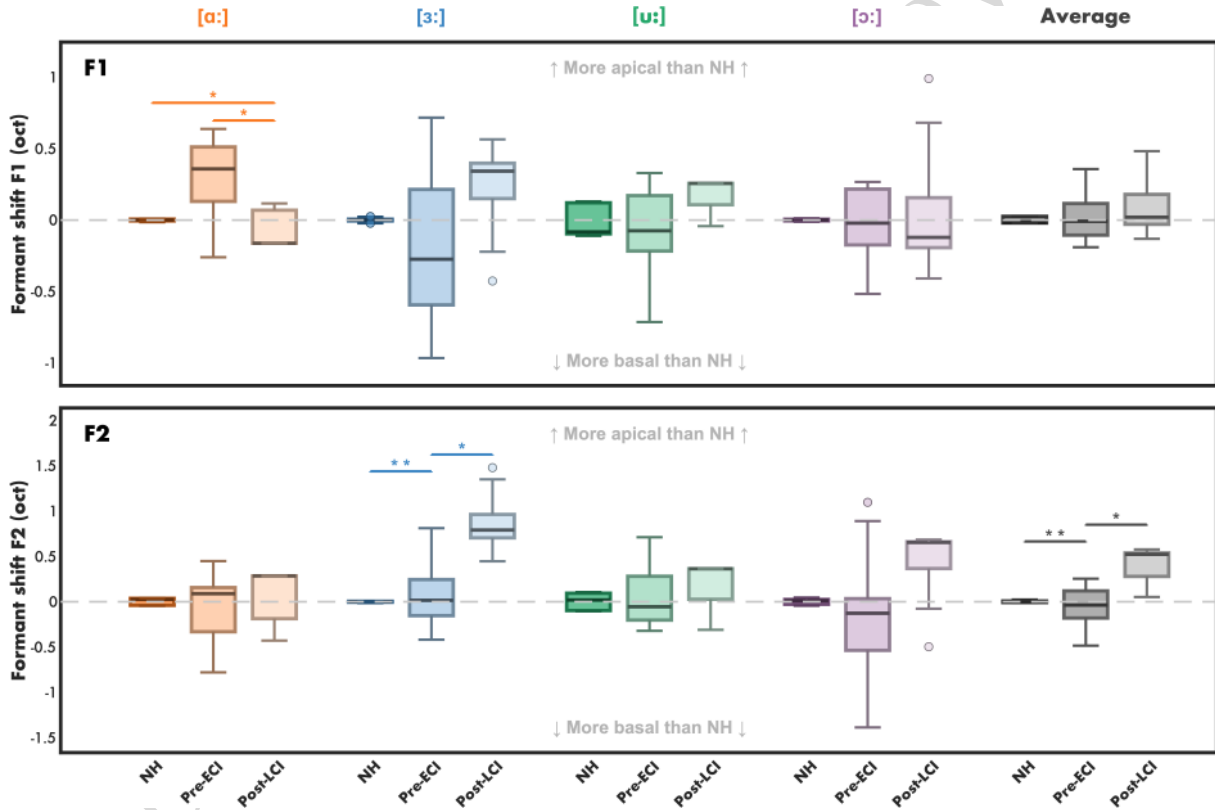


Figure 3.11: **Formant centroid shift relative to the NH reference, per vowel.** Shift = $\log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{CI}})$ in octaves; positive values indicate the CI user's distribution is centred at a lower frequency than NH (more apical). For NH, leave-one-out centroids are used. Significance bars indicate pairwise Wilcoxon rank-sum tests with Holm-Bonferroni correction.

For F1, both CI groups showed a slight tendency towards apical shift relative to NH, consistent with the known apical insertion bias of CI electrode arrays. For F2, a significant difference was found between the two CI groups and between the NH listener group and the Pre-ECI group, which exhibited a preference towards lower F2 frequencies. However,

this observation at the average level should be interpreted with caution because it was largely driven by a single vowel, namely [ɜ:].

F1/F2 Reliance Ratio

Figure 3.12 shows the F1/F2 reliance ratio for each group and vowel. A value of $R > 0.5$ indicates that a participant placed greater confidence in their F2 electrode responses than in their F1 electrode responses.

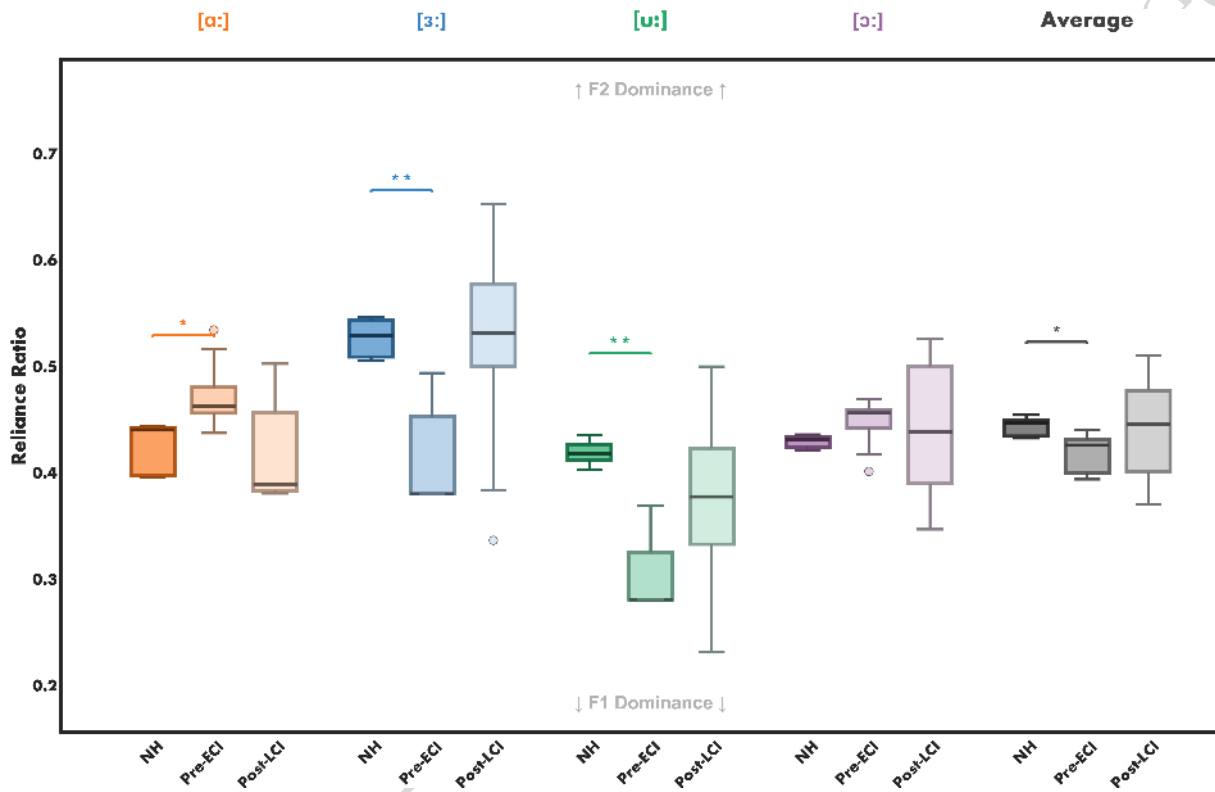


Figure 3.12: **F1/F2 reliance ratio per group and vowel.** $R = \text{peak}(\text{bell}_{F2}) / (\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$. Values above 0.5 indicate F2 dominance; values below 0.5 indicate F1 dominance. Significance bars indicate pairwise Wilcoxon rank-sum tests with Holm-Bonferroni correction (* $p < .05$, ** $p < .01$, ns: not significant).

Descriptively, NH listeners showed moderate F1 dominance on average ($R = 0.443 \pm 0.008$), consistent with the fact that most English vowels are differentiated primarily by F1 (Peterson and Barney, 1952; Hillenbrand et al., 1995). Post-LCI listeners showed a similar mean reliance ($R = 0.439 \pm 0.046$), whereas Pre-ECI listeners showed a lower ratio ($R = 0.418 \pm 0.019$), suggesting a stronger tendency towards F1 dominance in Pre-ECI users. The within-group standard deviations were substantially larger for the CI groups than for NH, reflecting the considerable inter-individual variability in formant reliance among CI users.

Pairwise comparisons revealed that Pre-ECI differed significantly from NH for [a:] (adjusted $p = .037$), [ɜ:] (adjusted $p = .001$), [u:] (adjusted $p = .004$), and at the mean level (adjusted $p = .026$), but not for [ɔ:] ($p = .203$). Post-LCI did not differ significantly from NH for any vowel or at the mean level (all adjusted $p \geq .149$), nor did Post-LCI

differ from Pre-ECI (all adjusted $p \geq .059$).

These results indicate that Pre-ECI users show a distinct pattern of spectral reliance: they weight F1 electrode responses more heavily than both NH listeners and Post-LCI users, particularly for vowels whose identity depends on high F2 frequency ([ɜ:], [u:]). This asymmetry was not present in Post-LCI users, suggesting that prior auditory experience with normal hearing may have enabled Post-LCI users to maintain a more balanced reliance across the frequency axis.

3.3.3 Study Question 2 - Do Frequency-to-Place Factors Predict the Displacement of Electrode Confidence Relative to NH Listeners?

Figure 3.13 displays the delta confidence as a function of each factor, pooled across all CI participants with factor data. At each frequency location, the delta confidence reflects the signed deviation of a CI user's perceptual confidence from the NH reference bell at the same frequency position. Univariate LMMs were fitted separately for each factor and formant, first on the pooled CI sample and then within each group. Results are summarised in Tables 3.3, 3.4, and 3.5.

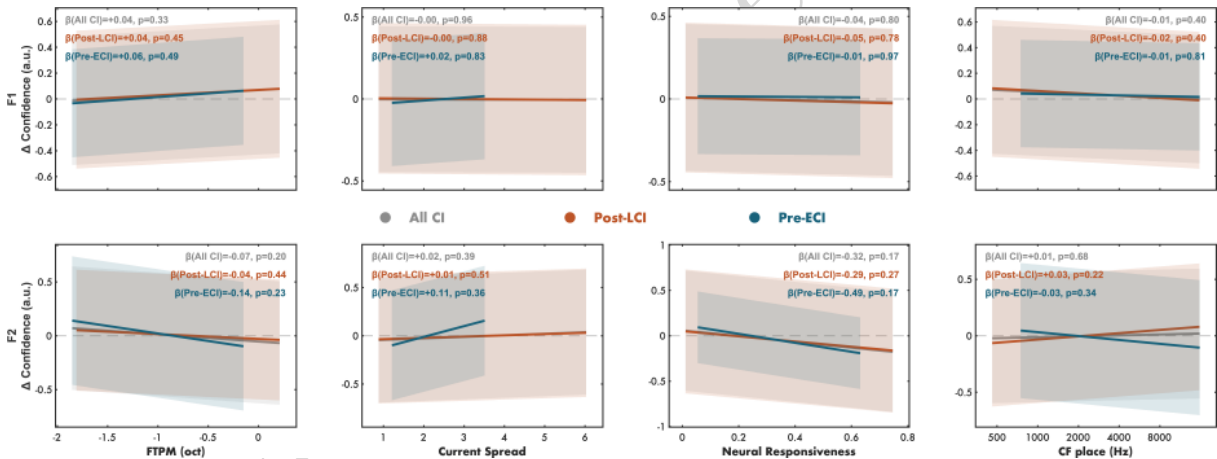


Figure 3.13: Local confidence displacement ($\Delta = \text{bell}_{CI} - \text{bell}_{NH}$) as a function of FTPM, CS, NR, and CF_{place} . Each panel shows the difference between the CI and NH-reference bell height at matched cochlear positions for F1 (top row) and F2 (bottom row). Grey: all CI participants; orange: Post-LCI; blue: Pre-ECI. A value of zero indicates no deviation from NH; negative values indicate that CI users place less confidence at that electrode than NH listeners. Shaded bands represent ± 1 residual SD around the regression line.

Pooled Sample

None of the electrode-level factors significantly predicted the delta confidence for either F1 or F2 in the pooled CI sample (all $p \geq .191$; Table 3.3). This null result held across FTPM, CS, NR, and CF_{place} , for both formants.

Table 3.3: **LMM fixed effects: delta confidence ($\Delta = \text{bell}_{\text{CI}} - \text{bell}_{\text{NH}}$) - pooled CI sample.** Model: $\Delta \sim \text{Factor} + \text{Vowel} + (1|\text{Participant}) + (1|\text{Electrode})$.

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	514	+0.040	0.045	+0.88	.380
F1	CS	388	-0.000	0.013	-0.00	.997
F1	NR	300	-0.040	0.151	-0.26	.793
F1	$\log_2(\text{CF}_{\text{place}})$	514	-0.011	0.019	-0.57	.568
F2	FTPM	514	-0.066	0.052	-1.27	.204
F2	CS	388	+0.016	0.019	+0.87	.385
F2	NR	300	-0.303	0.231	-1.31	.191
F2	$\log_2(\text{CF}_{\text{place}})$	514	+0.008	0.020	+0.41	.684

Post-LCI Group

No factor significantly predicted delta confidence in the Post-LCI group for either F1 or F2 (all $p \geq .289$; Table 3.4). Effect sizes and directions were broadly consistent with the pooled pattern, with the largest (non-significant) effect being a negative NR slope for F2 ($\hat{\beta} = -0.281$, $p = .289$), directionally consistent with the pooled result.

Table 3.4: **LMM fixed effects: delta confidence - Post-LCI group.** Model: $\Delta \sim \text{Factor} + \text{Vowel} + (1|\text{Participant}) + (1|\text{Electrode})$.

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	346	+0.048	0.055	+0.87	.386
F1	CS	308	-0.001	0.014	-0.06	.953
F1	NR	272	-0.044	0.168	-0.26	.796
F1	$\log_2(\text{CF}_{\text{place}})$	346	-0.014	0.026	-0.55	.581
F2	FTPM	346	-0.045	0.058	-0.78	.438
F2	CS	308	+0.015	0.021	+0.71	.477
F2	NR	272	-0.281	0.264	-1.06	.289
F2	$\log_2(\text{CF}_{\text{place}})$	346	+0.028	0.025	+1.12	.262

Pre-ECI Group

No factor significantly predicted delta confidence in the Pre-ECI group (all $p \geq .107$; Table 3.5). The largest effect was a positive FTPM slope for F1 ($\hat{\beta} = +0.156$, $SE = 0.096$, $p = .107$), which did not reach significance and was in the opposite direction to the prediction that greater mismatch would reduce confidence.

Table 3.5: **LMM fixed effects: delta confidence - Pre-ECI group.** Model: $\Delta \sim \text{Factor} + \text{Vowel} + (1|\text{Participant}) + (1|\text{Electrode})$.

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	168	+0.156	0.096	+1.62	.107
F1	CS	80	+0.018	0.080	+0.22	.823
F1	NR	28	-0.012	0.290	-0.04	.966
F1	$\log_2(\text{CF}_{\text{place}})$	168	-0.006	0.025	-0.24	.811
F2	FTPM	168	-0.140	0.116	-1.20	.231
F2	CS	80	+0.111	0.127	+0.88	.383
F2	NR	28	-0.495	0.541	-0.92	.370
F2	$\log_2(\text{CF}_{\text{place}})$	168	-0.034	0.036	-0.96	.338

3.4 Discussion

This study examined how experienced CI users represent vowels in the F1×F2 frequency space and whether frequency-to-place factors could account for deviations from the NH reference. Two groups of CI users were compared according to their hearing history and implantation timing: Pre-ECI and Post-LCI.

Three principal findings emerge. First, both CI groups showed electrode confidence distributions significantly displaced from the NH reference to a statistically equivalent degree. In terms of shift, F2 was shifted towards lower frequencies for Post-LCI users relative to NH and Pre-ECI; however, this shift was relatively moderate and not generalised across all vowels. Second, the two CI groups differed in the relative weighting they assigned to F1 and F2: Post-LCI users maintained a balanced F1/F2 reliance ratio comparable to NH, whilst Pre-ECI users showed F1 dominance on average, suggesting that the two groups differ in their reliance on spectral cues from different cochlear regions. Third, none of the frequency-to-place factors predicted a local confidence deficit relative to NH - a null result that mirrors the absence of systematic vowel category shifts reported in experienced CI users. These results may reflect complete group-level adaptation to the CI signal. Each finding is discussed in turn.

3.4.1 Experienced CI Users Differ Modestly from NH Listeners in Vowel Representation

A natural initial expectation, shared by Harnsberger et al. (2001), is that Post-LCI users - who underwent a remapping process since implantation - should show residual displacement from NH, whilst Pre-ECI users - who built their representations entirely around the CI input - should show a more NH-like pattern.

The present data do not diverge substantially from these expectations: even though the Wasserstein analysis reported a significant displacement between NH listeners and CI user responses, both the shift and reliance analyses clearly mitigate that displacement, with differences restricted to a subset of vowels and not systematic across the full set.

These modest differences are consistent with previous data indicating that the per-

ception of vowels in the frequency space does not differ substantially between experienced CI users and NH listeners. Harnsberger et al. (2001) examined vowel perception in the F1×F2 space of CI users using a method-of-adjustment paradigm and found that seven of eight experienced Post-LCI users showed no systematic shift in vowel category centres relative to NH, concluding that most users had successfully adapted to their device’s basalward spectral shift. The small differences between Harnsberger et al. (2001) and the present data are likely attributable to methodological differences rather than genuine perceptual divergence. In our task, participant responses are constrained to the electrode mapping: even though the NH F1×F2 space is individually rescaled to match each CI user’s electrode layout, the discrete nature of electrode positions limits the available response options and may partly account for the residual differences observed between NH listeners and CI users.

3.4.2 Pre-ECI Users Show Greater Reliance on Apical Electrodes than Post-LCI Users

Despite equivalent overall displacement from NH, the two CI groups differed in a clinically important respect. Whereas Post-LCI users maintained a balanced F1/F2 reliance ratio comparable to NH ($R_{\text{LCI}} = 0.439 \pm 0.046$ vs $R_{\text{NH}} = 0.443 \pm 0.008$), Pre-ECI users showed F1 dominance for [ɜ:], [u:], and on average (all adjusted $p \leq .037$). Two interpretations of this finding should be considered. The first is that Pre-ECI users, having developed their auditory system around electrical input from birth, learned to extract information from apical electrodes more effectively than Post-LCI users. The second - and arguably more parsimonious - interpretation is that greater F1 reliance reflects degraded F2 encoding rather than superior F1 processing: if the spectral quality of F2 cues is poorer for Pre-ECI users (for example, due to reduced neural survival or broader current spread in the basal cochlear regions encoding F2), compensatory reliance on the better-preserved F1 channel would produce the same pattern without implying any advantage in apical processing. Supporting this second interpretation, NH listeners - who have access to both formants at full fidelity - showed a balanced reliance ratio ($R = 0.443$), indicating that when F2 cues are reliably encoded, listeners do not preferentially weight F1. The present data cannot distinguish between these two accounts; the PECAP data available for the small subset of Pre-ECI participants with both CS and NR estimates did not reveal systematic differences in basal neural health between groups, though the sample was too small to draw firm conclusions. This finding illustrates one advantage of the electrode-level design adopted here: because each response is associated with a specific pair of electrodes, the task can reveal asymmetries in spectral reliance that would be invisible in conventional identification-only paradigms.

The difficulty of apical electrode coding has been well documented in the CI literature, and the relevant observations have been made predominantly in Post-LCI populations. At the electrode-array level, Gifford et al. (2022) showed that deeper insertion of a lateral wall array benefits word recognition scores (CNC words in quiet) up to approximately 600° of AID, with performance reaching a plateau thereafter, suggesting a limit on the perceptual utility of the most apical contacts. This plateau is consistent with physiological constraints specific to the apical spiral ganglion: unlike the basal and middle turns, the apical region contains ganglion cells of closely grouped characteristic frequencies, meaning that electrical stimulation from a single electrode activates neural populations that are

not well separated in frequency, reducing channel specificity (Faulkner, 2011). One study even reports that deactivating the most apical electrode in the Post-LCI population can result in an improvement of speech performance (Berg et al., 2025b). However, not all apical stimulation is necessarily detrimental: Berg et al. (2025b) speculated that some CI listeners may be able to exploit temporal interactions between channels, but only in lower-frequency (apical) regions where neural activation patterns overlap more extensively. This raises the possibility that Pre-ECI users, having developed their auditory system around electrical input from birth, may have learned to exploit these temporal interactions in a way that Post-LCI users - whose auditory cortex was shaped by acoustic hearing - cannot.

The present finding that Pre-ECI users relied more heavily on F1 than Post-LCI users is broadly coherent with previous observations concerning spectral cue utilisation in Pre-ECI versus Post-LCI populations, though the direction of the group differences does not map onto the cited literature in a straightforward way. DiNino and Arenberg (2020) directly compared acoustic cue weighting for speech contrasts in Pre-ECI and Post-LCI adults, and found that Pre-ECI users better utilised fine-grained formant cues, with formant cue weights correlating with vowel recognition. Giezen et al. (2010) reported that Pre-ECI users adopted cue-weighting strategies resembling those of NH children for phoneme contrasts, whereas Post-LCI users deviated from NH patterns and showed reduced weighting of spectral structure (Moberly et al., 2014). In the present study, it was Pre-ECI users who differed from NH, not Post-LCI users. This apparent discrepancy likely reflects differences in the perceptual dimension being measured: the cited studies assessed categorical cue weighting for phoneme identification, whereas the present task measures confidence-weighted spectral reliance across the electrode array, which conflates perceptual strategy with the quality of the signal available at each electrode. Nagels et al. (2024) further showed that Pre-ECI children weight both F0 and vocal-tract-length cues for voice gender categorisation similarly to NH children, in contrast to Post-LCI adults who rely predominantly on F0 and fail to use vocal-tract-length cues - which are encoded largely through formant frequencies - consistent with the reduced exploitation of fine-grained spectral cues in Post-LCI users suggested by the literature above.

These converging results support a common interpretation: whether through early cortical plasticity, the absence of competing acoustic representations, degraded encoding of higher-frequency spectral cues, or a combination of these factors, Pre-ECI users appear to have developed auditory strategies that rely more heavily on spectral structure in the low-frequency apical region than Post-LCI users.

3.4.3 Frequency-to-Place Factors do not Explain the Small Displacement

A further finding of the current study is the absence of any relationship between frequency-to-place factors and the small displacement in the perception of vowel representation between CI users and NH listeners. When the NH reference bell was subtracted at each matched electrode position, none of FTPM, CS, NR, or CF_{place} predicted the sign or magnitude of the local confidence displacement in the pooled sample (all $p \geq .191$), and this null result was consistent within both the Post-LCI (all $p \geq .289$) and Pre-ECI (all $p \geq .107$) subgroups.

This null result could reflect two distinct possibilities. First, the task may not be sen-

sitive enough or not adapted to probe individual differences in frequency-to-place factors. The task was designed to go beyond simple identification by incorporating a confidence rating as a proxy for perceptual clarity; however, speech-related tasks are known to be poor proxies for frequency-to-place factors, particularly after the adaptation period. Second, the CI users tested here may be fully adapted to these factors - a possibility supported by the fact that the differences from NH listeners are quantitatively small and restricted to specific vowels and conditions. Both interpretations are plausible and the present data cannot definitively distinguish between them. Chapter 5 may help clarify this: by testing a CI user longitudinally during the adaptation process with the same vowel task, it will be possible to examine whether the task can track changes in the perceptual process and reveal trends related to frequency-to-place factors during the period when adaptation is still ongoing.

3.4.4 Clinical Implications and Future Perspectives

Despite the small displacement of CI users and the absence of a relationship between frequency-to-place factors and vowel displacement, a few results stand out and may inform the clinical management of CI users and suggest directions for targeted mapping strategies.

First, we observed that, on average, the F2 formant was perceived at a lower frequency by Post-LCI users relative to Pre-ECI users and NH listeners. This finding converges with the growing literature on anatomy-based fitting: Creff et al. (2024) showed that tonotopic fitting - aligning electrode allocation to individual cochlear geometry via post-operative imaging - significantly improved speech recognition in noise without degrading performance in quiet, and Dessard et al. (2024); Venail et al. (2015b) found that larger mismatches between default and anatomy-based maps were associated with poorer speech performance and slower adaptation. Recent work by Sagi and Svirsky (2025) further suggests that the appropriate tonotopic target may itself depend on stimulation level, as cochlear tonotopy shifts basally at suprathreshold levels typical of conversational speech, raising questions about which frequency-place function should guide anatomy-based fitting. Taken together, these results suggest that the downward F2 shift observed here reflects a genuine perceptual preference for lower-frequency F2 allocation that could, in principle, be partially corrected through anatomy-informed remapping. Whether such remapping would be beneficial in experienced users - for whom the current allocation has already been adapted to - remains an open question; Svirsky et al. (2005) cautioned that changing an adapted map can temporarily disrupt performance even if the new map is ultimately better matched to cochlear tonotopy, and more recent work has confirmed that the costs of map changes can persist for weeks before adaptation restores performance.

For Pre-ECI users, the F1 dominance finding argues for restraint rather than intervention. If apical F1 electrodes are already well integrated into the perceptual strategy of Pre-ECI users, clinical interventions that deactivate these electrodes or shift the allocation away from the F1 region - motivated by their high FTPM values - may disrupt a well-adapted strategy.

3.4.5 Limitations

Several limitations qualify the conclusions drawn here. The electrode-level factor analyses involved small subsamples (eight participants for FTPM and CF_{place} , six for CS and NR as underlying factors), severely limiting statistical power and precluding per-vowel or per-group breakdowns.

An important methodological limitation is that participants made their F1 and F2 judgements jointly. Consequently, attributing the response to one specific electrode is limited both by the paradigm itself and by the analyses performed, which may partly explain the absence of any effect of individual frequency-to-place factors considered independently for each formant. One potential solution would be to consider them jointly - for example, as a ratio between F1-electrode and F2-electrode factors - though any such analysis would remain constrained by the fact that the confidence rating integrates information from both formants simultaneously. An experimental design alternative is discussed in the Postface of this chapter.

3.5 Conclusion

Using a vowel identification and confidence rating task in the $F1 \times F2$ perceptual space, we characterised and compared the vowel representations of two groups of experienced CI users - Post-LCI and Pre-ECI - relative to NH listeners. Differences between the three groups were small overall, yet Post-LCI users showed a preference for lower F2 frequencies relative to the other two groups, and Pre-ECI users weighted F1 more heavily in their judgements. We examined whether these differences could be explained by three frequency-to-place factors at the electrode level, but none of them accounted for the displacement of CI users relative to their NH peers. This null result may reflect either successful long-term adaptation of CI users to these factors, or insufficient sensitivity of the present task to detect their residual influence.

Postface

The paradigm presented in this chapter aimed to characterise the optimal vowel representation of experienced CI users by presenting different $F1 \times F2$ electrode combinations and asking participants to identify the vowel perceived alongside a continuous confidence rating. The task was thought as a proxy for spectral clarity, which is suspected to be degraded even in experienced users. However, the small differences observed between both CI groups and the NH listeners - and the absence of any relationship between frequency-to-place factors and those displacements - point to two possible interpretations: either the CI users tested were fully adapted to their device's spectral distortions, or the task lacked sufficient sensitivity to detect their residual influence. As Chapter 5 will show, both explanations are likely to be true simultaneously.

Improving the Experimental Design: from Single Sounds to Paired Comparisons. The most fundamental limitation of the present implementation is the use of a single stimulus per trial. Presenting one sound at a time makes the task considerably more demanding for participants and, crucially, conflates certainty about $F1$ and $F2$ in a single confidence rating - making it impossible to attribute a response to one formant independently of the other. This is probably one reason why electrode-level frequency-to-place factors did not reach significance in the individual-level analyses: each trial yields a single confidence value for a joint $F1 \times F2$ combination, so the respective contributions of the two electrodes cannot be fully disentangled.

A natural solution would be to move to a paired-comparison format, as already used successfully in Chapter 4. Two sounds would be presented per trial - either varying both $F1$ and $F2$ simultaneously, or varying only one formant at a time - and participants would judge which sounds more like a unique target vowel. An example interface for this alternative design is shown in Figure 3.14. The single-vowel paired-comparison structure would be easier for participants, and the Bayesian posterior analysis applied in Chapter 4 could be directly adapted here to handle the full set of pairwise comparisons efficiently, even with an adaptive presentation paradigm that reduces the total number of trials required. Varying only one formant per trial would additionally allow independent $F1$ and $F2$ bells to be estimated without post hoc decomposition, directly resolving the primary analytical limitation of the current paradigm.

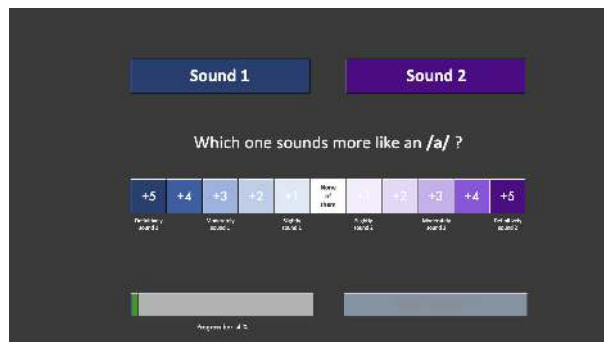


Figure 3.14: **Alternative interface for the vowel task.** In this design, two different $F1 \times F2$ combinations are presented per trial and the participant judges which sounds more like the target vowel. Although all pairwise combinations would in principle need to be tested, an adaptive presentation schedule would substantially reduce the number of trials required. The paired-comparison structure would be easier for participants and would allow Bayesian posterior estimation - as used in Chapter 4 - to be applied efficiently.

Beyond the current implementation and potential improvement of the design, the current paradigm opens several new scientific and clinical directions.

Personalised Remapping from Perceptual Data Alone. The formant centroid shift analysis provides, for each participant, a direct quantitative estimate of how far their $F1$ and $F2$ representations are displaced from the NH reference. This information could be used to derive a participant-specific frequency allocation adjustment without requiring post-operative CT imaging: if a user's $F2$ bell is centred 0.10 oct below NH, that directly quantifies the allocation shift needed to bring their perceptual representation into register. This approach would translate the research paradigm into a clinical tool accessible to any centre, bypassing the need for radiological data. It could be implemented during the adaptation procedure to target directly the best mapping for the participant from Day 1.

Cross-Population and Cross-Modality Extensions. We can imagine providing the test beyond the populations studied in the present study, for example with bimodal CI users, HA users, or people with specific hearing disorders.

Applied to electric-acoustic stimulation users, the current paradigm could identify the perceptual crossover position between acoustic and electric reliance - the cochlear location at which the listener transitions from acoustic low-frequency hearing to electrode-based encoding. Applied to HA users with progressive high-frequency loss, it could characterise how the effective perceptual bandwidth narrows as $F2$ information degrades, providing a continuum model that bridges the audiological and CI literatures.

In people with auditory neuropathy spectrum disorder, who have intact cochlear mechanics but disordered neural timing, the test presented here would reveal whether the displacement from NH reflects the same peripheral geometry as in sensorineural CI users or a qualitatively different pattern.

Online Testing and Large-Scale Normative Data. The most immediate practical extension of this work is an online implementation of the current paradigm, in addition with a version with the proposed modifications.

A standardised stimulus set - built around fixed log-spaced cochlear frequency positions rather than participant-specific electrode centre frequencies - would enable cross-participant and cross-device comparison without normalisation, and would remove the need to enter individual MAP data prior to testing, and even speed up the testing time.

We are currently developing such a platform, which would administer the task remotely, compute the metrics in real time, and benchmark each user's distribution against a normative group of NH reference data collected across age groups and languages.

This would transform the paradigm from a laboratory research tool into a scalable clinical and epidemiological instrument, enabling longitudinal monitoring of perceptual trajectories in large cohorts and the construction of population-level norms against which individual CI users could be evaluated.

Chapter 4

Long-term effects of FTPM, CS and NR on music

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Preface

This chapter presents a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **“Frequency importance function of CI users derived from a music-related task: relationship with frequency-to-place mismatch, neural responsiveness and current spread”**.

As described in Chapter 3 - Preface, the early pilots for both the vowel and music experiments followed a similar trajectory: a MUSHRA-based interface with realistic signals was initially adopted, and subsequently abandoned because, among others, the lack of control over which electrodes carried the perceptually relevant information made it impossible to draw meaningful conclusions with respect to the individual factors studied here (*i.e.*, FTPM, NR, and CS).

Critically, the limitations identified for speech stimuli in those earlier pilots were amplified in the music domain. Whereas speech pilots used a single signal type (*i.e.*, sentences from a female and male speaker), the music pilots employed stimuli drawn from different musical genres, with different levels of complexity (*e.g.*, different number of instruments) with and without lyrics, introducing additional variability in spectral content and temporal structure. Music signals are also inherently more complex than speech: they span a wider frequency range, contain richer harmonic structures, and engage overlapping perceptual dimensions (*e.g.*, pitch, timbre, rhythm, and harmony) that are difficult to disentangle.

As a result, the lack of electrode-level control that was already problematic for speech became untenable for music, reinforcing the need to move toward simpler, fully controlled stimuli.

The remainder of this preface summarises the pilot work that informed the final paradigm; the experiment reported in Sections 4.2–4.5 constitutes the main study. Note that the pilot numbering (Pilot 5b) continues the sequential numbering established in Chapter 3, where Pilots 1-5 covered the speech and early music development stages.

Pilot 5b - Inverted chords with real piano recordings

Approach: This pilot was our first attempt to exploit the concept of chord inversion. A first-inversion chord contains the same pitch classes as the root-position chord, but with the root note raised by one octave so that it becomes the highest-pitched note in the triad - for example, the C major triad C-E-G in root position becomes E-G-C in first inversion. This displacement redistributes spectral energy to a higher cochlear region whilst preserving the same set of notes. In a paired comparison between a root-position chord and its inversion, a listener with uniform cochlear sensitivity should perceive both versions as equally pleasant (pleasantness is operationally defined in Section 4.2.3 as the participant subjective preference in the forced-choice paired comparison). However, if one version places energy in a cochlear region with distorted perception - for instance, due to FTPM or degraded ENI quality - it may be judged as less pleasant. By testing different combinations of inverted chords spanning the electrode array, this paradigm was intended to produce a perceptual profile reflecting the relative contribution of each cochlear region. Real piano recordings, bandpass-filtered to isolate target harmonics, were used as stimuli. It is worth noting that, because the stimuli were derived from real piano recordings, the resulting chords were not always traditional major or minor triads: the bandpass filtering sometimes isolated harmonics that did not correspond to conventional chord tones, and the harmonic spread within each chord depended on the richness of the original recording.

Evaluation:

- **Realistic signals:** *Partially satisfied* - the use of real piano recordings offered ecological validity, as chords embedded in natural timbral content are more representative of everyday music listening than synthetic tones. However, this approach faced the same challenge as the speech pilots: real chords selected according to each participant's electrode mapping did not always align well with individual frequency allocation tables, and consequently did not allow complete coverage of all electrode positions. Synthetic stimuli generated directly from each participant's map were also trialled in a subsequent iteration, and ultimately provided full electrode coverage, motivating the final design.
- **Clinical suitability:** *Satisfied* - the paired-comparison paradigm combined with a Bayesian model was well understood by participants and was considerably simpler and more enjoyable than a subjective rating of a single sound (as in the vowel task).
- **Electrode-level control:** *Partially satisfied* - coverage was achieved for most electrode positions, and the gap between covered regions never exceeded one electrode. However, coverage was not always complete for all participants, and the harmonic content delivered to each electrode was not fully controlled due to the variability in spectral spread across real piano recordings.

Decision: The paired-comparison paradigm and the chord inversion concept were retained, but real piano recordings were replaced by synthetic chords generated dynamically according to each participant's individual electrode mapping. This afforded full control over which electrodes were stimulated in each comparison, over the spectral content delivered to each electrode, and over the harmonic relationships between chord notes.

The final task presented in this chapter is the result of a progressive refinement - from a MUSHRA-based paradigm using realistic recordings, through a real-chord paired-comparison pilot, toward a fully synthetic and precisely controlled chord paradigm designed for electrode-level investigation.

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Frequency importance function of cochlear implant users derived from music-related task: relationship with frequency-to-place mismatch, neural responsiveness and current spread

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Abstract

Music quality and enjoyment are perceptual areas that CI users commonly report as priorities for improvement, even after years of device experience.

Whilst frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) are suspected to contribute to these difficulties, their influence on music-specific perception has rarely been examined at the level of individual electrodes. This chapter presents an original chord paired-comparison paradigm designed to probe electrode-level sensitivity to spectral distortion during music listening.

Twenty-four CI users (fourteen late-implemented, Post-LCI; ten early-implemented, Pre-ECI) rated the pleasantness of chord pairs, each composed of a root-position chord and its first inversion, synthesised to probe specific pairs of electrodes.

Three analyses were performed. Analysis 0 characterised perceptual trends across the full sample: electrode-level frequency importance functions (FIFs) were derived for all participants using a Bayesian paired-comparison model, revealing a systematic apical-to-basal preference gradient that was equivalent across groups. Root-position preference, distortion sensitivity, and cue integration were also compared between Post-LCIs and Pre-ECIs. Analysis 1 examined the effect of FTPM-induced interval distortion on chord-level preference in the subset of participants with available FTPM estimates ($n = 7$). Analysis 2 related the electrode-level FIFs to individual FTPM, CS, and NR estimates in the subset of participants with FTPM estimates ($n = 7$) and PECAP data ($n = 5$).

Results showed a dominant preference for root-position chords across all participants, irrespective of implantation history, most plausibly explained by the acoustic structure of the stimuli rather than individual FTPM profiles. Post-LCI and Pre-ECI users differed in the nature of their distortion sensitivity: Post-LCI users showed inversion-conditional distortion avoidance consistent with a cue integration account, whilst Pre-ECI participants were heterogeneous, with one exhibiting inversion-unconditional avoidance and one showing no distortion sensitivity. In the exploratory factor analyses, CS was the most consistent electrode-level predictor of perceptual preference weight, whilst FTPM showed

a significant positive association in Post-LCI users that was partly confounded with electrode position. Post-LCI users showed significantly greater electrode-level preference differentiation than Pre-ECI users, and a significant $\text{FTPM} \times \text{Group}$ interaction confirmed that the two groups relate to their individual mismatch profiles in opposing directions. These findings provide the first evidence that FTPM-induced spectral distortion affects chord quality perception in CI users, and that this effect depends on both implantation history and the acoustic structure of the stimuli presented.

4.1 Introduction

Music quality and enjoyment are perceptual areas that CI users commonly report as priorities for improvement, even after years of device experience (Drennan et al., 2015). These two aspects of music are, however, fundamentally complex phenomena that cannot be reduced to acoustic fidelity alone. Musical enjoyment is shaped by cognitive and emotional processes - including expectation, cultural familiarity, and reward-circuit dynamics - rather than solely by the precision of spectral encoding (Huron, 2006; Zatorre and Salimpoor, 2013). The clearest evidence for this dissociation comes from congenital amusia, in which severe deficits in pitch discrimination coexist with broadly preserved emotional responses to music (Hyde and Peretz, 2004), demonstrating that spectral accuracy is neither necessary nor sufficient for musical pleasure in general. Consistent with this view, many CI users report continued enjoyment of music despite profound alterations to its spectral and timbral properties (Gfeller et al., 2008), and the relationship between measurable perceptual degradation and subjective musical pleasure remains poorly understood. It is important to note, therefore, that the present study does not address global musical pleasure - a question that implicates high-level cognitive and affective mechanisms largely independent of peripheral spectral coding. Instead, it targets a more constrained perceptual judgement: the relative quality of simultaneously presented chord versions, where quality is determined directly by how faithfully the CI conveys harmonic frequency relationships. This distinction is important, because spectral fidelity - unlike global pleasantness - is potentially addressable through targeted fitting strategies.

Unlike speech, for which accurate perception mainly relies on temporal envelope cues that CI devices transmit relatively well, music perception - specifically its harmonic and timbral dimensions - requires the accurate conveying of substantially more complex acoustic components - including fine spectral structure, pitch accuracy, and harmonic relationships - that current CI technology conveys poorly or may alter (McDermott, 2004; Limb and Rubinstein, 2012).

As a consequence, music quality - and in particular the perception of melody, harmony, and chord structure - is consistently rated among the most impaired aspects of auditory experience by adult CI users (Drennan et al., 2015). Moreover, it has been observed that such degradation is highly variable between individuals: some CI users report meaningful musical enjoyment, while others are profoundly affected (Gfeller et al., 2008), suggesting that individual factors play a significant role in shaping music perception outcomes.

Different categories of individual factors have been identified as contributing to this variability, including auditory history (*e.g.*, duration of deafness (Gfeller et al., 2008), residual acoustic hearing at the time of implantation (Gfeller et al., 2007), duration of CI use (Gfeller et al., 2008), and age at testing (Gfeller et al., 2008)), cognitive abilities (*e.g.*,

working memory, processing speed, and attention (Gfeller et al., 2008)), and musical background and lifestyle (*e.g.*, pre-implant and post-implant musical training (Gfeller et al., 2008; Fuller et al., 2018), and post-implant musical habits (Gfeller et al., 2010)). These same factors have been shown to relate not only to speech perception but also to chord discrimination ability and music enjoyment, though not always in the same pattern as for speech (Lam et al., 2022).

An additional source of variability that cuts across all these factors is the timing of implantation relative to auditory experience. Post-LCIs developed auditory representations through normal acoustic hearing prior to their hearing loss, and must subsequently adapt those pre-existing representations to the constraints of the CI signal. Pre-ECIs, by contrast, never acquired such representations, their auditory representations having been shaped entirely under the CI modality. Although music listening questionnaires report no difference in music sound quality or appreciation between these groups (Moran et al., 2016), the dissociation described above - degraded performance alongside higher appreciation in Pre-ECIs (Bruns et al., 2016; Fuller et al., 2019) - suggests that the two groups relate to their CI signal in fundamentally different ways, possibly reflecting the absence of a spectral reference frame in users who have never heard acoustically.

Of particular interest here is a separate source of variability that, unlike the factors described above, offers genuine perspectives for improving music perception outcomes: the frequency-to-place factors. These factors were previously distinguished as FTPM and ENI quality¹. They are of clinical relevance precisely because they are, at least in theory, addressable through individualised fitting strategies.

FTPM is possibly a key contributor to music degradation among these factors, and a major source of inter-individual variability. Its effects on music perception arise mainly through a shift of the entire frequency spectrum information along the cochlea, a shift which has been evaluated at approximately 1.5 octaves (towards high frequencies) on average and which is highly variable between CI users (Canfarotta et al., 2020a). Crucially, this shift is not uniform: FTPM appears to be non-linear along the electrode array, with greater mismatch typically present at the most apical and basal electrodes than at electrodes located in the middle of the array (Landsberger et al., 2015; Stakhovskaya et al., 2007a). This non-linearity is important because, whilst the auditory system appears capable of adapting to a relatively uniform spectral shift (Baskent and Shannon, 2004), an inconsistent shift along the array might impose qualitatively different demands, disrupting the harmonic organisation of sound rather than merely transposing it. Such non-linear distortion has two particularly detrimental effects for music: first, it may disrupt the harmonic relationships between notes, an aspect which is critical for melody and chord quality perception (a mechanism demonstrated in the study presented in Chapter 2); secondly, the distortion concentrated at the most apical electrodes is especially problematic for music, since crucial musical cues - including the fundamental frequency, melodic contour, lower harmonics, and timbral richness - are encoded by these apical electrodes, and it has been extensively shown that low-frequency information plays a substantial role in sound quality perception in CI users (Galvin et al., 2016; Caldwell et al., 2017a).

An important question is therefore whether, and for how long, FTPM continues to

¹ENI encompasses CS (current spread) and NR (neural responsiveness). Detailed definitions are provided in the preceding chapters. In this chapter, we focus specifically on how these factors contribute to music degradation.

affect music perception in experienced users. For speech comprehension (which constitutes to date the majority of interest in the research literature), the available evidence suggests that the perceptual consequences of FTPM diminish as users adapt over the first year of device use: Mertens et al. (2022) found that FTPM correlated with speech-in-noise performance at six months but that this effect was no longer significant at twelve months, consistent with the central adaptation reported in earlier work (Canfarotta et al., 2020a). The apparent capacity for adaptation to spectral shift is also supported at the level of pitch and channel discrimination, which Mathew et al. (2017) showed improves over the first six months of device use and correlates with speech perception ability. For music perception - specifically melody recognition, pitch discrimination, and chord quality - however, no such adaptation appears to take place. Longitudinal studies in experienced CI users demonstrate that tasks relying on these musical dimensions do not improve spontaneously over time (Gfeller et al., 2010; Sulibhavi et al., 2022).

Whether central adaptation similarly compensates for the perceptual consequences of FTPM for music, as it appears to for speech, therefore remains unknown. A likely explanation for this dissociation is that speech understanding and sound quality are not equivalent: CI users may achieve sufficient speech intelligibility through adaptation to a spectrally distorted signal, yet sound quality - which reflects the fidelity of the perceived frequency representation rather than the ability to extract meaningful sense - remains substantially degraded even in experienced users (Berg et al., 2025b; Caldwell et al., 2017a). Music perception is fundamentally dependent on this sound quality dimension: accurate pitch relationships, harmonic coherence, and timbral fidelity are all determined by the precision of spectral encoding, and are thus directly vulnerable to FTPM-induced distortion in ways that speech understanding is not. Even if there is no strong direct evidence that FTPM is a persistent contributor to music degradation, some recent findings suggest that reducing FTPM through CT-based tonotopic fitting acutely improves music perception outcomes in new CI users (Creff et al., 2024) - whether a similar effect persists in experienced users remains to be established.

In addition to the absence of strong evidence linking music perception to FTPM, existing studies relating FTPM to performance share a common limitation: they characterise FTPM as a single value - typically the mean across the array or related to a specific electrode range - and link this average measure to speech performance (Mertens et al., 2022; Canfarotta et al., 2020a; Venail et al., 2015a) or music scores (Creff et al., 2024; Jiam et al., 2019). Such an approach cannot specifically identify which electrodes are most responsible for music degradation. The electrode-level relationship between FTPM and music perception therefore remains uncharacterised - a gap in knowledge that is of direct clinical relevance, since targeted fitting interventions would require knowledge of which electrodes drive the degradation.

Beyond FTPM, the quality of the ENI (*i.e.*, the functional coupling between each electrode and the local neural population it targets) may represent another source of music perception variability that may interact with FTPM. As part of this study, we examine two aspects of ENI quality, namely CS and NR. CS broadens the spatial extent of neural excitation, reducing spectral selectivity and increasing inter-channel interaction, thereby degrading the place-coding resolution on which pitch and timbre perception depend (Friesen et al., 2001); greater CS at a given electrode may thus amplify the perceptual consequences of any FTPM at that location. NR - broadly reflecting the health and density of surviving spiral ganglion neurons - modulates the effective site of stimula-

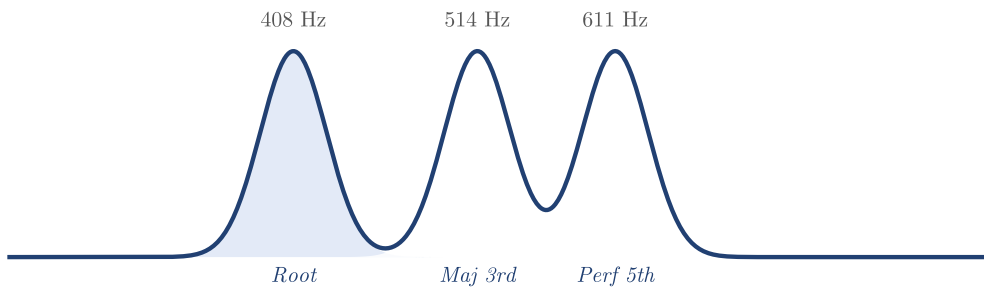
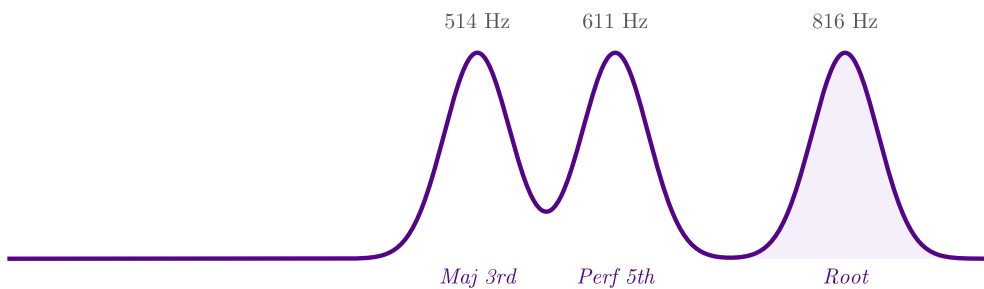
Root position**First inversion**

Figure 4.1: **Spectral structure of a root-position and first-inversion major chord.** Each panel shows the frequency components of a major triad built on a root of 408 Hz, tuned in 12-tone equal temperament. *Top*: root-position chord, comprising the root (408 Hz), major third (514 Hz), and perfect fifth (611 Hz). *Bottom*: first-inversion chord, in which the root note is displaced to the top of the chord (816 Hz), while the major third and perfect fifth remain unchanged. The two chords share two of three note frequencies; only the position of the root note differs. Comparing these two chords in a paired-comparison paradigm isolates the perceptual contribution of the electrodes stimulated by the displaced root note.

tion relative to the targeted cochlear location and the quality of information transmitted by a given electrode; where NR is degraded, neural activation may shift, broaden, or change in ways that compound the effects of both FTPM and CS (Garcia et al., 2021; Lambriks et al., 2023b). Despite the mechanistic plausibility of both these contributions, direct behavioural evidence linking ENI quality to music perception outcomes remains limited (Drennan and Rubinstein, 2008; Başkent et al., 2016), representing a further gap in knowledge in our understanding of what drives variability in music perception.

Together, these gaps motivate the development of an electrode-level perceptual tool capable of identifying which electrodes and associated cochlear regions are most affected by the frequency-to-place factors in terms of music perception - and that could ultimately be used to inform targeted remapping strategies. Achieving such electrode-level probing, however, would require a paradigm that affords precise control over which electrode carries the perceptually relevant information at any given moment - a requirement that realistic musical stimuli, with their associated temporal and spectral complexity, cannot satisfy.

To meet this requirement while preserving musical relevance, the present study uses chords as a stimulus unit. Chords occupy an intermediate position in the hierarchical organisation of Western music, situated between individual notes and complete musi-

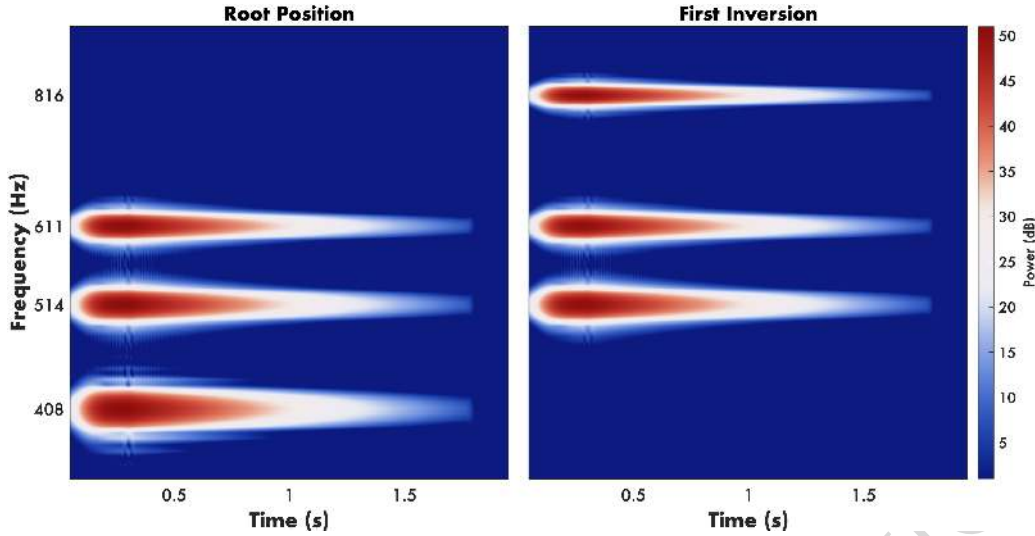


Figure 4.2: **Spectrogram representation of a root-position and first-inversion major chord.** Time-frequency representation of the same major triad as Figure 4.1 (root = 408 Hz, 12-tone equal temperament). *Left:* root-position chord (408, 514, 611 Hz). *Right:* first-inversion chord (514, 611, 816 Hz). The two chords share two frequency components (514 and 611 Hz); only the root note differs in octave position. The sigmoid attack and exponential decay envelope are visible in the temporal structure of each component. Colour scale indicates power in dB.

cal pieces. A chord can be defined as the simultaneous combination of notes at specific harmonic intervals - most commonly thirds in Western music - that together produce a perceptually coherent and pleasant harmonic unit (Plomp and Levelt, 1965). Throughout this study, stimuli consist exclusively of major triads, as major chords are among the most widely recognised harmonic structures and have been extensively used in the psychoacoustic literature on consonance and chord quality perception (Plomp and Levelt, 1965; Lahdelma and Eerola, 2022). The use of chords as a probe stimulus is further motivated by evidence that chord discrimination ability in CI users is systematically related to auditory history and device experience (Lam et al., 2022), suggesting that it captures clinically meaningful individual variability in the harmonic dimension of music perception.

Crucially, chord quality judgements are well suited to addressing the spectral fidelity question precisely because they are not reducible to global musical pleasure. Unlike musical enjoyment - which is shaped by expectation, cultural familiarity, and reward-circuit dynamics largely independent of peripheral spectral coding (Huron, 2006; Zatorre and Salimpoor, 2013) - the pleasantness of a chord version in a paired comparison task is determined directly by whether the harmonic frequency relationships between notes are faithfully conveyed to the auditory system. Chord quality therefore provides a tractable and ecologically grounded proxy for spectral fidelity, one that is sensitive to FTPM-induced distortion while remaining orthogonal to the broader and less tractable question of why humans find music enjoyable in the first place.

More specifically, we use the concept of first-inversion chords to probe music perception at the electrode level. In a first-inversion chord, the root note is displaced from the lowest to the highest position within the triad, effectively moving the root note up by one octave - for example, the C major triad C-E-G in root position becomes E-G-C in first inversion. This displacement redistributes the lowest spectral energy of the chord to a higher cochlear

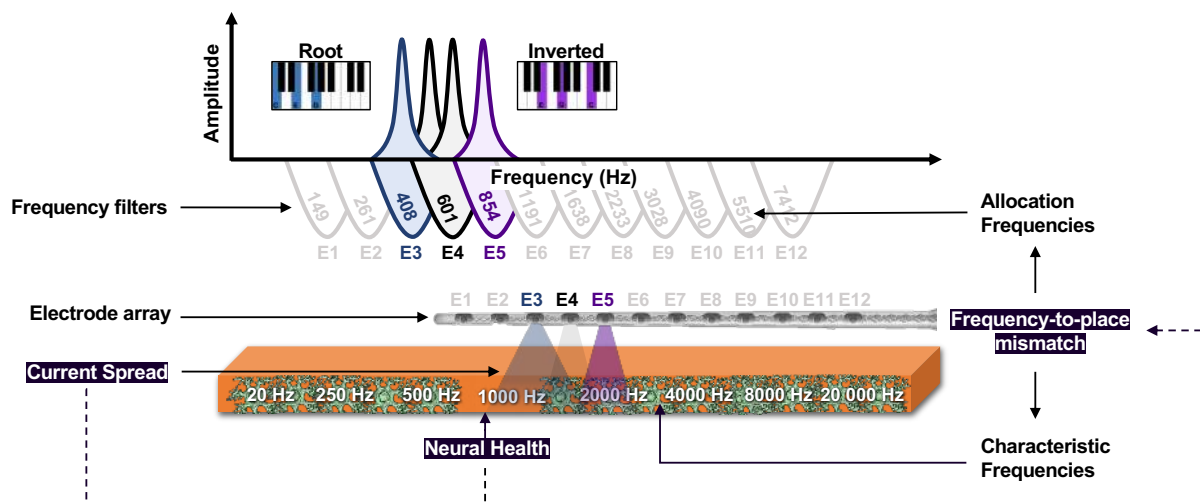


Figure 4.3: **Schematic illustration of the chord inversion paradigm and its potential relationship with the individual factors investigated.** An original chord and its inversion share the same notes, but differ in the octave position of the root note, thereby targeting different cochlear regions. The incoming signal is decomposed into frequency bands and assigned to electrodes E1–E12 according to the processor’s frequency allocation table. Within the schematic representation of the cochlea (bottom), three individual factors determine the relationship between the allocated and the effectively perceived frequency. First, electrode array insertion depth sets the baseline FTPM by imposing a shift between characteristic frequencies and allocation frequencies. Second, current spread (coloured triangles) extends stimulation beyond the target electrode, recruiting neurons tuned to adjacent frequencies; its spatial extent is influenced by local neural health. Third, neural health varies along the cochlea; where spiral ganglion neurons are absent, the effective site of stimulation may shift to more distant surviving neural populations. The dotted lines between current spread and neural health indicate that these factors are suspected to modulate the primary FTPM effect rather than constituting independent sources of mismatch. In the example shown, the root chord’s displaced note targets electrode E3, while the inverted chord’s displaced note targets electrode E5, isolating the perceptual contribution of those two regions.

region whilst preserving the same three pitch classes. Figures 4.1 and 4.2 illustrate this principle for a major triad built on a root frequency of 408 Hz. The root-position chord comprises three simultaneous tones - the root (408 Hz), major third (514 Hz), and perfect fifth² (611 Hz). The first-inversion chord shares the major third and perfect fifth but displaces the root to the top of the chord (816 Hz), thereby redistributing spectral energy to a higher cochlear region whilst preserving the same three pitch classes.

For a listener with uniform sensitivity along the cochlea, both versions should be perceived as equally pleasant; however, if frequency-to-place factors place the root note of one version into a cochlear region where it disrupts the harmonic organisation or the proper conveying of the root note, that version may be judged as less pleasant than the other (see Figure 4.3 for a schematic overview of how these factors interact within the

²The term “perfect” denotes the interval class in standard music-theoretic nomenclature, not acoustic purity; in 12-tone equal temperament the perfect fifth is defined as 7 semitones ($2^{7/12} \approx 1.498$), which approximates but does not equal the just ratio of $3/2$.



Figure 4.4: **Experimental interface for the chord paired comparison task.** On each trial, participants listened to two chords by pressing the Chord A (blue) and Chord B (purple) buttons at the top of the screen. After listening to both chords, participants indicated which they found more pleasant and the strength of their preference using an 11-point bipolar rating scale. The left side of the scale (blue) indicates a preference for Chord A and the right side (purple) a preference for Chord B, with labels ranging from “slightly better” (+1) through “moderately better” (+3) to “significantly better” (+5). The central “=” position indicates no preference. A progress bar at the bottom displayed the proportion of trials completed. Replay was permitted at any point during the trial.

paradigm used in this study).

On each trial, participants listened to both chord versions - the major chord triad and its first inversion - assigned to randomised positions across trials, with replay permitted at any point before responding. After listening to both chords, participants indicated their preference using an 11-point scale anchored at ± 5 (*significantly better*), with intermediate ratings of ± 3 (*moderately better*) and ± 1 (*slightly better*), and a central position indicating no preference (=). The interface is illustrated in Figure 4.4.

Using this paradigm with multiple chord inversion comparisons probing all electrodes, carefully constructed according to each participant’s individual electrode mapping, we address three complementary levels of analysis.

The first level of analysis (Analysis 0) characterises perceptual trends across the full sample of 24 participants (14 Post-LCI, 10 Pre-ECI), independently of individual frequency-to-place factor measurements. As part of this analysis, we derive electrode-level frequency importance functions (FIFs) for each participant using a Bayesian paired-comparison model, and examine whether the resulting profiles differ between Post-LCI and Pre-ECI groups in terms of which frequency regions are associated with higher or lower electrode-level sound quality. We also describe the overall pattern of chord preferences - including root-position preference, no-preference rate, and cue integration - and examine whether these perceptual tendencies differ between groups.

The second level of analysis (Analysis 1) examines, in the subset of participants with available FTPM estimates (5 Post-LCI and 2 Pre-ECI), whether chord preferences at the trial level can be explained by FTPM-induced disruption of harmonic organisation. Precisely, we compare at the trial level the degree of interval distortion introduced by FTPM between harmonics within each chord and relate this to the participant’s choice (an illustration of the paradigm is proposed in Figure 4.5). We assume that, if FTPM-related

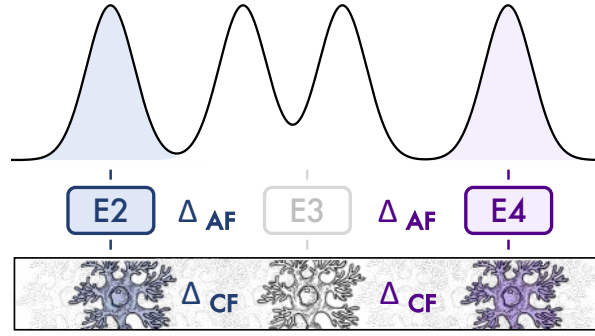


Figure 4.5: **Illustration of interval distortion introduced by FTPM and studied in Analysis 1.** A paired comparison opposes two chords sharing a common electrode; interval distortion arises when the cochlear-place interval Δ_{CF} differs from the allocated-frequency interval Δ_{AF} , *i.e.* when FTPM varies across electrodes.

distortion subsists, the chord chosen should be the one in which FTPM-related distortion is less pronounced. If confirmed, this would provide direct evidence that FTPM-related spectral distortion affects harmonic perception even after long-term experience with the device, suggesting that adaptation does not fully compensate the perceptual consequences of FTPM for music.

The third level of analysis (Analysis 2) relates the electrode-level FIFs derived in Analysis 0 to individual frequency-to-place factor estimates (FTPM, CS, and NR), in the subset of participants with FTPM estimates and PECAP data (see Table 3.1). This exploratory analysis examines whether electrodes associated with greater absolute FTPM magnitude, broader CS, or lower NR receive reduced perceptual weight in the FIF profiles.

Across all analyses, we hypothesise that FTPM-induced distortions carry a greater and more persistent perceptual cost for Post-LCIs, whose internal reference retains a memory of normal acoustic tonotopy, leading to more systematic down-weighting of spectrally distorted electrode regions in their FIF profiles and stronger preference effects in the paired comparison analysis. Pre-ECIs, having never developed auditory representations through normal acoustic hearing, may exhibit perceptual profiles shaped more directly by peripheral deficits (*i.e.*, CS and NR), with less sensitivity to FTPM-related distortions. Given the limited prior literature on this specific comparison, these are treated as directional hypotheses rather than established predictions.

4.2 Methods

4.2.1 Participants

14 Post-Post-LCI users (8 female; mean age = 45.4 years, SD = 13.2; mean duration of CI use = 5.6 years, SD = 2.1) and 10 Pre-Pre-ECI users (6 female; mean age = 25.4 years, SD = 3.4; mean duration of CI use = 20.9 years, SD = 1.0) participated in the study. CI participants were recruited from the Emmeline Centre at Addenbrooke's Hospital (Cambridge, UK). All participants were native speakers of British English. The study was approved by the NHS ethical committee (IRAS number: 335150), and all participants gave informed consent before the experiment.

4.2.2 Stimuli

Stimuli consisted of synthetic chords in which the three composite tones were related by a major triad interval structure, tuned in 12-tone equal temperament (12-TET; frequency ratios of $1 : 2^{4/12} : 2^{7/12} \approx 1 : 1.260 : 1.498$). Unlike conventional musical chords built on standard pitches (*e.g.*, middle C, D, E), the fundamental frequencies of the chord tones were determined algorithmically from each participant's individual frequency allocation map, ensuring that they aligned with the centre frequencies of specific electrodes in the participant's array. The spectral structure and time-frequency representation of two example chords are illustrated in Figures 4.1 and 4.2, respectively.

Chord Generation

Chord stimuli were constructed algorithmically for each participant based on their clinical frequency allocation map.

Electrode Pairing. Electrodes were paired such that the centre frequencies of the two electrodes in each pair were approximately one octave apart (frequency ratio ≈ 2). More precisely, each pair consisted of two electrodes whose centre frequencies differed by a factor close to 2, so that the lower electrode of each pair served as the root of a root-position chord whilst the upper electrode served as the root of the corresponding first-inversion chord. The pairing algorithm iterated over a tolerance range from 5% to 30% (in 1% steps) until every active electrode appeared in at least one pair, with gaps between covered electrode regions never exceeding one electrode position. The final tolerance was participant-dependent, reflecting differences in frequency allocation across devices and maps. Because the number of active electrodes varied across participants and devices (12 for MED-EL, 16 for Advanced Bionics, 20–22 for Cochlear), the number of electrode pairs - and therefore the number of unique chord comparisons - differed between participants.

Chord Construction. For each electrode pair, two chords were generated: a root-position major triad whose root frequency was the centre frequency of the lower electrode, and a first-inversion major triad in which the root was displaced up by one octave to the centre frequency of the upper electrode. Note frequencies were computed in 12-TET. Given a root frequency f , the root-position chord comprised three simultaneous tones at:

$$f_{\text{root}}, \quad f_{\text{root}} \times 2^{4/12}, \quad f_{\text{root}} \times 2^{7/12} \quad (4.1)$$

corresponding to the root, major third (4 semitones), and perfect fifth (7 semitones). The first inversion retained the third and fifth at their original frequencies and relocated the root note up by one octave:

$$f_{\text{root}} \times 2^{4/12}, \quad f_{\text{root}} \times 2^{7/12}, \quad f_{\text{root}} \times 2 \quad (4.2)$$

The two chords in each pair thus shared two of three note frequencies (the major third and perfect fifth), differing only in the octave position of the root note. Critically, this means that each chord version activated a different pair of electrodes: the root-position chord stimulated the lower electrode (carrying the root) and the electrode assigned to the

third and fifth, whilst the first-inversion chord stimulated the upper electrode (carrying the displaced root) and the same shared electrode.

Note-to-Electrode Mapping and Collapse

An important consequence of the coarse electrode spacing in CIs is that the three notes of a major chord do not resolve onto three separate electrodes. Because the frequency separation between adjacent chord notes (a major third ≈ 4 semitones, a perfect fifth ≈ 7 semitones) is smaller than the bandwidth assigned to most electrodes, two of the three notes consistently mapped to the same electrode. For root-position chords, the root occupied one electrode while the third and fifth collapsed onto the adjacent electrode; for first-inversion chords, the third and fifth shared one electrode while the octave-displaced root occupied another. Each chord was therefore effectively rendered as a two-electrode stimulus, with the two active electrodes in each pair determining which cochlear regions were compared across a given trial.

4.2.3 Experimental Procedure

The experiment used a two-alternative forced-choice paired comparison paradigm. On each trial, participants heard two chords - one in root position and one in first inversion - built around different electrode positions along the array. Each electrode pair was presented five times, yielding a total of $5 \times n_{\text{pairs}}$ trials per participant. Short breaks were offered every eight trials. Replay was allowed at any point during a trial.

Each chord was synthesised as the sum of three pure tones at the specified note frequencies, each shaped by a piano-like amplitude envelope comprising a sigmoid attack (300 ms) and an exponential decay (1.5 s), and normalised to prevent clipping. Chords were 2 s long at a sampling rate of 48 kHz. On each trial, the two chords were presented successively, separated by an inter-stimulus interval of 500 ms, and a visual indicator on screen showed which chord was playing at any given moment. Chords were presented monaurally at 65 dB SPL, via a MacBook Pro, an RME Fireface UCX II sound card, and a Sennheiser HD 600 headset positioned over the clinical processor of each CI user.

Response Format

After listening to both chords, participants indicated which chord they found more pleasant and how strongly they preferred it, using a single bipolar rating scale (Figure 4.4). The scale ranged from +5 (*Chord A is significantly better*) through = (*no preference*) to -5 (*Chord B is significantly better*), with an intermediate anchor at ± 3 (*moderately better*), yielding an 11-point response range. This format captures both the direction and the strength of preference in a single response.

The assignment of root-position and inverted chords to the two listening positions was randomised independently on each trial. Trial order was fully randomised within each session.

For the purpose of the analyses described below, the bipolar rating was used differently at each level of analysis, in alignment with the corresponding hypotheses. For Analysis 0,

the full signed grade was used directly to characterise perceptual trends across the sample, and the full graded response was used as input to the Bayesian paired-comparison model (Section 4.2.5) to derive electrode-level FIFs. For Analysis 1, the rating was decomposed into a binary preference (direction of the response) and a preference strength (absolute value of the rating, 1–5); trials on which participants selected the “=” position were excluded from the binary sensitivity check but retained in the graded models (coded as a signed grade of zero), as the zero value carries information - it reflects the absence of preference - and discarding these trials would systematically reduce the effective sample size for participants with high no-preference rates. For Analysis 2, the FIF profiles derived in Analysis 0 were related to individual factor estimates.

Trial and Session Structure

Each electrode pair was tested five times per session, with the assignment of chords to the two listening positions randomised independently on each repetition. The total number of trials per session was $5 \times n_{\text{pairs}}$, where n_{pairs} depended on the number of active electrodes in the participant’s map. Some participants completed two sessions; their data were combined for analysis.

4.2.4 Estimation of Frequency-to-Place Factors

Three electrode-level individual factors were estimated for each CI participant: FTPM, CS, and NR. The estimation procedures are described in detail in Chapter 3 (Section 3.2.4); only a brief summary is provided here.

FTPM

FTPM was estimated from post-operative X-ray images using a semi-automatic algorithm combining electrode localisation, cochlear spiral fitting, and the Stakhovskaya frequency-position function (Stakhovskaya et al., 2007b). For each electrode, the mismatch was expressed in octaves as:

$$\text{FTPM}_e = \log_2 \left(\frac{\text{CF}_{\text{place}, e}}{\text{CF}_{\text{assigned}, e}} \right) \quad (4.3)$$

where $\text{CF}_{\text{place}, e}$ is the characteristic frequency at the cochlear location of electrode e (estimated from post-operative imaging) and $\text{CF}_{\text{assigned}, e}$ is the centre frequency of the allocation band assigned to that electrode in the clinical map. Negative values indicate a basal shift (*i.e.*, the electrode stimulates a cochlear region tuned to a higher frequency than the frequency allocated to it).

CS and NR

CS and NR were estimated using the PECAP method (Garcia et al., 2021), as described in Sections 3.2.4–3.2.4.

4.2.5 Data Analysis

The data were analysed at three complementary levels, corresponding to the sub-goals described in the Introduction. Analysis 0 characterised perceptual trends across the full sample of 24 participants, independently of frequency-to-place factor measurements, and derived electrode-level FIFs from the paired-comparison responses. Analysis 1 examined FTPM-induced harmonic distortion effects on chord preference at the trial level, restricted to the subset of participants with available FTPM data. Analysis 2 related the electrode-level FIFs derived in Analysis 0 to individual factor estimates (FTPM, CS, NR) and listener group, in the subset of participants with available factor data (see Table 3.1).

Analysis 0 - Full-Sample Perceptual Trends and Frequency Importance Function

Analysis 0 was conducted on all 24 participants (14 Post-LCI, 10 Pre-ECI) with chord preference data, regardless of whether individual frequency-to-place factor estimates were available. It addressed four descriptive questions: (i) whether certain frequency regions were systematically associated with higher or lower electrode-level sound quality, and whether the resulting FIF profiles differed between Post-LCI and Pre-ECI groups; (ii) whether participants showed a systematic preference for root-position chords over their inversions; (iii) whether the rate of no-preference responses and the strength of preference grades differed between groups; and (iv) whether the root-preference magnitude differed between Post-LCI and Pre-ECI.

Electrode-Level Frequency Importance Functions. For each participant, paired comparison responses were analysed using `PairedCompCalc` (Leijon et al., 2019), which estimates latent quality parameters on an interval scale from ordinal graded judgements via maximum *a posteriori* (MAP) estimation with a Bayesian prior. The full 11-point bipolar rating was used as input; tied responses (“=”, Grade = 0) were included and handled natively by the model, contributing a symmetric likelihood that provides no discriminating information between the two electrodes. For each participant, the model estimated a latent quality parameter π_e for each electrode e :

$$P(\text{prefer } i \text{ over } j) = \frac{\pi_i}{\pi_i + \pi_j} \quad (4.4)$$

The resulting $\log(\pi_e)$ values were normalised to yield preference scores, with higher values indicating electrodes more consistently and strongly preferred. The most apical electrode served as the Bayesian reference and was fixed at zero by construction; all preference scores were therefore expressed relative to this reference electrode.

To compare FIF profiles across participants with different devices and electrode counts, electrode positions were normalised to a common axis (0 = most apical, 1 = most basal) within each participant. LMMs with a random slope for normalised electrode position were fitted to test for a systematic preference gradient along the array and for group differences in FIF shape:

$$\text{Preference} \sim \text{ElecPos_norm} \times \text{Group} + (1 + \text{ElecPos_norm} \mid \text{Participant}) \quad (4.5)$$

A random slope was included because the Bayesian reference electrode anchors every participant's profile at zero, collapsing between-participant intercept variance to near zero; the between-participant variability is therefore in the gradient of the profile rather than its overall level.

Root-Position Preference. For each participant, a root-preference rate was computed as the proportion of decided trials ($\text{Grade} \neq 0$) in which the root-position chord was chosen, correcting for the random assignment of root and inversion to the two listening positions. A binomial test against chance (50%) was applied per participant. At the group level, root-position preference was modelled using LMMs with signed grade as the dependent variable and S1_IsRoot as the fixed effect, where S1_IsRoot is a binary indicator for whether Sound 1 was in root position (1) or first inversion (0). Models were fitted separately for each group and with a group-by-inversion interaction term in the pooled model:

$$\text{SignedGradeS1} \sim \text{S1_IsRoot} \times \text{Group} + (1 \mid \text{Participant}) \quad (4.6)$$

No-Preference Rate and Grade Strength. The proportion of trials with $\text{Grade} = 0$ was computed per participant as the no-preference rate. The mean absolute grade on decided trials was computed as a measure of preference strength. Both metrics were compared between groups using LMMs:

$$\text{NoPrefNum} \sim \text{Group} + (1 \mid \text{Participant}) \quad (4.7)$$

$$\text{AbsGrade} \sim \text{Group} + (1 \mid \text{Participant}) \quad (4.8)$$

where NoPrefNum is a binary indicator (1 = no preference, 0 = decided) and AbsGrade is the absolute value of the grade on decided trials. Group effects were extracted from the GroupCat_ECI coefficient in each model, providing a direct LMM-based test of Post-LCI versus Pre-ECI differences.

Analysis 1 - Chord-Level Interval Distortion

Analysis 1 addressed whether experienced CI users prefer chord versions whose harmonic structure is less distorted by their individual FTPM profile. It was restricted to the seven participants with available FTPM data (five Post-LCI, two Pre-ECI) and operated at the level of individual paired comparison trials. Graded LMMs predicting the signed preference rating were the primary analysis; a binary GLLM served as a sensitivity check. The rationale for using graded models as primary is twofold: they retain all trials including no-preference responses (coded as 0), maximising statistical power; and they capture preference strength, not just direction, which provides more information about the magnitude of distortion sensitivity.

Electrode Mapping and Note Collapse. Each chord was mapped to two electrodes following the collapse described in Section 4.2.2.

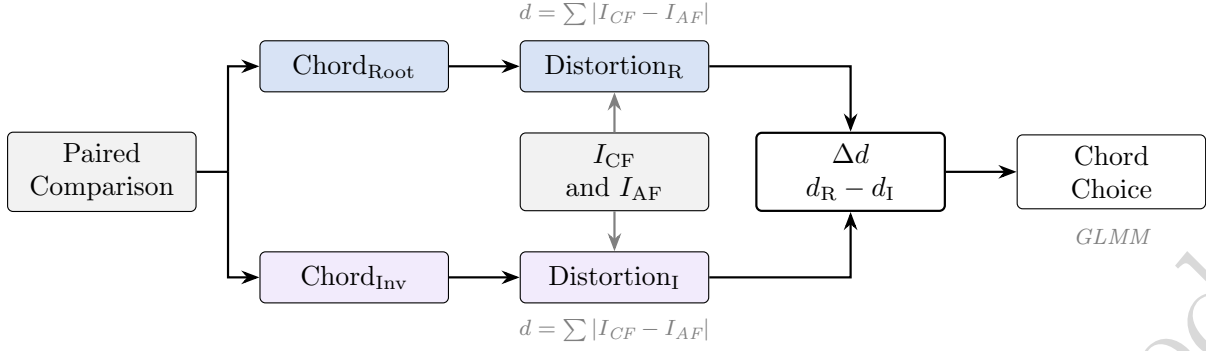


Figure 4.6: **Analysis pipeline for Analysis 1** Each paired comparison trial presents two chords. For each chord, the intended interval I_{AF} (based on allocated frequencies) is compared to the physical interval I_{CF} (based on cochlear place frequencies) to compute interval distortion ($d = \sum |I_{CF} - I_{AF}|$). Distortion reduces algebraically to the FTPM gradient across the two chord electrodes ($I_{CF} - I_{AF} = \text{FTPM}_j - \text{FTPM}_i$), such that a uniform mismatch across electrodes leaves the interval structure intact.

Interval Distortion Computation. For each chord, interval distortion was computed as:

$$d = \sum |I_{CF} - I_{AF}| \quad (4.9)$$

which reduces algebraically to the differential mismatch between the two chord electrodes:

$$I_{CF} - I_{AF} = \text{FTPM}_j - \text{FTPM}_i \quad (4.10)$$

A chord in which both electrodes have identical FTPM values has zero interval distortion; only gradients in mismatch across electrodes disrupt harmonic structure.

Statistical Models. The model set is summarised in Table 4.1. All models are graded LMMs with a random intercept for participant, predicting the signed preference grade (range -5 to $+5$; Grade = 0 retained for no-preference trials). The models are organised hierarchically around a single narrative. M-Base (inversion only) establishes the dominant root-position preference and serves as the reference against which all other models are compared via AIC. M-Grad (FTPM gradient only) tests the mechanistically appropriate predictor directly: because uniform mismatch preserves interval structure whilst gradients across electrodes destroy it, the inter-electrode FTPM difference is the theoretically motivated predictor rather than raw distortion alone. M-Add (distortion + inversion) controls for the root-position confound additively, testing whether distortion exerts a preference effect beyond chord type - this is the model in which the Pre-ECI distortion effect is detectable. M-Int (distortion $\times$ inversion) is the key model: it relaxes the assumption of a common distortion slope across inversion conditions and tests directly whether distortion sensitivity is conditional on chord type. Two group comparison models then examine whether distortion sensitivity differs between groups: M-Grp adds group membership additively, and M-Grp $\times$ tests the group-by-distortion interaction. Finally, M-Int_{bin} is the binary GLLM equivalent of M-Int, fitted on decided trials only (Grade $\neq 0$) as a sensitivity check to confirm that results are not driven by the handling of tied responses.

Table 4.1: **Model set for the chord-level interval distortion analysis (Analysis 1).** All primary models are graded LMMs predicting the signed preference grade (range -5 to $+5$; Grade = 0 retained for no-preference trials) with a random intercept for participant. Δd : difference in interval distortion between Sound 1 and Sound 2 (Equation 4.9); $\Delta \text{FTPM}_{\text{grad}}$: difference in FTPM gradient (absolute inter-electrode FTPM difference) between the two chords; S1_IsRoot: binary indicator for whether Sound 1 was in root position (1) or first inversion (0); Group: Post-LCI vs Pre-ECI. Models with ‘ $\times$ ’ include both main effects and their interaction. The binary sensitivity check uses a GLLM with a logit link on decided trials only (Grade $\neq 0$).

Model	Fixed effects	Role	Research question
M-Base	S1_IsRoot	Baseline	<i>Is preference explained by chord type alone?</i>
M-Grad	$\Delta \text{FTPM}_{\text{grad}}$	FTPM gradient mechanism	<i>Does the FTPM gradient between chord electrodes predict preference?</i>
M-Add	$\Delta d + \text{S1_IsRoot}$	Distortion + inversion control	<i>Does distortion predict preference after controlling for chord type?</i>
M-Int	$\Delta d \times \text{S1_IsRoot}$	Inversion interaction	<i>Does the distortion effect differ between root-position and inverted chords?</i>
M-Grp	$\Delta d + \text{Group}$	Group additive	<i>Do Post-LCI and Pre-ECI differ in overall distortion sensitivity?</i>
M-Grp $\times$	$\Delta d \times \text{Group}$	Group interaction	<i>Does the magnitude of the distortion effect differ between groups?</i>
<i>Sensitivity check (binary GLLM, decided trials only):</i>			
M-Int _{bin}	$\Delta d \times \text{S1_IsRoot}$	Binary equivalent of M-Int	<i>Does the interaction replicate when excluding no-preference trials?</i>

Analysis 2 - Exploratory Electrode-Factor Associations

This analysis related the electrode-level FIF profiles derived in Analysis 0 to individual factors FTPM, CS, and NR via a structured set of LMMs (M0–M9; Table 4.2). It is explicitly exploratory, constrained by the small number of participants with available factor data (see Table 3.1).

Relating Preference to Individual Factors. LMMs were fitted with preference score as the dependent variable, individual factors as fixed effects, and a random intercept for participant. All predictors were standardised (zero mean, unit variance) prior to model fitting to allow direct comparison of coefficient magnitudes. The model set (Table 4.2) comprised single-predictor models (M1–M3), additive models (M4–M5), interaction models between FTPM and ENI quality (M6–M7), and group comparison models (M8–M9). Degrees of freedom were estimated using the Satterthwaite approximation. It should

Table 4.2: **Model set for the exploratory electrode-level factor analysis (Analysis 2).** All predictors standardised (zero mean, unit variance). FTPM: frequency-to-place mismatch (oct); CS: current spread; NR: neural responsiveness. Models M8 and M9 were fitted to the pooled sample only. All models include a random intercept for participant.

Model	Scope	Formula (fixed effects)	Research question
M0	All	Preference ~ 1	<i>Is there systematic variability in electrode preference to be explained?</i>
M1	All	Preference $\sim \text{NR}$	<i>Does neural responsiveness predict electrode preference?</i>
M2	All	Preference $\sim \text{CS}$	<i>Does current spread predict electrode preference?</i>
M3	All	Preference $\sim \text{FTPM}$	<i>Does FTPM predict electrode preference?</i>
M4	All	Preference $\sim \text{NR} + \text{CS}$	<i>Do NR and CS together predict preference beyond each alone?</i>
M5	All	Preference $\sim \text{NR} + \text{CS} + \text{FTPM}$	<i>Do NR, CS, and FTPM each independently predict electrode preference?</i>
M6	All	Preference $\sim \text{FTPM} \times \text{NR}$	<i>Does the FTPM effect on preference depend on neural responsiveness?</i>
M7	All	Preference $\sim \text{FTPM} \times \text{CS}$	<i>Does the FTPM effect on preference depend on current spread?</i>
M8	Pooled Group	Preference $\sim \text{FTPM} + \text{Group}$	<i>Does FTPM predict preference after controlling for listener group?</i>
M9	Pooled Group	Preference $\sim \text{FTPM} \times \text{Group}$	<i>Does the FTPM effect on preference differ between Post-LCI and Pre-ECI?</i>

be noted that electrode-level observations within a participant are not independent; the LMM random intercept accounts for between-participant differences in overall preference level but does not model within-participant spatial autocorrelation across the electrode array. Pearson correlations reported in the collinearity diagnostics are likewise computed on pooled electrode-level observations and should be interpreted as descriptive summaries rather than formal tests of association.

Group Comparison. Between-group differences in the relationship between FIF shape and FTPM were assessed via models M8 and M9, which included Post-LCI versus Pre-ECI group membership as an additive and interactive fixed effect alongside FTPM, respectively.

Data Completeness. Not all individual factors were available for all participants (Table 3.1). FTPM was available for seven participants, CS and NR for a partially overlapping subset. All three factors were simultaneously available for only two participants,

constraining M5–M7 to exploratory status. Models fitted on different subsets are not directly comparable via model selection criteria such as AIC.

Collinearity and Electrode Position Confound. Pairwise correlations among predictors (NR-CS, NR-FTPM, CS-FTPM) were computed on the pooled sample prior to model fitting as descriptive summaries of the degree of overlap between predictors. A structural confound exists between FTPM and electrode position: because electrode arrays do not reach the most apical cochlear regions, FTPM tends to be largest at apical electrodes, introducing a systematic gradient that may be collinear with electrode index (defined here as the sequential integer label assigned to each electrode from apex to base, with electrode 1 being the most apical). Within-participant Pearson correlations between FTPM and electrode index were computed to quantify this confound for each participant.

Random Slopes. A random-slope variant of M3 was attempted. Where convergence failed or slope variance was estimated at near zero, the random-intercept model was retained.

4.2.6 Statistical Analysis

All analyses used LMMs or GLLMs following the general modelling framework of Chapter 3. For Analysis 0, group differences were tested via the `GroupCat_ECI` coefficient in each LMM, and FIF profiles were derived using the Bayesian paired-comparison model and compared across groups using LMMs with random slopes. For Analysis 1, graded LMMs predicting the signed grade were the primary analysis; the binary GLLM `M-Intbin` served as a sensitivity check on decided trials only. For Analysis 2, electrode-level LMMs predicted preference scores from individual factors, with group comparison restricted to the pooled dataset. The significance threshold was set at $\alpha = .05$ throughout. For the single-predictor models in Analysis 2 (M1–M3), the Holm-Bonferroni correction was applied within this family of three tests. All analyses were conducted in MATLAB R2025a; FIF estimation used the `PairedCompCalc` Python package (Leijon et al., 2019).

4.3 Results

The same CI participants described in Chapter 3 were tested; full demographic details and factor availability are given in Table 3.1. For Analysis 0, all 24 participants (14 Post-LCI, 10 Pre-ECI) with chord preference ratings were included. For Analysis 1, seven participants with complete FTPM data were included: five Post-LCI and two Pre-ECI. For Analysis 2, individual factors were present for only a subset of participants; all three factors were simultaneously available for only three participants (Post-LCI 3, Post-LCI 5, and Pre-ECI 1), constraining the full model M5 to these cases.

4.3.1 Analysis 0: Full-Sample Perceptual Trends

Electrode-Level Frequency Importance Functions

Do experienced CI users show systematic electrode-level differences in sound quality, and do Post-LCI and Pre-ECI groups differ in the frequency regions associated with higher or lower preference?

For each participant, a FIF was derived from the paired-comparison responses using the Bayesian model described in Section 4.2.5.

Figure 4.7 shows the group-averaged FIF profiles on a common frequency axis. Both groups showed a similar pattern: preference scores were highest in the apical and mid-frequency regions (below approximately 1.5 kHz) and declined progressively towards the basal end of the array. The LMM with a random slope for normalised electrode position confirmed a significant preference gradient ($\hat{\beta} = -3.19$, $p < .001$), with preference declining from apex to base. The quadratic model provided the best fit by AIC, suggesting a flat then declining profile rather than a uniform linear gradient: preference was relatively stable across apical and mid-array electrodes before dropping in the basal third. The categorical frequency region model confirmed this pattern, with mid-array electrodes significantly lower than apical ($\hat{\beta} = -0.75$, $p = .013$) and basal electrodes substantially lower ($\hat{\beta} = -2.18$, $p < .001$).

No significant group difference was observed in either the overall level of preference differentiation (Group: $p = .39$) or the shape of the preference gradient (Elec_Pos_norm $\times$ Group: $p = .17$), indicating that Post-LCI and Pre-ECI participants showed comparable electrode-level quality profiles.

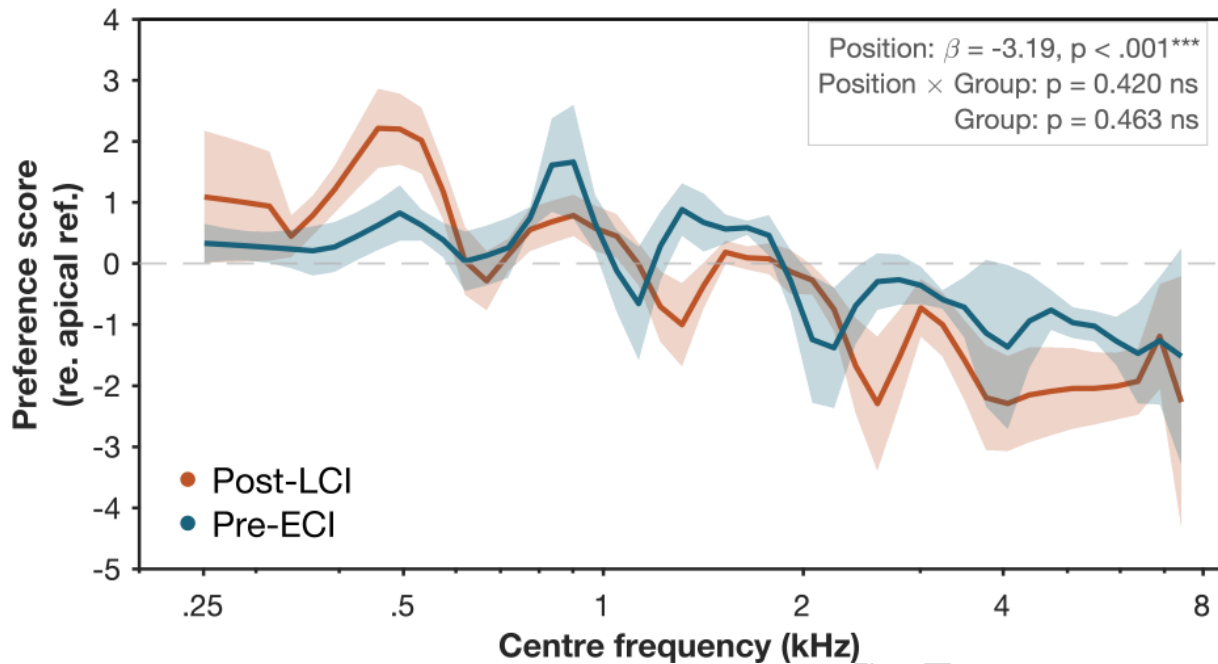


Figure 4.7: **Group-averaged frequency importance functions (Analysis 0).** Group mean FIF profiles interpolated onto a common log-frequency axis. Shaded bands represent ± 1 SE across participants. The most apical electrode serves as the Bayesian reference (fixed at zero) for each participant; all preference scores are expressed relative to this reference. Both groups show a comparable pattern of declining preference from apical to basal regions, with no significant group difference in gradient or overall level.

Root-Position Preference

Do experienced CI users show a systematic preference for root-position chords over their inversions, and does this differ between Post-LCI and Pre-ECI groups?

Across the full sample of 24 participants, a strong and consistent preference for root-position chords was observed in both groups. The LMM fitted to Post-LCI participants yielded a significant positive S1_IsRoot coefficient ($\hat{\beta} = +1.16$, $p < .001$), as did the model for Pre-ECI participants ($\hat{\beta} = +1.15$, $p < .001$). The pooled model with a Group $\times$ S1_IsRoot interaction showed no significant group difference in root-preference slope ($\hat{\beta}_{\text{interaction}} = -0.007$, $p = .980$) and no significant group main effect ($\hat{\beta}_{\text{ECI}} = +0.43$, $p = .271$), confirming that root-position preference was equivalent in magnitude across Post-LCI and Pre-ECI groups. At the individual level, all but one participant exceeded the 50% chance level, and per-participant OLS regression of signed grade on root-position indicator yielded positive coefficients across the large majority of participants in both groups, with broadly comparable magnitudes. These findings establish that root-position preference is a robust perceptual phenomenon that is not contingent on prior acoustic hearing experience.

No-Preference Rate and Grade Strength

Do Post-LCI and Pre-ECI groups differ in their tendency to express no preference, and in the strength of their graded ratings when they do choose?

The LMM for no-preference rate showed no significant group difference ($\hat{\beta}_{\text{ECI}} = -0.031$, $p = .619$), indicating that the tendency to respond with no preference was equivalent across groups. The LMM for mean absolute grade on decided trials, however, revealed a significant group effect ($\hat{\beta}_{\text{ECI}} = +0.796$, $p = .043$): Pre-ECI participants expressed stronger preference ratings than Post-LCI participants when they did choose one chord over the other. Figure 4.8 summarises these group-level comparisons.

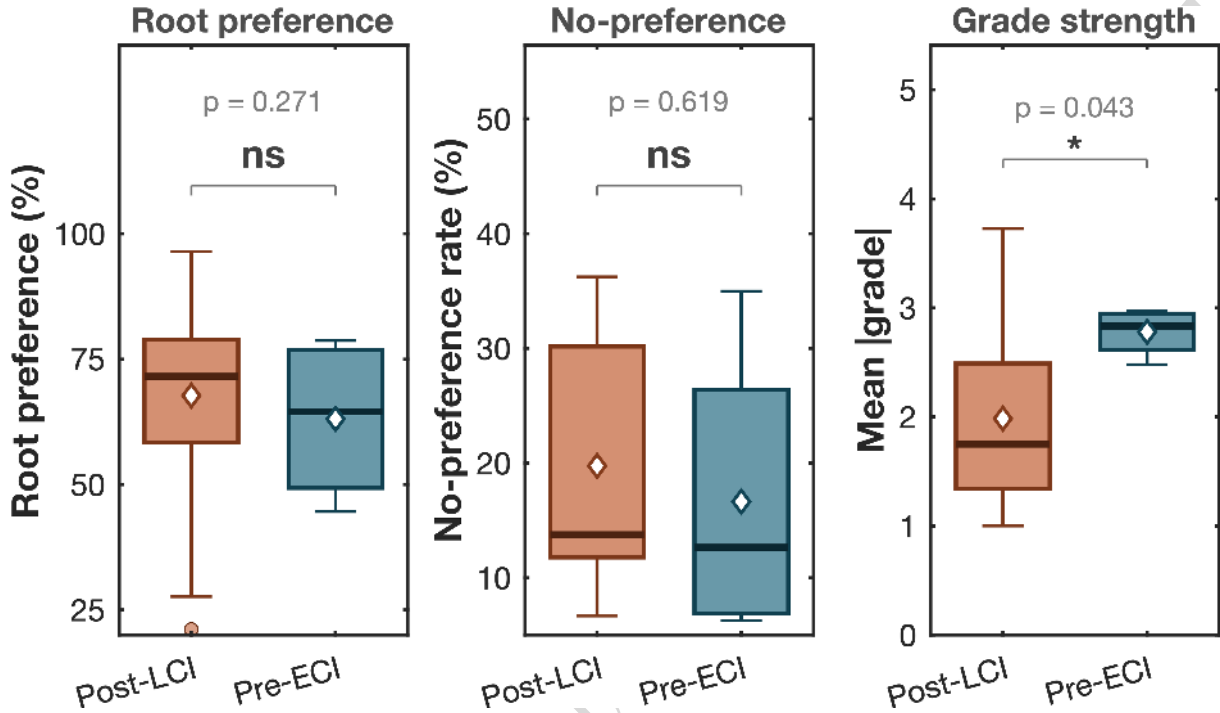


Figure 4.8: **Group comparison of key perceptual metrics (Analysis 0; $N = 24$: 14 Post-LCI, 10 Pre-ECI).** Boxplots show the distribution of per-participant values for root-preference rate (left), no-preference rate (centre), and mean absolute grade on decided trials (right), for Post-LCI and Pre-ECI groups. The diamond marker indicates the group mean. Significance annotations derive from the LMM GroupCat_ECI coefficient: ns = not significant; * $p < .05$.

Summary of Analysis 0

The electrode-level FIF profiles revealed a consistent apical-to-basal preference gradient across both groups, with basal electrodes systematically down-weighted relative to apical regions. No significant group difference was observed in FIF shape or overall preference differentiation, indicating that the electrode-level quality landscape is comparable regardless of implantation history. Both Post-LCI and Pre-ECI participants also showed a strong and equivalent root-position preference, with no significant group difference in preference rate, slope, or no-preference rate. The only significant group difference was in preference strength: Pre-ECI users expressed stronger grades when they did choose one chord over the other ($p = .043$). These findings motivate the treatment of root-position preference as a structural confound in the subsequent distortion analyses, and suggest that the full-sample perceptual profile is dominated by an acoustic cue - the structural advantage of root-position chords - rather than by

frequency-to-place factor differences between groups.

4.3.2 Analysis 1: Interval Distortion and Chord Preference

Descriptive Overview

What proportion of trials resulted in selection of the chord with less FTPM-induced interval distortion, and does this differ from chance across the sample?

The seven participants with FTPM data contributed a total of 620 trials (460 Post-LCI, 160 Pre-ECI). Of these, 488 (78.7%) were decided trials ($\text{Grade} \neq 0$) and 132 (21.3%) were no-preference trials ($\text{Grade} = 0$). The no-preference rate varied considerably across participants (range: 6.2–36.2%), with Post-LCI 5 and Pre-ECI 1 showing the highest rates (36.2% and 35.0%, respectively).

Across all 620 trials, the root-position chord was chosen in 67.4% of decided trials ($p < .001$, binomial test), consistent with the dominant root-position preference observed in Analysis 0. The proportion of decided trials in which the chord with less interval distortion was chosen was 51.4%, not significantly different from chance ($p = .556$, binomial test). A complementary paired t -test on continuous distortion values across the pooled decided sample did not reveal a significant difference between chosen and rejected chords (chosen $M = 0.260$ oct, rejected $M = 0.271$ oct; $t(487) = -1.05$, $p = .292$). Within the Post-LCI subgroup alone, the same test was likewise non-significant (chosen $M = 0.261$ oct, rejected $M = 0.273$ oct; $t(360) = -1.03$, $p = .304$).

The descriptive pattern at the individual level was nonetheless directionally consistent with distortion avoidance in most Post-LCI participants: four of the five showed lower mean distortion for chosen than rejected chords (deltas ranging from +0.006 to +0.074 oct), with Post-LCI 5 as the exception. Among Pre-ECI participants, Pre-ECI 2 showed notably strong binary distortion avoidance (65.3% less-distorted chosen, $p = .011$, binomial), whilst Pre-ECI 1 showed a reversed pattern (38.5%, $p = .126$, ns).

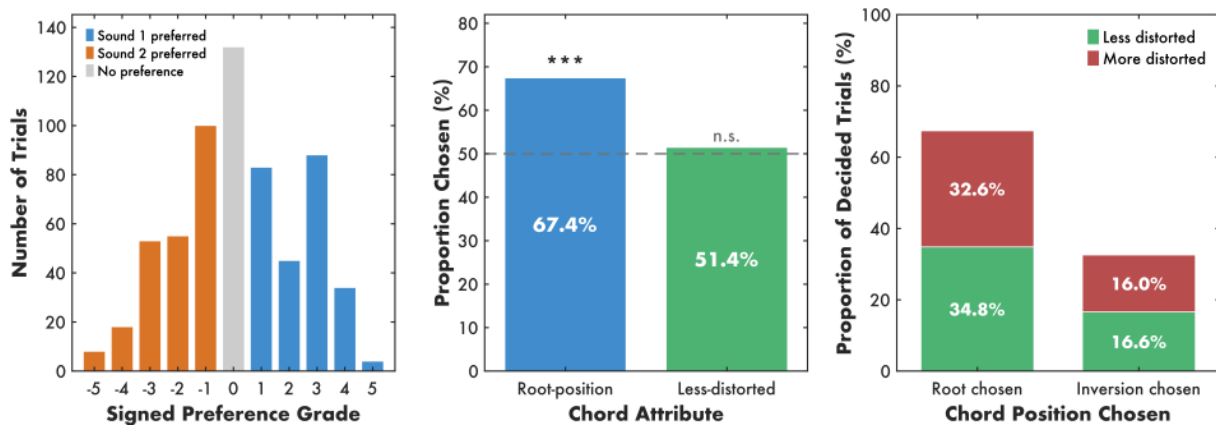


Figure 4.9: **Descriptive overview of chord preference trials (Analysis 1; $N = 7$: 5 Post-LCI, 2 Pre-ECI).** **Left:** Distribution of signed preference grades (negative = inversion preferred; positive = root preferred). Bars coloured by direction of preference; grey bar indicates grade 0 (no preference). **Centre:** Proportion of decided trials in which the root-position chord and the chord with less interval distortion were chosen, relative to chance (dashed line). Significance assessed by two-tailed binomial test against 50%. **Right:** Stacked breakdown of decided trials by chord position chosen (root vs. inversion) and distortion status (less vs. more distorted), expressed as a percentage of all decided trials.

Root-Position Preference

Is the root-position preference consistent across participants with FTPM data, and does its magnitude vary with group membership?

A root-preference grade was computed for each trial by sign-flipping the signed grade on trials in which the inversion was in the reference position, such that positive values always indicate a preference for the root-position chord. The distribution was markedly right-skewed across the pooled sample, confirming a consistent bias towards root-position chords. Per-participant OLS regression of signed grade on root-position indicator yielded positive coefficients for all seven participants; Post-LCI 5 showed the smallest root-preference coefficient among Post-LCI participants, though it remained positive and significant in the baseline model (M-Base: $\hat{\beta} = +0.51$, $p = .003$). Pre-ECI participants showed root-position coefficients as large as, or larger than, those of most Post-LCI participants, confirming that root-position preference was not contingent on prior musical experience with tonal harmony.

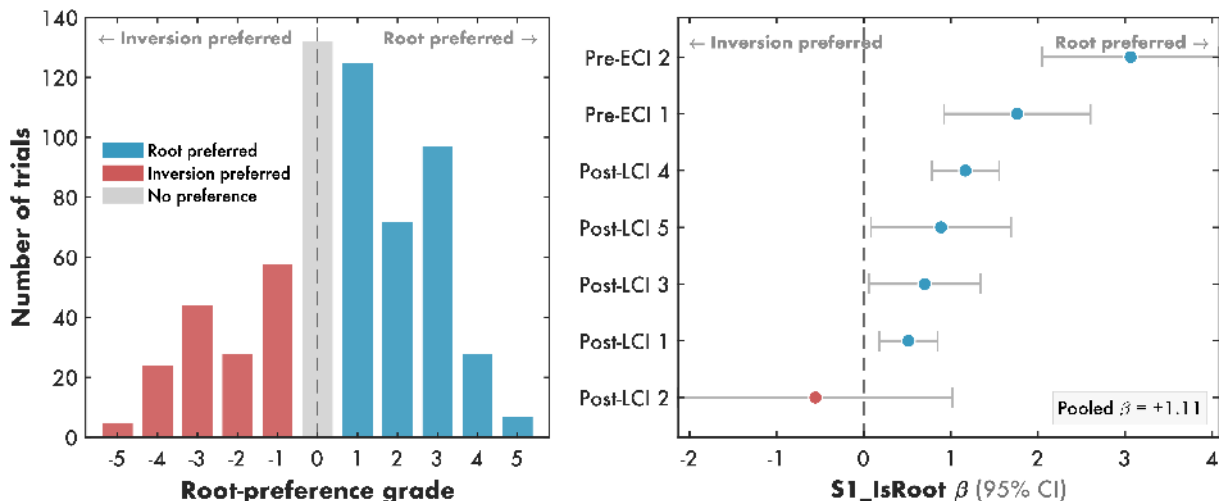


Figure 4.10: **Root-position preference across participants (Analysis 1; $N = 7$).** **Left:** Distribution of root-preference grades, computed by sign-flipping trials in which the inversion chord was in the reference position. Blue bars indicate trials in which the root-position chord was preferred; red bars indicate trials in which the inversion was preferred; grey bar indicates no preference (grade 0). **Right:** Per-participant forest plot of the root-preference regression coefficient ($\hat{\beta}$), estimated by OLS. Horizontal lines represent 95% confidence intervals.

Distortion Effects: M-Grad, M-Add, and M-Int

Does FTPM-induced interval distortion predict chord preference, and does this effect depend on chord type?

To visualise the distortion-preference relationship in a consistent frame of reference, all trials were recoded relative to the root-position chord: the distortion advantage of the root chord ($\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$, negative when root is less distorted) and preference for the root chord (positive grade indicating root preferred) were computed for every trial, eliminating the position-assignment coding artefact. Figure 4.11 shows the resulting trial-level scatter with the M-Int simple slope for root-position trials annotated.

The FTPM gradient predictor (M-Grad) was significant in the pooled sample ($\hat{\beta} = -1.53$, $p = .026$) and in the Post-LCI subgroup ($\hat{\beta} = -1.59$, $p = .042$), consistent with the theoretical prediction that gradients in mismatch across chord electrodes - rather than uniform shifts - are the primary driver of harmonic distortion. When chord inversion was added as an additive covariate (M-Add), the distortion effect remained non-significant at the pooled level ($\hat{\beta} = -0.55$, $p = .136$). However, when the Pre-ECI subgroup was analysed in isolation, M-Add revealed a significant additive distortion effect ($\hat{\beta} = -2.62$, $p = .005$), indicating that when inversion is controlled, Pre-ECI participants showed clear distortion avoidance.

The key model M-Int confirmed that the distortion-preference slope differed significantly depending on chord inversion status (Table 4.3; interaction $\hat{\beta} = -3.03$, $p < .001$, Post-LCI subgroup). Simple slopes derived analytically from the model variance-covariance matrix revealed two contrasting patterns: when the root chord was in the reference position, the distortion slope was -2.03 (distortion avoidance), whereas when the inversion was in the reference position, the slope was $+1.00$ (ns, $p = .078$), indi-

cating that root-position preference dominated regardless of distortion level. The same model fitted to the full pooled sample yielded an equivalent pattern ($\hat{\beta}_{\text{interaction}} = -2.41$, $p = .003$; AIC = 2608.3, best-fitting pooled model). The binary sensitivity check M-Int_{bin} replicated the interaction on decided trials only ($p < .001$), confirming that the result was not driven by the retention of no-preference trials.

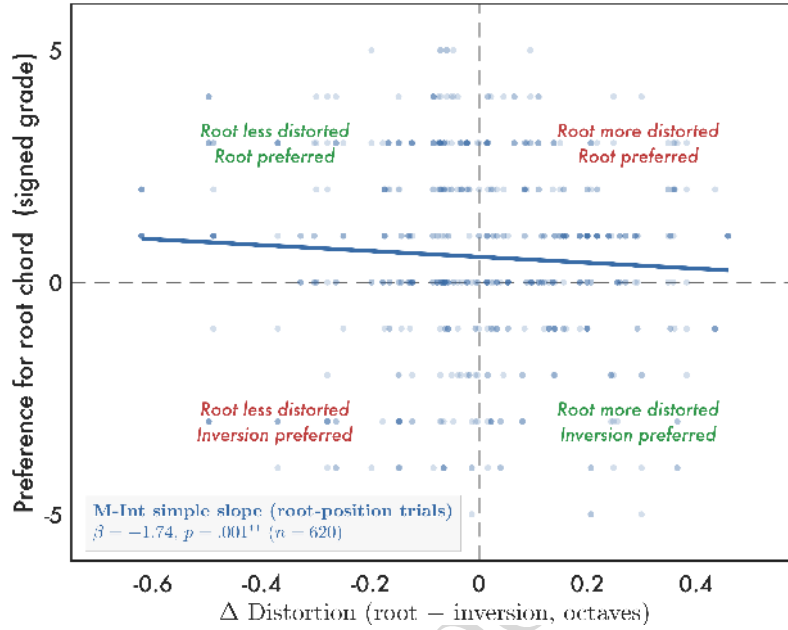


Figure 4.11: **Distortion avoidance in root-position trials (Analysis 1; $N = 7$; 620 trials)**. Each point represents one trial, recoded so that the x -axis reflects the distortion advantage of the root-position chord ($\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$, negative when root is less distorted) and the y -axis reflects preference for the root chord (positive = root preferred). The OLS regression line is annotated with the M-Int simple slope for root-position trials ($\hat{\beta} = -2.03$, $p < .001$): when the root chord carries less interval distortion than its inversion, participants prefer it more strongly. Green quadrant labels indicate the two quadrants consistent with distortion avoidance (top-left: root less distorted and preferred; bottom-right: root more distorted and not preferred); red labels indicate the opposite pattern. First-inversion trials (not shown) yielded a non-significant near-zero slope ($\hat{\beta} = +1.00$, $p = .078$), indicating that root-position preference dominated regardless of distortion level in those trials.

Table 4.3: Post-LCI key model M-Int: SignedGrade $\sim \Delta d \times \text{ChordType} + (1|\text{Participant})$.

Parameter	$\hat{\beta}$	SE	df	p	
Intercept	-0.21	0.34	5.6	.566	
Root-position chord	+0.66	0.17	455.1	<.001	***
Δd	+1.00	0.56	459.7	.078	
$\Delta d \times$ Root-position chord	-3.03	0.85	459.8	<.001	***
Random intercept SD = 0.72; Residual SD = 1.81					
AIC = 1875.4; BIC = 1900.2; LogLik = -931.7					

Individual-Level Interaction Effects

Is the distortion-avoidance effect consistent across individual Post-LCI participants, and does its magnitude and sign differ between Post-LCI and Pre-ECI listeners?

Table 4.4 and Figure 4.12 report the M-Int interaction term for each participant.

Table 4.4: Individual-level Distortion $\times$ Chord-type interaction (M-Int) per participant.

Participant	Group	$\hat{\beta}_{\text{interaction}}$	SE	p	
Post-LCI 1	Post-LCI	-1.38	0.68	.046	*
Post-LCI 2	Post-LCI	-6.06	2.58	.020	*
Post-LCI 3	Post-LCI	-2.84	1.97	.153	
Post-LCI 4	Post-LCI	-11.22	5.80	.058	.
Post-LCI 5	Post-LCI	-1.39	0.80	.086	.
Pre-ECI 1	Pre-ECI	+0.15	3.01	.961	
Pre-ECI 2	Pre-ECI	-3.28	3.73	.391	

Among Post-LCI participants, all five showed negative interaction coefficients, consistent with the group-level pattern. Post-LCI 2 showed the strongest effect ($\hat{\beta} = -6.06$, $p = .020$), with a clear distortion slope reversal between chord-type conditions. A within-chord-type correlation analysis confirmed a significant relationship between electrode-level distortion and preference for root-position chords ($r = 0.76$, $p = .004$) but not for first-inversion chords ($r = 0.28$, $p = .374$). Post-LCI 1 showed a significant interaction ($\hat{\beta} = -1.38$, $p = .046$) and was the only participant for whom the paired t -test on trial-level chosen versus rejected distortion reached significance ($t(93) = -2.47$, $p = .016$). Post-LCI 4 showed a marginally significant interaction ($\hat{\beta} = -11.22$, $p = .058$) despite the smallest trial count and the highest no-preference rate among Post-LCI participants (33.3%). Post-LCI 3 showed a non-significant interaction in the expected direction ($\hat{\beta} = -2.84$, $p = .153$); notably, the FTPM gradient was a stronger predictor than distortion for this participant in the M-Grad model ($\hat{\beta} = -3.00$, $p = .025$). Post-LCI 5 showed a marginal interaction in the expected direction ($\hat{\beta} = -1.39$, $p = .086$), though this participant also showed a reversed binary pattern - chosen chords were on average more distorted than rejected chords ($t(50) = 2.77$, $p = .008$) - warranting caution.

Among Pre-ECI participants, the picture differed qualitatively. Pre-ECI 1 showed no evidence of distortion sensitivity ($\hat{\beta} = +0.15$, $p = .961$), with preference dominated entirely by root-position preference. Pre-ECI 2 showed the strongest binary distortion avoidance of any participant (65.3% of decided trials less-distorted, $p = .011$, binomial) and a significant additive distortion effect once chord type was controlled (M-Add: $\hat{\beta} = -3.88$, $p = .001$). However, the M-Int interaction was not significant for this participant ($\hat{\beta} = -3.28$, $p = .391$), indicating that Pre-ECI 2's distortion avoidance was not conditional on chord type - in contrast to the Post-LCI pattern.

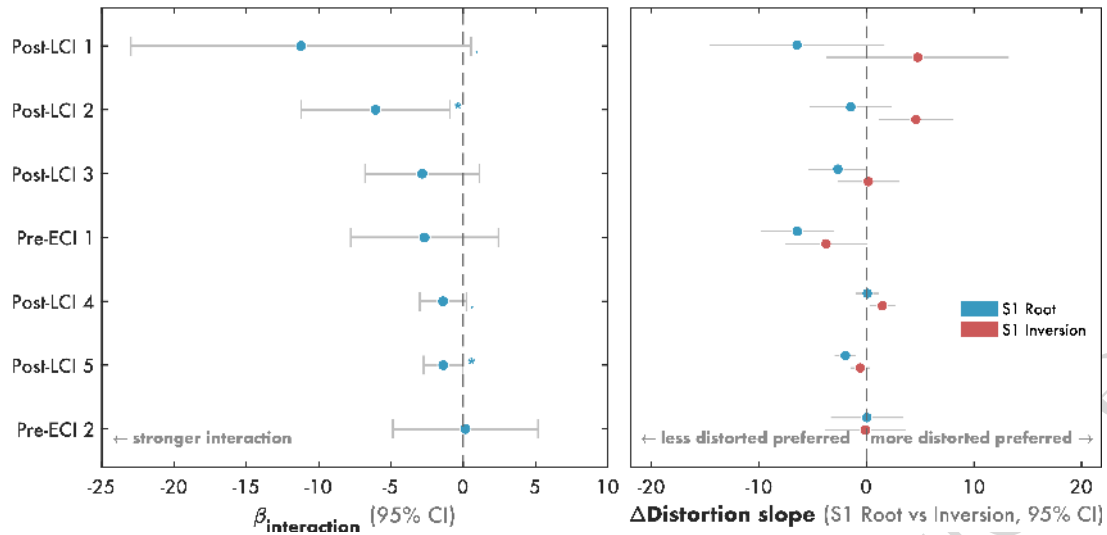


Figure 4.12: **Individual-level Distortion $\times$ Chord-type interaction (M-Int; $N = 7$).** **Left:** Per-participant M-Int interaction coefficient ($\hat{\beta}_{\text{interaction}}$, 95% CI). Negative values indicate that distortion avoidance is stronger in root-position trials than in first-inversion trials. Significance markers: * $p < .05$; . $p < .10$. **Right:** Simple slopes ($\Delta d \hat{\beta}$, 95% CI) per participant, split by chord-type condition. Teal dots: root-position trials; red dots: first-inversion trials.

Group Comparison: M-Grp and M-Grp $\times$

Do Post-LCI and Pre-ECI groups differ in their overall sensitivity to interval distortion (M-Grp), or in the magnitude of the distortion effect on preference (M-Grp $\times$)?

The Distortion $\times$ Group interaction in the pooled graded model M-Grp $\times$ was not significant ($\hat{\beta} = +0.21$, $p = .828$). The binary sensitivity check M-Int_{bin} fitted with a group interaction likewise showed no significant group effect ($p = .558$). The Pre-ECI subgroup paired t -test was non-significant ($t(126) = -0.30$, $p = .765$), though this reflects the averaging of Pre-ECI 2's strong avoidance with Pre-ECI 1's null pattern rather than genuine group equivalence.

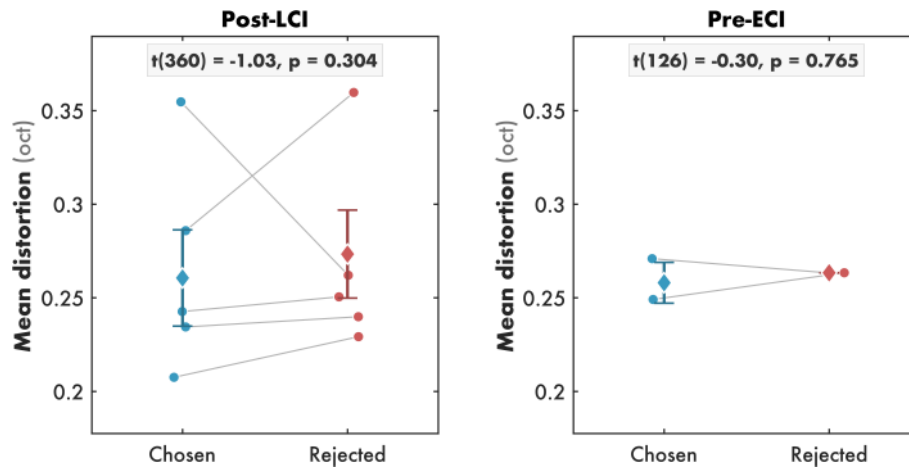


Figure 4.13: **Chosen versus rejected chord distortion by group (Analysis 1).** Per-participant mean distortion for chosen and rejected chords (circles), connected by lines. The diamond marker shows the group mean; whiskers show ± 1 SE. Blue: chosen chord; red: rejected chord. **Left:** Post-LCI subgroup ($n = 5$, 361 decided trials). **Right:** Pre-ECI subgroup ($n = 2$, 127 decided trials). Paired t -test results (trial-level) are shown per panel.

Structural Confound Assessment

Root-position and first-inversion chords did not differ systematically in mean distortion across participants. In the pooled Post-LCI data, root-position chords had mean distortion of 0.250 oct versus 0.269 oct for first-inversion chords. Although inversion chords tended to have slightly higher distortion, the difference was small relative to within-condition variance and did not reach significance at the individual level for most participants. M-Int addresses this confound directly by estimating separate distortion slopes for each chord-type condition.

Summary of Analysis 1

Chord quality preference was primarily driven by root-position preference, with participants favouring root-position chords in approximately 67% of decided trials. Interval distortion exerted a secondary influence that was detectable through M-Int in Post-LCI participants: distortion sensitivity manifested in root-position trials ($\hat{\beta} = -2.03, p < .001$), but was overwhelmed by root-position preference in first-inversion trials ($\hat{\beta} = +1.00, ns$). M-Int provided the best fit to the Post-LCI subgroup ($p < .001$) and the pooled sample ($p = .003$), and the result was replicated in the binary sensitivity check M-Int_{bin}. All five Post-LCI participants showed negative interaction coefficients (two significant, two marginal, one non-significant in the expected direction). The Pre-ECI pattern was heterogeneous: Pre-ECI 2 showed strong distortion avoidance via M-Add that was not conditional on chord type, whilst Pre-ECI 1 showed no distortion sensitivity under any model.

4.3.3 Analysis 2: Exploratory Electrode-Factor Associations

The goal of this analysis was to relate the FIF profiles derived in Analysis 0 to individual frequency-to-place factors, in order to quantify the contribution of FTPM, CS, and NR to electrode-level sound quality. This analysis is exploratory, constrained by the small number of participants with available factor data.

Collinearity Diagnostics

Are the three individual factors (NR, CS, FTPM) sufficiently independent to be interpreted separately?

Among the pooled electrode-level observations (Figure 4.14), NR and CS were moderately correlated ($r = -0.44$, $p < .001$), indicating that electrodes with higher neural responsiveness tended to have narrower current spread. NR and FTPM showed a weaker, non-significant correlation ($r = +0.24$, $p = .124$). CS and FTPM showed a marginal negative correlation ($r = -0.24$, $p = .054$). The NR-CS correlation warranted caution in interpreting joint models M4 and M5. It should be noted that these correlations are computed on pooled electrode-level observations across participants; electrode-level data points within a participant are not independent, and these values should therefore be interpreted as descriptive summaries of the degree of overlap between predictors rather than as formal tests of association.

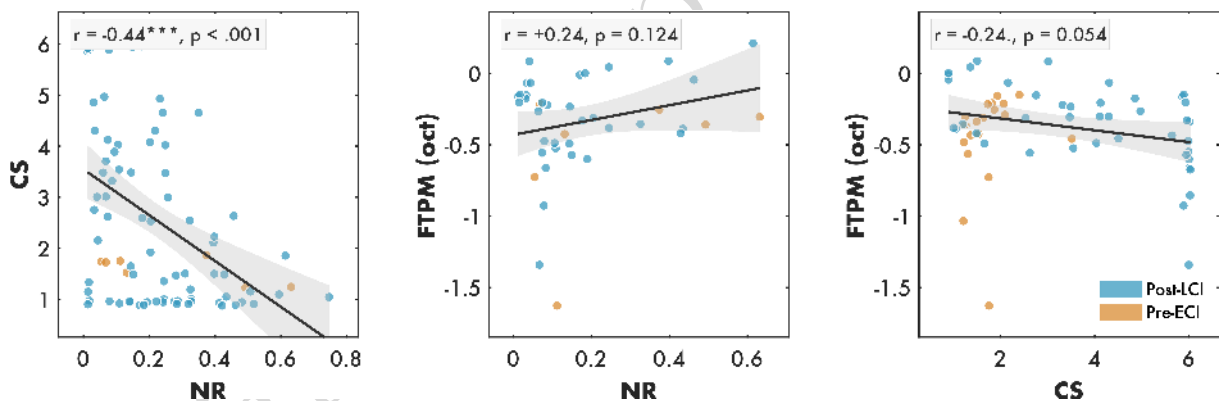


Figure 4.14: **Pairwise collinearity diagnostics for electrode-level factors (exploratory).** Each panel shows electrode-level observations (Post-LCI: blue; Pre-ECI: orange) with a pooled OLS regression line and 95% confidence interval. Pearson r and p -value are annotated per panel. Note: data points within a participant are not independent. **Left:** NR vs CS ($r = -0.44$, $p < .001$). **Centre:** NR vs FTPM ($r = +0.24$, $p = .124$). **Right:** CS vs FTPM ($r = -0.24$, $p = .054$).

Post-LCI: Single-Predictor Models

Does NR (M1), CS (M2), or FTPM (M3) independently predict electrode preference scores in the Post-LCI subgroup?

In the Post-LCI subgroup (Figure 4.15), FTPM was the only significant single predictor of electrode-level preference (M3: $\hat{\beta} = +0.45$, $p = .035$). The positive sign was in the predicted direction: electrodes with less mismatch tended to receive higher preference

scores. However, FTPM covaried with electrode position in this sample (Section 4.3.3), so the effect may partly reflect a preference gradient along the cochlea. CS showed a marginal trend in the expected direction (M2: $\hat{\beta} = -0.40$, $p = .077$). NR was not significant as a single predictor (M1: $\hat{\beta} = +0.01$, $p = .959$).

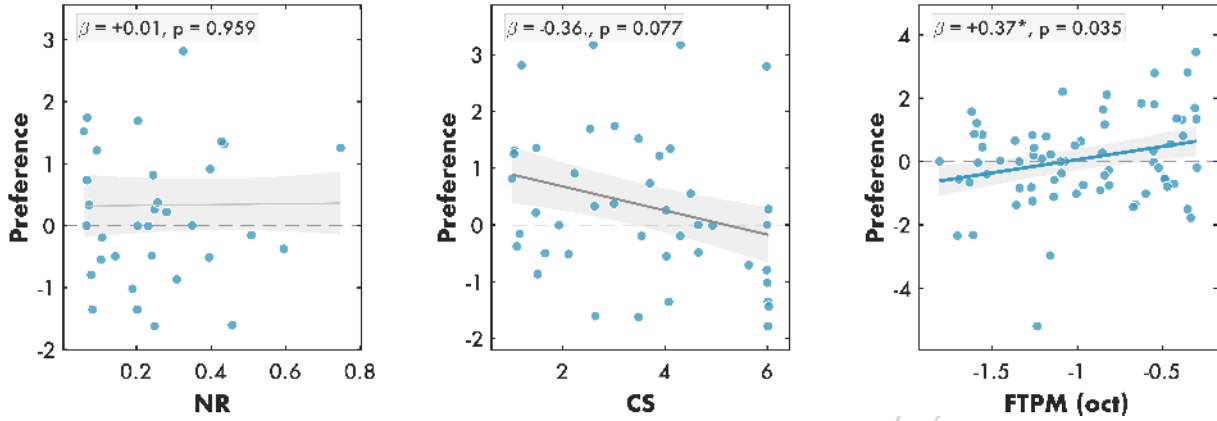


Figure 4.15: **Single-predictor LMM results for the Post-LCI subgroup (exploratory; $n = 5, 42$ electrode observations).** Each panel shows electrode-level preference scores as a function of one factor (NR, CS, FTPM). Dots represent individual electrodes; the line shows the LMM fixed-effect prediction with 95% CI shading. Line weight and colour encode significance: thick blue = significant ($p < .05$); medium grey = marginal ($p < .10$); thin light grey = non-significant. Standardised $\hat{\beta}$ and p -values are annotated per panel.

Post-LCI: Additive and Full Models

Do NR and CS each independently predict electrode preference when entered jointly (M4), and do all three factors contribute independently in the full model (M5)?

When NR and CS were entered jointly (M4; Figure 4.16), CS became significant ($\hat{\beta} = -0.69$, $p = .025$) whilst NR remained non-significant ($\hat{\beta} = -0.34$, $p = .182$). The sign reversal for NR between M1 ($\hat{\beta} = +0.01$) and M4 ($\hat{\beta} = -0.34$) is consistent with suppression arising from the NR-CS collinearity. The full model M5, constrained to the two participants with all three factors available, retained the CS effect ($\hat{\beta} = -0.65$, $p = .037$), whilst FTPM was non-significant ($\hat{\beta} = -0.39$, $p = .467$) and NR marginal ($\hat{\beta} = +0.59$, $p = .118$). The small sample renders M5 exploratory at best.

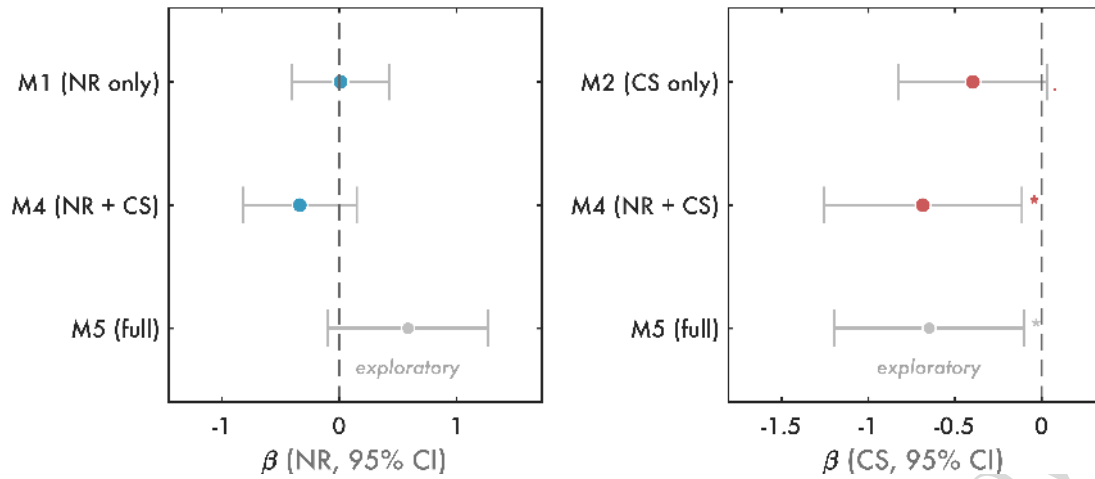


Figure 4.16: **NR and CS coefficients across additive models (Post-LCI; exploratory).** **Left:** NR $\hat{\beta}$ in M1, M4, and M5. The sign reversal from M1 to M4 reflects suppression driven by NR-CS collinearity. **Right:** CS $\hat{\beta}$ across the same models, showing strengthening once NR variance is partialled out. Error bars show 95% CI. Significance markers: * $p < .05$; . $p < .10$. The M5 dot is shown in grey to indicate its exploratory status.

Post-LCI: Interaction Models (Exploratory)

Does the FTPM effect on electrode preference depend on local NR (M6) or local CS (M7)?

The FTPM $\times$ NR interaction model M6 produced significant coefficients for all terms (NR: $\hat{\beta} = -3.76$, $p = .003$; FTPM: $\hat{\beta} = +6.15$, $p < .001$; interaction: $\hat{\beta} = +6.78$, $p < .001$). The FTPM $\times$ CS interaction (M7) was also significant (CS: $\hat{\beta} = +1.46$, $p = .093$; FTPM: $\hat{\beta} = +5.14$, $p = .014$; interaction: $\hat{\beta} = -3.27$, $p = .014$). Both interaction models were fitted on very small samples and are reported as exploratory.

Group Comparison

Does FTPM predict electrode preference after controlling for group membership (M8), and does the FTPM effect differ between Post-LCI and Pre-ECI listeners (M9)?

The group comparison models were fitted to the pooled sample with FTPM data available for both groups. M8 revealed a significant group effect: Post-LCI participants had higher mean electrode-level preference scores than Pre-ECI participants ($\hat{\beta}_{\text{Group, LCI}} = +0.68$, $p = .046$), consistent with greater perceptual differentiation across the electrode array in post-lingually implanted users. The FTPM $\times$ Group interaction (M9) was significant ($\hat{\beta} = +0.81$, $p = .043$; Figure 4.17). Decomposing the interaction, the simple slope for Post-LCI was $+0.37$ (less mismatch associated with higher preference, the predicted direction) and for Pre-ECI was approximately -0.35 (counter to predictions). The opposing slopes explain why the pooled M3 showed a near-zero FTPM effect ($\hat{\beta} = +0.06$, $p = .744$).

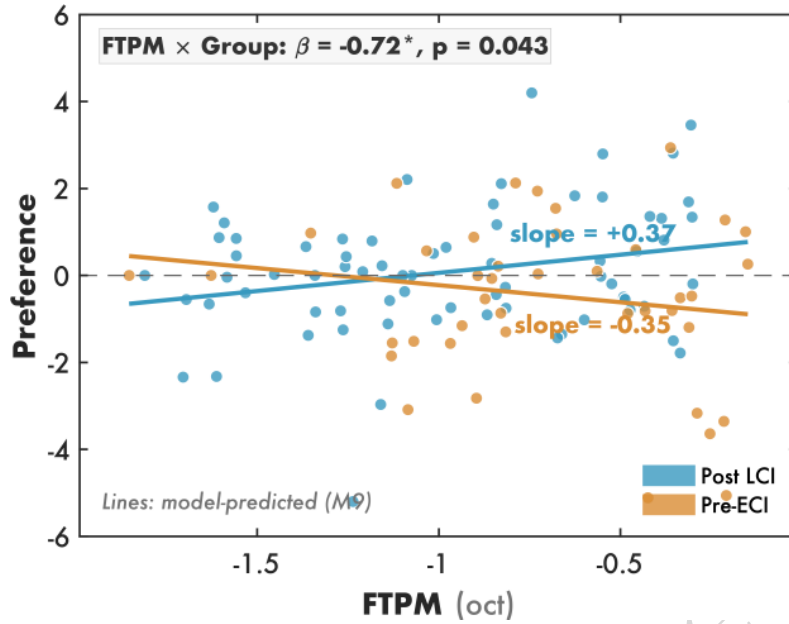


Figure 4.17: **FTPM $\times$ Group interaction on electrode preference (M9; exploratory; $N = 7$).** Electrode-level preference scores as a function of FTPM (oct) for Post-LCI (blue) and Pre-ECI (orange) participants. Lines show model-predicted simple slopes from M9. The opposing slopes ($+0.37$ for Post-LCI, ≈ -0.35 for Pre-ECI) illustrate how the two groups' effects cancel in the pooled model without a group term.

Pre-ECI Subgroup

Do NR (M1), CS (M2), or FTPM (M3) predict electrode preference in the Pre-ECI subgroup?

No single predictor reached significance in the Pre-ECI subgroup (M1: $\hat{\beta} = +0.43$, $p = .593$; M2: $\hat{\beta} = +0.05$, $p = .918$; M3: $\hat{\beta} = -0.31$, $p = .317$; Figure 4.18). In the full model M5, constrained to the single participant with all three factors available, FTPM reached significance ($\hat{\beta} = -1.21$, $p = .043$), but the direction was counter to predictions and the sample size precludes generalisation.

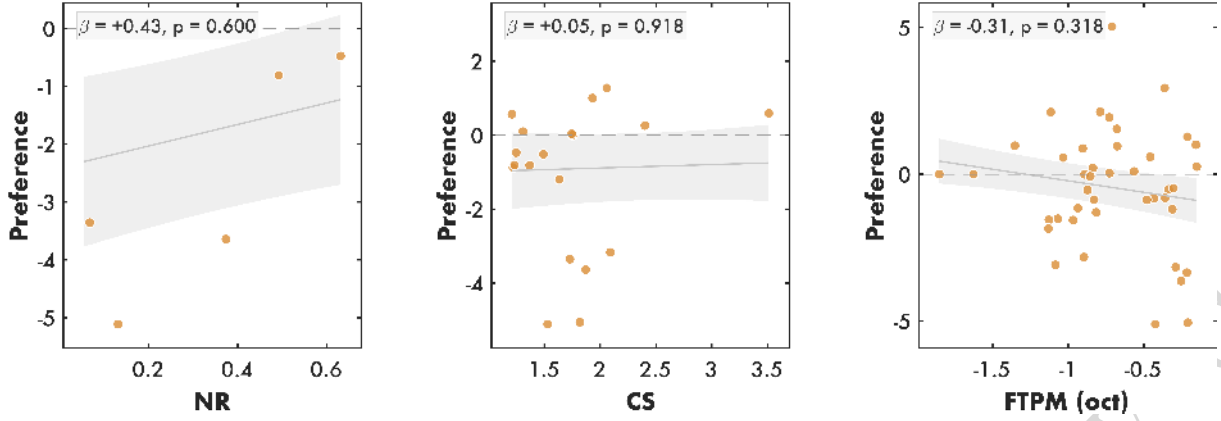


Figure 4.18: **Single-predictor LMM results for the Pre-ECI subgroup (exploratory; $n = 2$).** Layout identical to Figure 4.15. No predictor reached significance in this subgroup. Note: data points are drawn from two participants only; individual profiles are shown in Appendix [X].

Electrode Position Confound

Is FTPM systematically correlated with electrode position along the array, and could this confound the FTPM-preference association observed in M3?

FTPM was significantly correlated with electrode index in the pooled data ($r = +0.25$, $p = .004$). At the individual level, the correlation was significant for four of the seven participants with FTPM data, with Post-LCI 2 as the notable exception showing a negative slope ($r = -0.71$). This structural relationship complicates the interpretation of M3: the positive Post-LCI coefficient ($\hat{\beta} = +0.45$) is in the predicted direction but may be inflated by a preference for basal cochlear regions that coincides with less mismatch.

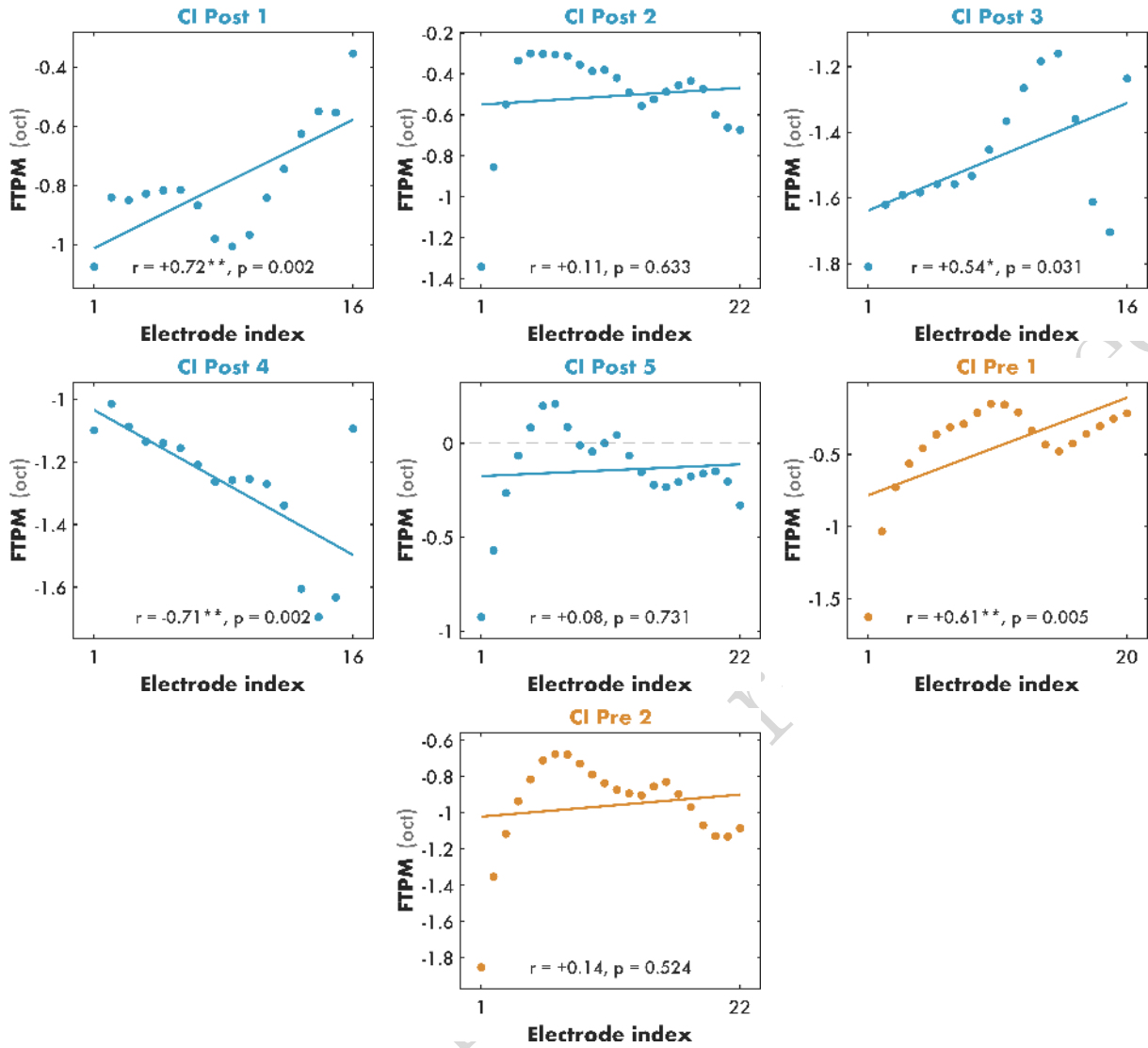


Figure 4.19: **FTPM as a function of electrode index per participant (exploratory; $N = 7$).** Each panel shows electrode-level FTPM values (oct) plotted against electrode index, with an OLS regression line. Pearson r and p -value are annotated per panel; note that electrode-level observations within a participant are not independent. A positive slope indicates that apical electrodes have greater mismatch than basal electrodes.

Random Slopes

Do participants differ systematically in their FTPM-preference slope?

The random-slope variant of M3 estimated a near-zero slope variance in all groups, and the fixed-effect estimate was unchanged from the random-intercept model, indicating either homogeneous FTPM-preference slopes across participants or insufficient power to estimate between-participant slope variability.

Summary of Analysis 2

The exploratory electrode-level factor analysis yielded a preliminary pattern. CS was the most consistent single predictor: electrodes with narrower current spread tended to be preferred, an effect that reached significance in the joint NR-CS model (M4: $\hat{\beta} = -0.69$, $p = .025$) and was maintained in the full model M5 ($\hat{\beta} = -0.65$, $p = .037$). FTPM showed a significant positive association with preference in the Post-LCI group (M3: $\hat{\beta} = +0.45$, $p = .035$), consistent with the prediction that less mismatch leads to higher preference, though the electrode position confound means this cannot be interpreted unambiguously. The $\text{FTPM} \times \text{Group}$ interaction (M9) was significant ($p = .043$), with opposing FTPM slopes in the two listener groups (Post-LCI $\approx +0.37$; Pre-ECI ≈ -0.35), confirming that the absence of a pooled FTPM effect reflects cancellation rather than a genuine null. The overall group effect on electrode-level preference was significant (M8: $\hat{\beta} = +0.68$, $p = .046$), with Post-LCI participants showing higher mean preference differentiation across the electrode array than Pre-ECI participants, consistent with greater sensitivity to electrode-level spectral quality in users with prior acoustic hearing experience. Interaction models M6 and M7 hinted that the FTPM-preference relationship may be modulated by local NR and CS, but these results are constrained to a small number of participants and must be considered provisional.

4.4 Discussion

This chapter examined whether FTPM, CS, and NR affect chord perception in experienced CI users, and whether those effects differ between Post-LCI and Pre-ECI participants. Three complementary analyses addressed this question at different levels of the paired-comparison task: Analysis 0 characterised perceptual trends across the full sample independently of factor data; Analysis 1 examined chord-level preference between root and inverted versions in relation to FTPM-induced distortion; and Analysis 2 examined electrode-level perceptual weight through FIFs derived from the same task, comparing these weights with all three frequency-to-place factors.

The first sub-goal was to describe the overall pattern of chord preferences, to derive electrode-level FIFs across the full sample, and to examine whether perceptual tendencies differed between Post-LCI and Pre-ECI groups. Analysis 0 revealed a systematic apical-to-basal preference gradient in the FIF profiles that was equivalent across groups, confirmed that root-position preference was equally strong in both groups, and showed that Pre-ECI participants expressed stronger preference grades on decided trials.

The second sub-goal was to determine whether experienced CI users systematically prefer chord versions whose harmonic structure is less distorted by their individual FTPM profile. Analysis 1 confirmed this in part: a detrimental effect of within-chord FTPM-induced distortion was observed, but with different expressions across groups. In Post-LCI users the effect was conditional on chord inversion, whilst in Pre-ECI 2 distortion avoidance was inversion-unconditional.

The third sub-goal was to characterise FIFs across the electrode array and to examine whether electrodes associated with greater absolute FTPM magnitude receive reduced perceptual weight. Analysis 2 confirmed this in the Post-LCI group, with electrodes having larger absolute mismatch associated with lower preference weight.

The fourth sub-goal extended this to ENI quality: CS showed a clear effect (narrower spread associated with greater electrode preference) whereas NR had no detectable influence. The group comparison across both factor analyses (Analysis 1 PG3a; Analysis 2 M8, M9) further supported differences between Post-LCI and Pre-ECI users, though the qualitative nature of those differences was more nuanced than originally anticipated.

All these points are discussed in turn, beginning with Analysis 0, then Analysis 1, then Analysis 2, and finally both factor analyses conjointly.

4.4.1 Analysis 0: Full-Sample Perceptual Trends

Electrode-Level Frequency Importance Functions The FIF profiles derived from the full sample revealed a systematic preference gradient along the electrode array, with apical and mid-array electrodes receiving higher preference weights than basal electrodes. This gradient was equivalent across Post-LCI and Pre-ECI groups, and the best-fitting model was quadratic, suggesting a relatively flat profile across apical and mid-frequency regions followed by a decline in the basal third of the array. An immediate interpretation of this pattern would be that low-frequency electrodes contribute more to music-related sound quality than high-frequency electrodes, consistent with the literature demonstrating the importance of low-frequency content for sound quality perception in CI users (Roy et al., 2012; Mo et al., 2022; Demirel et al., 2025). However, we consider this interpretation unlikely in the present context, for reasons discussed in detail in Section 4.4.2: because the Bayesian model derives electrode-level weights from paired comparisons in which root-position chords systematically concentrate energy on apical electrodes, the strong root-position preference observed across both groups would inflate the preference weights assigned to those apical electrodes independently of any genuine frequency-region advantage. The apical preference gradient in the FIF profiles is therefore more parsimoniously explained as a consequence of the root-position confound than as evidence for a frequency-dependent perceptual asymmetry. This interpretation is further supported by the absence of a group difference in the gradient: if the apical advantage reflected sensitivity to FTPM-induced distortion - which is typically greatest at apical electrodes - one would expect Post-LCI users to show a different profile from Pre-ECI users, which was not observed.

Root-Position Preference and Grade Strength The dominant finding was that root-position preference was strong, consistent, and equivalent across Post-LCI and Pre-ECI groups. Root-position chords were chosen on approximately two thirds of decided trials in both groups, and the LMM group interaction on root-preference slope was near-zero ($p = .980$). This equivalence is important for two reasons. First, it establishes that root-position preference is not contingent on prior acoustic hearing experience, and rules out accounts of the effect based on stored acoustic harmonic templates. Second, it motivates treating root-position preference as a structural confound in the subsequent distortion analyses rather than as a group-specific perceptual tendency.

The only significant group difference in Analysis 0 was in preference strength: Pre-ECI participants expressed stronger grades on decided trials ($\hat{\beta}_{\text{ECI}} = +0.796$, $p = .043$). This is an unexpected finding. One interpretation is that Pre-ECI users, who have never adapted an acoustic template to the CI signal, are less uncertain about their perceptual

judgements when they do perceive a difference, and thus commit to stronger ratings. Alternatively, the Pre-ECI group may be more responsive to the specific acoustic cues that drive root-position preference - such as low-frequency energy concentration or interval structure - without the competing influence of FTPM-induced distortion that may introduce ambiguity for Post-LCI users. These interpretations remain speculative in the absence of a larger sample and should be treated accordingly.

The no-preference rate did not differ between groups ($p = .619$), indicating that willingness to express a preference was equivalent, and that the grade strength difference reflects genuine response intensity rather than a lower threshold for committing to a non-neutral response in one group.

4.4.2 Analysis 1: Chord-Level Paired-Comparison

Root-Position Preference Dominates Chord Selection

The most robust finding was the strong preference for root-position chords, observed in six of the seven participants regardless of implantation timing. Participants chose the root-position chord on approximately 67% of decided trials, and the inversion-only model (PG5) outperformed all single-predictor distortion models by a wide margin. Different explanations, mostly driven by the paradigm design itself, may be drawn.

Low-Frequency Content and Music Quality. Root-position chords place the chord fundamental frequency at the lowest frequency, concentrating more energy in the low-frequency region compared to their inverted counterparts, and these root-position chords were judged as more pleasant by CI participants in both groups and in a majority of trials. To contextualise this preference, it is first necessary to consider what role frequency content plays in music perception more broadly. The frequency content of a musical signal has a well-established effect on its perceived quality: Moore and Tan (2003) demonstrated that limiting the bandwidth of music reduces its perceived naturalness, with both low- and high-frequency regions contributing to sound quality judgements. Beyond global bandwidth, different frequency regions carry distinct musical information: low frequencies primarily convey rhythmic and beat-related content, with Hove et al. (2014) showing that the auditory system detects timing deviations more accurately in lower-pitched streams, and Nozaradan et al. (2018) demonstrating at the cortical level that low-frequency sounds selectively boost neural entrainment to the musical beat; high frequencies, by contrast, carry melodic and pitch-related information as well as timbral cues associated with instrument attack transients and upper harmonics (*e.g.*, Moore and Tan, 2003). This frequency-region asymmetry is particularly relevant for CI users: rhythm and beat are among the best-preserved musical attributes in CI listeners, whereas melody recognition and pitch discrimination are severely impaired relative to NH peers (McDermott, 2004). Accordingly, several studies have demonstrated that the availability of low-frequency content influences sound quality ratings in CI users: Roy et al. (2012) showed that bass frequency removal degrades sound quality perception, Mo et al. (2022) showed that increasing gain in low and mid-range frequency bands improved sound quality ratings whereas manipulating gain above 2 kHz had no significant effect, and Demirel et al. (2025) reported that sound quality ratings declined monotonically as the high-pass filter cutoff frequency increased. These findings would suggest a straightforward explanation for the root-position prefer-

ence observed here: root chords concentrate energy in the low-frequency region, and CI users are known to rely on low-frequency content for sound quality judgements. However, such findings were obtained using real pieces of music, which is not the case in the present study where single, isolated chords were presented. Single chords do not carry temporal information - neither rhythm, beat, nor melody - and the conventional justification for low-frequency superiority in CI music perception therefore does not apply directly to the present paradigm. To our knowledge, this distinction has not been made explicitly in the CI music literature, where low-frequency preference has mainly been attributed to the rhythm and beat advantage without consideration of contexts in which static harmonic information is the primary perceptual target. Alternative mechanism must therefore be considered.

Root Salience and Interval Structure Discriminability. Such root-preference may be accounted for by the difference in interval structure between root-position and inverted chords, and how these intervals interact with the CI mapping of each participant. In root position (*e.g.*, C4-E4-G4), the bass note is separated from the note immediately above by a major third (4 semitones, ≈ 0.33 octaves), reinforcing its perceptual salience as the harmonic root and supporting a coherent harmonic template match (Parncutt, 1988; Terhardt, 1984); furthermore, the upper two notes are sufficiently separated in frequency to span distinct electrode bands. In first inversion (*e.g.*, E4-G4-C5), the two lowest notes are only a minor third apart (3 semitones, ≈ 0.25 octaves), which may reduce the distinctiveness of the bass note; more critically, given the coarse frequency resolution of CI processing (Friesen et al., 2001), these two closely spaced notes may fall within the same electrode frequency band and be encoded by the same or adjacent electrodes (Shannon et al., 1995), thereby compromising the harmonic coherence of the chord.

Therefore, the root-position preference is more likely a structural consequence of the paradigm design than a reflection of individual FTPM-related distortion. Root-position chords are acoustically privileged in multiple converging ways - lower spectral energy concentration, better root salience, and a more coherent harmonic template - all of which are independent of the listener's implantation timing history. Moreover, Pre-ECI 1, who has no prior acoustic hearing experience, showed an equally strong preference, arguing against an experience-dependent or FTPM-driven account.

FTPM-Distortion Sensitivity is Conditional on Chord Inversion in Post-LCI Users

A second finding of Analysis 1 was that FTPM-related distortion significantly predicted chord preference in Post-LCI users, but only through an interaction with chord inversion (PG3a: $\hat{\beta}_{\text{interaction}} = -3.03$, $p < .001$). When the chosen chord was in root position, reduced distortion increased preference (simple slope = -2.03), consistent with the hypothesis that FTPM-induced spectral distortion degrades perceived chord quality. In contrast, when the chosen chord was in first inversion, the distortion slope was non-significant and positive ($+1.00$), indicating that root-position preference dominated behaviour regardless of distortion level.

This pattern can be interpreted within a *cue integration framework*, in which listeners combine and weight multiple sources of information to guide perceptual decisions (Ernst

and Banks, 2002; Knill and Pouget, 2004). Root-position information constitutes a highly salient and reliable cue that receives greater weight in the decision process. When the two cues are misaligned (*e.g.*, when the root-position chord carries greater FTPM-related distortion than the inverted chord), root-position salience dominates and distortion sensitivity is attenuated. When the two cues are aligned (*e.g.*, when the root-position chord carries less distortion), both cues reinforce the same choice and Post-LCI users become sensitive to distortion, modulating their decisions using the rating scale.

The interaction was consistent in direction across all five Post-LCI participants, with negative interaction coefficients ranging from -1.38 to -11.22 (two significant: Post-LCI 1 and Post-LCI 2; two marginal: Post-LCI 4 and Post-LCI 5; one non-significant in the expected direction: Post-LCI 3). Among Pre-ECI participants, the pattern differed qualitatively: Pre-ECI 2 showed a significant additive distortion effect in the graded models once inversion was controlled (PG3: $\hat{\beta} = -3.88$, $p = .001$) but not a significant interaction (G3a: $p = .391$), suggesting that this participant avoided distortion regardless of chord inversion rather than conditionally upon it. Pre-ECI 1 showed no distortion effect under any model specification. This dissociation - inversion-conditional distortion sensitivity in Post-LCI versus inversion-unconditional avoidance in Pre-ECI 2 - may reflect a qualitative difference in the perceptual mechanism: Post-LCI users may require the root-position chord to act as a harmonic anchor before distortion becomes salient, whilst Pre-ECI 2 avoidance appears more global and less contingent on harmonic structure. To our knowledge, this is the first demonstration that FTPM-induced interval distortion affects chord quality perception in CI users.

4.4.3 Analysis 2: FIFs and Frequency-to-Place Factors

This analysis was the original objective of the paradigm developed here: create a task in which individual electrodes could be compared using a music-related signal, and in which electrode-level perceptual weights could be inferred and related to individual frequency-to-place factors.

FTPM Magnitude and the Electrode Position Confound

The electrode-level analysis found that electrodes with higher absolute FTPM tended to receive higher preference scores in the Post-LCI group (M3: $\hat{\beta} = +0.45$, $p = .035$). The direction is consistent with predictions, though the interpretation is complicated by a structural confound between FTPM magnitude and electrode position. In the current sample, FTPM was positively correlated with electrode index ($r = +0.25$, $p = .004$ pooled), meaning that apical electrodes tended to have greater mismatch than basal electrodes. Any preference gradient across the cochlea could therefore inflate the apparent FTPM effect, and the Analysis 2 result should be interpreted with this bias in mind.

CS and NR

CS was a more interpretable predictor in Analysis 2. Electrodes with narrower current spread tended to be preferred (M4: $\hat{\beta} = -0.69$, $p = .025$), an effect that survived in the full model. This is consistent with the idea that more spatially focused stimulation produces a

clearer spectral representation, which may lead to more distinct encoding of the harmonic structure probed by the paradigm. Crucially, CS is not systematically confounded with electrode position in the same way as FTPM, making it a more interpretable predictor.

NR, by contrast, showed no relationship with electrode-weighted preference (M1: $p = .959$). The absence of an NR effect may be explained by the use of monopolar stimulation, which induces broad CS and makes it difficult to isolate the contribution of NR at individual electrode sites. The null NR result is therefore not surprising in this context, though it remains to be established whether a focused stimulation paradigm would reveal a relationship between NR and electrode preference in this task.

4.4.4 Convergence of Analysis 1 and Analysis 2

The two factor analyses shared a common question - does FTPM orient chord preference? - but addressed it at different levels of analysis. The results were directionally consistent: Analysis 1 found that less interval distortion within each chord was associated with higher preference (interaction model PG3a), and Analysis 2 found that less absolute mismatch was associated with higher electrode-level preference weight (M3: $\hat{\beta} = +0.45$). Both analyses therefore point towards a benefit of reduced mismatch for chord pleasantness perception.

However, the two analyses differ in the strength of their causal inference. Analysis 1 is the more interpretable, because the paired-comparison design naturally controls for electrode position: on each trial, both chords are built from the same electrode pair, and only the interval distortion differs between them. Analysis 2, by contrast, compares preference scores across electrode positions within a participant and is therefore vulnerable to any systematic preference gradient along the array. The FTPM-position confound ($r = +0.25$) means that the Analysis 2 effect could be partly or wholly driven by electrode position rather than FTPM-related distortion, and should therefore be interpreted with greater caution.

4.4.5 Post-LCI Versus Pre-ECI Differences

Pre-ECI 1 showed no distortion sensitivity under any model specification, consistent with the hypothesis that sensitivity to FTPM-related distortion requires a stored spectral template built through prior acoustic hearing experience. Pre-ECI 2 showed strong binary distortion avoidance and a significant additive distortion effect once inversion was controlled, but no inversion-conditional interaction. The formal Distortion $\times$ Group interaction was not significant at the pooled level ($p = .828$), reflecting the averaging of these two qualitatively different Pre-ECI profiles rather than evidence for group equivalence.

The pattern of results across the two Pre-ECI participants is more interpretable when considered individually. Pre-ECI 1, who has no prior acoustic hearing experience, showed no distortion sensitivity and preference dominated entirely by inversion status. Pre-ECI 2 showed strong distortion avoidance that was not contingent on chord inversion. This inversion-unconditional avoidance may reflect a more global sensitivity to spectral distortion that does not require the harmonic anchoring provided by root-position chords - consistent with a listener who attends primarily to low-level spectral quality rather than to harmonic template matching.

The group comparison in Analysis 2 provided cleaner evidence for a group difference. Post-LCI participants showed significantly higher overall electrode-level preference scores than Pre-ECI participants (M8: $\hat{\beta} = +0.68$, $p = .046$), suggesting greater perceptual differentiation across the electrode array. The significant FTPM $\times$ Group interaction (M9: $p = .043$) indicated opposing FTPM slopes: positive for Post-LCI users (less absolute FTPM associated with higher preference, the predicted direction) and negative for Pre-ECI users (less absolute FTPM associated with lower preference, counter to predictions). Whether this counter-directional Pre-ECI slope reflects a genuine inverse relationship or a statistical artefact of the electrode position confound in a small Pre-ECI sample ($n = 2$) cannot be determined from the present data.

To our knowledge, this is the first evidence at the electrode level that FTPM impacts CI users differently depending on whether they developed hearing acoustically or electrically.

4.4.6 Limitations

Sample Size. The factor analyses reported here are based on five Post-LCI and two Pre-ECI participants with FTPM data. Whilst the within-participant repeated-measures design provides reasonable statistical power for detecting trial-level effects, the between-participant inferences - particularly the group comparison - rest on a very small sample, with the Pre-ECI subgroup reduced to two participants following the exclusion of one participant with erroneous radiological FTPM estimates. The individual-level heterogeneity within Pre-ECI (one non-responder, one strong responder) further limits the interpretation of group-level models, and additional pre-lingual participants are being recruited. Data collection is ongoing, and the patterns reported here may change as additional participants are tested. The interaction models M6 and M7 in Analysis 2 were fitted on 13 and 22 observations respectively and should be treated as hypothesis-generating rather than confirmatory.

Ecological Validity. One goal was to build a task that would serve as a proxy for how each electrode is used and potentially affected by frequency-to-place factors during music perception. However, as described in Section 4.4.2, the use of single isolated chords limits the interpretability of how CI users perceive music in an ecologically valid sense, because single chords do not carry the temporal information - rhythmic structure, melodic contour, harmonic progression - that characterises real pieces of music. This limitation motivates the development of future paradigms that retain the electrode-level sensitivity of the present approach while incorporating richer musical material.

Task-Related Biases. Several caveats of the present paradigm were discussed in Section 4.4.2 and were partly reflected in the results through the dominant root-position preference, which was largely independent of FTPM-related distortion. A further bias concerns the structure of the comparisons: the paradigm only allows direct comparison between electrodes separated by exactly one octave. Although the Bayesian analysis reconstructs a complete picture of electrode-level preference weights across the array, the direct one-octave constraint may introduce noise into the derived FIFs, visible in some participants as local anomalies at electrodes separated by one, two, and three octaves.

4.4.7 Clinical Implications

The finding that FTPM likely disrupts harmonicity perception in experienced Post-LCI users has potential clinical relevance. Although the present task does not encompass all aspects of music perception, it suggests that even modest remapping approaches - designed to preserve the harmonicity relationship between allocation frequency and the frequency place effectively stimulated - could improve the perceptual coherence of harmonic material without requiring prolonged adaptation. Such targeted interventions would complement the growing body of evidence supporting anatomy-based fitting strategies (Creff et al., 2024; Jiam et al., 2019; Lambriks et al., 2023c).

The finding that CS predicted electrode-level preference suggests that stimulation strategy may also matter for music perception. Focused stimulation techniques such as tripolar configurations (Bierer, 2010) could improve spectral resolution at the electrode level and potentially benefit chord discrimination; however, the interaction between stimulation focality and NR in this context remains to be established.

4.5 Conclusion

This chapter presented the first investigation of FTPM-induced distortion as a predictor of chord quality perception in CI users, using a paired-comparison paradigm designed to probe electrode-level music perception. Four main findings emerged.

First, root-position chords were strongly preferred over their inversions across all 24 participants irrespective of implantation history ($p = .980$ for group comparison), establishing root-position preference as a structural acoustic phenomenon and motivating its treatment as a confound in subsequent analyses. The sole group difference was in preference strength, with Pre-ECI participants expressing stronger grades on decided trials ($p = .043$).

Second, FTPM-induced interval distortion affected chord preference differently across groups. In Post-LCI users, distortion sensitivity was inversion-conditional, emerging only when root-position and less-distorted cues were aligned, consistent with reliability-weighted cue integration. In Pre-ECI 2, distortion avoidance was inversion-unconditional, suggesting a more global sensitivity that does not require harmonic anchoring; Pre-ECI 1 showed no distortion sensitivity, consistent with the absence of a stored acoustic template.

Third, at the electrode level, both FTPM magnitude and CS predicted preference weights in Post-LCI users, whilst NR did not, likely due to monopolar stimulation. Post-LCI users showed higher electrode-level preference differentiation than Pre-ECI users (M8: $p = .046$), and the significant FTPM $\times$ Group interaction (M9: $p = .043$) confirmed opposing relationships to individual FTPM profiles between groups.

Fourth, these findings converge to suggest that FTPM is a genuine determinant of music quality perception in experienced CI users, and that both frequency remapping and focused stimulation represent plausible clinical intervention targets.

Postface

The study presented in this chapter investigated whether FTPM, CS, and NR predict chord pleasantness perception in experienced CI users, using a paired-comparison paradigm designed to probe electrode-level music perception. The chapter was strongly experimental in its construction: we developed a completely new paradigm and implemented two separate and independent analyses, tracking the effect of spectral distortions in complementary ways. Whilst some unanticipated biases emerged - most notably the strong root-position preference that dominated the paired-comparison signal - several aspects of the design performed well. Participants consistently appreciated the paired-comparison format and found the continuous pleasantness scale intuitive; the Bayesian Bradley-Terry framework was both computationally efficient and sensitive to electrode-level differences.

These strengths suggest that the paradigm, with targeted modifications, has genuine potential beyond its current implementation. The following paragraphs outline the directions we consider most promising.

The Root-Position Bias: a Structural Limitation. The dominant root-position preference was the most consequential unanticipated finding of this experiment, and it is worth reflecting honestly on whether it can be eliminated in a future design. We hypothesise that the preference may be related to the acoustical structure of the chords - specifically the wider and more discriminable bottom interval of root-position voicing - rather than to individual FTPM profiles. Any paradigm that uses two-note or three-note chords with variable voicing would therefore likely reproduce this bias. At the time of writing, we do not see any obvious alternative that would remove it entirely.

A Fitting Tool for Hearing Aid Frequency-Lowering Algorithms. Whilst FTPM is conventionally considered specific to CI users, one condition directly transposes the phenomenon to another population: HA users for whom a frequency transposition or compression algorithm has been applied to compensate for a cochlear dead region. In both cases, frequency information normally processed at one cochlear location is displaced to another - which is, by definition, FTPM. Adapting the chord paradigm for acoustic stimulation would allow direct measurement of whether these algorithms introduce perceptible harmonic distortion, and whether the degree of distortion predicts music quality ratings in hearing aid users.

To our knowledge, no current clinical tool quantifies the perceptual cost of frequency-lowering at the level of individual spectral intervals. A HA version of the paradigm - using acoustic chord stimuli delivered through the device - could identify the frequency-lowering configuration that minimises perceived distortion for an individual user, providing a perceptually grounded complement to other fitting measures.

Interaural FTPM in Bilateral CI users. Bilateral CI users have two independent electrode arrays, and the FTPM profiles of the two implants are rarely identical: insertion depth, array design, and surgical approach differ between ears, producing different electrode-to-cochlear position relationships on each side. When a chord is presented binaurally, the physical interval between two notes is therefore perceived simultaneously

through two arrays with different tonotopic mappings, introducing an interaural spectral asymmetry that compounds the within-ear interval distortion measured in the present study.

The techniques and framework developed here could be transposed and adapted to characterise this interaural mismatch. Several experimental variants are conceivable: presenting the same chord to both ears simultaneously to study binaural fusion, splitting the notes of a chord across ears to probe inter-aural interval coherence, or systematically varying the relative FTPM profiles of the two arrays to quantify the perceptual cost of interaural asymmetry independently of within-ear distortion.

Personalised Music Design from Electrode Preference Profiles. The FIFs derived from the present paradigm identify, for each individual, which electrode produce the most and least preferred chord percepts. Rather than using this information only to characterise or remediate distortion, it could be inverted: musical material could be composed or arranged to favour electrodes that attract the highest preference weights for that individual. This would also be a way to address the most fundamental limitation of the present paradigm - the use of isolated static chords that carry no temporal information - by confronting the FIF results with real music. If FIFs identify certain electrode positions as consistently preferred, melodic lines and harmonic progressions could be written to concentrate interval relationships within those positions, effectively designing music around the perceptual topology of each user electrode array. This approach - personalised music composition informed by electrode-level preference data - would represent a fundamentally new model of music accessibility for CI users, moving beyond signal processing interventions that aim to reduce distortion towards creative interventions that work within the constraints of the electrode array rather than against them.

Neural Correlates of FTPM-Driven Distortion. The present paradigm relies entirely on subjective preference judgements, which introduces response bias and limits the interpretability of null results. An objective complement would be to record EEG during the chord task and examine the mismatch negativity (MMN) response to each chord pair. If FTPM-induced distortion is perceptually registered at a pre-attentive level, the MMN amplitude should be correlated with the degree of interval distortion imposed by the participant's individual FTPM profile, providing a physiological validation of the behavioural FIF and a measure that does not depend on explicit pleasantness judgements.

Online Implementation and Large-Scale Normative Data. The most immediate practical extension is an online implementation of the chord paradigm. In contrast to the vowel task described in Chapter 3, the simplicity of the chord stimuli, the brevity of the session, and the efficiency of the Bayesian post-processing make a fully remote deployment more feasible. We are currently developing such a platform. The HA extension described above could be incorporated into the same framework, enabling a unified, device-agnostic tool for measuring frequency-to-place distortion across the full spectrum of users with music hearing impairment.

Chapter 5

Adaptation to FTPM, CS and NR

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Preface

This chapter is a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **“Adaptation to frequency-to-place factors in cochlear implant users: longitudinal case studies of vowel and chord perception.”**.

Chapters 3 and 4 examined whether individual differences in FTPM, CS, and NR shape the perception of vowels and chords in CI users who have been listening with their device for years.

In the vowel identification task (Chapter 3), we found that both experienced CI groups showed a displacement of their vowel perception relative to NH listeners. However, the displacement was quantitatively small and restricted to a shift of F2 towards low frequencies for the Post-LCIs and a greater reliance on F1 for the Pre-ECIs. None of the frequency-to-place factors studied here explained these small differences. Such absence of relationship could have two meanings: first, it could indicate that experienced users had fully adapted to these factors; second, it could indicate that these factors do not affect perception within the frame of the present task, regardless of adaptation.

In the chord preference task (Chapter 4), interval distortion - corresponding to Analysis 1 and a disruption of the harmonic organisation introduced by the FTPM profile - significantly predicted chord preference, but only through an interaction with chord inversion ($p < .001$ for Post-LCIs; $p = .003$ pooled): distortion sensitivity was detectable only on root-position trials, where the root chord occupied the reference position, and was absent on first-inversion trials, where root-position preference dominated regardless of distortion level. This effect was consistent across Post-LCIs (all five showing negative interaction coefficients) but absent in Pre-ECIs, for whom distortion avoidance was either unconditional (one participant) or absent entirely (one participant). Similarly in Analysis 2 (at the electrode level), FTPM predicted perceptual weighting in the expected direction - less absolute mismatch associated with higher electrode preference for the Post-LCIs. CS was statistically significant for Post-LCIs, with narrower CS associated with higher electrode preference. NR was not significant in either group.

Together, these findings suggest that, for the vowel task, experienced CI users have either adapted to these frequency-to-place factors or that the task is not sufficiently sensitive to detect their perceptual influence (or both). For the chord task, they suggest that even after years of CI use, perception remains constrained by some of these factors - particularly for Post-LCIs.

The study presented in this chapter aims to clarify the adaptation process during the first months following device activation. This work constitutes a preliminary observation of how perception changes with brain adaptation following CI activation.

For the vowel task, the central question is whether the task is genuinely sensitive to these factors: if it is not, we should observe no relationship between the factors and vowel task performance even during the active adaptation period, confirming that the paradigm is simply not suited to tracking them.

For the chord task, we are interested in whether sensitivity to distortion changes over the months following activation. If it does not change, it would suggest that the perceptual constraints imposed by these factors are persistent, and that only a clinical remapping intervention could alleviate them. If sensitivity decreases over time, it would

suggest that some capacity for adaptation exists, with the auditory system progressively accommodating the spectral distortions imposed by the factors studied.

This chapter addresses these questions through a single longitudinal case study of a newly implanted adult Post-LCI user, tested repeatedly during the first weeks and months post-activation using the same vowel identification and chord preference tasks as Chapters 3 and 4. FTPM was estimated from post-operative X-ray imaging, and CS and NR were estimated from PECAP recordings, providing electrode-level frequency-to-place factor data for all three measures.

Only one case study is presented here; three additional longitudinal participants are currently being tested and will be incorporated into the final version of this work. The goal is therefore not to draw population-level conclusions from a single individual, but to characterise the temporal dynamics of adaptation and to generate hypotheses about the mechanisms that determine whether, and how quickly, CI users adapt to individual frequency-to-place factors.

Adaptation to frequency-to-place factors in cochlear implant users during the first months of cochlear implant use: longitudinal case study of vowel and chord perception

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Abstract

Understanding how CI users adapt to their device in real time requires longitudinal paradigms sensitive enough to detect perceptual change at the electrode level. This chapter tracks the perceptual representations of a single newly implanted Post-LCI across four sessions spanning Days 6 to 134 post-activation, using the vowel identification and confidence task from Chapter 3 and the chord preference paradigm from Chapter 4. In the vowel task, F1 and F2 confidence representations co-varied across sessions for most vowels, with F2 drifting monotonically away from the NH reference towards more basal cochlear positions - opposite to the convergence expected from adaptation. One vowel, [ɜ:], showed independent F1 movement consistent with genuine electrode-to-category learning. The F1/F2 reliance ratio showed no temporal trend, with the Day 6 spectral weighting profile already characteristic of a Post-CLIs, suggesting that spectral reweighting is not a mechanism of early adaptation. Electrode-level factor analyses revealed a coherent temporal trend at F2, whereas no such temporal trend was seen for F1. At F2, electrodes with more absolute FTPM, narrower CS and smaller NR attracted more confidence in the early time of the adaptation, and trend reversed gradually overtime. In the chord task, FTPM distortion sensitivity was present at Days 6 and 20 but had disappeared by Day 41, giving way to electrode-level FTPM preference by Day 134. NR facilitated preference formation in the early sessions, whilst FTPM emerged as the dominant organisational factor in the extended phase. Across both tasks, the period between Days 20 and 41 marked a qualitative transition in the representational basis of perception, from acoustic template matching to electrode-level organisation. These findings characterise a two-phase model of early CI adaptation and demonstrate that the chord paradigm is more sensitive than the vowel task for tracking this process in real time.

5.1 Introduction

The preceding chapters characterised how experienced CI users represent vowels and chords at the individual electrode level, and identified which frequency-to-place factors

shape that representation after years of device use. In Chapter 3, both Post-LCIs and Pre-ECIs showed only small displacements in their electrode confidence distributions from the NH reference. Critically, the delta analysis revealed that this displacement reflects a global reorganisation of spectral weighting rather than electrode-by-electrode degradation: at matched frequency positions, none of the frequency-to-place factors predicted the local deviation from NH, and this null held within both groups independently. The distinction is important: a spectral weighting account implies that the entire confidence profile has been uniformly shifted or rescaled relative to NH, whereas an electrode-by-electrode account would predict that specific electrodes - those with the greatest mismatch or poorest neural interface - would show localised deficits. The absence of electrode-specific effects in Chapter 3 is consistent with the former but does not rule out the latter during the early adaptation period. In Chapter 4, FTPM-induced interval distortion affected chord quality judgements in Post-LCIs, and FTPM predicted perceptual preference weights at the electrode level; CS was marginally significant for Post-LCIs; NR was not significant for affecting quality judgements in either group.

Together, these findings might characterise the perceptual endpoint of the adaptation process - the state at which experienced CI users distribute confidence across the electrode array and weight chords after years of exposure to a spectrally distorted input. However, two important ambiguities remain that cross-sectional designs cannot resolve. For the vowel task, the null effect of frequency-to-place factors on the delta confidence profile could reflect either genuine completion of adaptation - meaning electrode-level factors no longer drive any local deficit - or simply that the paradigm is insensitive to residual effects. For the chord task, it is unclear whether the residual influence of FTPM (and marginally CS) on preference reflects a fundamental limit on adaptation, or a transient state that would resolve with further experience. Disentangling these possibilities requires testing users before adaptation has taken place, at repeated intervals across the early post-activation period, before adaptation has finalised.

Indeed, this early post-activation period is the most dynamic phase of CI use. Most perceptual improvement occurs within the first three to six months following device activation, with gains progressively flattening thereafter (Fu and Galvin, 2007a; Holden et al., 2013a). At the level of electrode discrimination, Mathew et al. (2018) showed that both electrophysiological and behavioural measures of channel selectivity improve over the first six months, with the rate of improvement correlating with speech perception gains - suggesting that electrode-level perceptual changes track the broader adaptation trajectory. Despite its clinical importance, the mechanisms at the electrode level driving this improvement remain poorly understood: standard clinical management focuses on adjusting stimulation levels and collecting subjective feedback, with electrode-level factors not yet systematically integrated into fitting practice (Holden et al., 2013a).

Of the three frequency-to-place factors examined in this thesis, FTPM is the best characterised in the adaptation context. The foundational study by Fu and Galvin (2003) demonstrated that CI users can partially adapt to experimentally imposed frequency shifts over three months, but performance with the shifted processor never recovered to clinical-map levels, and subsequent computational modelling confirmed that none of the participants achieved complete adaptation (Svirsky et al., 2005). The general consensus is that moderate FTPM can be largely compensated by auditory plasticity when daily use is sufficient, but severe mismatch may substantially prolong the adaptation time course and may never be fully resolved over the first six to twelve months (Fu and Galvin,

2007a; DeFreese et al., 2025). Critically, however, all existing FTPM adaptation studies reduce mismatch to a single subject-level value - a mean across electrodes - making it impossible to determine which electrodes are driving or limiting the adaptation process at the individual level.

For CS and NR, longitudinal perceptual data are essentially absent. CS is largely fixed following implantation as a property of the electrode-neural interface, and the implicit assumption in the literature is therefore that its perceptual consequences are static throughout adaptation (Brown et al., 2000). This assumption has never been directly tested. NR is also largely stable after the initial post-surgical period (Brown et al., 2000), but its perceptual consequences may not be equally static: unlike CS, which acts as a fixed spectral resolution constraint, inter-electrode variation in NR requires the central auditory system to learn to reweight signals of unequal neural fidelity across the array - an adaptive process whose direction and magnitude may themselves evolve over time. In particular, electrodes with higher NR produce sharper and more consistent signals that may initially be perceived as unfamiliar relative to the internal representation established during the period of deafness, such that the perceptual advantage of high-NR electrodes may only emerge after central recalibration - predicting a non-monotonic NR-perception trajectory that would be invisible to cross-sectional designs.

This chapter addresses three questions that the cross-sectional designs of Chapters 3 and 4 could not answer. The following metrics, introduced in Chapters 3 and 4, are used to track the adaptation trajectory across sessions: the *Wasserstein distance* (overall magnitude of displacement from NH), the *formant centroid shift* (direction of displacement along the frequency axis), the *F1/F2 reliance ratio* (balance of confidence between formant regions), and the electrode-level *delta bell* (signed deviation from the NH reference at each electrode position) for the vowel task; and the *frequency importance function* (electrode-level preference weight) and *interval distortion* (FTPM-induced harmonic disruption) for the chord task.

Question 1: What does the adaptation trajectory look like for the vowel task? As a newly implanted CI user gains experience with their device, does their electrode confidence distribution converge towards the NH reference, towards the experienced Post-LCI average, or along another trajectory? The Wasserstein distance, formant centroid shift, and F1/F2 reliance ratio provide a framework for tracking this representational change across sessions in a way that is directly comparable to the experienced-user baseline established in Chapter 3.

Question 2: Do frequency-to-place factors predict the electrode-level adaptation trajectory for the vowel task? Beyond the group-level trajectory, do individual electrodes characterised by greater absolute FTPM, broader CS, or poorer NR show systematically different patterns of confidence change over time? This question extends the delta confidence analysis of Chapter 3 - which revealed a null effect in experienced users, interpreted as evidence of global reorganisation - to ask whether that null is present from the very first session, or whether factor-driven local deficits are visible during the early post-activation window before global reorganisation has consolidated.

Question 3: Do frequency-to-place factors predict the chord preference adaptation trajectory? Chapter 4 established that FTPM-induced interval distortion affects chord quality judgements in experienced Post-LCIs, and that FTPM magnitude predicts electrode-level perceptual preference weights. Whether these relationships are

present from the beginning of CI use, or whether they emerge progressively as adaptation consolidates, is unknown. Tracking the distortion-preference relationship and the FIF profiles across sessions in a newly activated user provides a longitudinal complement to the cross-sectional findings of Chapter 4, and may reveal whether early targeted remapping - designed to reduce interval distortion at specific electrode pairs - would be clinically beneficial before chord preferences have consolidated around a suboptimal map.

This chapter presents a longitudinal case study of a newly implanted post-lingually deaf adult CI user, assessed across four sessions at Days 6, 20, 41, and 134 post-activation using the same vowel identification and chord preference tasks developed in Chapters 3 and 4. The design is explicitly exploratory: with a single participant, the analyses are descriptive rather than inferential, and the purpose is to generate theoretically grounded hypotheses for a larger prospective longitudinal study rather than to draw confirmatory conclusions.

5.2 Methods

This chapter presents longitudinal data from a single newly implanted CI user (Case Study 1) who completed the vowel identification task and the chord preference task at four time points during the first months following device activation. The experimental tasks, stimuli, response formats, and estimation of individual factors were identical to those described in Chapters 3 and 4; only aspects specific to the longitudinal protocol are described here.

5.2.1 Participant

Case Study 1 was a post-lingually deafened adult recruited from the Emmeline Centre for Hearing Implants at Addenbrooke's Hospital (Cambridge, UK) prior to CI activation, under the same ethics protocol as the cross-sectional cohort (Section 3.2.1).

Table 5.1: **Clinical details for Case Study 1.**

Sex	Female
Age at activation	26 years
Age at onset of deafness	11 years
Duration of deafness	15 years
Aetiology	Unknown
Device manufacturer and model	Cochlear
Number of active electrodes	22
Implanted ear	Left
Contralateral ear status	Hearing Aid

5.2.2 Longitudinal Design

Case Study 1 was tested at four sessions following device activation, scheduled so as to capture early adaptation (first six weeks) and later consolidation (beyond two months),

as summarised in Table 5.2.

Table 5.2: **Longitudinal testing schedule for Case Study 1.** Days post-activation, approximate period and tasks completed.

Session	Days post-activation	Approx. period	Tasks
S1	6	Week 1	Vowel, Chord
S2	20	Week 3	Vowel, Chord
S3	41	Week 6	Vowel, Chord
S4	134	Month 4.5	Vowel, Chord

5.2.3 Tasks

The vowel identification task was identical to that described in Chapter 3 (Sections 3.2.2–3.2.3): at each session, Case Study 1 heard synthetic vowel stimuli whose F1 and F2 formant frequencies were set to the centre frequencies of their allocated electrodes, identified the perceived vowel from five response options, and rated their confidence on a five-point Likert scale. The same F1/F2 electrode combinations were presented at every session, enabling within-electrode tracking of confidence change over time. The chord preference task was identical to that described in Chapter 4 (Sections 4.2.2–4.2.3), with the same electrode pairs tested at every session. Using identical stimuli and procedures across sessions was essential to ensure that any observed change in performance could be attributed to perceptual adaptation rather than to differences in task demands or stimulus content.

5.2.4 Estimation of Frequency-to-Place Factors

The three electrode-level factors - frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) - were estimated following the same procedures as Chapter 3 (Section 3.2.4). FTPM was computed from a single post-operative X-ray and treated as constant across sessions, on the assumption that electrode position does not change following implantation. CS and NR were estimated from a single PECAP recording obtained at Session 1 (6 days post-activation) and likewise treated as constant across sessions, consistent with the established stability of ECAP-derived measures after the initial post-surgical period (Brown et al., 2000). Both CS and NR are per-electrode values derived from the PECAP decomposition (see Section 3.2.4), not averages across the array; each electrode therefore has its own CS and NR estimate, enabling the electrode-level analyses reported below. Any apparent change in the factor-perception relationship across sessions therefore reflects central adaptation to a stable peripheral input rather than a change in the peripheral measure itself. A representation of these factors for Case Study 1 is provided in Figure 5.1.

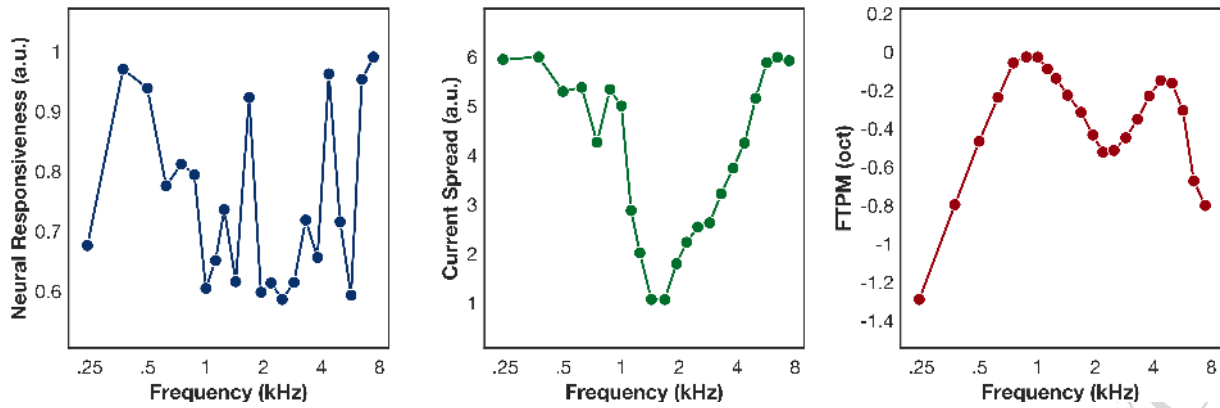


Figure 5.1: **Electrode-level factor profiles for Case Study 1.** Each panel shows the raw value of one factor as a function of electrode centre frequency (kHz, log scale). Each point represents one electrode. Left: neural responsiveness (NR; higher values indicate healthier neural populations). Centre: current spread (CS; higher values indicate broader stimulation and poorer selectivity). Right: frequency-to-place mismatch (FTPM, in octaves; more negative values indicate greater basal shift).

5.2.5 Data Analysis

Exclusion of [i:] responses

As in Chapter 3 (Section 3.2.5), responses for [i:] were excluded from all vowel analyses. All results therefore concern four vowel categories: [ɑ:], [ɜ:], [ɔ:], and [u:].

Electrode Confidence Distribution

The electrode confidence distribution - the *bell* - was computed at each session following the identical procedure described in Section 3.2.5: confidence-weighted kernel density estimation in log-frequency space, with bandwidth $h = 0.10 \log_{10}\text{-Hz}$ and a grid of 300 points spanning 200–9000 Hz. The same NH reference bell computed from the cross-sectional cohort of Chapter 3 was used as the reference throughout, held constant across all four sessions.

5.2.6 Statistical Analysis

All statistical analyses were conducted in MATLAB R2025a. Given the single-participant design, trial-level or grid-point level pooling across sessions was used to provide adequate degrees of freedom within the mixed-effects framework. It should be noted that electrode-level observations within a session are not independent, as electrodes within the same array share a common peripheral and central processing context; the random intercept for electrode in the LMMs accounts for stable between-electrode differences but does not model within-array spatial autocorrelation. Similarly, Spearman correlations reported below are computed on electrode-level observations that are not strictly independent; these are presented as descriptive summaries rather than formal inferential tests. No blanket correction for multiple comparisons was applied; a convergence-of-evidence approach was

adopted instead, with effects consistent across formant dimension, vowel, and task treated as robust and isolated significant results flagged cautiously. This approach is appropriate for an exploratory single-case design where the goal is hypothesis generation rather than confirmatory inference. All analyses were conducted at $\alpha = .05$. Throughout, results are interpreted against two reference benchmarks: the NH group (representing the perceptual target) and the Post-LCIs from Chapter 3 (representing the asymptotic endpoint of adaptation).

Question 1: What Does the Adaptation Trajectory Look Like for the Vowel Task?

The same three complementary analyses applied to group differences in Chapter 3 were applied here longitudinally, with session replacing group as the primary variable of interest.

Wasserstein Distance Trajectory. The Wasserstein distance from the NH reference bell was computed for each session, vowel, and formant dimension following Equation 3.4, using the same leave-one-out NH baseline. Distances were plotted as a function of days post-activation, separately per vowel and formant, and compared to the NH baseline and the Post-LCI distribution from Chapter 3. A Spearman rank correlation between Wasserstein distance and days post-activation was computed per vowel and formant as a descriptive summary of the trajectory direction.

Formant Centroid Shift Trajectory. The formant centroid shift - $\log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{session}})$ in octaves, positive values indicating a more apical representation than NH - was computed per session, vowel, and formant dimension following Equations 3.5–3.6. Trajectories were plotted and a Spearman correlation with days post-activation was computed per vowel and formant.

F1/F2 reliance ratio trajectory. The reliance ratio $R = \text{peak}(\text{bell}_{F2})/(\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$ was computed per session and per vowel following Equation 3.7. The trajectory was plotted per vowel and on average, and compared to the NH reference and Post-LCI values from Chapter 3.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta Confidence Analysis. The delta confidence measure $\Delta_{ejv} = \text{bell}_{\text{session}}(f_e, v) - \text{bell}_{\text{NH}}(f_e, v)$ was computed per session, electrode, and vowel following Equation 3.3, quantifying how far above or below the NH reference each electrode’s confidence lies at each session. The same univariate LMM structure was applied with Δ_{ejv} as the outcome, testing per session whether FTPM, CS, NR, or $\log_2(\text{CF}_{\text{place}})$ predicted the direction and magnitude of the local deviation from NH. A significant negative coefficient at early sessions that diminishes or disappears at later sessions would indicate that factor-driven local deficits are present during the adaptation window and resolve over time - the key prediction motivating this analysis. This per-session delta analysis is the longitudinal

complement to the group-level delta null result reported in Chapter 3: where the Post-LCIs showed no factor-driven local deficits on average, the present analysis asks whether Case Study 1 shows such deficits early in adaptation before they resolve.

Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Analysis 1 - Interval Distortion and Chord Preference. For each trial, interval distortion was computed following Equation 4.9 from Chapter 4, defined as the summed absolute difference between the cochlear-place interval and the allocated-frequency interval across the two chord electrodes. All trials were recoded relative to the root-position chord following the same sign-flip convention used in Chapter 4: $\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$ (negative when root is less distorted) and preference grade positive when root chord preferred.

The models mirror those of Chapter 4, applied per session rather than pooled across participants, all fitted as LMMs with a random intercept for electrode pair. M-Base (*chord type only*) establishes root-position preference at each session. M-Grad (*FTPM gradient only*) tests whether the inter-electrode FTPM gradient (the absolute difference in FTPM between the two chord electrodes) predicts preference strength. M-Add (*distortion + chord type*) controls for root-position preference additively. M-Int (*distortion $\times$ chord type*) tests whether distortion avoidance is conditional on chord type:

$$\text{Grade}_{\text{root}} \sim \Delta d_{\text{root}} \times \text{ChordType} + (1 \mid \text{PairCat}) \quad (5.1)$$

Longitudinal change was assessed by pooling across sessions with session as a continuous covariate:

$$\text{Grade}_{\text{root}} \sim \Delta d_{\text{root}} \times \text{ChordType} \times \text{Session} + (1 \mid \text{PairCat}) \quad (5.2)$$

Three time intervals were analysed: Early (Sessions 1-2, Days 6-20), Extended (Sessions 3-4, Days 41-134), and Full (Sessions 1-4).

Analysis 2 - Electrode Factor Prediction of Chord Preference. This analysis examined how electrode-level factors (FTPM, CS, NR) modulated the longitudinal trajectory of chord preference at the electrode level ($n = 88$: 22 electrodes $\times$ 4 sessions). The outcome variable was the electrode-level Bradley-Terry preference score, estimated from the full set of pairwise comparisons involving each electrode using `PairedCompCalc` (Leijon et al., 2019). Each predictor was z-scored prior to model fitting.

Two interaction models were fitted per predictor per time interval - a linear model testing monotonic change over time and a non-linear model allowing session-specific interactions relative to Session 1 (Day 6) as reference:

$$\text{Preference} \sim \text{Time} \times \text{Factor}_{\text{sc}} + (1 \mid \text{ElectrodeCat}) \quad (5.3)$$

$$\text{Preference} \sim \text{SessionFactor} \times \text{Factor}_{\text{sc}} + (1 \mid \text{ElectrodeCat}) \quad (5.4)$$

where $\text{Factor}_{\text{sc}}$ is the z-scored predictor and `SessionFactor` is a categorical session variable. Model panels display only the interaction component, with all predictor terciles anchored at zero at Session 1, such that divergence between lines reflects purely the time $\times$ predictor interaction. The linear model was preferred when its AIC was lower than

the non-linear model; otherwise the non-linear model was reported. Three epoch subsets were analysed: Early (Sessions 1-3, Days 6-41), Extended (Sessions 3-4, Days 41-134), and Full (Sessions 1-4).

5.3 Results

Results are presented across four testing sessions (S1: Day 6, S2: Day 20, S3: Day 41, S4: Day 134 post-activation), organised around the two tasks administered at each session. Throughout, Case Study 1's trajectories are benchmarked against the NH reference and the Post-LCI group from Chapters 3 and 4.

5.3.1 Vowel Identification Task

Descriptive Overview of Responses

Figure 5.2 shows the perceptual maps for each vowel across the four sessions, alongside the NH reference (right column). Each perceptual map displays the distribution of responses across the $F1 \times F2$ electrode grid, with colour intensity reflecting the Bayesian confidence at each grid point. Figure 5.3 shows the corresponding electrode confidence bells, plotting the confidence-weighted kernel density estimate across the frequency axis for each vowel and session. The NH and Post-LCI reference bells are also reported.

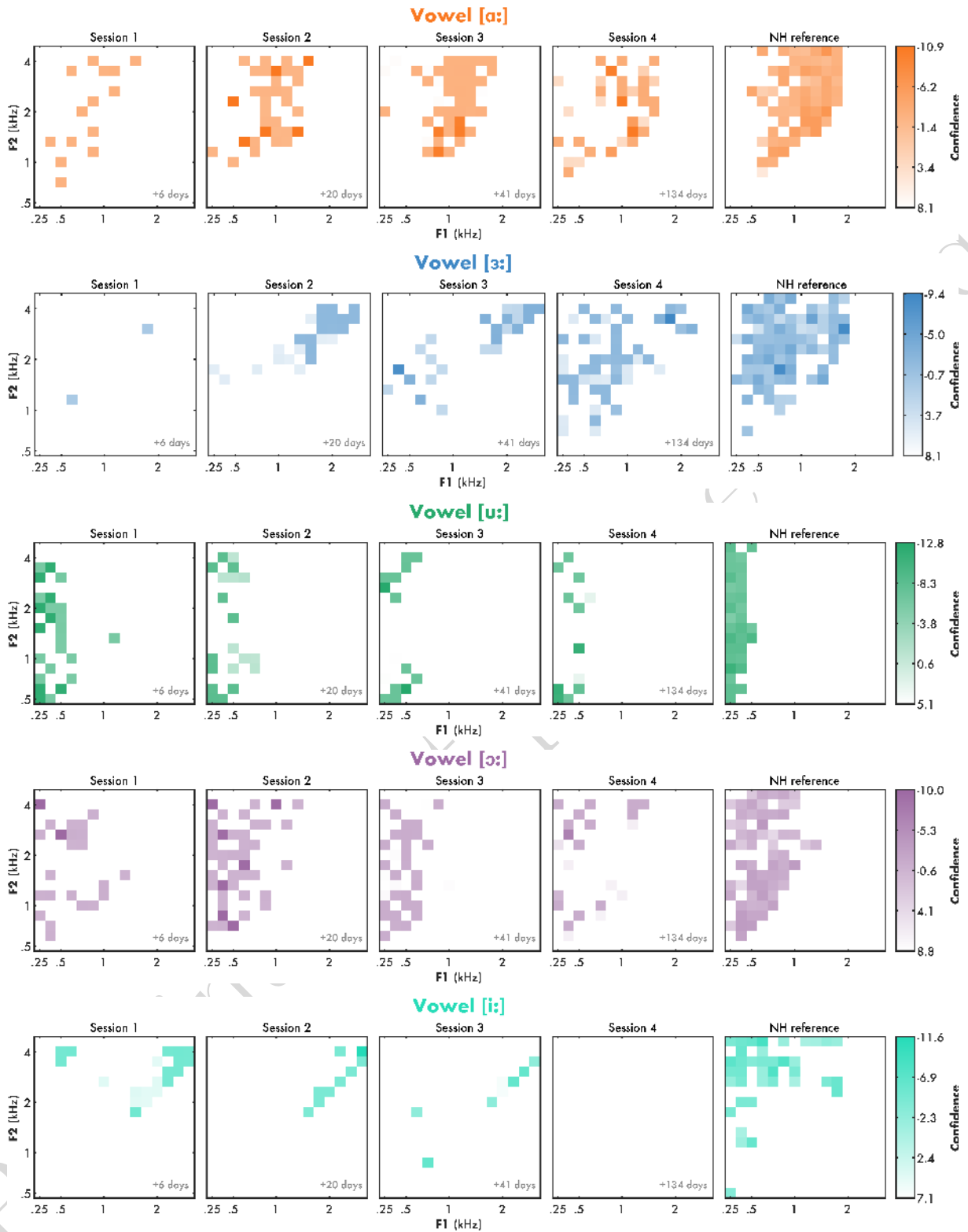


Figure 5.2: **Perceptual maps per vowel across sessions (Case Study 1), alongside the NH reference maps.** Each row shows one vowel (including [i:]) across the four sessions (columns: Day 6, Day 20, Day 41, Day 134). The last column corresponds to the NH group. Colour intensity reflects the confidence of responses at each F1×F2 electrode grid point.

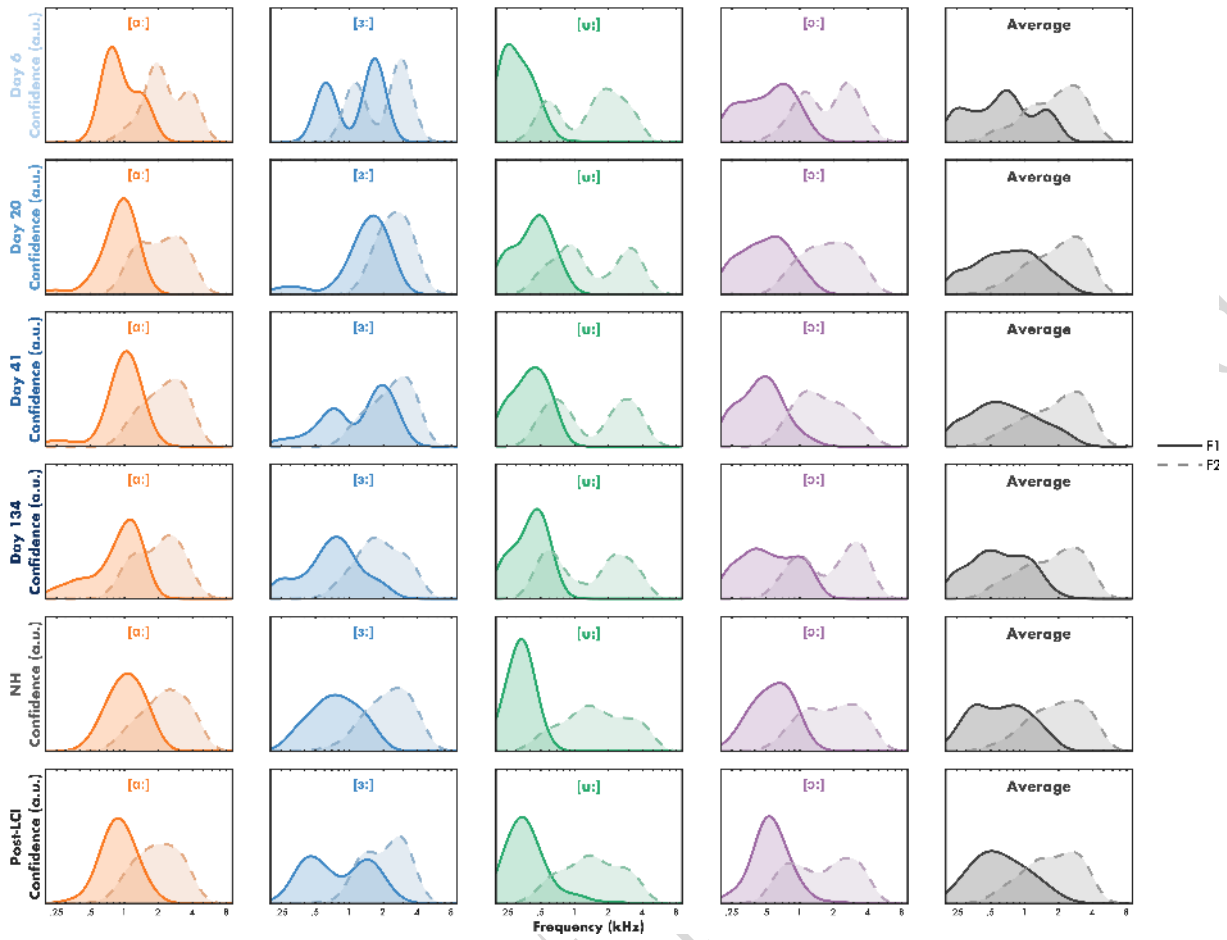


Figure 5.3: **Electrode confidence bells per vowel and session (Case Study 1).** Each panel shows the confidence-weighted kernel density estimate across the frequency axis for one vowel (rows) at each session (coloured lines, light = Day 6 to dark = Day 134). The NH and Post-LCI reference bells are reported in the last two rows.

At S1 (Day 6), responses were broadly distributed across the grid for all vowels, with no clear concentration around the target position. By S4 (Day 134), the response distributions had sharpened for most vowels, with a more clearly defined mode, though the degree of sharpening was vowel-specific. [ɜ:] showed the most dramatic change, with a highly dispersed distribution at S1–S2 that consolidated substantially by S4. [ɔ:] and [u:] remained relatively diffuse throughout.

Question 1: What Does the Adaptation Trajectory Look Like at the Representational Level for the Vowel Task?

Wasserstein Distance from the NH Reference. Figure 5.4 shows the Wasserstein distance trajectory per vowel and formant, and Table 5.3 reports numerical values.

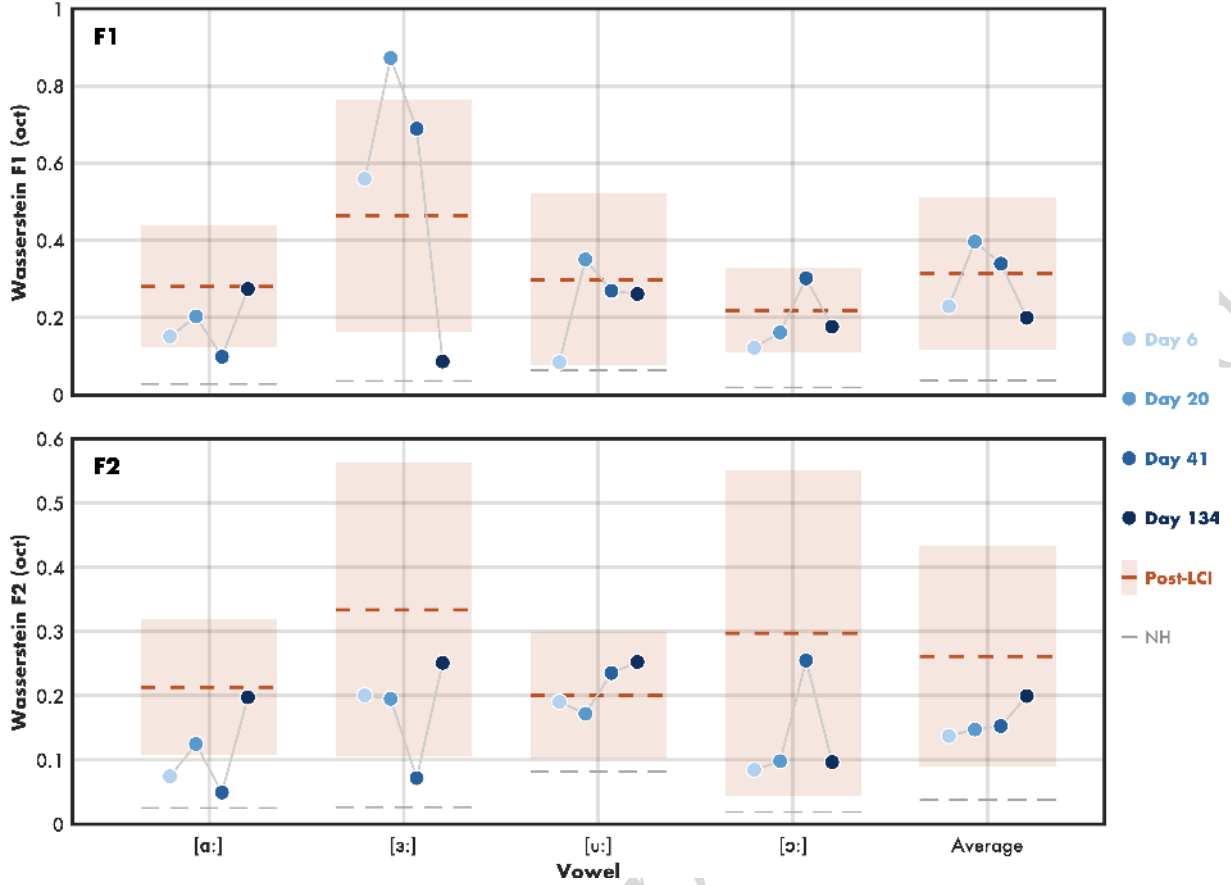


Figure 5.4: **Wasserstein distance from the NH reference bell across sessions (Case Study 1).** F1 (top) and F2 (bottom) distances per vowel and session. The grey dashed line shows the NH baseline; the red dashed line with shaded band shows the Post-LCI mean $\pm$ SD. Dots are shaded from light (Day 6) to dark (Day 134).

Table 5.3: **Wasserstein distances from the NH reference bell (oct), Case Study 1.** Values shown per formant, vowel, and session.

Formant	Vowel	S1 (d6)	S2 (d20)	S3 (d41)	S4 (d134)
F1	[a:]	0.151	0.203	0.099	0.274
	[ɜ:]	0.559	0.872	0.689	0.086
	[u:]	0.085	0.351	0.270	0.261
	[ɔ:]	0.122	0.162	0.302	0.177
	Average	0.229	0.397	0.340	0.200
F2	[a:]	0.075	0.125	0.049	0.198
	[ɜ:]	0.201	0.195	0.072	0.251
	[u:]	0.190	0.172	0.235	0.253
	[ɔ:]	0.085	0.098	0.255	0.097
	Average	0.138	0.148	0.153	0.200

F1 Wasserstein distances were dominated by [ɜ:], which showed the largest displacement from NH across all sessions, peaking at 0.872 oct at Day 20 before converging sharply to 0.086 oct at Day 134 - the lowest F1 value recorded at any session. The F1 average peaked at S2 (0.397 oct) and returned to its lowest value at S4 (0.200 oct), but

this apparent convergence is driven almost entirely by [3:]; the remaining vowels showed no consistent trend. F2 distances showed no convergence toward the NH reference: the F2 average increased monotonically from S1 (0.138 oct) to S4 (0.200 oct), moving further from rather than closer to NH. No Spearman correlation reached conventional significance (all $p \geq .083$), with the exception of the F2 average ($\rho = +1.00$, $p = .083$). For most vowels, F1 and F2 Wasserstein distances tended to co-vary across sessions: when F1 displacement increased, F2 followed, and vice versa, suggesting global vowel shifts rather than formant-specific reorganisation. [3:] was the clear exception, with a highly vowel-specific F1 trajectory that was largely independent of its F2 trajectory. Throughout, Case Study 1’s distances remained broadly comparable to the Post-LCI reference, though with substantially greater session-to-session variability, consistent with the noisier individual trajectory expected early in adaptation.

Formant Centroid Shift. Figure 5.5 shows the formant centroid shift trajectory and Table 5.4 reports numerical values.

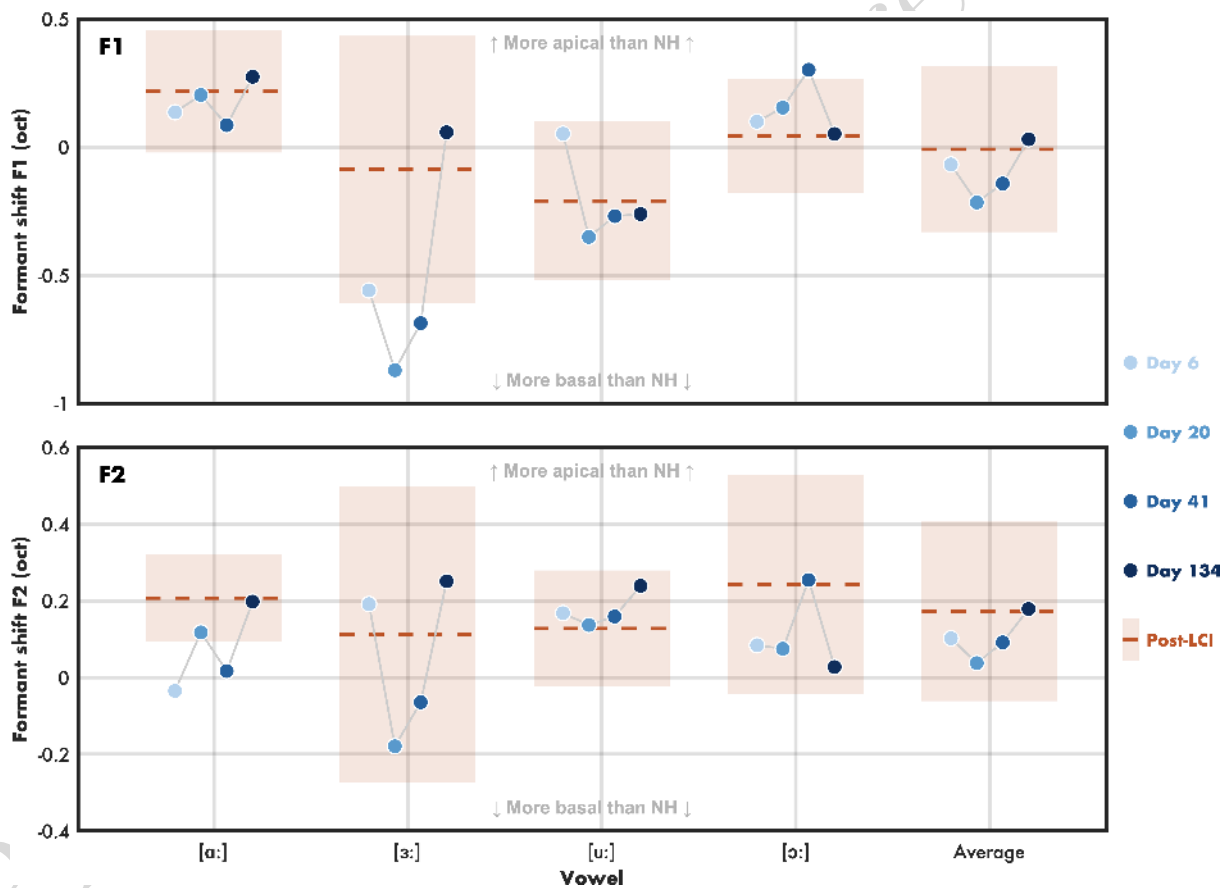


Figure 5.5: **Formant centroid shift relative to the NH reference across sessions (Case Study 1).** Shift = $\log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{session}})$ in octaves; positive values indicate the session centroid is perceived at a more apical place than for NH (lower frequency). Zero marks the NH reference; the red dashed line with shaded band shows the Post-LCI mean $\pm$ SD.

Table 5.4: **Formant centroid shift (oct), Case Study 1.** Positive = perceived at a more apical place than NH (lower frequency); negative = more basal (higher frequency) than NH.

Formant	Vowel	S1 (d6)	S2 (d20)	S3 (d41)	S4 (d134)
F1	[ɑ:]	+0.136	+0.203	+0.086	+0.274
	[ɜ:]	-0.559	-0.871	-0.687	+0.058
	[u:]	+0.052	-0.351	-0.270	-0.261
	[ɔ:]	+0.099	+0.154	+0.302	+0.051
	Average	-0.068	-0.216	-0.142	+0.031
F2	[ɑ:]	-0.035	+0.118	+0.017	+0.198
	[ɜ:]	+0.191	-0.180	-0.065	+0.251
	[u:]	+0.168	+0.137	+0.159	+0.239
	[ɔ:]	+0.084	+0.075	+0.254	+0.027
	Average	+0.102	+0.037	+0.091	+0.179

The shift analysis sharpens the co-movement pattern observed in the Wasserstein trajectories. For most vowels, F1 and F2 shifts evolved in the same direction across sessions: for example, for [ɜ:] between S3 and S4, F1 shifted from -0.687 to $+0.058$ oct (moving from basal to slightly apical relative to NH) and F2 moved in the same direction from -0.065 to $+0.251$ oct, both shifting towards more apical representations simultaneously. For F2, values remained consistently positive across sessions (average: $+0.102$ at S1, $+0.037$ at S2, $+0.091$ at S3, $+0.179$ at S4), indicating that F2 representations were persistently perceived at a more apical place than for NH - that is, at a lower perceived frequency than the NH reference - with a slight increase over the observation window. For F1, the average trajectory moves from slightly negative (-0.068 oct at S1) through a trough at S2 (-0.216 oct) and back towards zero at S4 ($+0.031$ oct), superficially suggesting convergence; however, this trend is driven almost entirely by [ɜ:], and given that F2 is simultaneously drifting away from NH there is no basis to conclude that F1 convergence will continue beyond Day 134. The Post-LCI reference band indicates that the experienced group's F2 centroid is consistently more apical than NH (positive values), consistent with the persistently positive F2 values observed in Case Study 1 from Day 6 - suggesting that the characteristic apical F2 shift is established from the very first session rather than developing progressively, and that both Case Study 1 and the Post-LCI group are displaced in the same direction relative to NH.

F1/F2 Reliance Ratio. Figure 5.6 shows the reliance ratio trajectory per vowel and session.

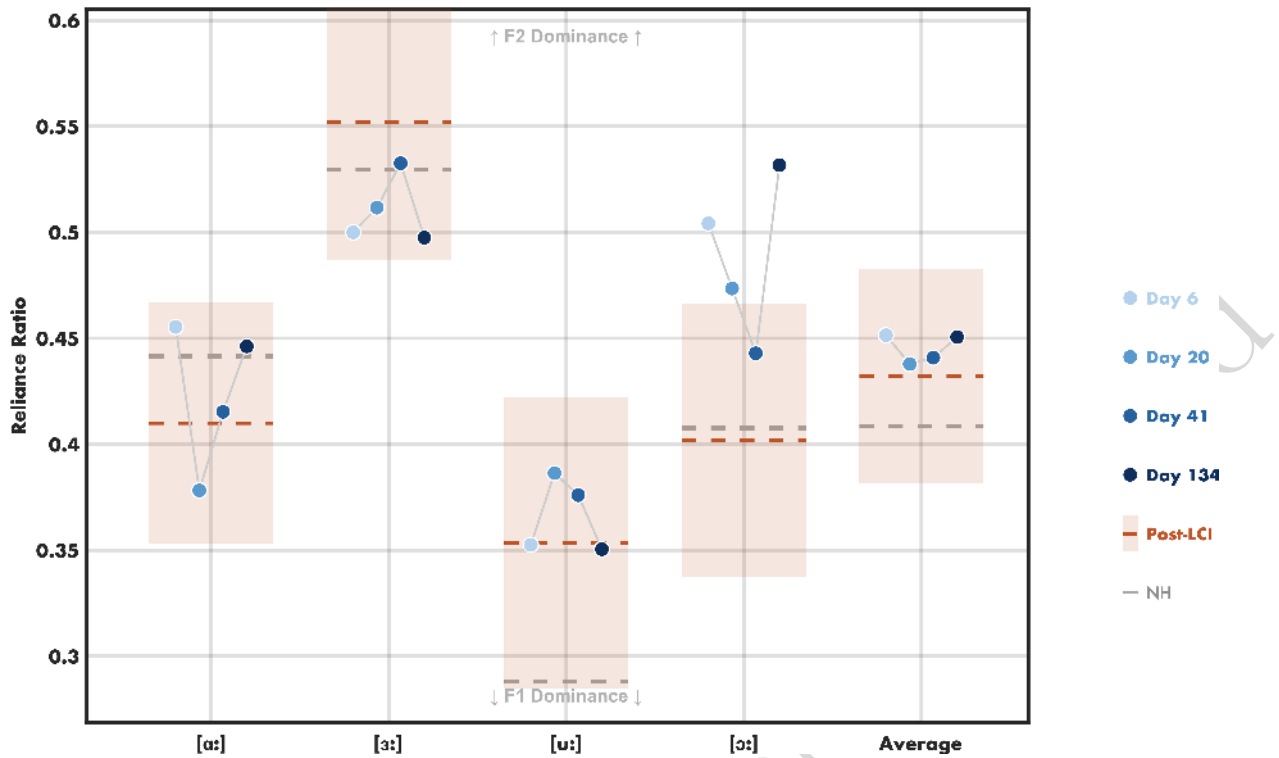


Figure 5.6: **F1/F2 reliance ratio per vowel and session (Case Study 1).** $R = \text{peak}(\text{bell}_{F2}) / (\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$. Values below 0.5 indicate F1 dominance. The grey dashed line shows the NH reference per vowel; the red dashed line with shaded band shows the Post-LCI mean $\pm$ SD.

No clear temporal trend was observed in the reliance ratio for any vowel. The mean ratio ranged narrowly from 0.438 to 0.451 across sessions (NH: 0.409), indicating mild and stable F1 dominance throughout. The profile at Day 6 was already comparable to that at Day 134, and no vowel showed a clear drift toward or away from the NH reference over the observation window. The Post-LCI mean reliance ratio was 0.432 (average across vowels), closely matching Case Study 1's values at every session and confirming that the balanced F1/F2 weighting strategy is established from Day 6 at a level consistent with the Post-LCI group. The relative weighting of F1 and F2 information was therefore established from the very first session and was not the primary locus of adaptation.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta confidence analysis. Figure 5.7 shows the delta confidence as a function of each factor per session, and Table 5.5 reports LMM results.

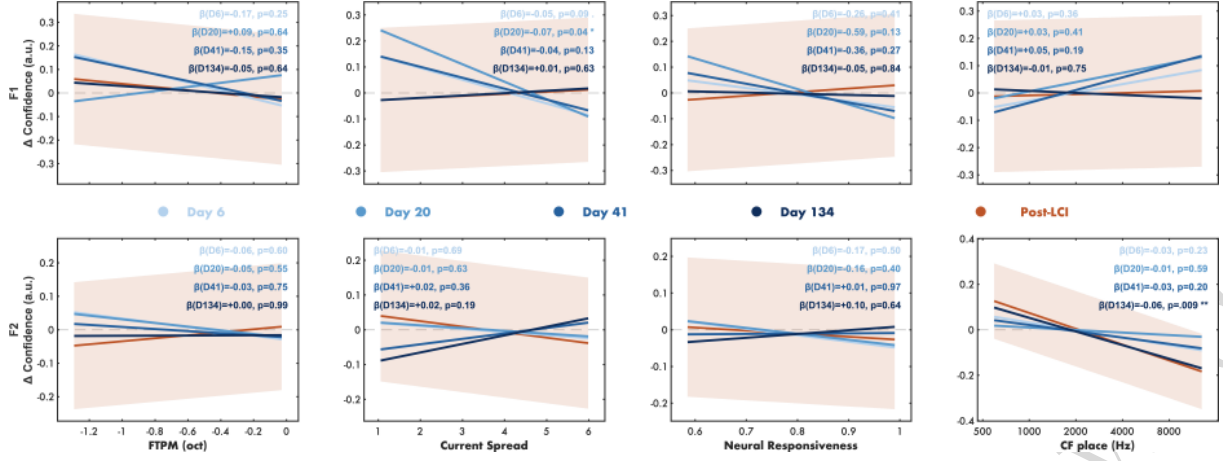


Figure 5.7: **Local confidence deficit** ($\Delta = \text{bell}_{\text{session}} - \text{bell}_{\text{NH}}$) as a function of **electrode-level factors across sessions (Case Study 1)**. Each panel shows one factor (columns: FTPM = frequency-to-place mismatch in octaves; CS = current spread; NR = neural responsiveness; CF_{place} = characteristic frequency of the stimulated cochlear location) and one formant (rows: F1, F2) across sessions (light to dark shading = Day 6 to Day 134). A value of zero indicates no deviation from the NH reference at that frequency location. The Post-LCI group mean trend and ± 1 SD band are shown in red.

Table 5.5: **LMM $\hat{\beta}$ per session for each factor and formant (Case Study 1)**. Model: $\Delta \sim \text{Factor} + \text{Vowel} + (1|\text{Electrode})$. FTPM = frequency-to-place mismatch (oct); CS = current spread; NR = neural responsiveness; CF_{place} = characteristic frequency of stimulated cochlear location. Significance: $p < .10$, $*p < .05$, $**p < .01$.

Formant	Factor	Day 6	Day 20	Day 41	Day 134
F1	FTPM	-0.172	+0.088	-0.148	-0.049
	CS	-0.046	-0.067*	-0.042	+0.009
	NR	-0.258	-0.593	-0.365	-0.045
	CF_{place}	+0.030	+0.034	+0.046	-0.007
F2	FTPM	-0.064	-0.055	-0.031	+0.002
	CS	-0.009	-0.008	+0.016	+0.025
	NR	-0.174	-0.162	+0.009	+0.103
	CF_{place}	-0.033	-0.011	-0.028	-0.060**

Only two effects reached significance across the full delta analysis: CS predicted F1 delta at Day 20 ($\hat{\beta} = -0.067$, $p = .037$) and CF_{place} predicted F2 delta at Day 134 ($\hat{\beta} = -0.060$, $p = .008$). Despite this near-complete null in formal significance, the F2 delta trajectories showed a consistent and interpretable pattern across all three primary factors. For FTPM, electrodes with greater absolute mismatch were initially associated with lower F2 confidence relative to NH ($\hat{\beta} = -0.064$ at Day 6), but this deficit progressively attenuated and had fully resolved by Day 134 ($\hat{\beta} = +0.002$). For CS, electrodes with broader current spread were initially associated with lower F2 confidence ($\hat{\beta} = -0.009$ at Day 6), and this relationship reversed over time, with broader-current-spread electrodes attracting slightly more relative confidence by Day 134 ($\hat{\beta} = +0.025$), consistent with a progressive recalibration of spectral resolution expectations. For NR, electrodes

with higher neural responsiveness were associated with lower F2 confidence at the earliest sessions ($\hat{\beta} = -0.174$ at Day 6), a pattern that reversed substantially by Day 134 ($\hat{\beta} = +0.103$), suggesting that the perceptual advantage of high-NR electrodes emerges only after initial central recalibration rather than being immediately available from activation.

All three factors showed monotonic Spearman trends for CS and NR ($\rho = +1.00$, $p = .083$ each), and the convergent pattern across three independent factors suggests a real trajectory of early F2 deficit followed by progressive resolution, even if individual session effects fall short of significance in this single-participant design. For F1, no coherent temporal pattern was observed: estimates alternated in sign with no consistent direction for any factor, consistent with no structured factor effects on F1 confidence beyond CF_{place} .

5.3.2 Chord Preference Task

Descriptive Overview of Responses

Figure 5.8 shows the frequency importance functions (FIFs) across the four sessions, and Figure 5.9 shows the distribution of preference grades per session.

At S1 (Day 6), the FIF was dominated by a strong mid-frequency preference: electrode E16 (999 Hz) was the most preferred (median = +3.93) and the four most basal electrodes (E1–E4, >4988 Hz) were the most strongly rejected (median = -3.52 to -1.71), yielding a total preference span of 7.45 units across the 22-electrode array. A broadly similar pattern was observed at S2 (Day 20), with E16 remaining the most preferred electrode (median = +2.85) and E1 the most rejected (median = -3.67), though the overall span had compressed to 6.52 units. By S3 (Day 41), the preference peak had shifted slightly apically to E17 (871 Hz, median = +2.64), and the most rejected electrode changed from the most basal (E1) to E4 (4988 Hz, median = -2.09), indicating a reorganisation of the electrode-level preference profile alongside a further reduction in span to 4.73 units. By S4 (Day 134), the FIF had become substantially less structured: the most preferred electrode was E11 (1933 Hz, median = +1.34), and the most rejected was E19 (621 Hz, median = -3.55) - an apical electrode that had been near-neutral at earlier sessions. The mean preference across electrodes declined progressively from +0.12 at S1 to -0.32 at S4, and the peak-to-trough span stabilised at approximately 4.9 units by S4, reflecting both a compression and a qualitative reorganisation of the preference profile rather than a simple flattening.

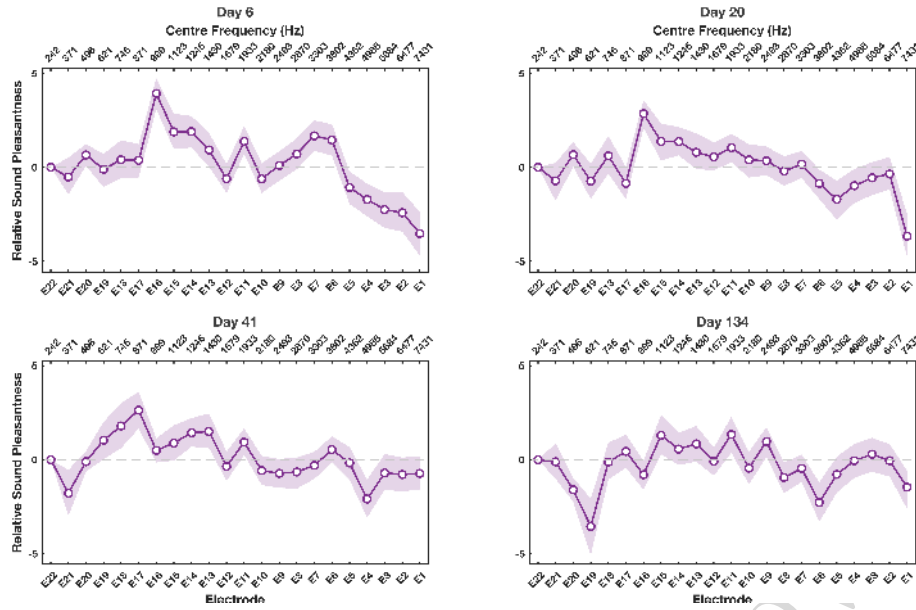


Figure 5.8: **Frequency importance function (FIF) across sessions (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). The y-axis shows relative sound pleasantness (median with 16th–84th percentile shaded band); the bottom x-axis shows electrode labels (E = electrode number, E1 = most basal, E22 = most apical) and the top x-axis shows centre frequencies in Hz. The y-axis scale is shared across all four panels to allow direct comparison.

Grade distributions are centred around zero at all sessions, with a progressive compression of the range from S1 (grades spanning approximately ± 3) to S4 (grades spanning approximately ± 1), reflecting a reduction in the overall strength of preference with experience. The no-preference rate (grade = 0) accounted for 16, 21, 17, and 13 trials at S1–S4 respectively, remaining broadly stable whilst the non-zero grades became less extreme.

Root-position chords were preferred on 67.2%, 59.3%, 60.3%, and 53.7% of decided trials across sessions, and the less-distorted chord was preferred on 65.6%, 69.5%, 60.3%, and 56.7% of trials, both declining monotonically across sessions.

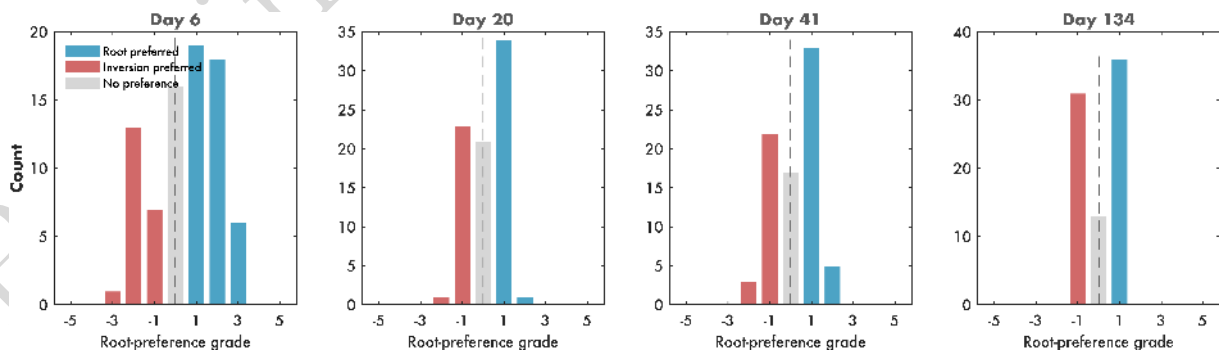


Figure 5.9: **Distribution of chord preference grades across sessions (Case Study 1).** Positive grades indicate preference for the root-position chord; negative grades indicate preference for the first-inversion chord. Grade = 0 indicates no preference. The progressive compression of the distribution from S1 to S4 reflects declining preference strength over the adaptation window.

Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Analysis 1 - Interval Distortion and Root-Position Preference

Per-Session Models. At S1, interval distortion (defined as the summed absolute difference between cochlear-place and allocated-frequency intervals; Equation 4.9) strongly predicted preference in the LMM (M-Add: $\hat{\beta}_{\Delta d} = -1.99$, $p = .003$) and root-position chords were significantly preferred (M-Base: $\hat{\beta}_{\text{root}} = +0.89$, $p = .012$), though the paired t -test on chosen versus rejected chord distortion was only marginal ($t(63) = -1.93$, $p = .058$). At S2, distortion sensitivity was maintained (M-Add: $\hat{\beta} = -1.23$, $p < .001$) and the paired t -test confirmed significantly lower distortion in chosen chords ($t(58) = -3.24$, $p = .002$), though root preference was no longer significant (M-Base: $p = .115$). At S3, neither distortion ($p = .306$) nor root preference ($p = .125$) reached significance in any model, and the paired t -test was null ($p = .662$). At S4, distortion sensitivity partially recovered (M-Add: $\hat{\beta} = -0.83$, $p = .027$) but root preference remained absent (M-Base: $p = .617$); instead, the FTPM gradient (the absolute inter-electrode FTPM difference) emerged as a significant predictor (M-Grad: $\hat{\beta}_{\Delta \text{FTPM}_{\text{gradient}}} = +1.92$, $p = .017$), with the paired t -test again marginal ($p = .060$). Figure 5.10 shows the per-session model coefficients and Figure 5.11 shows the corresponding chosen versus rejected chord distortion distributions.

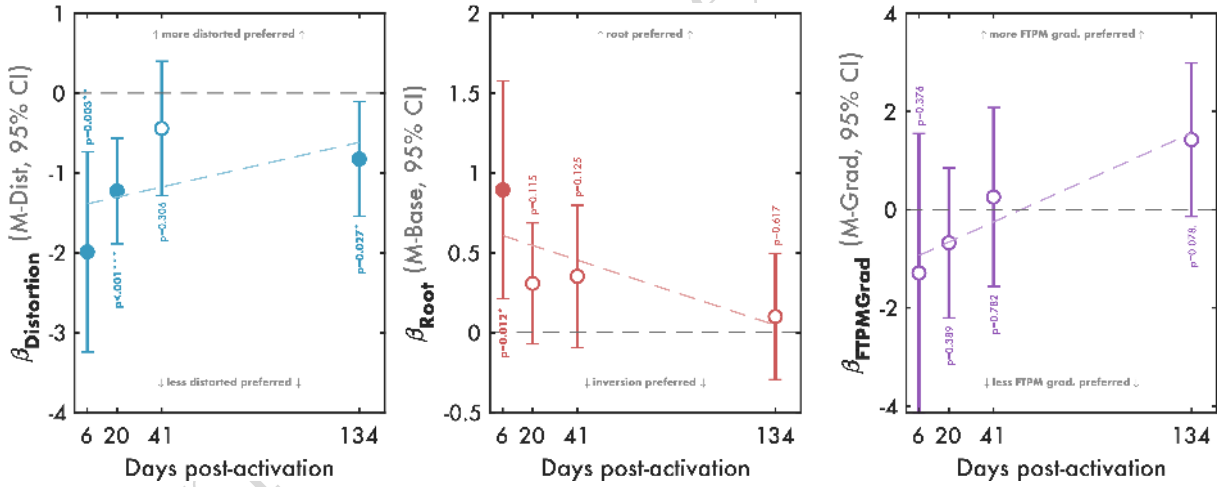


Figure 5.10: **Analysis 1: per-session regression coefficients for distortion sensitivity, root preference, and FTPM gradient (Case Study 1).** Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals (Equation 4.9); FTPM gradient = absolute inter-electrode FTPM difference. Error bars show 95% confidence intervals. Coefficients are drawn from the best-fitting LMM at each session (M-Add at S1–S2, M-Base at S3, M-Grad at S4).

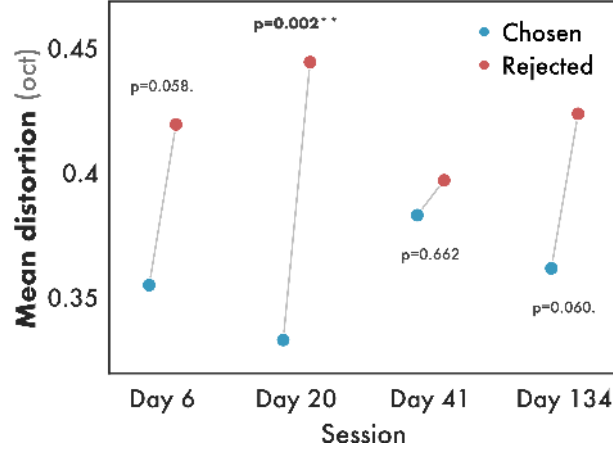


Figure 5.11: **Distortion of chosen versus rejected chords per session (Case Study 1).** Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals. Each pair of dots shows the session mean distortion for chosen (teal) and rejected (red) chords, connected by a grey line. Chosen chords had systematically lower distortion than rejected chords at S1 ($p = .058$) and S2 ($p = .002$), but not at S3 ($p = .662$) or S4 ($p = .060$). Significance assessed by paired t -tests.

Temporal Change. The M-Add $\times$ Session LMM confirmed declining distortion sensitivity over time ($\hat{\beta}_{\text{Session} \times \Delta d} = +0.42$, $p = .040$) and the M-Base $\times$ Session LMM confirmed declining root preference ($\hat{\beta}_{\text{Session} \times \text{root}} = -0.23$, $p = .039$). FTPM gradient sensitivity increased significantly over time in the M-Grad $\times$ Session LMM ($\hat{\beta}_{\text{Session} \times \Delta \text{FTPM}_{\text{gradient}}} = +0.91$, $p = .049$). The combined M-Add $\times$ Session model confirmed both declining distortion sensitivity ($\hat{\beta} = +0.42$, $p = .039$) and declining root preference ($\hat{\beta} = -0.23$, $p = .034$) simultaneously.

Epoch Analysis. In the Early epoch (S1–S2), both distortion ($\hat{\beta} = -1.49$, $p = .001$) and root preference ($\hat{\beta} = +0.59$, $p = .002$) were significant in the M-Int LMM. In the Extended epoch (S3–S4), neither reached significance (distortion $p = .380$; root $p = .106$). In the Full epoch, both distortion ($\hat{\beta} = -0.93$, $p = .002$) and root preference ($\hat{\beta} = +0.41$, $p < .001$) were significant in the pooled M-Int model, though their interaction was not ($p = .372$). Figure 5.12 shows the trial-level distortion-preference relationship per session in the recoded frame, with the Post-LCI reference slope overlaid.

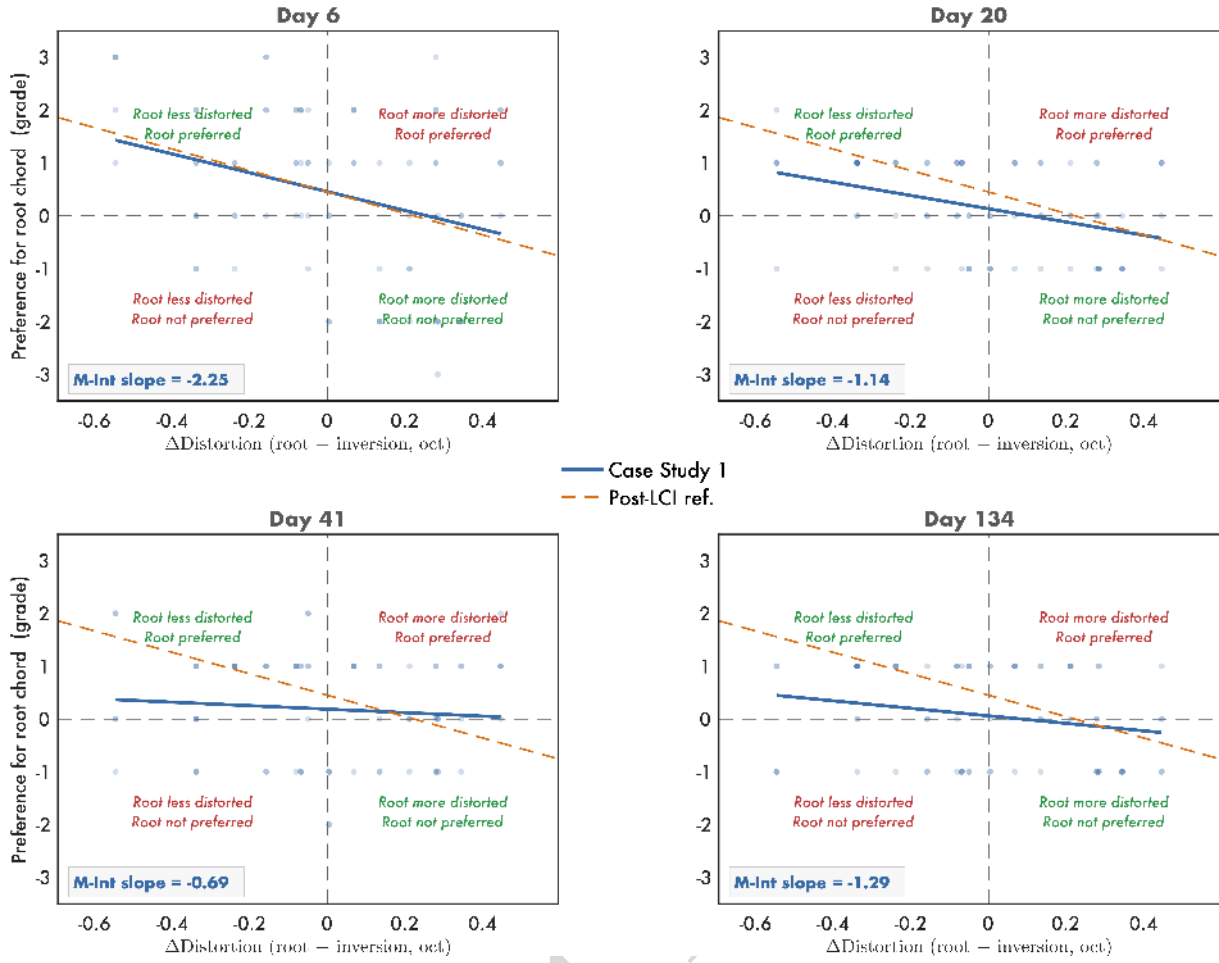


Figure 5.12: **Analysis 1: distortion-preference relationship per session (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). All trials are recoded relative to the root-position chord: $x = \Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$ (negative = root less distorted); y = preference for root chord (positive = root preferred). Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals (Equation 4.9). The blue line is the per-session OLS fit; the dashed orange line shows the Post-LCI reference slope from Chapter 4 (M-Int simple slope, $\hat{\beta} = -2.03$, intercept = +0.45). The M-Int simple slope for each session is annotated bottom-left. Green quadrant labels indicate the two quadrants consistent with distortion avoidance; red labels indicate the opposite pattern.

Analysis 2 - Electrode Factor Prediction of Chord Preference

The analysis was conducted at the electrode level ($n = 88$: 22 electrodes $\times$ 4 sessions). The linear LMM was preferred for all three predictors in the Full epoch (AIC: NR 287.6, CS 288.4, FTPM 286.8) and is treated as the primary test. Figure 5.13 summarises the epoch-level interaction coefficients for all three predictors. In addition to the longitudinal interaction analysis, per-session cross-sectional OLS slopes are reported alongside the Post-LCI group reference slope to directly benchmark Case Study 1 against the experienced-user asymptote at each session.

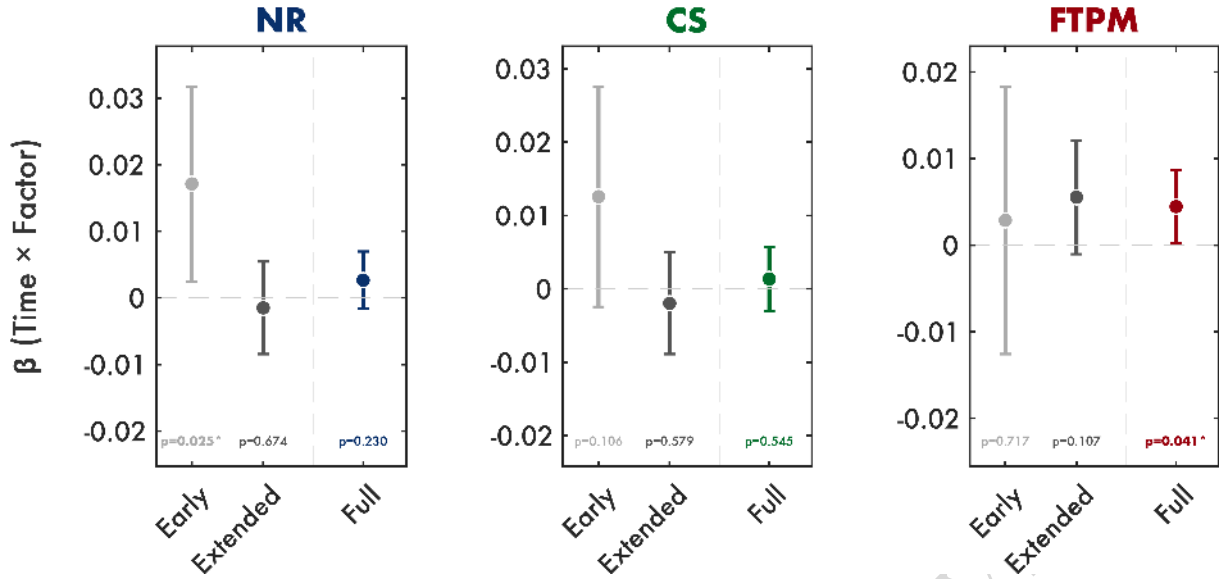


Figure 5.13: **Analysis 2: epoch-level forest plot of predictor×Time interaction coefficients (Case Study 1).** Each row shows the $\hat{\beta}$ (day^{-1}) and 95% CI for one predictor in one epoch (Early: Days 6–41; Extended: Days 41–134; Full: Days 6–134). NR = neural responsiveness; CS = current spread; FTPM = frequency-to-place mismatch. A positive coefficient indicates that preference for chords associated with higher values of that predictor increased over time within that epoch. Filled markers denote $p < .05$.

NR. In the Full epoch, the NR×Time interaction was not significant ($\hat{\beta} = +0.0027/\text{day}$, $SE = 0.0022$, $p = .230$). In the Early epoch (Days 6–41, $n = 66$), the interaction was significant and positive ($\hat{\beta} = +0.0171/\text{day}$, $SE = 0.0075$, $p = .025$): electrodes with higher neural responsiveness attracted progressively stronger preference from S1 to S3. In the Extended epoch (Days 41–134), the interaction reversed in sign and was non-significant ($\hat{\beta} = -0.0015/\text{day}$, $p = .674$), indicating that the early NR-driven preference increase did not persist into the later phase. The per-session OLS slopes reveal that the NR-preference relationship was initially strongly negative at S1 ($\hat{\beta} = -5.61$, $r = -.483$, $p = .023$), meaning that electrodes with higher NR were the least preferred at the earliest session. This relationship converged monotonically towards the Post-LCI reference ($\hat{\beta}_{\text{ref}} = +0.067$, $p = .959$) - which itself is indistinguishable from zero - across Sessions S2 ($\hat{\beta} = -4.20$) and S3 ($\hat{\beta} = -1.45$, $p = .421$), before diverging slightly at S4 ($\hat{\beta} = -2.43$, $p = .173$). The convergence towards the Post-LCI null is therefore the primary longitudinal signal: Case Study 1’s early negative NR effect dissolves progressively over the first 41 days, consistent with the auditory system learning to exploit high-NR electrodes.

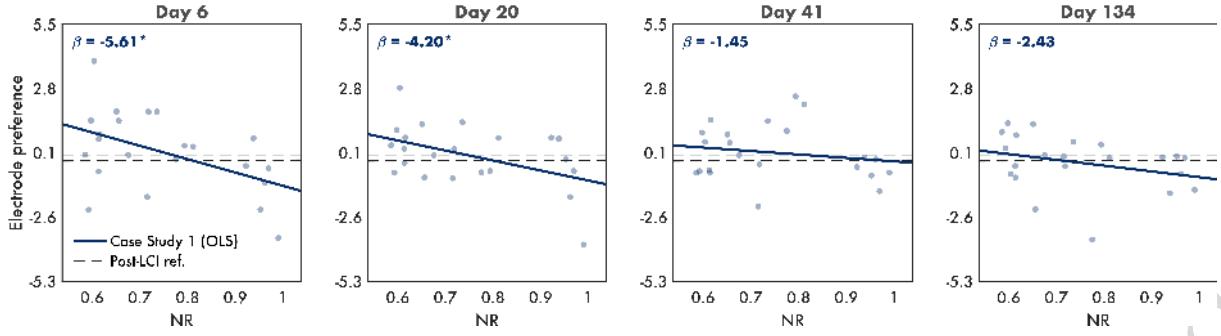


Figure 5.14: **Analysis 2: neural responsiveness (NR) and electrode preference across sessions (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). Each dot represents one electrode (E = electrode); the solid blue line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = +0.067$, Chapter 4).

CS. CS showed no significant interaction in any epoch: Full ($\hat{\beta} = +0.0013/\text{day}$, $p = .545$), Early ($\hat{\beta} = +0.0126/\text{day}$, $p = .106$), and Extended ($\hat{\beta} = -0.0020/\text{day}$, $p = .579$). The per-session OLS slopes nonetheless show a coherent convergence pattern: the CS-preference slope at S1 ($\hat{\beta} = -0.411$, $p = .053$) exceeded the Post-LCI reference ($\hat{\beta}_{\text{ref}} = -0.212$, $p = .077$) in magnitude by a factor of approximately two, and converged steadily towards it across sessions, reaching $\hat{\beta} = -0.344$ at S2, $\hat{\beta} = -0.155$ at S3, and $\hat{\beta} = -0.263$ at S4 (delta from reference = -0.051). CS therefore shows the closest convergence to the Post-LCI asymptote of all three factors by Day 134, despite the absence of a significant longitudinal interaction, and the direction of the effect is consistent across all four sessions.

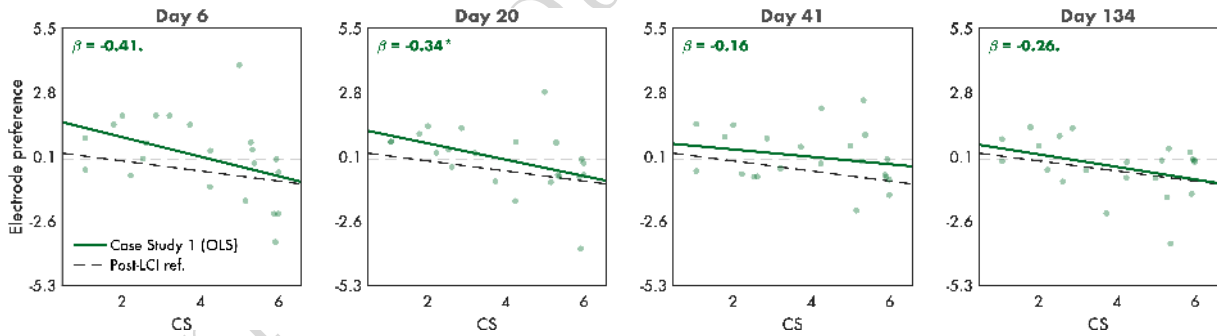


Figure 5.15: **Analysis 2: current spread (CS) and electrode preference across sessions (Case Study 1).** Same layout as Figure 5.14. The solid green line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = -0.212$, Chapter 4). No significant $\text{CS} \times \text{Time}$ interaction was observed in any epoch (all $p > .10$), yet the per-session slopes track the Post-LCI reference closely by Day 134.

FTPM. FTPM showed a significant positive interaction in the Full epoch ($\hat{\beta} = +0.0045/\text{day}$, $SE = 0.0022$, $p = .041$): electrodes with greater absolute FTPM were associated with progressively increasing chord preference across the full 134-day window. This effect was absent in the Early epoch ($\hat{\beta} = +0.0029/\text{day}$, $p = .717$) and marginal in the Extended epoch ($\hat{\beta} = +0.0055/\text{day}$, $p = .107$), indicating that the FTPM effect accumulated gradually. The per-session OLS slopes reveal a non-monotonic trajectory relative to the Post-LCI reference ($\hat{\beta}_{\text{ref}} = +0.835$, $p = .035$): Case Study 1 began well above the reference

at S1 ($\hat{\beta} = +2.22$, $p = .061$, $\text{delta} = +1.39$), converged substantially by S2 ($\hat{\beta} = +1.35$, $\text{delta} = +0.51$), then diverged at S3 where FTPM was the only factor to reach significance ($\hat{\beta} = +1.80$, $p = .026$, $\text{delta} = +0.96$), before collapsing substantially below the reference at S4 ($\hat{\beta} = +0.11$, $p = .902$, $\text{delta} = -0.73$). This non-monotonic pattern - initially overshooting the Post-LCI asymptote and then undershooting it - is consistent with a system in which mismatch-driven electrode contributions are initially over-weighted, briefly re-emerge as the electrode-based representation consolidates at S3, and then recede as the preference profile loses its FTPM gradient by Day 134.

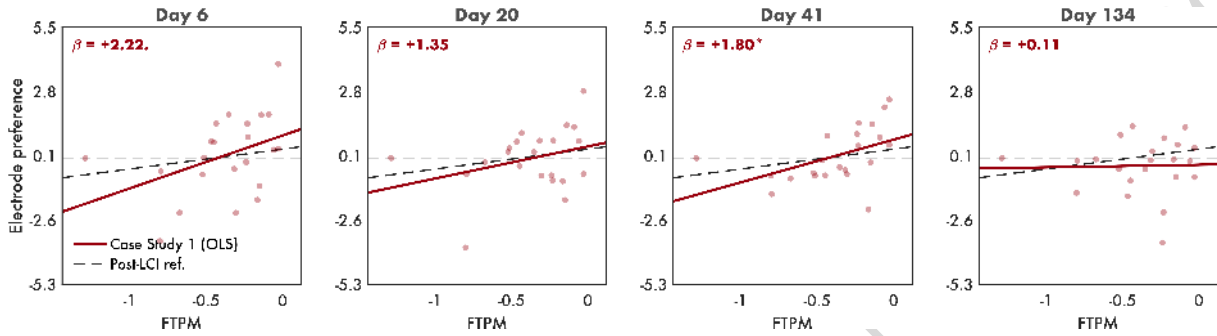


Figure 5.16: **Analysis 2: frequency-to-place mismatch (FTPM) and electrode preference across sessions (Case Study 1).** Same layout as Figure 5.14. The solid red line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = +0.835$, Chapter 4). A significant positive $\text{FTPM} \times \text{Time}$ interaction in the Full epoch ($\hat{\beta} = +0.0045/\text{day}$, $p = .041$) reflects a non-monotonic trajectory: Case Study 1 overshoots the Post-LCI reference at S1 and S3, converges at S2, and drops below the reference at S4.

5.4 Discussion

This chapter tracked the progression of the perceptual representations of a newly implanted post-lingual adult CI user across four sessions spanning 6 to 134 days post-activation, using the vowel identification task from Chapter 3 and the chord preference paradigm from Chapter 4. The two tasks were administered at the same sessions, enabling a direct comparison of how the adaptation process unfolds across two fundamentally different perceptual domains. The vowel and chord findings are discussed separately before being brought together in a comparative interpretation.

5.4.1 Vowel Task

Question 1: What does the Adaptation Trajectory Look Like for the Vowel Task?

Conjoint F1/F2 Movement and Formant Displacement A striking feature of the Wasserstein and shift trajectories is that F1 and F2 distances tend to co-vary across sessions for most vowels. For $[\alpha]$, $[u]$, and $[\sigma]$, increases in F1 displacement were accompanied by increases in F2 displacement, and decreases similarly paired, with both formants drifting in the same direction across sessions. This co-movement is inconsistent with a

formant-specific learning account, in which adaptation would be expected to affect the F1 and F2 regions independently. An alternative interpretation is that the co-movement is driven by one formant specifically - with the other following - rather than reflecting genuinely global recalibration. This possibility is partly supported by the factor analyses reported in Question 2: if electrode-level factors at a specific formant region are guiding the temporal trajectory, the co-movement of the other formant may be a consequence rather than a primary adaptation signal. What can be said from the present data is that the two formants do not adapt independently, and that the primary centre of change appears to be F2 given its structured relationship with individual factors across sessions.

Notably, the reliance ratio showed no temporal trend across any vowel, with the Day 6 profile already indistinguishable from Day 134. Case Study 1 arrived at implantation with a balanced F1/F2 weighting strategy - F1 slightly dominant on average, consistent with Post-LCI users in Chapter 3 - suggesting that spectral weighting is pre-existing and is not a mechanism of early adaptation.

F2 is Persistently Perceived at a More Apical Place than for NH The F2 centroid shift was consistently positive across all sessions (average +0.102 to +0.179 oct), indicating that F2 representations were persistently perceived at a more apical place than for NH throughout the observation window. Importantly, the Post-LCI reference band shows that the experienced group F2 centroid is also more apical than NH, suggesting that Case Study 1's apical F2 placement is not a transient early feature but is present at Day 6. The F2 trajectory therefore describes progressive consolidation of an apical bias already present at activation rather than a drift away from NH. This is broadly consistent with the view that post-lingual CI users adapt their perceptual frequency map towards the actual cochlear stimulation sites over the first months of device use, with the direction and magnitude of the shift depending on the individual electrode placement and frequency allocation.

The F1 average trajectory (from -0.068 oct at S1 through a trough of -0.216 oct at S2 back towards zero at S4) superficially suggests convergence, but this is driven almost entirely by [3:]. Its large and non-monotonic F1 trajectory - reaching -0.871 oct at Day 20 before recovering to near zero at Day 134, whilst its F2 trajectory remained comparatively flat - is the signature of genuine electrode-to-category learning for a single vowel category, independent of the global session effect. The fact that [3:] - which has the highest F1 frequency of the four retained vowels - was the vowel undergoing this correction is consistent with mid-array electrodes being the most dynamically reorganised during early adaptation. Excluding [3:], there is no convergence trend for F1, and whether F1 would continue converging beyond Day 134 remains unknown.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta Confidence: Early F2 Deficits and Progressive Resolution The delta confidence analysis reveals a structured temporal pattern for F2 despite failing to reach statistical significance, and constitutes the most informative finding of the vowel experiment. Only two effects reached significance across the full analysis: CS predicted F1 delta at Day 20 ($\hat{\beta} = -0.067$, $p = .037$) and CF_{place} predicted F2 delta at Day 134 ($\hat{\beta} = -0.060$,

$p = .008$). Despite this near-complete null, a convergent pattern emerged across all three primary F2 factors.

For FTPM, electrodes with greater absolute mismatch were initially associated with lower F2 confidence relative to NH ($\hat{\beta} = -0.064$ at Day 6), but this deficit progressively attenuated and had fully resolved by Day 134 ($\hat{\beta} = +0.002$): the participant placed progressively more confidence towards electrodes with less absolute mismatch over time, consistent with a refinement of spectral coding around lower-FTPM regions as the frequency map is progressively recalibrated (Fu and Galvin, 2007b). For CS, electrodes with broader current spread were initially associated with lower F2 confidence ($\hat{\beta} = -0.009$ at Day 6), and this relationship reversed over time ($\hat{\beta} = +0.025$ at Day 134): rather than avoiding broader-current-spread electrodes, the listener progressively learned to extract useful information from them, suggesting a dynamic rather than static consequence of CS on perception that is not captured in cross-sectional analyses of experienced users. For NR, the pattern is also striking: electrodes with higher neural responsiveness were associated with lower F2 confidence at the earliest sessions ($\hat{\beta} = -0.174$ at Day 6), which reversed substantially by Day 134 ($\hat{\beta} = +0.103$). NR, as estimated by PECAP, reflects the relative synchronous responsiveness of the auditory nerve at each electrode compared with other array positions (Garcia et al., 2021): its paradoxical early negative relationship with confidence may reflect the fact that high-NR electrodes produce more salient but initially unfamiliar percepts, and that their perceptual advantage only becomes exploitable after central recalibration. This is consistent with broader evidence that the quality of the electrode-neuron interface is a strong predictor of speech perception outcomes but that this advantage takes time to be expressed behaviourally (He et al., 2024; Pfingst et al., 2015). Monotonic Spearman trends were observed for both CS and NR ($\rho = +1.00$, $p = .083$ each).

This convergent pattern across three independent factors is the strongest longitudinal signal in the vowel data, and suggests that it is F2 - rather than F1 - that is primarily structured by electrode-level factors across sessions. Whether this finding generalises across participants, and whether the reliance ratio proves sensitive to such reorganisation over a longer adaptation window, will need to be examined in the ongoing longitudinal cohort.

5.4.2 Chord Task

Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Early Sensitivity: Pre-Existing Acoustic Templates Drive Preference At S1 and S2 (Days 6 and 20), Case Study 1 showed clear sensitivity to interval distortion and strong root-position preference, both consistent with the application of pre-existing acoustic spectral templates built through prior NH experience (Reiss et al., 2007). Chords whose physical interval structure most closely matched the intended acoustic intervals were preferred, and root-position chords were preferred over inversions. These effects declined rapidly: by S3 (Day 41), neither distortion nor root preference reached significance in any model.

The disappearance of these effects by 41 days does not necessarily reflect improved

adaptation in the conventional sense. A more plausible account is that the acoustic spectral templates driving early sensitivity were progressively suppressed as the central auditory system began to represent chord stimuli in terms of the actual electrode positions being stimulated rather than the acoustic frequencies they nominally represented. Once chord identity is encoded primarily at the electrode level, interval distortion defined in acoustic frequency space becomes less perceptually relevant. This transition from acoustic to electrode-based coding is consistent with the broader literature on perceptual adaptation to FTPM: post-lingual CI users who begin with strong acoustic templates must progressively recalibrate their perceptual frequency map before the electrode-level representation can organise perception (Fu and Galvin, 2007b; Svirsky et al., 2010). The six-week transition point identified here is at the early end of the 3–6 month window typically reported for speech adaptation, which is consistent with the chord paradigm being more sensitive to early changes than standard speech tests. This is likely related to the special nature of speech perception: the cortical mechanisms underlying phoneme categorisation involve complex, language-specific boundary learning and dynamic adaptation to different speakers, making the speech perception system inherently resilient to small spectral changes (Hillenbrand et al., 1995; Mathew et al., 2018). This resilience means that speech-based measures are likely insensitive to the kind of electrode-level spectral reorganisation detected here by the chord paradigm; however, when speech-based differences are detected, they are likely to be perceptually important precisely because they have exceeded this high tolerance threshold. An alternative interpretation is that the increased ability to perceptually distinguish between electrodes - rather than adaptation per se - underlies the loss of template-driven preference: as electrode-level representations become more distinct, global acoustic-based judgements give way to more fine-grained local comparisons.

Late Emergence: Electrode-Level FTPM Takes Over By S4 (Day 134), the FTPM gradient (the absolute difference in FTPM between the two chord electrodes) emerged as the primary predictor of chord preference ($\hat{\beta} = +1.92$, $p = .017$), indicating a shift in the perceptual basis of preference: the listener was no longer evaluating chords against an acoustic template but against an electrode-level representation in which apically displaced electrodes contributed more salient percepts. This is consistent with recent evidence that FTPM-induced interval distortion is the primary mechanism by which music perception is degraded in CI users, and that this distortion operates specifically through the compression of pitch intervals at the cochlear stimulation sites (Lewis et al., 2025).

The electrode-level trajectory analysis confirmed this two-phase picture. NR (neural responsiveness) showed a significant positive interaction in the Early epoch (S1–S3, $\hat{\beta} = +0.017/\text{day}$, $p = .025$): electrodes with better neural health attracted progressively stronger preference in the first six weeks, consistent with NR being associated with the formation of reliable chord percepts as the system transitions from acoustic to electrode-based representation (Garcia et al., 2021; Pfungst et al., 2015). FTPM (frequency-to-place mismatch) showed a significant positive interaction in the Full epoch ($\hat{\beta} = +0.0045/\text{day}$, $p = .041$), with no significant early effect, confirming that the FTPM effect accumulated gradually and only became detectable across the full 134-day window. CS (current spread) showed no significant effect at any stage.

The dissociation between NR (Early) and FTPM (Late) is theoretically informative. In the early weeks, the most reliable perceptual representations are formed at electrodes

with the best neural health, providing the auditory system a stable baseline around which to organise chord percepts (He et al., 2024). As electrode-based representation consolidates in the Extended phase, FTPM becomes the dominant organisational factor, with electrodes with greater absolute FTPM gaining the most preference improvement over time.

When compared against the cross-sectional Post-LCI group, the per-session slopes suggest a global trend of convergence towards the experienced-user asymptote, though the three factors follow distinct trajectories. For NR, Case Study 1 began with a strongly negative per-session slope ($\hat{\beta} = -5.61$ at S1), meaning high-NR electrodes were the least preferred; this converged progressively towards the Post-LCI reference of approximately zero ($\hat{\beta}_{\text{ref}} = +0.067$, $p = .959$) across S2 and S3, before diverging slightly at S4 - consistent with the sign reversal observed in the delta analysis. For CS, convergence was the most systematic: the per-session slope moved steadily from -0.411 at S1 towards the Post-LCI reference of -0.212 , reaching -0.263 at S4 (delta = -0.051), the closest alignment of any factor. For FTPM, the trajectory was non-monotonic: Case Study 1 initially overshoot the Post-LCI reference ($\hat{\beta} = +2.22$ vs $\hat{\beta}_{\text{ref}} = +0.835$ at S1), converged at S2, diverged at S3 when FTPM was the only factor to reach significance ($\hat{\beta} = +1.80$, $p = .026$), then dropped substantially below the reference at S4 ($\hat{\beta} = +0.11$). This non-monotonic pattern is reminiscent of the incomplete adaptation described in modelling studies of severely shifted frequency maps (Svirsky et al., 2010), and suggests that the FTPM effect does not follow a simple monotonic trajectory but instead reflects a system that transiently amplifies mismatch-driven contributions before gradually stabilising. Taken together, these comparisons suggest that the experienced-user endpoint is not approached uniformly across factors or across sessions, but through a structured sequence in which NR effects resolve first, CS converges steadily, and FTPM follows a more complex trajectory that reflects its dual role as both a spatial organising principle and a source of perceptual instability during early adaptation.

5.4.3 Cross-Task Comparison

Placing the vowel and chord results alongside each other reveals both convergences and divergences that are theoretically informative.

Convergence: a Two-Phase Temporal Structure. Both tasks show evidence of a transition between an early phase governed by prior acoustic experience and a later phase governed by electrode-level properties. In the vowel task, F2 delta deficits are present early and attenuate progressively; in the chord task, acoustic template sensitivity is present early and gives way to electrode-level FTPM preference. In both cases, the transition occurs between S2 and S3 (Days 20 and 41), suggesting that approximately six weeks post-activation marks a shift in the perceptual representation across both speech and music domains. This timing is consistent with the early adaptation window identified in the broader literature on Post-LCI users, where perceptual learning is most rapid in the first months and eventually reaches an asymptote (Fu and Galvin, 2007b).

Convergence: NR is associated with early adaptation, FTPM organises late. For the prediction of the vowel adaptation trajectory based on electrode-level factors

(FTPM, CS, NR), the delta analysis showed that all three factors imposed early F2 deficits at Day 6 that progressively attenuated, whilst the Post-LCI group showed NR as the sole residual predictor in the cross-sectional delta null. For the prediction of the chord adaptation trajectory based on the same electrode-level factors, the NR-Early and FTPM-Late pattern was more clearly delineated: NR was associated with the formation of reliable chord percepts in the first six weeks, and as the auditory system integrated more electrodes into its representation, FTPM - as a spatial mapping property - became the dominant organising principle. This parallel across tasks suggests a common underlying mechanism: NR provides the quality of the initial neural baseline (Garcia et al., 2021), and FTPM progressively structures the electrode-level representation once that baseline is established.

Divergence: the Direction of the FTPM Effect. At Day 134, electrodes with greater absolute FTPM simultaneously attracted higher chord preference (Analysis 2), yet in the delta analysis the same electrodes showed lower F2 confidence relative to NH. These two findings from the same electrodes at the same session can be reconciled by what each measure captures: chord preference is a relative evaluation rewarding perceptual intensity, whilst delta reflects local deficit relative to NH. Apical electrodes stimulate cochlear regions the auditory system finds reliable in absolute terms and perceptually vivid in relative terms, but because NH listeners would place even more confidence at those cochlear positions, the delta remains negative. This dissociation describes tolerance of mismatch rather than correction: the listener learns to use mismatched apical electrodes productively without correcting their frequency-to-place error (Svirsky et al., 2010).

Divergence: Task Sensitivity to Adaptation. The chord task showed clear and statistically significant changes across sessions, whilst the vowel task produced only sub-threshold trends. This difference in sensitivity is likely intrinsic to the paradigm designs rather than to the underlying perceptual process: the paired-comparison format of the chord task is inherently more sensitive than the closed-set identification format of the vowel task, and the chord outcome variable - preference grade - is a direct relative judgement that does not require formant estimation, Bayesian posterior computation, or confidence weighting. This is consistent with the superior sensitivity of the chord paradigm observed in the Post-LCI group from Chapter 4, where distortion effects that were marginal or absent in pooled models became detectable once the interaction with chord type was modelled. The impaired perception of harmony and chord quality in CI users is well established (Zimmer et al., 2019; Gilbert et al., 2025), but the present longitudinal data are the first to show that this impairment is dynamically structured by electrode-level factors across time. These observations suggest that the chord paradigm is the more appropriate tool for tracking adaptation in real time, whilst the vowel bell framework provides a richer mechanistic characterisation of what has changed at the electrode level.

Limitations. The single-participant design is the primary limitation of this chapter. All findings are presented as theoretically motivated observations requiring replication in a larger longitudinal cohort with denser session sampling in the first six weeks. The formant shift metric is likely insensitive at early sessions when posterior confidence is near zero, and the delta null at Day 6 should be interpreted cautiously for this reason.

5.5 Conclusion

This study tracked the perceptual representations of a newly implanted post-lingual CI user across four sessions spanning 6 to 134 days post-activation, using the vowel and chord paradigms developed in Chapters 3 and 4. Three main conclusions emerge.

Regarding how the vowel adaptation trajectory, F2 representations were persistently more apical than NH from Day 6 and showed no convergence across sessions, consistent with the asymptotic state already observed in Post-LCIs rather than a progressive trajectory towards it. F1 convergence was driven entirely by [3:], reflecting vowel-specific categorical learning rather than a global adaptation process. The F1/F2 reliance ratio was stable throughout, confirming that spectral weighting is a pre-existing property that does not require recalibration.

Regarding whether electrode-level factors predict the vowel adaptation trajectory, all three factors imposed early local F2 deficits relative to NH that attenuated progressively across sessions, contrasting with the complete null observed in Post-LCIs from Chapter 3. Case Study 1 was therefore at an intermediate stage of the adaptation resolution process at Day 134, with the deficit-attenuation pattern providing a longitudinal precursor to the experienced-user endpoint.

Regarding whether electrode-level factors predict the chord preference adaptation trajectory, the task revealed a clear two-phase structure. In the early phase (Days 6-20), preference was driven by acoustic spectral templates inherited from prior normal-hearing experience, with per-session factor slopes markedly exceeding the Post-LCIs reference. By Day 41 these effects had alleviated and by Day 134 all three factor slopes had converged towards the Post-LCIs asymptote, with NR scaffolding initial reliable percepts in the early phase and FTPM becoming the dominant spatial organising principle as electrode-based representation consolidated. These findings describe a process of tolerance rather than correction: the listener learns to use mismatched electrodes productively through a structured two-phase process that is substantially complete within the first six weeks of device use.

Postface

The case study presented here is the first of a planned series of four newly implanted post-lingual adult CI users, for whom longitudinal data collection is ongoing. The findings reported above should therefore be read as hypothesis-generating rather than confirmatory: the two-phase temporal structure, the NR-Early and FTPM-Late dissociation, and the convergent F2 delta attenuation are theoretically motivated patterns that require replication across participants before inferential conclusions can be drawn.

What further participants will enable. With four participants, the session-level data can be pooled within a mixed-effects framework with participant as a random intercept, transforming the descriptive trajectories reported here into inferential estimates with proper uncertainty quantification. Crucially, between-participant variation in FTPM magnitude and pattern will allow a direct test of whether individuals with greater cochlear mismatch show slower or qualitatively different adaptation trajectories - a prediction that the frequency-to-place framework developed in this thesis motivates directly but that a single case cannot test. Whether the six-week transition point identified here (between S2 and S3) is a reliable marker of the acoustic-to-electrode representational shift, or an individual feature of Case Study 1, will only become clear across the larger cohort.

Limitations of the current paradigms. Both tasks carry limitations that the larger longitudinal study should address. For the vowel task, the use of a single signal at each trial and the conjoint estimation of F1 and F2 still limit the interpretation at the electrode level. Additionally, the reliance ratio remained stable from Day 6, suggesting that F1/F2 spectral weighting is established from activation in post-lingual users and is not a sensitive marker of early adaptation. This absence of temporal trend may partly reflect the way the ratio is calculated here, and alternative formulations should be considered in future analyses. For the chord task, structural differences between the root and the inverted chord presented at every trial also limit the interpretation. Other paradigms without these intrinsic acoustical constraints should be explored in the future.

Clinical implications and future directions. A key strength of the paradigms developed here is that they can be administered and interpreted at the individual level, yielding electrode-level profiles that are directly clinically actionable. For Case Study 1, the chord preference data at Day 134 suggest that the most basal electrodes (E1–E4) are consistently the least preferred, that the mid-frequency region (around E11–E16) carries the most perceptual weight, and that FTPM-driven preference has largely resolved by 4.5 months post-activation. On the basis of these data, no remapping intervention would be recommended for this participant at the current stage: the preference profile has stabilised, distortion sensitivity has attenuated, and the electrode-level factor associations are converging towards the experienced-user asymptote. However, had the chord task revealed persistent distortion sensitivity at Day 134 - particularly at specific electrode pairs with large FTPM gradients - targeted frequency reallocation at those pairs would have been a reasonable clinical recommendation.

More broadly, if the two-phase structure replicates across the cohort, it has a direct clinical implication: the acoustic template phase (Days 6–20) represents the window dur-

ing which distortion avoidance and root-position preference are most salient, and during which targeted remapping - for example, reducing interval distortion at specific electrode pairs - would be most likely to interact with the listener's existing perceptual framework. By Day 41, the listener has begun to encode chords in electrode-level terms, and map adjustments at this stage may be less effective at altering the emerging electrode-based preference structure.

The chord paradigm, which proved more temporally sensitive than the vowel task, is a practical candidate for serial clinical assessment: it requires no metalinguistic training, produces reliable preference data within a single session, and yields electrode-level preference maps that can be compared directly to the individual factor profiles estimated from PECAP and post-operative imaging. Integrating these measures into a clinically deployable monitoring tool - in which adaptation trajectories are tracked against normative reference ranges derived from the full cohort - represents the primary translational goal of the larger longitudinal study.

Chapter 6

General Discussion, Perspectives, and Conclusion

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6.1 Overview of Findings

The main goal of this thesis was to investigate how three frequency-to-place factors - FTPM, CS, and NR - shape perception in CI users across complementary experimental approaches, and to develop tests which would be suitable to evaluate these effects at the electrode level. By “at the electrode level”, we mean that each electrode in the array is treated as an independent observation site: the perceptual response associated with a given electrode can be related to the specific FTPM, CS, and NR values measured at that contact, enabling identification of which individual electrodes along the array drive or limit perception. The stimuli used in Chapters 3–5 were designed to activate specific electrodes so that each response could be attributed to a known stimulation site.

The comparison between Pre-ECI and Post-LCI groups is central to the thesis design. Post-LCI users acquired language with normal hearing and must remap stored acoustic representations onto the CI signal, whereas Pre-ECI users built their auditory representations entirely through electrical stimulation; any differential sensitivity to frequency-to-place factors between these groups therefore isolates the contribution of prior acoustic experience to adaptation.

- In Chapter 2, we characterised the relationship between FTPM, signal type, and perceived sound quality using acoustical CI simulations with NH listeners.
- In Chapter 3, we presented an original vowel identification and confidence task and evaluated how experienced CI users represent vowels in the $F1 \times F2$ perceptual space, and whether those representations could be accounted for by implantation history (Pre-ECIs versus Post-LCIs) and electrode-level frequency-to-place factors (FTPM, CS, and NR). Vowels were selected because their identity is determined by two formant frequencies ($F1$ and $F2$) that map directly onto electrode positions, providing a natural probe for place-dependent spectral distortions.
- In Chapter 4, we presented an original chord preference task, designed to probe sensitivity to electrode-level FTPM, CS, and NR in a music-related task, again comparing Pre-ECIs and Post-LCIs. Chords were selected because their perceived quality depends on the relationships between simultaneously activated electrodes, making them sensitive to inter-electrode distortion profiles that vowels - which activate only two electrodes per trial - cannot capture.
- In Chapter 5, we applied both tasks longitudinally in a newly implanted adult Post-LCI, evaluating over time how perceptual sensitivity to these factors evolved across the first 134 days of device use, and whether the tasks were sufficiently sensitive to detect the distortions induced during adaptation.

The trajectory from Chapter 2 to Chapter 5 was shaped as much by what we learnt along the way as by what we set out to investigate.

Chapter 2 was conceived as a rigorous preliminary study under highly controlled conditions, with the goal of identifying which acoustic properties of sounds interact most strongly with FTPM. Whilst this original goal - finding a signal whose perceived quality would allow us to infer the degree of FTPM or residual adaptation to it - was not achieved, the study produced three important lessons.

First, it confirmed that FTPM has a real and measurable effect on perceived sound quality in acoustic simulation in an acute situation (*i.e.*, without adaptation), that depended on both the type of signal used and the shape of the FTPM profile. This last point confirmed our initial intuition that investigating FTPM at the electrode level was necessary. In addition, deeper investigations revealed that the degree of inter-electrode variability in FTPM was also associated with reduced sound quality. This latter finding was a direct source of inspiration for the chord interval distortion analysis (Analysis 1) developed in Chapters 4 and 5.

Second, it made clear that the MUSHRA paradigm, despite its psychometric robustness, could not be transposed to a clinical context: its duration, cognitive demands, and analysis complexity make it impractical for both CI users and clinicians to be able to conduct the task or intervene meaningfully based on the results.

Third, we realised that using real music samples or speech signals would be very challenging to interpret at the electrode level in the absence of a clear understanding of how acoustical content interacts with FTPM. Using synthetic stimuli was necessary to be able to control the parameters and understand which electrodes were being stimulated. This is the main reason why we used vowels and chords in the final versions of the tests presented here.

The two tasks developed were conceived as the best compromise available: quick enough to complete within a clinical appointment, containing sufficiently simple stimuli to permit electrode-level analysis, and informative enough to guide clinical decision-making.

In hindsight, the relative strengths and limitations of the two paradigms became apparent through the course of the thesis. The vowel task, whilst analytically rich, is inherently limited by two aspects. First, the paradigm has been judged complicated by CI participants: making a choice using only one sound, over a closed-set list of vowels, with spectral information limited to F1 and F2, was found to be highly challenging. Second, the conjoint presentation of two spectral dimensions - namely F1 and F2 - limits interpretations at the electrode level. These two limitations can explain - at least partly - the absence of significant trends in experienced users. The longitudinal study mitigates these limitations somewhat: the pattern of progressive factor-driven F2 deficits attenuating across sessions suggests that the task is indeed able to track electrode-level effects, at least during the adaptation period.

The chord task, by contrast, benefited from its paired-comparison structure, which is inherently more sensitive to small perceptual differences. It proved more reactive to electrode-level factors across both experienced users and the newly implanted case study, and was consistently better received by participants, who found it engaging and easy to perform. For these reasons, we believe that the chord task - despite its intrinsic limitations such as the strong root preference - has the stronger clinical future of the two. However, modifications related to the acoustical biases between the root and inverted chords should be seriously considered.

A further consideration when interpreting the Chapter 3 null result is that speech perception may be inherently resilient to the kind of electrode-level spectral distortions probed here. Speech has been argued to occupy a privileged perceptual status in the sense that listeners dynamically normalise F1 and F2 boundaries across speakers, accents, and speaking rates, accommodating substantial acoustic variability within stable phoneme categories (Hillenbrand et al., 1995). This flexibility means that small shifts in formant

representation - of the magnitude induced by FTPM or CS at individual electrodes - may fall within the tolerance of the speech categorisation system and produce no measurable change in identification or confidence. The broad category boundaries inherent to speech would make it difficult for small discriminable spectral differences to shift phoneme identity, even if those differences are perceptually detectable in a non-speech task such as chord preference. This interpretation is consistent with the dissociation observed across Chapters 3 and 4: the same electrode-level factors that produced no measurable effect on vowel identification shaped chord quality judgements, suggesting that the null result for speech reflects the robustness of phoneme categories rather than the absence of a perceptual distortion.

Across the four studies, a coherent picture emerges, and each chapter informs the others.

- FTPM shapes both sound quality and music perception, with effects that are real but centrally modulated: they are detectable in controlled simulations (Chapter 2) and emerge selectively in Post-LCIs who retain acoustic spectral templates (Chapter 4), and show sub-threshold attenuation trends over the first months of device use (Chapter 5).
- NR plays a qualitatively distinct role from the other two factors: rather than a perceptual distortion to be compensated, it represents the neural substrate that determines how well such compensation is possible, it defines the initial scaffold around which electrode-based representations are organised in new users (Chapter 5), and its consequences cannot be resolved by central remapping alone.
- CS acts as a consistent peripheral constraint on spectral resolution - significant as the most reliable electrode-level predictor of chord preference (Chapter 4), present in the longitudinal vowel factor analysis with the earliest significant effect across all factors (sig. at Day 20, Chapter 5), but not detectable in experienced users in the vowel task (Chapter 3), consistent with central compensation having resolved its perceptual consequences for speech by that stage.

The primary finding of the delta confidence analysis (Chapter 3) is that none of the three factors - FTPM, CS, or NR - predict the local deviation of electrode-level confidence from the NH reference at matched cochlear positions in experienced users. This is a null result with a specific and limited scope: it does not explain why CI users are globally displaced from NH, nor does it imply that the factors have no perceptual consequences. It says only that, among the factors tested, none account for why one frequency position deviates more from NH than another in experienced users. Whether the global displacement reflects insertion depth, tonotopic remapping, task design, or some combination of the above remains an open question.

It is also worth considering whether the group-level analysis may obscure individual effects that cancel across participants. If some users show positive factor-preference relationships and others show negative ones - as was observed for FTPM in the Pre-ECI group in Chapter 4, where opposing slopes between the two Pre-ECI participants cancelled the pooled effect - then the group null could potentially mask genuine individual-level sensitivity. The individual-level analyses reported in Chapters 4 and 5 partially address this concern, but the small subsamples available for factor analyses limit the power to detect such heterogeneity systematically.

Table 6.1: **Summary of perceptual trends and frequency-to-place factor effects across Chapters 2-5.** Each row corresponds to one experimental chapter, task, and population. Columns show the direction and significance of FTPM, CS, and NR effects where tested. × indicates the factor was not studied in that analysis.

Ch.	Population	Task	Perceptual trend	FTPM	CS	NR
2	NH (CI sim.)	MUSHRA quality	Degree-dep. degradation; worse at low-freq. electrodes; variability harmful	Degrades quality; worse at apical; variability matters	×	×
3	Exp. Post-LCIs & Pre-ECIs	Vowel ID & confidence	Post-LCIs: F2 more apical than NH; Pre-ECIs: higher F1 weight; small NH displacement	No effect; compensated	No effect; compensated	No effect; compensated
4	Exp. Post-LCIs & Pre-ECIs	Chord A1 (chord)	Strong root pref. (both groups); Pre-ECIs grades stronger	Cue-conditional in Post-LCIs; unconditional in Pre-ECI 2; no effect Pre-ECI 1	×	×
		Chord A2 (electrode)	Post-LCIs higher differentiation; FIFs highly individual	Less FTPM → higher pref. (Post-LCIs); opposing slope Pre-ECIs	Narrower CS → higher pref. in Post-LCIs, no effect in Pre-ECIs	No effect; compensated
5	New Post-LCI (Days 6–134)	Vowel ID & confidence	F2 persistently more apical than NH from Day 6 (consistent with Post-LCIs); F1 convergence for [3:] only; no F1/F2 ratio change	F2: more confidence initially for High FTPM electrodes; reversed over time	Narrow CS initially preferred; sig. at Day 20; Reversal by Day 20; opposite trend compared to Post-LCIs	High NR initially not preferred; reversed over time
		Chord A1 (chord)	Root pref. and distortion avoidance both decline over time; two-phase structure; per-session slopes converge to Post-LCIs reference by Day 134	Distortion avoidance drives preference in early phase (Days 6–20); FTPM gradient dominant at S4	×	×
		Chord A2 (electrode)	NR early scaffold; FTPM late organiser; representational transition ~6 weeks post-activation	Non-monotonic: exceeds Post-LCIs reference at S1 and S3; converges below reference at S4; sig. Full epoch interaction	No sig. effect in any epoch; per-session slopes track Post-LCIs reference closely by Day 134	Strongly negative at S1 (high-NR electrodes less preferred); dissolves to Post-LCIs null by S3; slight divergence at S4

6.2 What we Now Know About Frequency-to-Place Factor Effects

6.2.1 FTPM: Real but Centrally Modulated

FTPM has traditionally been treated as a single value per participant - a mean across electrodes - limiting the ability to identify which electrodes drive or limit perception. The electrode-level approach developed in this thesis reveals a more nuanced picture, beginning with acoustical simulations in NH listeners and extending to experienced and newly implanted CI users.

Chapter 2 provided controlled evidence that FTPM degrades perceived sound quality differently according to the acoustic environment and the shape of the FTPM profile. Critically, degradation was particularly marked when FTPM was simulated at electrodes conveying low-frequency information, and inter-electrode variability in FTPM - independently of overall mismatch degree - was additionally associated with reduced sound quality.

This latter finding is particularly novel: it suggests that the organisation of the allocation frequencies across the array matters, not just their absolute displacement from cochlear place. These effects were observed in an acute situation (*i.e.*, without any adaptation), in a NH listener group, under controlled conditions, establishing a causal baseline before introducing the complexity of real CI users and real adaptation.

In experienced users, none of the three frequency-to-place factors - including FTPM - accounted for the electrode-level deviations of CI vowel representations from the NH reference (delta analysis, Chapter 3), suggesting that experienced users were fully adapted to their individual FTPM profiles at the time of testing, or that the inherent breadth of speech categories renders the vowel task insufficiently sensitive to detect their residual influence. This null result is consistent with the broader literature showing that Post-LCIs largely adapt to their frequency allocation over the first 3 to 6 months of device use when mismatch is not severe, but that adaptation may be incomplete or much longer when mismatch is large (Fu and Galvin, 2007b; Svirsky et al., 2010). By contrast, FTPM shaped chord quality judgements in Post-LCIs who retain acoustic spectral templates (Chapter 4), confirming that FTPM effects persist in the music domain even after apparent compensation in speech. This is consistent with recent evidence that FTPM-induced interval distortion is particularly critical for music perception because relative pitch - the basis of melody and harmony - is distorted in a way that speech cues are not (Lewis et al., 2025). In the longitudinal case study, FTPM showed an early F2 delta pattern at Day 6 - electrodes with higher absolute mismatch were associated with higher delta confidence relative to NH - which attenuated monotonically across sessions towards the Post-LCIs null (Chapter 5). The chord task also revealed a capacity for progressive adaptation towards the FTPM response pattern of experienced users: the per-session slope began well above the Post-LCIs reference at Day 6, followed a non-monotonic trajectory across sessions, and had converged towards the Post-LCIs asymptote by Day 134, suggesting that the auditory system progressively learns to use mismatched electrodes productively, even if that process is neither linear nor complete within 134 days. This partially mirrors the finding from modelling studies that adaptation to a severe frequency shift is incomplete after 3 months of continuous use (Svirsky et al., 2010), and extends it by showing that

electrode-level FTPM continues to organise perceptual representations progressively even when global adaptation has largely stabilised.

Taken together, these findings are consistent with a centrally modulated account of FTPM: the peripheral mismatch is fixed at implantation, but its perceptual expression depends on the state of central learning. The Chapter 2 simulations confirm that the distortion is real and detectable in the absence of any adaptation; subsequent chapters show that this distortion is progressively compensated in the speech domain but remains detectable - and differently expressed - in music and sound quality domains even after years of experience. In early adaptation, the auditory system has not yet incorporated the output of mismatched electrodes, and electrode-level factor effects are detectable in both tasks. In experienced users, the effects have been partially compensated by central remapping, becoming undetectable in the vowel delta analysis but persisting in chord quality perception. In the intermediate period - around 134 days post-activation - FTPM is still actively shaping both speech and music representations, with consequences that differ across perceptual tasks. The important clinical message is that FTPM effects are not static: they are maximal at an intermediate stage of adaptation and may be partly resolved later on. Chapter 2 also motivates a new approach to FTPM alleviation that goes beyond anatomy-based fitting: reducing inter-electrode variability whilst preserving organisation across the array could provide perceptual benefit independently of reducing overall mismatch degree.

6.2.2 NR: a Distinct Peripheral Constraint

NR was not directly addressable in the acoustical simulations of Chapter 2: the noise vocoder used a fixed spatial spread parameter that did not vary across simulated electrodes. NR therefore enters the picture only from Chapter 3 onwards, where its effects were assessed at the electrode level in real CI users.

In experienced users, the delta confidence analysis of Chapter 3 showed that NR - like FTPM and CS - did not produce a significant electrode-level deviation from NH, consistent with central adaptation having resolved its perceptual consequences in the speech domain. In the chord task of Chapter 4, NR showed no significant effect on electrode-level preference, likely masked by the monopolar stimulation mode, which limits the electrode-specificity of neural excitation independently of local neural health. The most distinctive NR signal emerged in the longitudinal case study of Chapter 5: in the vowel delta analysis, electrodes with higher NR were initially associated with a larger F2 deficit relative to NH at Day 6, with the trend reversing over time as confidence at high-NR electrode positions increased. In the chord task, NR was the primary predictor of electrode-level preference formation in the Early epoch (Days 6-41): the relationship was initially paradoxical - high-NR electrodes were less preferred at Day 6 - before inverting as the auditory system learned to exploit them, a dynamic not observed for FTPM or CS at any stage.

These findings are situated within a broader literature demonstrating the importance of the electrode-neuron interface for perceptual outcomes. NR, as estimated by PECAP, reflects the relative synchronous responsiveness of the auditory nerve at each electrode site compared to other locations along the array (Garcia et al., 2021); it does not directly represent spiral ganglion neuron survival, but is validated as a functional indicator of

local neural integrity. The literature consistently shows that the quality of this interface predicts speech perception outcomes across participants: studies using ENI as a measure of how effectively each electrode stimulates the targeted auditory nerve fibres have found it to be the strongest single predictor of speech perception, particularly in noise (He et al., 2024). A recent systematic review with meta-analysis confirmed that re-mappings informed by ENI measures yield a moderate and significant improvement in speech-in-noise perception (pooled Cohen’s $d = 0.48$), with site-selection interventions showing the strongest effects (Lien et al., 2025). Pre-implant measures of cochlear-neural health have similarly been shown to account for approximately half the variance in speech perception in quiet independently of all other audiological and demographic predictors (Röbler et al., 2022). The site-specific nature of neural health - and the associated limitations of monopolar stimulation in exploiting electrode-level differences - is well documented: deactivating electrodes with poor neural interfaces improves speech perception for some participants, particularly poorer performers, but not consistently at group level (Bierer and Litvak, 2016).

What distinguishes the present findings from this body of work is the level of analysis: whereas the existing literature links NR to speech outcomes *across participants*, this thesis is the first to link PECAP-derived electrode-level NR to *within-participant, electrode-specific* perceptual differences in both a speech and a music related task. The paradoxical negative NR-preference relationship at Day 6 - high-NR electrodes less preferred, inverting within six weeks - has no direct precedent and suggests that newly implanted users initially penalise perceptually salient electrodes before learning to exploit them.

The distinctiveness of NR is therefore more mechanistic than statistical. Unlike FTPM, which represents a spatial mapping error that the auditory system can learn to correct, NR reflects the density and health of the spiral ganglion population at each electrode site. The longitudinal data suggest that some degree of central adaptation to NR does occur: the paradoxical negative NR-preference relationship at Day 6 inverts within the first six weeks, and the Early epoch interaction in the chord task is significant. However, this adaptation appears to be rapid and early-bounded: once the auditory system has reorganised around the initial neural substrate in the first weeks post-activation, NR ceases to drive further changes in either task. This is qualitatively different from FTPM, whose effects continue to evolve across the full 134-day window. NR therefore represents the neural foundation on which adaptation is built - one that can be partially accommodated in the earliest weeks but which ultimately sets a ceiling on how far both speech and music representations can be refined. The clinical implication is that PECAP-based NR estimation, already available as a clinical tool (Garcia et al., 2021), could predict the speed of this early adaptation window and identify electrodes where the neural substrate is too impoverished to support it - guiding electrode management decisions particularly for contacts in the F2 frequency region.

6.2.3 CS: a Stable Spectral Resolution Constraint

Like NR, CS was not directly addressable in the acoustical simulations of Chapter 2. However, Chapter 2 provided indirect motivation for studying CS in real CI users: the finding that inter-electrode FTPM variability degrades sound quality is mechanistically related to CS, since broad CS effectively broadens the frequency band stimulated by each electrode and compounds spectral overlap between adjacent channels.

In the vowel task of Chapter 3, CS was not a significant predictor of the electrode-level deviation from NH in experienced users, consistent with central compensation having resolved its perceptual consequences for speech at that stage. By contrast, CS was the most consistent electrode-level predictor of chord preference in Post-LCI (Chapter 4): narrower CS was associated with higher perceptual preference weight, an effect that survived in the full model alongside FTPM and NR. In the longitudinal case study (Chapter 5), CS showed an early F2 delta pattern consistent with the chord direction - narrower CS was initially associated with higher confidence relative to NH at Day 6 - but this trend reversed over time, with broader CS electrodes showing slightly higher delta confidence by Day 134. This reversal mirrors the pattern observed for FTPM and NR: an initial factor-aligned deficit that progressively attenuates and inverts as the auditory system recalibrates, with CS being the only factor to reach significance during this process (sig. at Day 20). No significant Time $\times$ CS interaction was found in the chord electrode-level trajectory analysis at any epoch, indicating that CS does not drive temporal changes in chord preference, even though it reliably predicts absolute preference levels in experienced users.

These findings are consistent with the broader literature on the perceptual consequences of CS. Experimental increases in channel interaction via spectral blurring have confirmed that apical channels - precisely the region where F1 is encoded and where our CS effects are most relevant - are the most critical for speech perception, with blurring applied to apical electrodes causing the greatest degradation (Goehring et al., 2021). A large-scale study of channel interaction, electrode placement, and speech perception further confirmed that electrode-to-modiolus distance - a key determinant of CS - significantly predicts vowel recognition outcomes (Berg et al., 2024), directly supporting our finding that narrower CS predicts better electrode-level perceptual performance. Focused stimulation strategies designed to reduce channel interaction have shown benefits for some participants, particularly poorer performers (Bierer and Litvak, 2016), consistent with our suggestion that targeting high-CS electrodes via deactivation or partial tripolar stimulation could provide perceptual benefit.

This pattern of results is interpretable in terms of the peripheral and largely fixed nature of CS: CS is largely determined at implantation and is not subject to central compensation in the way that FTPM might be. This stability makes CS a more straightforward clinical target than FTPM: if an electrode has persistently high CS, its perceptual consequences will not resolve over time, and deactivation or alternative stimulation strategies (*e.g.*, with tripolar or partial tripolar stimulation) may provide more reliable benefit.

6.3 Methodological Synthesis

6.3.1 The Vowel Task

Paradigm Limitations. The closed-set identification format is probably not the optimal design for detecting subtle perceptual differences. Two-alternative and three-alternative forced-choice paradigms yield more sensitive and less biased estimates of spectral discrimination, and in pilot sessions general agreement was reached that 2AFC and 3AFC formats produced more consistent responses. The practical challenge is translating these formats to the electrode-level grid structure: with 15-22 F1/F2 combinations per participant,

across 4 different vowels, a full 2AFC design would require thousands of trials, which is not clinically feasible. Developing a reduced-set 2AFC version of the task - covering a targeted subset of electrode combinations selected based on FTPM or NR values - is a priority for future work.

Another fundamental limitation is that the single confidence rating per trial conflates F1 and F2 certainty. When a participant hears a stimulus activating electrode e_1 at F1 and electrode e_5 at F2 and rates their confidence in the vowel perceived, that rating reflects their combined certainty about F1 identity, F2 identity, and their conjunction. We attempted to address this by computing separate F1 and F2 bell curves and the reliance ratio, but the initial confidence score is not formant-selective. This means that the electrode-level factor analyses cannot fully isolate the contribution of F1 factors from F2 factors in the confidence rating, even though the bell curves are constructed to do so post hoc. The logical next step to disentangle the specific contribution of F1 and F2 would be to analyse the ratio between F1 and F2 responses for each electrode and compare this to the ratio of the frequency-to-place factors studied.

What the Bell Analysis Saved. The electrode confidence bell was introduced relatively late in the analysis pipeline, and its impact on interpretability was transformative. By distributing confidence across the full acoustic frequency axis rather than restricting it to discrete F1 and F2 estimates, the bell makes the analysis continuous, removes the dependence on formant frequency extraction, and enables direct comparison with electrode-level factor values that are themselves expressed in frequency units. The bell framework also resolves a persistent problem in the earlier analysis: the sparsity of responses at specific F1/F2 combinations, which created gaps in the perceptual map that made summary statistics unreliable. The KDE smoothing fills these gaps by distributing confidence across neighbouring frequency positions, yielding a complete and continuous profile even where data are sparse. Perhaps most importantly, the bell framework enables a new approach to stimulus design. Rather than constructing stimuli from the participant's individual mapping, a standardised stimulus set could be designed that systematically samples across the frequency axis in log-frequency steps, independent of electrode assignment. The bell would then align responses from different participants onto the same frequency grid, enabling cross-participant comparisons without normalisation. This is the principle behind the online testing platform currently in development.

Ecological Validity. The task presents isolated two-formant stimuli in silence, without lexical context, prosody, or co-articulation. CI users in everyday life identify vowels embedded in continuous speech with extensive top-down support. The stimuli used here may therefore probe a more fundamental and decontextualised level of spectral processing than what drives everyday communication. This is both a limitation and a feature: it is precisely this decontextualised nature that makes the task sensitive to electrode-level factors, which would be masked by top-down compensation in natural speech. The tradeoff is that results may not generalise directly to speech perception scores in quiet or noise, and future work should directly test the correlation between bell-derived metrics and standard audiological outcomes.

6.3.2 The chord Task

Practical Advantages. Sessions were shorter and better appreciated by participants. The paired-comparison format is inherently more sensitive than closed-set identification: by directly contrasting two stimuli, participants provide relative judgements that are less susceptible to response bias and more stable across repeated testing. The electrode-level Bradley-Terry preference scores derived from pairwise comparisons are also more robust than per-electrode confidence aggregates, being estimated from the full set of comparisons involving each electrode rather than a sparse subset of tested positions. The chord task is therefore better positioned as a clinical screening instrument than the vowel task, and its session duration (between 5 and 20 minutes, depending on the number of electrodes compared) is more compatible with a standard audiology appointment.

Sensitivity to Distortion. The task was explicitly sensitive to FTPM-induced interval distortion from the very first session in the longitudinal case study (Chapter 5), and FTPM was still a significant predictors of chord quality in Post-LCIs (Chapter 4). This sensitivity to spectral distortion - which is central to the subjective experience of music quality in CI users (Caldwell et al., 2017a) - makes the chord paradigm a natural candidate for music rehabilitation assessment as well as electrode-level monitoring. The difficulty CI users experience with chord and harmony perception is well-established (Zimmer et al., 2019; Gilbert et al., 2025), but the present work is the first to attribute part of this difficulty to individual electrode-level FTPM profiles rather than treating it as a global device limitation. Recent work using musical pitch interval comparisons to assess CI frequency-to-place maps (Lewis et al., 2025) further confirms that interval distortion - the mechanism underlying our chord distortion analysis - is the relevant perceptual dimension for music quality in CI users.

The Root-Position Confound. The structural difference between root-position and inverted chords introduces an acoustic confound that cannot be fully eliminated: root-position chords concentrate energy in the low-frequency region and have a wider bottom interval, both of which are perceptually salient independent of FTPM. This was partly addressed through the interaction model framework, but the root-position preference was large enough to mask distortion sensitivity in the majority of trials. Future versions of the task should explore whether varying FTPM-induced distortion continuously within a fixed root-position structure - removing inversion entirely - provides greater sensitivity without the need for an interaction model.

6.4 Clinical Implications and Test Integration

6.4.1 When to Use these Tests

The longitudinal data from Chapter 5 suggest that the first six weeks post-activation represent the most dynamic and diagnostically informative window. The most dramatic perceptual changes - in chord distortion sensitivity, root-position preference and F2 delta resolution - all occurred between Days 6 and 41. A testing protocol designed for early

clinical monitoring should therefore prioritise this window, with assessments at approximately Day 7, Day 21, and Day 42 post-activation, rather than waiting for the standard three-month review. By Day 134, most of the dynamic changes have stabilised and the patient is approaching an asymptotic perceptual configuration that is better evaluated at the group level than individually.

The tasks are also informative in the context of MAP changes. When a clinical MAP adjustment is made - for example, deactivating apical electrodes or changing the frequency allocation table - the bell framework provides a direct before-and-after measure of how the confidence distribution shifts in response. A MAP change that reduces the Wasserstein distance from NH and normalises the F2 delta could be quantitatively validated using the tools developed here.

6.4.2 How to Integrate these Tests Clinically

Both tasks were designed with clinical accessibility in mind, but modifications are needed before routine deployment. The vowel task, as currently implemented, generates a different number of trials per participant depending on device type and number of active electrodes: users with 12-electrode devices (approximately 33 trials per session) have substantially shorter sessions than those with 22-electrode devices. The bell analysis framework, which operates on a continuous KDE rather than discrete electrode counts, might remove this dependency: it is possible to design a fixed stimulus set with a constant number of trials regardless of electrode count, enabling standardised session durations across devices.

We are currently developing an online testing platform that implements the bell framework with a standardised stimulus set, accessible to any CI user without the need to enter an individual mapping. The platform would deliver the stimuli, collect responses, compute the bell and Wasserstein metrics in real time. This would transform the research paradigm into a scalable clinical tool, deployable at outpatient appointments without audiologist involvement in the data collection phase.

The chord task, with its shorter session duration and higher appreciation among participants, is a strong candidate for a screening instrument. A 5-minute chord preference session - feasible within a standard clinical appointment - could identify users with significant FTPM-driven perceptual distortion, flagging them for more detailed vowel assessment or audiological review.

Benchmarking against standard audiological outcomes. A critical next step - prioritised for post-submission work - is to correlate the electrode-level metrics developed here (Wasserstein distance, formant centroid shift, reliance ratio, FIF profiles) with standard clinical speech perception scores obtained from the same participants. If bell-derived displacement from NH predicts word or sentence recognition scores in quiet and noise, this would validate the paradigm as a clinically meaningful tool rather than a research-only measure.

6.4.3 Remapping as a Direct Clinical Target

The formant shift analysis provides, for the first time, a quantitative and individually derived target for frequency remapping. If a Post-LCI F2 centroid is displaced relative to NH - meaning they perceive F2 at a different frequency than the current allocation places it - then shifting the frequency allocation table to align F2 electrodes with the individual's actual perceptual centroid could, in principle, bring the perceptual representation closer to NH. This is a more principled approach to remapping than the current practice of adjusting maps empirically based on subjective feedback, and it could be derived directly from a 5 to 10 minute vowel session using the bell framework.

The Pre-ECI versus Post-LCI dissociation in reliance ratio (Chapter 3) also has clinical implications that are not yet reflected in standard programming practice. Pre-ECI showed F1 dominance - a pattern that might reflect efficient use of apical electrodes - whilst Post-LCI matched the NH balance. Remapping strategies that target F2 reliability are likely to benefit Post-LCIs more than Pre-ECIs, for whom the F1 electrode configuration already supports a well-adapted perceptual strategy. Clinically differentiating between these groups at the MAP level - rather than applying a uniform frequency allocation strategy - is a practical recommendation supported by the present findings.

6.5 Future Directions

Larger longitudinal cohort. The most direct extension of this thesis is a prospective longitudinal study of 4 newly implanted CI users, assessed at the same four time points as Chapter 5 with both the vowel and chord tasks. With adequate sample size, the electrode-level factor effects on the adaptation trajectory - currently sub-threshold in a single participant - could be formally tested, and the generalisability of the two-phase model could be assessed. Particular focus should be placed on the first six weeks, where the present data suggest the most dynamic changes occur.

Ideal mapping from the bell framework. The formant shift analysis provides an empirical foundation for deriving individually optimised frequency allocation maps. The concept of an ideal map - one that minimises the displacement of each user's confidence distribution from the NH reference - has been explored theoretically but rarely derived from individual perceptual data. The bell framework, applied in a short pre-fitting session, could yield a participant-specific Wasserstein-minimising map that is empirically grounded in the individual's actual perceptual responses rather than inferred from population-level assumptions.

Pre-surgical prediction from PECAP. The consistent predictive value of NR across studies raises the question of whether pre-surgical PECAP assessment - now increasingly feasible with intraoperative recording - could predict the early perceptual trajectory. Users implanted with better baseline neural health across the array may show faster resolution of F2 delta deficits, faster preference formation, and ultimately better long-term outcomes, consistent with the literature showing that pre-implant cochlear-neural health accounts for a large proportion of variance in post-implant speech perception (Röbler et al., 2022).

If confirmed, this would provide a physiological basis for targeted post-operative rehabilitation - focusing training and monitoring resources on users with poor baseline NR who may require longer or more intensive support.

New paradigms. The limitations highlighted for both tests might be alleviated with some targeted interventions. For the vowel task, the priority should be to transform the single-sound presentation into a paired-comparison task in which the user estimates which of two sounds best corresponds to the presented vowel. Such a strategy raises concerns about test duration, as it would focus on one vowel per trial; however, responses would be expected to be more reliable and better suited to clinical use. For the chord task, modifications should aim to remove the bias related to root-position preference introduced by chord inversion. Even if such bias is partially alleviated through the construction of FIF profiles, other types of chord stimuli should be considered in future work to eliminate that confound entirely whilst preserving the electrode-level analysis.

6.6 Conclusion

This thesis set out to characterise how frequency-to-place factors - FTPM, CS, and NR - shape auditory perception in CI users, at the level of individual electrodes, across multiple perceptual domains, and across the time course of adaptation. Four studies were conducted, each targeting a different perceptual dimension and a different population.

The central scientific contribution of the FTPM and sound quality study is that both the degree and the inter-electrode variability of FTPM affect sound quality independently, at least in an acute situation without adaptation. It further establishes that electrode-level analysis of FTPM is necessary to capture its perceptual consequences, and that the confound between signal content and spectral distortion precludes the use of naturalistic stimuli for this purpose.

The central scientific contribution of the vowel paradigm is that none of the three factors - FTPM, CS, or NR - produced a significant electrode-level deviation from the NH reference in experienced users, a null result whose scope is specific: it does not explain why CI users are globally displaced from NH, but it establishes that individual electrode quality differences do not create additional local deviations on top of that global shift. In the newly implanted case study, early factor-driven F2 deficits were detectable at Day 6 across all three factors, with progressive attenuation across sessions - providing a longitudinal precursor to the experienced-user null, and consistent with adaptation being the mechanism by which local factor-driven deficits resolve over time.

The central scientific contribution of the chord paradigm is the first direct evidence that FTPM-induced spectral distortion affects music quality perception in Post-LCIs at the electrode level, with effects that depend on implantation history and emerge progressively over the adaptation window. CS was marginally significant in Post-LCIs, and NR was not significant in the cross-sectional analysis. The longitudinal data revealed an NR-Early / FTPM-Late dissociation: high-NR electrodes scaffold early preference formation, and high-mismatch apical electrodes attract increasing preference as electrode-based representation consolidates, though with a non-monotonic trajectory across sessions.

The central clinical contribution is the demonstration that both tools, applied early

in the post-activation window, can detect the dynamic process of adaptation in real time, identify which electrodes and factors are driving or limiting it, and ultimately provide a quantitative target for frequency remapping decisions. The online testing platform in development translates these research findings into a scalable clinical workflow.

Much remains to be done. The single longitudinal case study, the small electrode-level factor subsamples, and the task limitations described in this chapter all constrain the strength of the conclusions that can be drawn. But the framework is in place, the tools are validated, and the findings across four studies converge on a coherent account of how frequency-to-place factors constrain perception, how the auditory system adapts to them, and how that adaptation can be monitored and guided.

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